Diet and Cancer 1

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Dietary Omega-3 Polyunsaturated Fatty Acids and Cancer

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Introduction: Omega-3 PUFAs, Why Do We Speak About Them?

Gabriella Calviello and Simona Serini

Abstract In spite of the fact that few dietary components are so widely recognized as able to improve human health such as ω-3 polyunsaturated fatty acids (PUFAs), so that their sector in the nutritional market has been increasingly growing worldwide, many unresolved questions still remain about them. In particular, there is urgent need for better understanding their possible role as anti-neoplastic agents. First of all, the chemical structure, the intracellular metabolism, and the dietary sources and bioavailability of these dietary fatty acids will be described in this introductive chapter to make easier the comprehension of the remaining parts of the book. Afterward, a brief outline of ω-3 PUFA reported benefits in different fields of human health will be provided. In this introductive part we will tackle also the problem of the discrepancies occurring between the results of most experimental studies on animals and cultured cells, which, almost univocally, suggest the beneficial anti-neoplastic effects of these fatty acids, and the outcome of several of the epidemiological observational studies, which, conversely, shows a scarce or null positive association between high intake of fish or fish oil at high content in ω-3 PUFAs and prevention of different kinds of cancer. Finally, a brief outline of the organization of the present book will be provided.

Keywords ω-3 PUFA · Metabolism · Dietary sources · Bioavailability · Anti-neoplastic effects

Introduction

There are few dietary components that are so widely recognized as able to improve human health like ω-3 polyunsaturated fatty acids (PUFAs), and whose sector in the nutritional market has been increasingly growing worldwide. However, despite

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their large fame and easy commercial availability, many unresolved questions still remain about them. For instance, may their intake represent an actual preventive strategy against cancer? Is their ingestion safe for healthy people? Do they have the potential to act as chemotherapic agents in combination or not with other conventional anti-cancer therapies? Which kind of cancer patients could actually benefit from treatments with ω-3 PUFAs? At what stages of cancer growth and progression their intake could be considered really fruitful? The present book has been thought in order to collect the findings and opinions of several influential scientists in the fields and thus to help answering these and other questions. The beneficial effect of ω-3 PUFAs on hormone-related cancers (breast and prostate cancer) and colon cancer has received most of the attention, and for this reason they will be treated in separate chapters. ω-3 PUFAs, however, appear to exert their positive influence also toward a number of other kinds of cancers, including leukemias/lymphomas, melanoma, neuroblastoma, liver, and lung cancer. However, before treating these specific topics, a general introduction will be furnished in this chapter with the aim to clarify some basic aspects regarding ω-3 PUFAs, such as their chemical structure and metabolism, their sources and bioavailability, and the other diseases whose incidence or progression can be favorably affected by these fatty acids.

Chemistry and Metabolism of ω-3 PUFAs

Fatty acids (FA) are constituted by carbon chains of various lengths in which carbons are bound by single or double bonds. A methyl group is present at one end (the "*n*" or "ω" end) and a carboxyl group at the other end. The lack or presence of double bonds between the carbons defines the two classes of saturated and unsaturated fatty acids (UFA). Depending on the presence of one or more double bonds in the carbon chain, UFAs are divided into monounsaturated fatty acids (MUFAs) and PUFAs. The prevalent PUFAs found in nature belong to the ω-3 and ω-6 classes of PUFAs, whose first double bond is, respectively, placed either three carbons (in the ω-3 or *n*–3 position) or six carbons (in the ω-6 or *n*–6 position) from the methyl end of the carbon chain. The three main dietary ω-3 PUFAs are α-linolenic acid (C18:3 *n*–3, all-*cis*-9,12,15-octadecatrienoic acid, ALA), eicosapentaenoic acid (C20:5 *n*–3, all-*cis*-eicosa-5,8,11,14,17-pentaenoic acid, EPA), and docosahexaenoic acid (C22:6 *n*–3, all-*cis*-docosa-4,7,10,13,16,19-hexaenoic acid, DHA) (Fig. 1). ALA, together with linoleic acid (LA, 18:2 n:6), is considered the "essential fatty acid" (EFA), namely the diet must necessarily provide them, since the desaturase needed to place the double bond in position ω-3 or ω-6 in the PUFAs is lacking in mammals. On the other hand, this desaturase is present in vegetables, which, therefore, represent the main dietary source of ALA and LA for mammals. Many commonly used vegetables oils are enriched in LA (corn, safflower, and soybean oils), whereas canola oil, ground flaxseed, and walnuts contain high levels of ALA. PUFAs belonging to one of the two different classes (ω-3 or ω-6) are not interconvertible into PUFAs of the other class. In our tissues, ALA and LA can be metabolized by the sequential action of several desaturases and elongases to produce EPA and arachidonic acid (AA, 20:4 *n*–6), respectively. Further

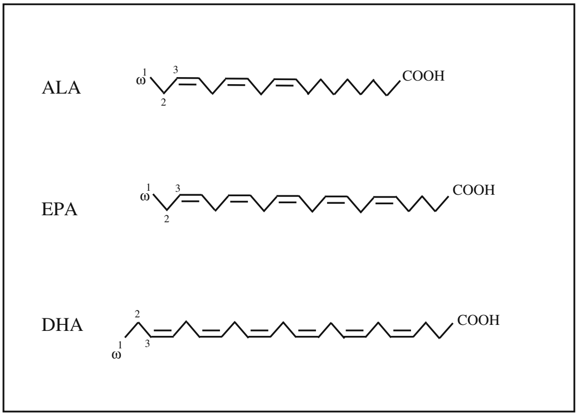

Fig. 1 Chemical structures of the major dietary ω-3 PUFAs. ALA: α-linolenic acid; EPA: eicosapentaenoic acid; DHA: docosahexaenoic acid

desaturase, elongase, and partial peroxisomal beta-oxidation steps are needed [1, 2] to generate ω-3 PUFAs longer than EPA, such as DHA (for a detailed description of all the reactions, see Fig. 7.1). It has been shown that the ALA affinity for Δ6-desaturase is higher than that of LA. However, if the Western diet is adopted, in which LA is present in higher amounts than ALA, LA becomes the EFA preferentially desaturated [3]. As a result, the endogenous production of EPA and DHA by the precursor ALA is not very efficient in humans, and the efficiency of the conversion of ALA to EPA or DHA becomes particularly low in preterm infants and may also decline with old age [4]. Consequently, the main sources of EPA and DHA are animal tissues deriving from poultry, fat fish, and seafood, which contain high levels of these fatty acids. However, the current dietary supplies of the majority of Western countries are able to furnish very low amounts of ω-3 PUFAs. It has been calculated that the dietary ratio of ω-6 to ω-3 PUFAs ranges from 15/1 to 16.7/1 in Western diets and, therefore, is much lower than the ratio of 1/1 present in wild animal's and probably also in our ancestor's diets [5]. In the short periods of time over the past 100–150 years an absolute and relative change of ω-3/ω-6 PUFA ratio in Western diets has occurred which could help to explain the increasing incidence of some kinds of human diseases [5]. Both ω-3 PUFAs and ω-6 PUFAs have the potential to influence gene expression and the unchanged dietary ratio between ω-6 PUFAs and ω-3 PUFAs of 1/1 over millions of years could have substantially influenced genetic modifications and human evolution. Now, a substantial decrease in EPA and DHA incorporated in cellular membranes and the concomitant increase in AA may have produced dangerous consequences for human health.

Intracellular Metabolism of ω-3 PUFAs and Competition with Arachidonic Acid

At this point, the description of the oxidative metabolism of AA and EPA, and, in particular, of the influence of ω-3 PUFAs on the oxidative metabolism of AA, seems particularly useful to understand the possible benefits deriving from the substitution of AA for these fatty acids in membranes. Following a series of cellular stimulations, AA is released from membranes by the action of phospholipase A_2 (PLA_2) and metabolized by cyclooxygenase (COX) and lipoxygenase (LOX) enzymes into the oxygenated metabolites prostaglandins (PG), thromboxanes (TX), leukotrienes (LT), and hydro fatty acids, collectively known as eicosanoids [6] (Fig. 2). The AA-derived eicosanoids are biologically highly efficient, acting at very small concentrations. They have the potential to influence key events of physiological and pathological processes, including proliferation, survival, and inflammation [7]. The formation of the AA products is normally controlled, but in some pathological conditions such as cancer excessive amounts are produced [8]. After ingestion of fish or fish oil, dietary EPA and DHA may induce a decreased eicosanoid production by AA and reduce all the molecular responses related to the oxidative metabolism of AA in different ways including (a) the partial replacement of AA in cell membranes, since they compete with it for acylation in position sn-2 of membrane phospholipids; (b) the direct competition of EPA and AA for

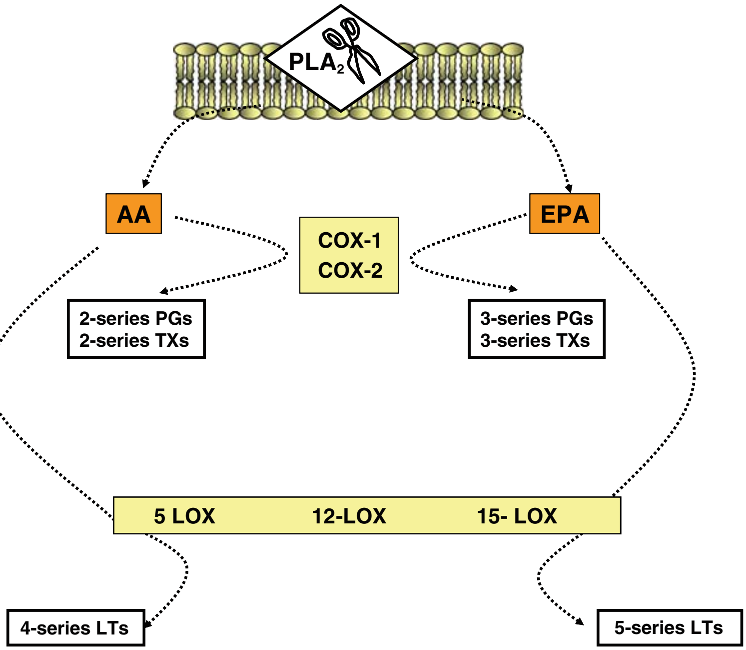

Fig. 2 Competition between arachidonic acid (AA) and eicosapentaenoic acid (EPA) for cyclooxygenases (COX) and lipoxygenases (LOX). Phospholipase A_2 (PLA_2) catalyzes the hydrolysis of membrane phospholipids to release free AA and EPA. Afterward, free AA and EPA are converted by the same enzyme COX and LOX to their oxygenated metabolites prostaglandins (PGs), tromboxanes (TXs), and leukotrienes (LTs), collectively named eicosanoids. AA- and EPA-derived eicosanoids possess different biological activities. Plenty of works have shown pro-inflammatory and pro-carcinogenic activities for AA-derived eicosanoids (see the text for further details)

COX and 5-LOX, and production of EPA-derived eicosanoids (3-series TX, 3-series PG, and 5-series LT), which show lower biological activity than AA-derived eicosanoids [9] (Fig. 2); (c) the EPA- and DHA-induced reduction of COX-2, the inducible form of COX, which is expressed mainly during inflammation and tumor growth [10, 11].

Moreover, recently it was found that both EPA and DHA may be metabolized to previously unknown potent bioactive (in the nM range) eicosanoid and docosanoid products with anti-inflammatory and protective properties [12]. They have been comprised in the classes of resolvins, docosatriens, and protectins. Resolvins derived from EPA and DHA are named resolvins E and D [13]. DHA is the parent compound for docosatrienes, containing conjugated triene structures [14]. "Neuroprotectins" indicate docosatrienes and D-series resolvins that have been shown to exert neuroprotective and anti-inflammatory actions [14]. Aspirin can trigger in vivo the synthesis of a further highly active series of these compounds (17 R–D-series resolvins and docosatrienes) [14] (Fig. 3).

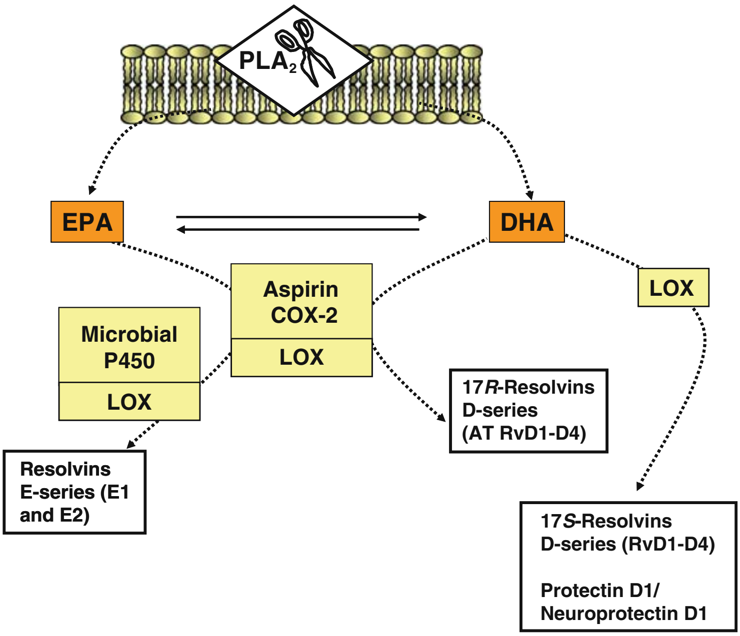

Fig. 3 Formation of novel discovered potent bioactive eicosanoids and docosanoids from eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA). The figure reports the recent acquisitions according to which EPA and DHA may be metabolized to previously unknown novel compounds (named resolvins and protectins) with high potency as anti-inflammatory and pro-resolving agents [see the text and Ref. [13] for further details]. PLA$_2$: phospholipase A$_2$; COX: cyclooxygenase; LOX: lipoxygenase

Dietary Sources and Bioavailability of ω-3 PUFAs

Fatty fish is a good natural source for the long-chain ω-3 PUFAs EPA and DHA. Fish oil supplements are other sources artificially added to the diet and include fish oil capsules containing high levels of EPA and/or DHA and food enriched with fish oils. Recently, given the possible contamination of marine fish, effort has been made to obtain purified ω-3 PUFAs from a source different from fish, and algal oils have been obtained from cultured microalgae, which could represent a quite safe and convenient source of non-fish-derived ω-3 PUFAs. Recently also walnuts have been considered a good source of ω-3 fatty acids [15], being rich in ALA. Even though the conversion of ALA into longer ω-3 PUFAs (see above) is generally considered low, it has been shown that a moderate consumption of walnuts (4 walnuts/day for 3 weeks) markedly increases the blood levels of ALA and of its metabolic derivative, EPA. Probably, as suggested by the authors, plant ALA, at the high levels found in appropriate food items, such as walnuts, may favorably affect the ω-3 long-chain PUFA status.

Bioavailability of ω-3 PUFAs is generally evaluated measuring their amount or concentrations in total lipids or lipid fractions (free fatty acids, triglycerides, phospholipids, cholesterol esters) in plasma, serum, lymph, platelets, and red blood cells as well as in the tissues under study. For instance, if we consider plasma total lipids, high amount of ω-3 PUFAs may be incorporated into them. For instance, starting from a basal serum total lipid level of about 20 μM EPA and 80 μM DHA in humans [16], a dietary fish oil supplementation (3.0 g/day EPA + DHA) or daily servings of salmon (1.2 g/day EPA + DHA) [17, 18] may allow an enrichment of total serum lipids ranging from 100 to 130% for EPA and from 25 to 45% for DHA. Even higher increases have been reported for total phospholipids after dietary supplementation with EPA + DHA ethyl esters (1.9 and 0.9 g/day, respectively) (250% for EPA and 40% for DHA) [17, 18]. However, a not completely clarified aspect of PUFA metabolism is what is the best ω-3 PUFA source to obtain an optimal absorption. It was recently shown that, irrespective of the source of ω-3 PUFAs present in formula supplements for infants (either egg PL or low EPA fish oil and fungal TG), the concentrations of EPA and DHA achieved in the different infant lipid plasma fractions (total PL, TG, and CE) were very similar [19]. Accordingly, the intake of equivalent doses of EPA and DHA given either as a mixture of EPA and DHA ethyl esters or as a natural fish oil (containing mainly ω-3 PUFAs esterified to TG) led to similar serum levels of EPA and DHA in adults [20]. However, recently, it was found that high concentrations of ω-3 PUFAs in plasma were achieved better if the dietary source of these fatty acids was fish (containing mainly ω-3 PUFAs esterified in glycerol lipids), rather than capsules containing ω-3 PUFA ethyl esters [18–22]. Recently it was also reported that algal-oil DHA capsules and cooked salmon were bioequivalent in providing DHA to plasma and red blood cells [23]. Even though levels of EPA and DHA in serum or plasma lipids may give important information regarding the bioavailability of these fatty acids, recently the enrichment of erythrocyte membranes (the so called ω-3 index) has been considered a better biomarker for ω-3 PUFAs [24], at least to establish the risk of coronary heart disease mortality, especially sudden cardiac death. Also in other pathologic fields this index may be useful, mirroring the incorporation of these FA in cell membranes of other body districts, where ω-3 PUFAs may actually produce their beneficial effects. The use of this index appears interesting, since it is just the membrane enrichment with ω-3 PUFAs which is often considered crucial to explain their beneficial effects. Moreover, according to many authors, the membrane ω-3/ω-6 PUFA ratio has been considered even a better index to explain their beneficial effects [25].

Beneficial Effects of ω-3 PUFAs on Human Health

Currently, the beneficial action of ω-3 polyunsaturated fatty acids (PUFAs) in cancer prevention, therapy, and cachexia is supported by plenty of evidence that will be examined in the following chapters with detail [26–30]. However, the first observation of a possible beneficial healthy effect of ω-3 PUFAs, dating back to about four decades ago, was the relationship existing between the low mortality from cardiovascular diseases of Greenland Inuit populations and their high consumption of fish [31]. Nowadays, the role of ω-3 PUFAs as nutritional factors with the potential to prevent the incidence as well as to lower the progression of different chronic pathologic conditions has been well established. Most of the results have been obtained in the cardiovascular field, and now it is well recognized that ω-3 PUFAs beneficially improve dyslipidemias, especially lowering plasma levels of triglycerides [32]. Moreover, it has been proven that they slightly decrease blood pressure [33], inhibit the formation of atherosclerotic plaque [34], and reduce the risk of sudden death [35], cardiac arrhythmias [36], and stroke [37] in individuals with established cardiac pathologies. Furthermore, they can be useful in preventing the pathological vascular complications of diabetes [38]. On this basis, many nutritionist and cardiologist agencies worldwide agree in recommending at least two or three fish portions/week for the primary and secondary prevention of cardiovascular diseases and supplementations of ω-3 PUFAs as fish oil extracts [39–41]. Consequently, the prescription of fish oil capsules has currently become common in clinical cardiology practice. The increased sales of drinks and food products fortified with ω-3 PUFAs worldwide also demonstrate the extreme popularity of the notion that ω-3 PUFAs exert various health effects. Recently, their dietary intake has been also recommended during pregnancy and lactation [4, 42] since it has been established that ω-3 PUFAs exert crucial effects on growth and neurological development of fetuses and newborn infants [43, 44]. Plenty of data have been published on the subject, and it has been shown that maternal plasma phospolipid (PL) concentration of PUFAs increases during pregnancy, probably mobilized from maternal stores [45]. Especially DHA increases in plasma PL, and this is related to the fact that fetus needs PUFAs, especially DHA, for the normal development of its brain and retina [46, 47]. It has been shown that during pregnancy women may become increasingly deficient in DHA [45], and probably the maternal capacity to meet the high fetal requirement for DHA [48] may work at its limits or even be inadequate. Moreover, observational and interventional studies have clarified the positive influence of ω-3 PUFAs on gestation length, birth weight, and risk for early premature birth [49]. Consensus recommendations and practice guidelines for pregnancy supported by different health agencies have been recently reviewed [50]. On the basis of all the recommendations published, the aim to be achieved by pregnant and lactating women should be an average daily intake of at least 200 mg DHA. In spite of that, as demonstrated by a study recently carried out [51], there is not enough awareness of the importance of ω-3 PUFA consumption during pregnancy among pregnant women, and the limited knowledge obtained by women derived mostly from popular books and magazines. Moreover, the importance of an increased dietary intake of ω-3 PUFAs has been recently recognized for the prevention of neurodegenerative pathologies [52, 53]. Epidemiological studies have indicated the possibility that dietary EPA and DHA may modify the risk and progression of Alzheimer's disease (AD). In particular, longitudinal prospective studies have shown the inverse relationship existing between fish intake and AD dementia [54–57] and cross-sectional analyses have linked low levels of DHA in plasma lipids or phospholipid DHA levels and a low ω-3 PUFA/ω-6 PUFA ratio in the erythrocyte membranes with cognitive decline, dementia, and AD in particular [58–61]. Moreover, a series of experimental studies on mouse models of AD have investigated the role of ω-3 PUFAs in the development of AD. These studies demonstrated that pre-administration of DHA to rats infused with the amyloid peptide Aβ1-40, whose formation in brain is considered crucial in the pathogenesis of AD, had profoundly beneficial effects in decreasing the decline of learning ability [62]. Also experiments with different transgenic rat models of AD showed univocally that the dietary supplementation with DHA decreased the levels of Aβ [63–66] improving the animal cognitive functions. Similarly, their benefits have been demonstrated in immunity and inflammatory disorders [67, 68]. They have shown to decrease colonic damage and inflammation, weight loss, and mortality in animal models of colitis [69]; to reduce joint inflammation; and to improve clinical symptoms in subjects affected by rheumatic diseases, in particular, rheumatoid arthritis (RA) [70]. They are thought to exert their action modifying the inflammatory lipid mediator profile, leukocyte chemotaxis, and inflammatory cytokine production [69].

Studies on ω-3 PUFAs and Cancer: Discrepancies Between the results of In Vitro and In Vivo Experimental Studies and Human Observational Studies

Very interesting results have been obtained with ω-3 PUFAs also in the oncology field, and they will represent the subject of the following chapters. However, on the basis of the results analyzed in this book, it will appear clear that many inconsistencies exist among the epidemiological observational studies examining the risk of different kinds of cancer in human populations ingesting variable amounts of food containing ω-3 PUFAs. This seems quite intriguing, considering that, on the contrary, the results of all the experimental studies clearly and univocally indicate the powerful anti-tumor effectiveness of ω-3 PUFAs. A number of reasons may probably concur to create this discrepancy. In Chapters 2 and 4, the limitations suffered by the epidemiological studies in defining the actual level of daily dietary intake of ω-3 PUFAs on the basis of the consumption of fish or other dietary sources of these fatty acids will be examined in detail. It should be underlined, however, that in many studies also the ω-3 PUFA concentration in red blood cells, serum, or adipose tissue is often used as an objective biomarker of fatty acid intake. To try to explain these discrepancies it may be also useful to consider the interesting recently prospected possibility that the anti-tumoral effect of these fatty acids among the human population may vary in dependence on the possession of individual genetic features, and that high intakes of ω-3 PUFAs would be associated with a lower risk of neoplasia only among those individuals with genetic variants associated with a particular type of cancer [71]. This possibility will be treated in Chapter 5 as far as prostate cancer is concerned.

Controlled and definite concentrations of ω-3 PUFAs are instead administered to cultured cells and animals in the experimental studies [26]. It is also true that, in a very few studies, the concentrations used in vitro are higher than those achievable in plasma of human populations, even of high fish consumers.

Moreover, it should not be underestimated that a long-term dietary intake of fishes may supply concomitantly both ω-3 PUFAs and carcinogenic compounds, which, as well known, often contaminate fish tissues. It has become clear that not only wild fish but also farmed fatty fish which are commonly bred and eaten, such as salmon, may be highly contaminated by carcinogens such as pesticides [72]. For instance, organochlorine pesticides may accumulate easily in fatty fishes and exert carcinogenic effects, particularly enhancing the risk of hormone-dependent forms of cancers [29]. Thus, the concomitant intake may complicate the interpretation of the epidemiological studies, especially those regarding breast and prostate cancers, which may be induced by the organochlorine pesticides. The lack of agreement among epidemiological observational results represents a big drawback and may help to explain why just a few number of clinical intervention trials with ω-3 PUFAs have been so far performed in patients at risk for cancer, especially if compared to those conducted in cardiovascular patients. In view of the possible carcinogen contamination of fish tissues, highly purified PUFAs, oils rich in ω-3 PUFAs or fish from uncontaminated sources should be used for intervention trials and prevention studies. To this aim, alternative and safer, but still natural, ω-3 PUFA sources, such as cultured microalgae have been proposed [73, 74]. Recently, we indicated an artificial lake, whose waters were subject to constant purification and almost free of pollutants, as a model of basin which could furnish uncontaminated fish, particularly indicated for pregnant women and infants after weaning [75].

Effect of ω-3 PUFAs in Cancer: Organization of the Present Book

The present book was designed to cover in an exhaustive way all the main aspects concerning the possible effects of ω-3 PUFAs against cancer. To this aim, the first chapter will analyze the mechanisms of ω-3 PUFA anti-tumoral action. Chapters 2 and 3 will treat the effects of ω-3 PUFAs against some of the most frequent cancers among Western population, namely colon cancer and hormone-related cancers (breast and prostate cancer), all of them proven to be sensitive to dietary chemoprevention. Separate sections inside Chapters 2 and 3 will cover experimental studies, including those performed on animal, cell culture models, human interventional studies, and human epidemiological observational studies. Chapter 4 will provide the available information regarding the effects of ω-3 PUFAs on other kinds of cancers. Afterward, a subject of great interest at the moment will be treated (Chapter 5), namely the possible use of ω-3 PUFAs in combination with conventional anti-cancer agents. Finally, the anticachetic potential of these dietary compounds will be analyzed.

References

1. Sprecher H. New advances in fatty-acid biosynthesis. *Nutrition* 1996; 12(1 Suppl):S5–S7.
2. Sprecher H, Luthria DL, Mohammed BS, Baykousheva SP. Reevaluation of the pathways for the biosynthesis of polyunsaturated fatty acids. *J Lipid Res* 1995; 36(12):2471–7.
3. Salem N Jr, Wegher B, Mena P, Uauy R. Arachidonic and docosahexaenoic acids are biosynthesized from their 18-carbon precursors in human infants. *Proc Natl Acad Sci USA* 1996; 93(1):49–54.
4. McCann JC, Ames BN. Is docosahexaenoic acid, an n-3 long-chain polyunsaturated fatty acid, required for development of normal brain function? An overview of evidence from cognitive and behavioral tests in humans and animals. *Am J Clin Nutr* 2005; 82(2):281–95.
5. Simopoulos AP. Evolutionary aspects of diet, the omega-6/omega-3 ratio and genetic variation: nutritional implications for chronic diseases. *Biomed Pharmacother* 2006; 60(9):502–7.
6. Samuelsson B, Dahlen SE, Lindgren JA, Rouzer CA, Serhan CN. Leukotrienes and lipoxins: Structures, biosynthesis, and biological effects. *Science* 1987; 237(4819):1171–6.
7. Funk CD. Prostaglandins and leukotrienes: advances in eicosanoid biology. *Science* 2001; 294(5548):1871–5.
8. Karmali RA. Eicosanoids in neoplasia. *Prev Med* 1987; 16(4) 493–502.
9. Lands WE. Biochemistry and physiology of n-3 fatty acids. *FASEB J* 1992; 6(8):2530–6.
10. Swamy MV, Cooma I, Patlolla JM, Simi B, Reddy BS, Rao CV. Modulation of cyclooxygenase-2 activities by the combined action of celecoxib and decosahexaenoic acid: Novel strategies for colon cancer prevention and treatment. *Mol Cancer Ther* 2004; 3(2):215–21.
11. Tapiero H, Ba GN, Couvreur P, Tew KD. Polyunsaturated fatty acids (PUFA) and eicosanoids in human health and pathologies. *Biomed Pharmacother* 2002; 56(5):215–22.
12. Serhan CN. Novel eicosanoid and docosanoid mediators: Resolvins, docosatrienes, and neuroprotectins. *Curr Opin Clin Nutr Metab Care* 2005; 8(2): 115–21.
13. Serhan CN, Chiang N. Endogenous pro-resolving and anti-inflammatory lipid mediators: a new pharmacologic genus. *Br J Pharmacol* 2008; 153(Suppl 1):S200–15.
14. Serhan CN, Arita M, Hong S, Gotlinger K. Resolvins, docosatrienes, and neuroprotectins, novel omega-3-derived mediators, and their endogenous aspirin-triggered epimers. *Lipids* 2004; 39(11):1125–32.

15. Marangoni F, Colombo C, Martiello A, Poli A, Paoletti R, Galli C. Levels of the n-3 fatty acid eicosapentaenoic acid in addition to those of alpha linolenic acid are significantly raised in blood lipids by the intake of four walnuts a day in humans. *Nutr Metab Cardiovasc Dis* 2007; 17(6):457–61.
16. Harper CR, Edwards MJ, De Filippis AP, Jacobson TA. Flaxseed oil increases the plasma concentrations of cardioprotective (n-3) fatty acids in humans. *J Nutr* 2006; 136(1); 83–7.
17. Cao J, Schwichtenberg KA, Hanson NQ, Tsai MY. Incorporation and clearance of omega-3 fatty acids in erythrocyte membranes and plasma phospholipids. *Clin Chem* 2006; 52(12):2265–72.
18. Elvevoll EO, Barstad H, Breimo ES, Brox J, Eilertsen KE, Lund T, et al. Enhanced incorporation of n-3 fatty acids from fish compared with fish oils. *Lipids* 2006; 41(12):1109–14.
19. Sala-Vila A, Castellote AI, Campoy C, Rivero M, Rodriguez-Palmero M, López-Sabater MC. The source of long-chain PUFA in formula supplements does not affect the fatty acid composition of plasma lipids in full-term infants. *J Nutr* 2004; 134(4):868–73.
20. Krokan HE, Bjerve KS, Mørk E. The enteral bioavailability of eicosapentaenoic acid and docosahexaenoic acid is as good from ethyl esters as from glyceryl esters in spite of lower hydrolytic rates by pancreatic lipase in vitro. *Biochim Biophys Acta* 1993; 1168(1):59–67.
21. Cobiac L, Clifton PM, Abbey M, Belling GB, and Nestel PJ. Lipid, lipoprotein, and hemostatic effects of fish vs. fish-oil n-3 fatty acids in mildly hyperlipidemic males. *Am J Clin Nutr* 1991; 53(5):1210–6.
22. Visioli F, Risé P, Barassi MC, Marangoni F, Galli C. Dietary intake of fish vs. formulations leads to higher plasma concentrations of n-3 fatty acids. *Lipids* 2003; 38(4):415–8.
23. Arterburn LM, Oken HA, Bailey Hall E, Hamersley J, Kuratko CN, Hoffman JP. Algal-oil capsules and cooked salmon: nutritionally equivalent sources of docosahexaenoic acid. *J Am Diet Assoc* 2008; 108(7):1204–9.
24. Harris WS. The omega-3 index as a risk factor for coronary heart disease. *Am J Clin Nutr* 2008; 87(6):1997S–2002S.
25. Hibbeln JR, Salem N Jr. Dietary polyunsaturated fatty acids and depression: when cholesterol does not satisfy. *Am J Clin Nutr* 1995; 62(1):1–9.
26. Calviello G, Serini S, Palozza P. n-3 polyunsaturated fatty acids as signal transduction modulators and therapeutical agents in cancer. *Curr Signal Transduct Ther* 2006; 1:255–71.
27. Calviello G, Serini S, Piccioni E. n-3 Polyunsaturated fatty acids and the prevention of colorectal cancer: molecular mechanisms involved. *Curr Med Chem* 2007; 14(29):3059–69.
28. Calviello G, Serini S, Piccioni E, Celleno L. Beneficial effects of n-3 PUFAs on UV-induced Skin Damage and tumorigenesis. In: Eetu P. Heikkinen, ed. Fish Oils and Health, Nova Science Publishers. New York, 2008, pp. 141–58.
29. Terry PD, Rohan TE, Wolk A. Intakes of fish and marine fatty acids and the risks of cancers of the breast and prostate and of other hormone-related cancers: a review of the epidemiologic evidence. *Am J Clin Nutr* 2003; 77(3):532–43.
30. Brown TT, Zelnik DL, Dobs AS. Fish oil supplementation in the treatment of cachexia in pancreatic cancer patients. *Int J Gastrointest Cancer* 2003; 34(2–3):143–50.
31. Dyerberg J. Coronary heart disease in Greenland Inuit: a paradox. Implications for western diet patterns. *Arctic Med Res* 1989; 48(2):47–54.
32. Kinsella JE, Lokesh B, Stone RA. Dietary n-3 polyunsaturated fatty acids and amelioration of cardiovascular disease: possible mechanisms. *Am J Clin Nutr* 1990; 52(1):1–28.
33. Mozaffarian D. Fish, n-3 fatty acids, and cardiovascular haemodynamics. *J Cardiovasc Med (Hagerstown)* 2007; 8(Suppl 1):S23–6.
34. De Caterina R, Massaro M. Omega-3 fatty acids and the regulation of expression of endothelial pro-atherogenic and pro-inflammatory genes. *J Membr Biol* 2005; 206(2):103–16.
35. Russo GL. Dietary n-6 and n-3 polyunsaturated fatty acids: From biochemistry to clinical implications in cardiovascular prevention. *Biochem Pharmacol* 2009; 77(6):937–46.
36. Xiao YF, Sigg DC, Leaf A. The antiarrhythmic effect of n-3 polyunsaturated fatty acids: modulation of cardiac ion channels as a potential mechanism. *J Membr Biol* 2005; 206(2):141–54.

37. Bouzan C, Cohen JT, Connor WE, Kris-Etherton PM, Gray GM, König A, et al. A quantitative analysis of fish consumption and stroke risk. *Am J Prev Med* 2005; 29(4):347–52.
38. Nettleton JA, Katz R. n-3 long-chain polyunsaturated fatty acids in type 2 diabetes: a review. *J Am Diet Assoc* 2005; 105(3):428–40.
39. Gebauer SK, Psota TL, Harris WS, Kris-Etherton PM. n-3 fatty acid dietary recommendations and food sources to achieve essentiality and cardiovascular benefits. *Am J Clin Nutr* 2006; 83(6 Suppl):1526S–35S.
40. Breslow JL. n-3 fatty acids and cardiovascular disease. *Am J Clin Nutr* 2006; 83(6 Suppl):1477S–82S.
41. Lombardo YB, Chicco AG. Effects of dietary polyunsaturated n-3 fatty acids on dyslipidemia and insulin resistance in rodents and humans. *J Nutr Biochem* 2006; 17(1):1–13.
42. Jensen CL. Effects of n-3 fatty acids during pregnancy and lactation. *Am J Clin Nutr* 2006; 83(6 Suppl):1452S–7S.
43. Makrides M, Gibson RA. Long-chain polyunsaturated fatty acid requirements during pregnancy and lactation. *Am J Clin Nutr* 2000; 71(1 Suppl):307S–11S.
44. Sattar N, Berry C, Greer IA. Essential fatty acids in relation to pregnancy complications and fetal development. *Br J Obstet Gynaecol* 1998; 105(12):1248–55.
45. Al MD, van Houwelingen AC, Kester AD, Hasaart TH, de Jong AE, Hornstra G. Maternal essential fatty acid patterns during normal pregnancy and their relationship to the neonatal essential fatty acid status. *Br J Nutr* 1995; 74(1):55–68.
46. Martinez M. Dietary poly-unsaturated fatty acids in relation to neural development in humans. *Progr Lipid Res* 1989; 28: 123–33.
47. Makrides M, Simmer K, Goggin M, Gibson RA. Erythrocyte docosahexaenoic acid correlates with the visual response of healthy, term infants. *Pediatr Res* 1993; 33(4):425–7.
48. Innis SM. Essential fatty acids in growth and development. *Prog Lipid Res* 1991; 30(1): 39–103.
49. Cetin I, Koletzko B. Long-chain omega-3 fatty acid supply in pregnancy and lactation. *Curr Opin Clin Nutr Metab Care* 2008; 11(3):297–302.
50. Koletzko B, Lien E, Agostoni C, Böhles H, Campoy C, Cetin I, et al. The roles of long-chain polyunsaturated fatty acids in pregnancy, lactation and infancy: review of current knowledge and consensus recommendations. *J Perinat Med* 2008; 36(1):5–14.
51. Sinikovic DS, Yeatman HR, Cameron D, Meyer BJ. Women's awareness of the importance of long-chain omega-3 polyunsaturated fatty acid consumption during pregnancy: Knowledge of risks, benefits and information accessibility. *Public Health Nutr* 2009; 12(4): 562–9.
52. Alessandri JM, Guesnet P, Vancassel S, Astorg P, Denis I, Langelier B,et al. Polyunsaturated fatty acids in the central nervous system: evolution of concepts and nutritional implications throughout life. *Reprod Nutr Dev* 2004; 44(6):509–38.
53. Friedland RP. Fish consumption and the risk of Alzheimer disease: is it time to make dietary recommendations? *Arch Neurol* 2003; 60(7):923–4.
54. Kalmijn S, Launer LJ, Ott A, Witteman JC, Hofman A, Breteler MM. Dietary fat intake and the risk of incident dementia in the Rotterdam Study. *Ann Neurol* 1997; 42(5): 776–82.
55. Barberger-Gateau P, Letenneur L, Deschamps V, Peres K, Dartigues JF, Renaud S. Fish, meat, and risk of dementia: Cohort study. *BMJ* 2002; 325(7370):932–3.
56. Morris MC, Evans DA, Bienias JL, Tangney CC, Bennett DA, Wilson RS, et al. Consumption of fish and n-3 fatty acids and risk of incident Alzheimer disease. *Arch Neurol* 2003; 60(7):940–6.
57. van Gelder BM, Tijhuis M, Kalmijn S, Kromhout D. Fish consumption, n-3 fatty acids, and subsequent 5-y cognitive decline in elderly men: the Zutphen Elderly Study. *Am J Clin Nutr* 2007; 85(4):1142–7.
58. Soderberg M, Edlund C, Kristensson K, Dallner G. Fatty acid composition of brain phospholipids in aging and in Alzheimer's disease. *Lipids* 1991; 26(6):421–5.

59. Conquer JA, Tierney MC, Zecevic J, Bettger WJ, Fisher RH. Fatty acid analysis of blood plasma of patients with Alzheimer's disease, other types of dementia, and cognitive impairment. *Lipids* 2000; 35(12):1305–12.
60. Schaefer EJ, Bongard V, Beiser AS, Lamon-Fava S, Robins SJ, Au R, et al. Plasma phosphatidylcholine docosahexaenoic acid content and risk of dementia and Alzheimer disease: the Framingham Heart Study. *Arch Neurol* 2006; 63(11):1545–50.
61. Heude B, Ducimetiere P, Berr C. EVA Study. Cognitive decline and fatty acid composition of erythrocyte membranes – The EVA Study. *Am J Clin Nutr* 2003; 77(4):803–8.
62. Hashimoto M, Hossain S, Shimada T, Sugioka K, Yamasaki H, Fujii Y, et al. Docosahexaenoic acid provides protection from impairment of learning ability in Alzheimer's disease model rats. *J Neurochem* 2002; 81(5):1084–91.
63. Cole GM, Frautschy SA. Docosahexaenoic acid protects from amyloid and dendritic pathology in an Alzheimer's disease mouse model. *Nutr Health* 2006; 18(3):249–59.
64. Calon F, Lim GP, Yang F, Morihara T, Teter B, Ubeda O, et al. Docosahexaenoic acid protects from dendritic pathology in an Alzheimer's disease mouse model. *Neuron* 2004; 43(5):633–45.
65. Lim GP, Calon F, Morihara T, Yang F, Teter B, Ubeda O, et al. A diet enriched with the omega-3 fatty acid docosahexaenoic acid reduces amyloid burden in an aged Alzheimer mouse model. *J Neurosci* 2005; 25(12):3032–40.
66. Oksman M, Iivonen H, Hogyes E, Amtul Z, Penke B, Leenders I, et al. Impact of different saturated fatty acid, polyunsaturated fatty acid and cholesterol containing diets on beta-amyloid accumulation in APP/PS1 transgenic mice. *Neurobiol Dis* 2006; 23(3):563–72.
67. Calder PC. n-3 polyunsaturated fatty acids, inflammation, and inflammatory diseases. *Am J Clin Nutr* 2006; 83(6 Suppl):1505S–19S.
68. Wong KW. Clinical efficacy of n-3 fatty acid supplementation in patients with asthma. *J Am Diet Assoc* 2005; 105(1):98–105.
69. Calder PC. Polyunsaturated fatty acids, inflammatory processes and inflammatory bowel diseases. *Mol Nutr Food Res* 2008; 52(8):885–97.
70. Proudman SM, Cleland LG, James MJ. Dietary omega-3 fats for treatment of inflammatory joint disease: efficacy and utility. *Rheum Dis Clin North Am* 2008; 34(2):469–79.
71. Poole EM, Bigler J, Whitton J, Sibert JG, Kulmacz RJ, Potter JD, et al. Genetic variability in prostaglandin synthesis, fish intake and risk of colorectal polyps. *Carcinogenesis* 2007; 28(6):1259–63.
72. Hites RA, Foran JA, Schwager SJ, Knuth BA, Hamilton MC, Carpenter DO. Global assessment of polybrominated diphenyl ethers in farmed and wild salmon. *Environ Sci Technol* 2004; 38(19):4945–9.
73. Sijtsma L, de Swaaf ME.Biotechnological production and applications of the omega-3 polyunsaturated fatty acid docosahexaenoic acid. *Appl Microbiol Biotechnol* 2004; 64(2):146–53.
74. Pulz O, Gross W. Valuable products from biotechnology of microalgae. *Appl Microbiol Biotechnol* 2004; 65(6):635–648.
75. Calviello G, Serini S, Piccioni E, Rinaldi C, Mostra D, Damiani G. Fish species living in an artificial lake with high quality waters as an optimal source of n-3 PUFAs. http://www.issfal.org.uk/pufa-recommendations.html

Part I
Possible Mechanisms

Chapter 1
Possible Mechanisms of ω-3 PUFA Anti-tumour Action

Michael B. Sawyer and Catherine J. Field

Abstract Cancer is a disease characterized by an imbalance between cell division and cell death. Although the molecular mechanisms which account for the biological effects of the ω-3 long chain polyunsaturated fatty acids (ω-3 PUFAs) are not completely understood, there is considerable evidence from animal tumours and human cell lines that providing docosahexaenoic acid (DHA) and/or eicosapentaenoic acid (EPA) will both increase apoptotic and other death pathways and decrease cell growth. ω-3 PUFAs appear to mediate these beneficial effects by affecting the expression and/or function of the lipids, proteins, and genes that regulate these processes. The current evidence supports a hypothesis that these anti-tumour effects are initiated by the ability of DHA and EPA to alter the lipid environment of the cell and in doing so modulate receptors, proteins, and lipid-derived signals originating from cell membranes. The evidence for the possible mechanisms for the beneficial effects of ω-3 PUFAs on tumour cell death and/or proliferation is reviewed in this chapter.

Keywords Apoptosis · Proliferation · Docosahexaenoic acid · Eicosapentaenoic acid · Cancer

Abbreviations

PUFA	Polyunsaturated fatty acids
AA	Arachidonic acid
ALA	Alpha linolenic acid
AOM	Azoxymethane
Apaf-1	Apoptotic peptidase activating factor 1
Bid	Bcl-2 interacting domain
CDK	Cyclin-dependent kinase

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CDKI	CDK inhibitors
COX-1 and -2	Cyclooxygenase 1 and 2
DAG	Diacylglycerol
DHA	Docosahexaenoic acid
DISC	Death-inducing signalling complex
DR	Death receptors
EGF	Epidermal growth factor
EGFR	Epidermal growth factor receptor
EPA	Eicosapentaenoic acid
FLIP	FLICE-inhibitory protein
GRB2	Growth factor receptor-bound protein
IAP	Inhibitor of apoptosis proteins
IGF	Insulin-like growth factor
IGFBP	IGF-binding protein
IP3	Inositol (1,4,5) triphosphate
IRS	Insulin receptor substrates
LA	Linoleic acid
LOX	Lipoxygenase
MAPK	Mitogen-activated protein kinase
MMPs	Matrix metalloproteinases
NFκB	Nuclear factor κB
PGE_2	Prostaglandin E_2
PI3K	Phosphatidylinositol-3-kinase
PIP2	Phosphatidylinositol (4,5) bisphosphate
PIP3	Phosphatidylinositol (3,4,5) triphosphate
PKC	Protein kinase C
PLA2 and C	Phospholipase 2 and C
PLC	Phospholipase C
PPAR	Peroxisome proliferator-activated receptors
pRB	Phosphorylated RB
RB	Retinoblastoma protein
ROS	Reactive oxygen species
SHC	Src homology and collagen domain
SMase	Sphingomyelinase
SREBP	Sterol regulatory element-binding protein
TNF	Tumour necrosis factor
TNFR1	TNF receptor 1
TRAIL-R1 and 2	TNF-related apoptosis-inducing ligand receptor 1 and 2

1.1 Introduction

Cancer is a disease characterized by an imbalance between cell division and cell death. There are many possible mechanisms for the beneficial effects of omega (ω-3) long chain polyunsaturated fatty acids (PUFAs) in cancer. ω-3 PUFAs in the context of this chapter will refer to the long chain ω-3 PUFAs, eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), unless otherwise indicated. This chapter will focus on mechanisms at the level of the tumour cell, specifically, the effect on pathways that regulate cell death and proliferation. This family of fatty acids has been demonstrated to promote anti-tumour immunity and inhibit cancer initiation, tumour angiogenesis, and metastasis. Excellent reviews are available on these mechanisms [1–5] but are beyond the scope of the current chapter. Since tissue homeostasis is the result of a subtle balance between proliferation and cell death, either unregulated cell proliferation or reduced cell death can promote tumour formation and progression. The possible mechanisms by which ω-3 fatty acids hinder the growth of tumour cells may involve an impairment of cell proliferation, an increase in cell death, or a combination of both. Cell proliferation and cell death are not mutually exclusive processes, as many signalling pathways are involved in both cell survival and cell death. In this chapter we will briefly review the pathways that are involved in regulating tumour cell death and growth and describe how they are disrupted/deregulated in tumour cells. This will be followed by a review of the evidence for a beneficial effect of ω-3 PUFAs on pathways of cell death and growth.

1.2 Cell Death

1.2.1 Non-apoptotic Forms of Cell Death

Cell death occurs via necrosis (including mitotic catastrophe and autophagy) which most often is the result of an insult or toxicity that triggers inflammation [6]. Necrotic cells are characterized by ATP depletion leading to plasma membrane blebs, cytoskeletal alterations, loss of volume control, mitochondrial permeabilization, cell swelling, and leakage of small molecules, which eventually leads to a fatal rupturing of the plasma membrane and the release of cytosolic enzymes (i.e. lactate dehydrogenase and aminotransferases) [7]. However, recent studies indicate that necrosis and apoptosis (described below) are not necessarily distinct and independent entities (reviewed by [7]). In these reviews, it is suggested that the route of cell death is determined by the change in the ATP supply of the cell. For example, when ATP is depleted, apoptosis is blocked, and pro-apoptotic signals induce necrotic cell death [7]. It appears that even after initiation of apoptosis, necrosis may supervene if ATP levels fall or a concurrent change leads to a breakdown of the plasma membrane barrier. On the other hand, partial ATP recovery can prevent necrotic cell death, with apoptosis developing. More recently the term “autophagy” has been coined to describe lysosome-based cellular degradation and this appears to be closely related to apoptosis (protease-based programmed cell death). Thus, it is likely that apoptosis and necrosis represent extremes on a continuum, and cells undergoing death display both apoptotic and necrotic patterns of cellular damage [7]. This paradigm has been termed “necrapoptosis” [7].

1.2.2 Apoptosis

The most common and well-defined evolutionarily conserved intrinsic programme of physical cell loss in most tissues of multicellular organisms is apoptosis [8]. Apoptosis is a tightly regulated and highly efficient process that regulates cell growth and homeostasis. It is characterized by typical morphological and biochemical changes in the cell that include cell shrinkage, nuclear DNA fragmentation, and membrane blebbing [9]. Unlike necrosis, membranes are well maintained and cells do not release their contents during this process. This influences the behaviour of adjacent cells or initiates inflammation [10]. The molecular events of apoptosis can be divided into three steps: (1) initiation/triggering by apoptosis-inducing agents; (2) activation of a family of cysteine proteases, known as the caspases, that activate a signal transduction cascade; and (3) proteolytic cleavage of cellular components [9]. The complex process of apoptosis is controlled by both external signals and genes. Although not mutually exclusive, two apoptotic signalling pathways exist: the death receptor (extrinsic) pathway and the mitochondrial (intrinsic) pathway.

Studies conducted on cultured cells have clearly demonstrated that DHA and EPA, alone or in combination, can increase apoptosis (at least in vitro) in a variety of different cancer cell lines including breast [11–14], colon [15–20], lung [21, 22], prostate [23, 24], lymphoma [25]; leukaemic [26, 27], hepatic [28], pancreatic [29–32], and larynx [33]. These studies have enabled researchers to identify the many mechanisms through which ω-3 PUFAs induce apoptosis. The pro-apoptotic effects of dietary ω-3 PUFA have been confirmed in a variety of animal models of cancer and have been recently reviewed [34]. Both the in vitro and animal studies are consistent with accumulating evidence in human studies where the intake of ω-3 PUFAs (EPA 100 mg/day and DHA 400 mg/day for 2 years) was found to promote apoptosis of colonic mucosa in humans [35], and supplementation with EPA (2 g/day for 3 months) significantly increased apoptosis in normal colonic mucosa in subjects with a history of colorectal adenomas [36]. Although not as well studied, there is some evidence that linolenic acid, the dietary precursor of EPA, has pro-apoptotic effects in some cancer cell lines (reviewed by [34]). Interestingly, most of the papers have reported that ω-3 fatty acids are pro-apoptotic in concentrations of 50–250 μM, serum concentrations that could be achieved in vivo through supplementation.

There are several reports that have investigated the ability of ω-3 PUFAs to regulate the apoptotic process in normal cells, with the conclusion that they result in beneficial health effects [37, 38]. In fact, ω-3 PUFAs have been shown to *prevent* apoptosis in heart, neuronal, and retinal tissues (reviewed by [39]). In these organs, ω-3 PUFAs appear to preserve function and exhibit anti-apoptotic properties through similar cellular signalling pathways that induce apoptosis in other organs. This suggests that the pro-apoptotic effects of ω-3 PUFAs on cancer cells are related to altered regulation of these pathways in the cancer cells.

1.2.2.1 Extrinsic Pathway of Apoptosis

The extrinsic pathway of apoptosis is activated by the engagement of death receptors (DR) on the cell surface. DR are members of the tumour necrosis factor (TNF) receptor gene superfamily. They have a broad range of biological functions, including regulation of both cell death and survival [40]. Triggering members of the DR family by death ligands results in the transduction of either apoptotic or survival signals [8]. The best-characterized death receptors include CD95 (APO-1/Fas), TNF receptor 1 (TNFRI), TNF-related apoptosis-inducing ligand-receptor 1 (TRAIL-R1) and -2 (TRAIL-R2) (reviewed by [40]). Binding of TRAIL receptors results in recruitment of the adaptor molecule Fas-associated death domain to form death-inducing signalling complexes (DISCs) [8] which activate caspase-8. Caspase-8 either directly stimulates executor caspases, such as caspase-3, or activates the mitochondrial death receptor pathway (intrinsic pathway) by cleaving Bcl-2 interacting domain (Bid) and increasing outer mitochondrial permeability. This ultimately results in the release of cytochrome c [41]. There are inhibitors of this pathway, including FLICE-inhibitory protein (FLIP) (which can block procaspase-8 activation at the DISC and the decoy membrane receptors) and TRAIL-R3 to -R5 (which bind TRAIL, but do not transmit a death signal) [8]. Further downstream, activation of inhibitors of apoptosis proteins (IAPs) inhibits effector caspase activation [8].

Binding of CD95, TNFR, or DR5/TRAIL (and a few others) results in the activation of acid sphingomyelinase (SMase) which hydrolyses membrane sphingomyelin to produce ceramide (reviewed by [41]). Sphingomyelin is a sphingolipid found in the cell membrane, predominantly in the microdomains called lipid rafts [41]. The generation of ceramide in the cell membrane has been shown to dramatically alter the physical structure of membrane rafts [41] which may explain the observation that ceramide can trigger apoptosis in almost any cell, including tumour cells. More recently, it has been demonstrated that in addition to changing membrane properties, ceramide activates a number of key proteins involved in apoptosis, including cathepsin D, which triggers cell death via Bid, Bax, and Bak (reviewed by [41]).

1.2.2.2 Intrinsic Pathway of Apoptosis

The second pathway, the intrinsic or mitochondrial pathway, is triggered by various extracellular and intracellular stresses that result in the permeabilization of the mitochondrial outer membrane through activation of pro-apoptotic members of the Bcl-2 family [10]. The Bcl-2 family includes over 30 proteins that can be either anti-apoptotic (i.e. Bcl-2, Bcl-XL) or pro-apoptotic (i.e. Bax, Bcl-XS, Bak, Bad, Bid). The activity of these proteins is regulated by homo- and heterodimerization, proteolysis, and phosphorylation. The balance of these proteins regulates apoptosis [39]. Upon disruption of the outer mitochondrial membrane, a set of proteins normally found in the space between the inner and outer mitochondrial membranes are released, including cytochrome *c*, second mitochondria-derived activator of caspase (Smac), direct inhibitor of IAP-binding protein (DIABLO), Omi/HtrA2, apoptosis-inducing factor (AIF), and endonuclease G [42]. The release of cytochrome *c* into the cytosol interacts with apoptotic peptidase activating factor 1 (Apaf-1) and dATP, leading to caspase-9 activation and hence downstream activation of the caspase-3 cascade [39]. Once activated, caspase-3 cleaves key substrates in the cell to produce the cellular and biochemical events of apoptosis [10]. Other released proteins facilitate caspase activation via inactivation of endogenous inhibitors of caspases and IAPs [10]. While activated executioner caspases can induce cell death through apoptosis, inhibition of these proteinases may only transiently protect cells. Once the mitochondria permeabilise, cell death will likely proceed regardless of caspase activation, suggesting that other toxic mediators released from the mitochondria act as caspase-independent death effectors [10].

1.2.2.3 Activation of Caspases

Caspases are a family of cysteine proteases that are activated via protein phosphatases [43]. These proteins play a critical role in the apoptotic pathway and are responsible for many of the biochemical and morphological changes associated with apoptosis (reviewed by [39]). The entire process, from the initial trigger to the destruction of the cell, can take hours or even days; however, the activation of caspases occurs within 10 min of stimulation. It is believed that once this pathway is initiated cell death is inevitable [10]. Caspase activation is initiated at the plasma membrane upon ligation of DR (receptor pathway) or at the mitochondria (mitochondrial pathway) [7]. They are categorized as initiator caspases (caspase-8, -9, -10, and -12), which cleave other caspases, or the executioner caspases (caspase-3, -6, and -7), which cleave a number of different substrates in the cytoplasm and nucleus, leading to many of the morphologic features of apoptotic cell death [6, 43]. Signalling in the intrinsic and extrinsic pathways merge at the level of the effector caspases, which results in the activation of the nuclear factor κB (NFκB) [43]. Executioner caspases initiate the proteolysis of cytoskeletal proteins such as actin or fodrin. This leads to the degradation of lamin which results in the characteristic nuclear shrinking associated with apoptotic death [43]. The cell regulates caspase inhibition either at the receptor level by FLIP, which blocks caspase-8 activation, or at the mitochondria by activation of the anti-apoptotic Bcl-2 family proteins or the IAPs (reviewed by [40]).

1.2.2.4 Disruptions of the Extrinsic and Intrinsic Pathways in Cancer Cells

A hallmark of human cancers is the ability of tumour cells to evade apoptosis [6]. Cancer cells have evolved strategies to resist cell death induction by decreasing anti-apoptotic molecules and/or by decreasing amounts or reducing the function of pro-apoptotic proteins (reviewed by [44]). For example, mutations in the tumour suppressor gene *p53*, the most common genetic defect in human cancers, reduce the ability of the cell to activate the mitochondrial cell death pathway [44]. Mutations in other genes directly involved in the regulation of the mitochondrial pathway are also commonly found in cancer cells. The expression pattern of the Bcl-2 family differs depending on the cell type or stage of differentiation and decreased expression of these proteins can result in deregulation of the intrinsic pathway [45]. Additionally in ovarian cancer, melanoma, and leukaemia decreased or absent activity of Apaf-1 has been reported (reviewed by [40]). Despite this, there is a growing body of evidence that cancer cells have an intrinsic drive to apoptosis as there are reported high basal levels of caspases-3 and -8. However, these proteins are held in check by high or aberrant expression of IAPs [44] and/or a low level of ceramide formation [46]. Anti-apoptotic alterations have also been reported in the extrinsic pathway (reviewed by [40]). Mutations or lower expression of the CD95 gene and other DR occur in a variety of haematological and solid tumours [47–50]. Absent or defective intracellular transport and expression and/or function of anti-apoptosis regulators such as decoy receptors and FLIP have also been reported in a number of tumours [40, 44, 48].

1.2.3 ω-3 Fatty Acids and Their Effects on Apoptosis

Most anti-cancer interventions (chemotherapy, irradiation, immunotherapy) activate apoptosis via targeting various molecules involved in apoptosis (reviewed by [40]). Some of these are also targets of ω-3 PUFAs (reviewed by [51]). Although the molecular mechanisms which account for the biological effects of ω-3 PUFAs are not completely understood, there is considerable evidence from in vitro models that ω-3 PUFAs can modulate apoptotic pathways and affect the expression and/or function of apoptosis regulating lipids and proteins. It has been hypothesized in several recent reviews that the effects of EPA and/or DHA on tumour cell apoptosis is likely due to their ability to alter the lipid environment and modulate receptors, proteins, and lipid-derived signals originating from cell membranes [1, 5, 34, 52]. In the following section we will use this hypothesis as a framework to review the considerable body of evidence for an effect of ω-3 PUFAs on the regulation of cancer cell death.

1.2.3.1 ω-3 PUFA-Induced Changes in Membrane Fluidity, Structure, and Composition in Tumour Cells

It is now well established that changes in the supply of ω-3 PUFAs alter the composition and function of membrane lipids [53]. It is also well established that ω-3 PUFAs are incorporated into the phospholipids of cell membranes of tumour cells when included in the diet or cell culture media [13, 54–58]. In support of the importance of changing membrane lipids, reducing the ability of tumour cells to incorporate EPA into membrane lipids reduces the apoptosis-inducing effects of this ω-3 PUFAs [32, 59]. The presence of ω-3 PUFAs in the plasma membrane results in unique physiochemical properties that affect numerous membrane characteristics including permeability [60], fluidity [61], lipid packing [60], fusion [61], deformability [60], and most recently lipid micro domain formation [58, 62, 63].

These ω-3-induced changes in membrane function, logically, could significantly alter signals and processes of cell death.

ω-3 PUFA Effects on Cellular Oxidative Stress and Related Molecular Signalling

Lipid peroxidation is initiated by hydrogen abstraction from an unsaturated fatty acid by reactive oxygen species (ROS). The resulting lipid radical reacts with oxygen to form a fatty acid peroxyl radical, which can attack adjacent fatty acid chains in the cell membrane, and thus propagate lipid peroxidation [64]. The mechanism by which lipid peroxidation products inhibit tumour growth is not entirely clear. Lipid peroxidation is reported to enhance the expression of the DR and suppress Bcl-2 expression, perhaps explaining some reports of peroxidation-induced apoptotic cell death [65]. ROS can induce necrosis through oxidative nucleic acid base modifications and DNA strand breaks [39]. ROS have also been demonstrated to directly transmit apoptotic signals by disrupting the mitochondrial permeability and by triggering the release of soluble inter-membrane proteins [57, 66–69]. Additionally, ROS are reported to alter gene expression, including upregulation of genes induced by p53 (Bax, p21Cip1/Waf1) and downregulation of Bcl-2 which would result in activation of the mitochondrial and receptor pathways of apoptosis (reviewed by [1]). Interestingly, highly undifferentiated tumour cells have a very low degree of lipid peroxidation which seems to be inversely related to their growth rate, such that when DNA synthesis is at a maximum, lipid peroxidation is suppressed, and vice versa [70]. Furthermore, rapidly growing normal tissues (e.g. testis, bone marrow, and intestinal epithelium) appear resistant to peroxidation, with low levels of peroxidation products [71], perhaps explaining why ω-3 PUFAs generally do not damage healthy cells.

ω-3 PUFAs are highly peroxidable and it is hypothesized that their incorporation into plasma and mitochondrial membrane phospholipids may sensitize cells to reactive oxygen species (ROS), inducing an oxidative stress [56, 72]. The peroxidation products generated from ω-3 PUFAs are considered crucial to explain the effect of these lipids on tumour cell death in many models of cancer [57, 73, 74]. DHA-derived oxidative products have been reported to reduce the protein level of the membrane anti-oxidant phospholipid hydroperoxide glutathione peroxidase [75]. In support of this, the addition of pro-oxidants augments the anti-cancer effect of ω-3 fatty acids [70, 76–78] and the addition of anti-oxidants reduces or abolishes the cytotoxic effect of DHA [18, 75]. Colon tumours are particularly susceptible to oxidative stress and recently it was demonstrated that dietary DHA is incorporated into mitochondrial membrane phospholipids of colon cancer cells, which sensitizes the membrane to enhance oxidative stress (induced by butyrate metabolism) [20]. Additionally, DHA was shown to induce lipid peroxidation which is correlated with changes in the molecular pathways involved in apoptosis and tissue invasion (i.e. matrix metalloproteinases (MMPs), caspase-3 and -9, Mcl-1 anti-apoptotic protein) in colon adenocarcinoma [79]. Again, these effects were reversed by treating the cells with an anti-oxidant [79]. However, this mechanism of cell death may be more specific to tumours that are more susceptible to oxidative stress, as we [13] and others [80] did not alter the pro-apopototic effects of ω-3 PUFAs on MCF-7 or MDA-MB-231 breast cancer cells with the addition of anti-oxidants.

DHA has been reported to preferentially accumulate in the phospholipid cardiolipin [20, 81]. Cardiolipin is only found in the mitochondria, where it is present in the inner membrane and at inter-membrane contact sites. It is required for mitochondrial structural integrity and for the proper function of the electron transport chain [82]. Cardiolipin is usually bound to the enzyme complexes of electron transport and ATP synthesis (i.e., cytochrome *c* oxidase) [82], suggesting that mitochondrial function is very much dependent on the proper amount of cardiolipin. Cardiolipin rich in ω-3 PUFAs is more susceptible to ROS, and peroxidation results in an altered composition and reduced mitochondrial membrane integrity [81]. This then compromises energy metabolism [81], which in turn initiates apoptosis [83]. Indeed, cardiolipin hydroperoxides have been shown to directly trigger the release of pro-apoptotic factors from mitochondria [84].

Although there is considerable evidence for membranes rich in ω-3 PUFAs to make cells more susceptible to the production of pro-apoptotic ROS, there are still many questions to answer. Most importantly, there is considerable evidence that ω-3 PUFAs reduced tumour cell growth by decreasing (not increasing) the intracellular oxidative stress [85]. In addition, it has been shown that DHA is also the precursor to substances known as docosanoids which have been shown not only to reduce inflammation (see section below) but also to inhibit oxidative stress-induced apoptosis [86].

ω-3 PUFA Changes in the Lipid Composition and Function of Membrane Microdomains

More recently, attention has focused on lipid rafts, glycolipid-enriched membrane domains that appear to "float" in the phospholipid bilayer of the membrane [87]. Cholesterol, sphingolipids, and phospholipids with saturated fatty acyl chain moieties are highly enriched in these lipid microdomains [(reviewed by [62])]. Lipid rafts are dynamic microenvironments in the exoplasmic leaflet of the phospholipid bilayer of plasma membranes, which are thought to preferentially group trans-membrane proteins according to their function [88]. The unique lipid raft environment attracts many of the important membrane receptors and key signalling proteins and lipids (reviewed by [62]), many of which are involved in apoptosis (including the DR and ceramide). Lipid rafts can include or exclude proteins to variable extents, leading to the hypothesis that rafts play a key role in signal transduction, perhaps functioning as platforms to concentrate signalling proteins [88]. Potentially, individual rafts may cluster together to connect interacting proteins into a signalling complex [88]. As well, rafts may protect signalling complexes from non-raft enzymes such as phosphatases that could otherwise affect the signalling process [88]. A subset of specialized rafts termed caveolae have also been described, which are flask-shaped structures in the membrane that are enriched in the protein caveolin-1, and mediate membrane functions including endocytosis, cholesterol transport, and signal transduction (reviewed by [62]).

Incubation with ω-3 PUFAs alters the composition/structure of Jurkat leukaemia cells [39], MDA-MB-231 breast cancer cells [58], and colon cancer cells [62, 89]. There is evidence that the incorporation of ω-3 PUFAs alters the regulation of raft-associated proteins [90–92]. In colon cancer cells, it was reported that dietary treatment of mice with ω-3 PUFAs was able to enrich the colonic caveolae phospholipid ω-3 content, reduce the levels of cholesterol and caveolin, and reduce the activity of the anti-apoptotic signal molecule H-Ras [90]. In MDA-MB-231 breast cancer cells we observed that incorporation of ω-3 PUFAs resulted in an increase in the pro-apoptotic lipid molecule ceramide and a decrease in raft-associated epidermal growth factor receptor [58]. Together, these observations offer a plausible explanation as to how ω-3 PUFAs might mediate apoptosis.

It is not clear how PUFAs regulate raft localization of proteins, but it has been proposed that the substitution of unsaturated fatty acid residues for saturated residues in rafts may disrupt the highly saturated and ordered lipid structure and modify the interaction of acylated (e.g. palmitoylated, myristoylated) proteins with nonpolar lipids resulting in selective displacement of proteins and altered lipid signalling (reviewed by [5, 92]). Although validation from in vivo studies is still required, evidence is mounting that the effect of ω-3 PUFAs on lipid microdomains may provide a common pathway to explain the beneficial effects of ω-3 PUFAs on tumour cell apoptosis.

1.2.3.2 Effect of ω-3 PUFAs on Enzyme and Receptor Activity and Location

Changes in membrane structure and composition have been shown to change the activity of specific integral membrane proteins which serve as ion channels, transporters, receptors, signal transducers, and enzymes (reviewed by [53, 93, 94]). The composition of dietary fat has also been reported to change the phospholipid fatty acid profile of the nuclear membrane, altering its function. Phospholipids in this membrane, especially if they contain unsaturated fatty acids, can regulate in vitro the activity of some DNA-binding proteins with which they interact. This affects functions such as DNA replication, transcription, and post-translational modification of proteins [95, 96]. The effects of ω-3 PUFAs on several key proteins involved in the regulation of apoptosis are reviewed below.

Bcl-2: There is considerable evidence to suggest that Bcl-2 family proteins may play a key role in ω-3 PUFA-induced cell death. Feeding fish oil significantly decreased Bcl-2 expression and increased apoptosis in the distal colon in a model of colon cancer [5]. Microarrays suggest that this may be due to a reduced expression of the genes involved in the anti-apoptotic Bcl-2 family [16, 97, 98] and an upregulation of apoptosis enhancing Bad [98]. In support of this, knockdown of Bad eliminated ω-3-induced cell death, and introduction of exogenous Bad restored the sensitivity to ω-3 fatty acids [98].

Ras: ω-3 PUFAs decreased the activation of Ras proteins, which are critical regulators of tumour cell function [99]. One study provided evidence that this occurred by interfering with post-translational modifications of the protein [99], while another done on colon cancer cells suggested it was via decreases in the Ras membrane to cytosol ratio [100].

Caspases: Upregulation of pro-apoptotic caspases (caspase-3, -8, and/or -9) was reported when colon cancer cells were incubated in the presence of DHA or EPA [16, 101, 102].

Cytochrome C: cDNA microarray analysis carried out on CaCo-2 human colon cancer cells cultured in the presence of DHA provided evidence for upregulation of genes involved in cytocrome C activation [16].

β-catenin: The majority of colorectal cancer cells in humans constitutively overexpress β-catenin protein [103]. β-catenin has two distinct functions: to maintain cell-to-cell adhesion and to mediate the Wnt/β-catenin signal transduction pathway. In some tumour cells Wnt/β-catenin signalling results in a relocation of β-catenin from the cell membrane to the nucleus, where it binds to T-cell factor and facilitates transcription of target genes that encode effectors for activation of cell proliferation, invasion, and inhibition of apoptosis [103]. Following incubations with DHA there was a reduction in the levels of the β-catenin protein (particularly in the nucleus) in a concentration-dependent manner. Also, there were increased products of T-cell factor target genes, such as peroxisome proliferator-activated receptors (PPAR), which are involved in regulation of apoptosis in HCT116 and SW480 human colon cancer cells [15].

1.2.3.3 Effect of ω-3 PUFAs on the Network of Lipid Molecular Second Messengers

Ceramide: The sphingolipid ceramide is derived from the hydrolysis of sphingomyelin by SMase enzymes. Changes in ceramide concentration (after activation of the acid SMase) have been reported to transform small "primary" rafts into large ceramide-enriched membrane platforms [104]. Ceramide-enriched membrane platforms cluster and re-organize receptors and signalling molecules, which result in the amplification of the signal for apoptosis. Increased intracellular ceramide concentrations have been shown to modulate apoptosis (via CD95 and CD40 activation [104]) by stimulating through key proteases, phosphatases, and kinases [41]. A role for ceramide in the induction of p21 via activation of NFκB and p53 has been established [105, 106]. Several recent studies indicate that ceramide is also present in mitochondria and may associate with procaspase-3 [41]. Increasing ceramide concentrations in cells is a target of many of the current anti-tumour treatments [40]. Incubation with ω-3 PUFAs increased ceramide levels in membrane rafts, indicative of the activation of the sphingomyelin–ceremide pathway in breast cancer cells [58, 107] and Jurkat leukaemia cells [108]. Conversely, a decreased level of ceramide was shown in normal T lymphocytes treated with ω-3 PUFAs [109–111], suggesting that this effect is tumour specific.

1.2.3.4 Molecules and Genes Involved in the Regulation of Inflammation

Eicosanoids and Docosanoids: A well-established action of ω-3 PUFAs is their ability to alter arachidonic acid (AA)-derived signalling by inhibiting the production of inflammatory lipid mediators derived from this fatty acid [112]. AA is one of the major ω-6 PUFAs esterified in membrane phospholipids and, following stimulation, it is released by the action of phospholipase A2 (PLA2) and C and becomes the substrate for cyclooxygenase (COX1 and 2) and lipoxygenase (LOX5-, 12-, and 15-) enzymes to form eicosanoids (prostaglandins, leucotrienes, hydroxyeicosatetraenoic acids, and thromboxanes) [113]. These short-lived bioactive compounds act locally to regulate key events in the inflammatory response and have been shown to upregulate the anti-apoptotic Bcl-2 and Bcl-Xl and to reduce Bax expression [114]. The relative proportions of PUFAs in cell membranes, as well as cell type, are the primary factors in regulating which eicosanoid will be generated. ω-3 PUFAs compete for acylation in the sn-2 position phospholipids, reducing the availability of AA for PLA [115, 116]. Compared with AA, EPA is the preferential substrate for both COX-2 and the LOXs; hence an increased EPA availability results in more of the EPA-derived lipoxygenase products at the expense of AA-derived lipoxygenase products [117]. EPA-produced eicosanoids (3-series prostanoids and 5-series leucotrienes) are less pro-inflammatory than those derived from AA [74].

Over-expression of COX-2 (the inducible isoform of the enzyme) has been detected in many types of cancer, including cancer of the breast, colon, and prostate (reviewed in [1, 2, 34, 74], and appears to confer resistance to apoptosis [118]). In a variety of tumours ω-3 PUFAs have been shown not only to alter the substrate availability but also to reduce the levels and activity of COX-2 [119, 120]. Recent work suggests that this may occur via an ω-3-mediated reduction in membrane Ras, an inducer of COX-2 activity [5]. However, alterations in COX-2 activity may only partially explain the anti-apoptotic effect of ω-3 PUFAs as DHA was shown to exert its pro-apoptotic effect in COX-2-negative colon cancer cells and in animal tissues [15, 80].

Both EPA and DHA have been recently recognized as the precursors of another group of lipids (resolvins, docosatriens, and protectins) that possess anti-inflammatory and protective properties, particularly evident in the resolution phase of inflammation [86]. The role of these EPA and DHA-derived docosanoids in anti-cancer activity remains to be explored. However, it was recently reported that DHA-induced apoptosis in human colorectal cancer cells not expressing COX-2 produced less of the anti-apoptotic lipid survivin [15]. In these cells, the change in survivin expression correlated with the changes induced by DHA in apoptosis, suggesting that this protein may exert a role in DHA-induced apoptosis observed in HCT116 and SW480 colon cancer cells [15].

Peroxisome Proliferator-Activated Receptors: ω-3 PUFAs may alter apoptotic signalling by acting directly as ligands for nuclear receptors, including PPAR [23, 98]. The PPAR family has at least three closely related members, PPARα (expressed in liver, kidney, heart, and muscle), PPARγ (βγ1, βγ2, and βγ3, expressed in fat cells, large intestine, monocytic cells, and recently identified in a number of breast cancer cell lines), and PPARβ/delta (expressed in nearly all tissues) [121]. PPARs have diverse actions that include the regulation of cell proliferation, cell differentiation, and the inflammatory response [96, 122, 123]. A substantial amount of data have accrued, primarily from in vitro studies, demonstrating that ω-3 PUFAs are important regulators of all of the PPARs (reviewed by [96]). EPA has been reported to bind directly to PPARs resulting in a conformation change that correlates with function [124]. DHA was found to induce apoptosis in prostate cancer cells through the activation of PPARγ [23]. The levels of PPARs in cells may also be modulated by ω-3 PUFAs [125].

NFκB: NFκB is a ubiquitously expressed transcription factor that regulates the expression of a number of different genes relating to cell survival and programmed cell death [126]. The expression and activation of the nuclear transcription factor NFκB has been found to be constitutively activated in many tumour cells and may, at least in part, protect cancer cells against apoptosis [44]. ω-3 PUFAs decrease NFκB activity/expression in a variety of cancer cells [13, 120, 127]. It is not clear how ω-3 PUFAs modulate NFκB activity; however, it has been suggested that these fatty acids modulate the activity of upstream signalling molecules involved in the activation of NFκB [13, 128–130].

1.2.4 Conclusions: ω-3 PUFAs and Their Effects on Cell Death and Apoptosis

In summary, ω-3 PUFAs affect many potential pathways that can promote apoptosis/cell death. This may be explained by both the multitude of changes in membrane

Table 1.1 Effects of ω-3 PUFAs on apoptotic pathways and signals

Plasma membrane	•↑ ω-3 PUFA incorporation into membrane phospholipids •↑ Unsaturation and susceptibility to peroxidation ○↓ phospholipids hydroperoxide glutathione peroxidase •Alter amount or activity of protein and lipids in or generated from the membrane (see below)
Rafts and caveolae	•↑ ω-3 PUFA content and saturation ○↑ Ceramide formation O↓ Epidermal growth factor receptor
Mitochondrial membrane	•↑ ω-3 PUFAs incorporated into phosphoplipids ○↑ Accumulation in cardiolipin
Alter proteins and enzymes involved in apoptosis	•↑ Amount and/or activity of apoptotic regulatory proteins ○↑ Pro-apoptotic proteins (Bad, Bak) ○↓ Anti-apoptotic proteins (Bcl-2) ○↓ Ras ○↑ Caspases (caspase 3, 8, and 9) ○↑ Cytochrome c ○↓ β-Catenin
Alter lipid mediators/signals involved in apoptosis	•↓ AA-derived eicosanoids •↑ EPA-derived eicosanoids ○Alter phospholipids substrate ○↓ Activity of COX-2 •↑ Survival and possibly resolvins and protectins •↑ Activation of PPAR •↓ NFκB activity/expression

lipid composition and the function that occur when ω-3 PUFAs are incorporated into tumours and to the diverse cellular models used. Table 1.1 summarizes the molecular pathways of apoptosis/cell death that appear to be regulated by EPA or/and DHA in cancer cells.

1.3 Cell Growth

1.3.1 Growth Factor Pathways

In addition to resistance to apoptosis, another characteristic feature of cancer is dysregulation of growth factor pathways. A great deal of research has focused on this feature of cancer cells and applying results of this research has led to the development of targeted therapies. Some growth factor pathways involved in cancer growth include: (1) epidermal growth factor receptor (EGFR) family, (2) Insulin-like Growth Factor-I Receptor, and (3) fibroblast growth factor receptor and mTOR/akt pathways.

1.3.1.1 Epidermal Growth Factor Receptor and Its Role in Tumour Growth

Dysregulation of the EGFR family plays a major role in many cancers; there are four members of the EGFR family: (1) EGFR/HER1/erbB1, (2) HER2/ErbB2/Neu, and (3) ErbB3/HER3, and ErbB4/HER4 [131–133]. There are several ligands for these receptors, including epidermal growth factor (EGF), transforming growth factor-alpha, heparin -binding-EGF, amphiregulin, betacellulin, and epiregulin. Members of the ErbB family weigh approximately 170 kDd and are composed of three domains: an extracellular domain that binds ligands, a hydrophobic domain that traverses the plasma membrane, and an intercellular domain that contains a tyrosine kinase domain and regulatory elements. Binding of the ligand to EGFR leads to homo- or heterodimerization of EGFR to itself or HER2, HER3, or HER4. Dimerization results in the phosphorylation of regulatory tyrosine residues and activation of the receptor.

EGFR was proposed as a target for cancer therapy by Mendelsohn in 1983 [134]. His laboratory demonstrated that very low concentrations of EGF stimulated growth of A431 epidermoid cancer cells. They further demonstrated that a small percentage of EGFR that had a high affinity for the EGF was responsible for the growth stimulation [134]. Subsequent investigators showed that EGFR was frequently overexpressed in malignancies such as oesophageal [135], stomach [136], colorectal [136, 137], pancreatic [138], and lung cancer [139]. Although most, if not all, diagrams of EGFR signalling draw the classical lipid bilayer as the backdrop for EGFR signalling pathways, this is probably inaccurate. It is now generally accepted that plasma membranes are not homogenous, but instead have significant areas of heterogeneity (see section above on lipid rafts). EGFR contains a caveolin-binding motif that is found in its kinase domain which binds to the caveolin scaffolding domain. Inactivated and activated EGFR can bind to caveolin through this caveolin -binding motif [140, 141]. It is believed that localization of EGFR to lipid rafts such as caveolae serves to position EGFR near essential signal transduction proteins. Examples of signal transduction proteins that are found in lipid rafts include H-ras [142], Src homology and collagen domain (SHC) [143, 144], and growth factor receptor-bound protein-2 (GRB2) [144].

1.3.1.2 Her-2 Receptor

Her-2, is a protein related to the EGF family, and was discovered originally as the causative oncogene in chemically induced rat neuroglioblastomas [145]. Her-2 is over-expressed in several cancers such as colorectal, ovarian, gastric, and approximately 25–30% of breast cancers [146–148]. The major mechanism by which Her-2 appears to cause carcinogenesis is over-expression by gene duplication rather than by mutations leading to constitutive activation [149]. In contrast to other members of the ErbB family there is no ligand for Her-2. Her-2 appears to be transphosphorylated by heterodimerizing with other ErbB family members [149, 150]. Similar to ErbB1/EGFR, over-expression of Her-2 was associated with a worse outcome in several malignancies. Slamon et al. [149, 151] studied Her-2 amplification in 100 node- positive breast cancer patients. They were able to extract DNA from 86 cases and determine the Her-2 gene copy number which was found to be elevated in 34/86 (40%). Her-2 amplification was related to time to distant relapse (p<0.0001) and overall survival (p=0.0011). Yonemura et al. [148, 149] studied 260 gastric cancer patients and found Her-2 expressed by immunohistochemistry in 31 out of the 260 gastric cancers (11.9%). In a Cox proportional hazards model incorporating known clinical prognostic factors and Her-2 expression, Her-2 was associated with a fivefold increased risk of death. Her-2 binds SHC and GRB2 and can activate the Ras/MAPK pathway. Her-2 and ErbB1 have a binding site for phospholipase C γ (PLC) and can activate it. Activated PLC generates inositol 1,4,5 triphosphate (IP_3) and diacylglycerol. Diacylglycerol activates protein kinase C (PKC) and IP_3 which can cause release of calcium ions from intracellular stores and promotes cell proliferation.

1.3.1.3 Effect of ω-3 PUFAs on the EGF Proteins

Several studies have been conducted demonstrating effects of ω-3 fatty acids on the membrane growth receptors in tumour cells. Schley et al. [152] studied effects of ω-3 PUFAs on growth of the MDA-MB-231 breast cancer cell line. Cells were exposed to varying concentrations of DHA, EPA, and linoleic acid (LA) with concentrations ranging from 75 μmol/L LA alone to EPA 60 μmol/L and DHA 40 μmol/L. Incubation of MDA-MB-231 cells with EPA 60 μmol/L and DHA 40 μmol/L decreased growth by 62% when compared to the control 75 μmol/L LA alone (p<0.05). The combination of EPA 45, DHA 30, and LA 75 decreased cell growth by 48% (p<0.05). EPA and DHA decreased EGFR protein levels in lipid rafts without decreasing whole membrane levels of EGFR. Menendez et al. [153] studied effects of various PUFAs on Her-2 expression in Sk-Br3 and BT-474 breast cancer cell lines that over-express Her-2. They incubated cells for 48 hours with 10 μM alpha linolenic acid (ALA), EPA, DHA, or control bovine serum albumin. In the SK-Br3 cells ALA decreased Her-2 expression by 28%, EPA decreased Her-2 expression by 50%, and DHA decreased Her-2 expression by 41% (p<0.05). Interestingly LA, an ω-6 fatty acid, increased levels of Her-2 by 24% (p<0.05). Menendez et al. [154] studied the effects of DHA on the cytotoxicity of taxanes against Sk-Br3 and BT-474 cells. They found when they combined both DHA and docetaxel with both SK-Br3 and BT-474 cells the resulting effect suggested synergy. Again in these experiments they found that exposure to DHA decreased Her-2 levels. A total of 40 μg/mL DHA exposure for 24 hours decreased Her-2 levels by 78% in BT-474 cells, and by 38% in SK-Br3 cells.

1.3.1.4 Insulin-Like Growth Factors

Although the major effect of insulin-like growth factors (IGF) is stimulation of growth, effects of IGFs on cancer cells were only studied in 1984. Myal et al. [155] studied effects of IGF-I and IGF-II on T-47D breast cancer cells. They found distinct binding sites for IGF-1 and IGF-2 and that insulin bound weakly to these sites. They found that 500 ng/mL of IGF-1 and IGF-2, numbers within the range of physiologic concentrations, stimulated the growth of T-47D breast cancer cells 2.5- and 4-fold, respectively. Relatively little work was done in this area for some time. Law et al. [156] studied effects of insulin-like growth factor receptor-1 (IGF-1R) on breast cancer survival in 438 cases of invasive breast cancer. They found that three breast cancer subtypes (luminal, triple negative, and Her-2) expressed IGF-1R to some extent. Breast cancers with phosphorylated IGF-1R/insulin receptors (IR) had a worse survival rate than breast cancers that did not have phosphorylated IGF-1R/IR (0.046). Breast cancers that expressed IR had a worse survival compared to breast cancers that did not (p=0.009). Fuchs et al. [157] studied effects of IGF-1, IGF-II, and IGF-binding protein 3 (IGFBP-3) levels on time to progression in a first-line chemotherapy study in metastatic colorectal cancer. In a multivariate analysis incorporating IGF-I, IGF-II, and IGRBP-3 and known prognostic factors, increased levels of IGFBP-3 were associated with a longer time to progression (p=0.03). IGFBP-3 binds circulating IGF-1 and thereby antagonizes its effects.

The IGF pathway is one of the more complicated growth factor pathways. There are three ligands, insulin, IGF-1, IGF-II, six binding proteins, and four different receptors including (1) IGF-1R, (2) IGF-2R, (3) insulin receptor A, and (4) insulin receptor B. IGF-2R binds IGF-II but has no activity. IGF-1R and IR are heterotetrameric protein complexes consisting of two extracellular alpha subunits and two beta subunits that span the cellular membrane and are coupled by disulphide bonds [158–160]. Further complicating the understanding of the receptors is that IRA, IRB, and IGF-1R homo- and heterodimerize. The following receptor combinations are therefore possible: IRA/IRA, IRB/IRB, IRA/IRB, IGF-1R/IGF-1R, IGF-1R/IRA, and IGF-1R/IRB [161–164]. Insulin only binds to IRA/IRA, IRB/IRB, and IRA/IRB receptors, whereas IGF-2 binds to homo- and heterodimers of IGF-1R and IRB/IRB [162–164]. IGF-I binds only to hetero- and homodimers of IGF-1R.

Binding of ligands to IGF-1R leads to autophosphorylation of tyrosine residues 1131, 1135, and 1136 in the kinase domain of IGF-1R [165]. Phosphorylating these tyrosines leads to binding sites for insulin receptor substrates (IRS) 1–4, although the major IRS proteins are IRS1 and IRS2. Phosphorylation creates binding sites for SHC which recruits GRB2 and initiates the Ras/MAPK pathway. IRS1 and IRS2 recruitment creates a binding site for the p84 regulatory subunit and for PI3K which recruits the catalytic subunit p110. The assembled PI3K then converts PIP_2 to PIP_3 leading to the activation of Akt/PKB.

1.3.1.5 Effect of ω-3 PUFAs on IGF

Epidemiologic studies have implicated ω-3 PUFAs in having a role in the IGF-1 pathway. Probst-Hensch et al. [166] studied levels of IGF-1 and IGFBP-3 in 312 men and 326 post-menopausal women in Singapore. ω-3 PUFA intake was associated with increased levels of IGFBP-3 (p=0.02); whereas intake of saturated fat was inversely related to IGFBP-3 levels (p=0.03). Bhathena et al. [167] studied effects of supplementing the diet with fish oil on hormones involved in carbohydrate metabolism. They enrolled 40 healthy men in their study, the first 10 weeks their diet was supplemented with 15 g of mixed fat/day, in the second 10 week period the subjects were treated with 15 g of fish oil concentrate/day, and in the last 8 weeks they ingested 200 mg of vitamin E/day. They found that fish oil increased glucose, decreased cholesterol, insulin, and somatomedian-c/IGF-1 (p<0.05). Ghoshal et al. [168] studied insulin-IGFBP-1, which is positively associated with carcinogenesis [169], in rats fed ω-3 or ω-6 PUFAs.

Female Sprague-Dawley rats were randomized to one of four diets: (1) 5% w/w corn oil, (2) 20% w/w corn oil, (3) 18% w/w safflower oil, or (4) 18% w/w menhaden oil. The safflower and menhaden oil diets were supplemented with 3% w/w soybean oil to make up for oils that were deficient in these two diets. Rats fed the 20% corn oil diet had a 6.5 -fold increases in IGFBP-1 compared to rats fed the 5% corn oil diet (p<0.05). Comparing the safflower (ω-6 PUFA)- fed rats to the menhaden (ω-3 PUFA)- fed rats there was a 4.5 -fold increase in IGFBP-1 (p<0.05). Chen et al. [170] studied effects of ω-3 PUFAs on IGF-R in MDA-MB-435 breast cancer cells xenografted into nude mice. Mice were randomized to either a high-fat diet based on the American Institute for Nutrition-93G formulation or a diet supplemented with 10% w/w flaxseed. In mice fed the flaxseed diet, the growth of tumours was decreased (p<0.05) and there were fewer metastases (p<0.05). When tumours were stained there were significant decreases for both IGF-I and EGFR (p<0.05) in rats fed the 10% flaxseed diet compared to rats fed the control diet. Kun et al. [171] studied effects of fish oil on rejection of intestinal transplant in male F344 rats. They found that a diet supplemented with fish oil had decreased levels of IGF receptor compared to rats fed a corn oil supplemented diet (p<0.05). Although only a few studies have examined the effects of ω-3 PUFAs on proteins involved in the

IGF pathway the overall impression is that ω-3 PUFAs decrease members that are associated with carcinogenesis and growth and increase protein members that are associated with negative regulatory effects on the IGF pathway such as IGFBP-3.

1.3.2 Effects of ω-3 PUFAs on Cell Proliferation Signals

1.3.2.1 Cyclins and Associated Proteins

Cellular replication, or the cell cycle, is essentially composed of four distinct phases. G_1 is an initial growth phase that leads to DNA synthesis (S phase), followed by a gap phase (G_2), and finally by mitosis (M phase), the actual segregation of chromosomes [172]. Cells may also exit the cell cycle into G_0 phase, a sustained resting phase that is characteristic of many terminally differentiated cells. Normal cells progress through the cell cycle after stimulation by exogenous agents such as growth factors, hormones, or cytokines [172]. Cancerous cells, however, appear to lose their dependency on these external signals and often progress, unregulated, through many cell cycles [173]. Multiple specific mutations in the genes encoding proteins that regulate cell cycle progression have been identified in tumour cells [173]. Two important families of regulatory molecules promote progression through the cell cycle: the cyclins and the cyclin-dependent kinases (CDKs). Cyclins are proteins whose levels oscillate throughout the cell cycle. Cyclins bind to and activate CDK molecules, such that the cell cycle is regulated primarily by the sequential activation and inactivation of CDKs through the periodic synthesis and destruction of cyclins. Negative regulators of the cell cycle include the retinoblastoma protein (RB), the p53 transcription factor, and the family of CDK inhibitors (CDKIs) consisting of $p21^{WAF1/CIP1}$, p27, p57, and the INK4 proteins. The RB protein inhibits cell proliferation in all tissues [172]. In its dephosphorylated state, it binds and inactivates proteins including transcription factors that are involved in cell cycle progression and DNA synthesis. P53 also induces the expression of members of the CDKI family, namely $p21^{WAF1/CIP1}$.

Studies have demonstrated that ω-3 PUFAs can slow or arrest the growth of cancer cells by affecting cyclins, the proteins that regulate movement of cells through the cell cycle. Albino et al. [174] studied effects of DHA on growth of melanoma cell lines. They studied DHA at concentrations ranging from 0.5 to 5 μg/mL for 3 days and found that DHA inhibited growth in 7 of 12 cells lines, whereas there was no effect on cell growth in 5 cell lines. They then dissected out the effects of DHA on various proteins involved in the cell cycle in one resistant (SK-Mel-110) cell line and one sensitive (SK-Mel-29) cell line. They found that DHA exposure increased the percentage of SK-Mel-110 cells in S phase with no effect on SK-Mel-29 cells. DHA increased phosphorylated RB (pRB) in SK-MEL-110 cells. The proportion of control SK-Mel-100 cells staining for pRB was 40% and this increased to more than 70% with exposure to only 0.5 μg/mL. There was no effect of DHA on pRB in SK-Mel-29 cells or of DHA on SK-Mel-29 cells on cyclin D1, E1, or the cyclin inhibitors $p21^{WAF1}$ or $p27^{KIP1}$. In contrast incubation with DHA increased cyclin D1 by 40% and increased $p27^{KIP1}$ levels by 300% with no effect on cyclin E1 or $p21^{WAF1}$. Similarly Khan et al. [175] found that DHA inhibited growth of the FM3A murine breast cancer cell line by affecting proteins involved in the cell cycle. They found that DHA increased $p27^{KIP1}$ levels and that this was due to stabilization of $p27^{KIP1}$ protein levels through inhibition of MAPK phosphorylation and not an increase in $p27^{KIP1}$ mRNA levels. In contrast to the studies by Albino et al. [174] they found that DHA decreased cyclin E and phosphorylated cyclin -dependent kinase 2 (CDK2) levels in FM3A cells and that DHA decreased pRB levels. Narayanan et al. [127] studied DHA effects on CaCo-2 colon cancer cells using microarrays and found that $p21^{WAF1}$ and $p27^{KIP1}$ levels were increased. Barascu et al. [176] studied effects of EPA and DHA on MDA-MB-231 cells and found that the duration of the G2/M phase was substantially increased by both ω-3 PUFAs (7 sevenfold and, respectively, by 50 μM EPA or DHA). Expressions of cyclin -dependent kinase 1 (CDK1), cyclin A, and cyclin B1, proteins involved in the progression from G2 to M, were all decreased. They further examined effects of PUFAs on these proteins and found that cyclin B1 phosphorylation was decreased. EPA and DHA inhibited growth of the SW1990 pancreatic cancer cell line in a time- and concentration- dependent manner. This effect appeared to be due to increased cyclin E mRNA [177].

In summary, studies have demonstrated that ω-3 PUFAs, particularly DHA, arrest malignant growth, usually in the S phase, and prevent G1/S progression in a variety of cell types (reviewed by [39]). In the cell lines that demonstrated reduced growth with ω-3 PUFAs, the fatty acid treatment appeared to alter the levels of cyclins, cyclin -dependent kinases, and pRB. However, the particular cyclin and related kinase that was affected appeared to differ between cell lines and experimental conditions.

1.3.2.2 TCF–β-Catenin and Effects on VEGF, MMP-7, and MT1-MMP

Many of the previous pathways discussed such as EGFR, Her-2, and IGF-1R share similar features and belong to the receptor tyrosine class of signal transduction. Typically these signalling pathways follow a classical receptor tyrosine kinase → SOS → Ras → Raf → ERK pathway. Another pathway exists for the Wnt ligands; Wnt ligands bind to a membrane receptor called Frizzled which in turn activates Dishevelled. Dishevelled then inhibits glycogen synthase kinase-3β (GSK-3β). GSK-3β is important because it phosphorylates β-catenin which causes its rapid destruction. GSK-3β works in concert with two other proteins Axin and Apc to degrade β-catenin. When GSK-3β is inhibited β-catenin levels rise and it moves to the nucleus where it upregulates several proteins such as myc and cyclin D1. β-catenin is involved in the activation of genes involved in survival and is overexpressed in tumours, particularly those of the colon. Many gene products with transcription activated by β-catenin (VEGF, MMP-7, and MT1-MMP) are implicated in the regulation of cell growth, invasion, and angiogenesis in colorectal cancer [15]. The clinical importance of this pathway is demonstrated by mutations in the Apc gene which result in the hereditary colon cancer syndrome familial adenomatous polyposis coli [178–180]. β-catenin is reported to be a target of ω-3 PUFAs in a number of colon tumour cell lines.

Calviello et al. [181] studied the effects of DHA on β-catenin in SW480 and HCT116 colorectal cancer cell lines. These cell lines were chosen because HCT116 carries an activating mutation for β-catenin and SW480 cells have a mutation in adenomatous polyposis coli. The cell lines were incubated with DHA (2.5–10 μM) for 48 hours. They found significant decreases in β-catenin protein at 24 and 48 hours in both cell lines without a decrease in β-catenin mRNA. They subsequently showed that likely the decrease in β-catenin was due to increased degradation as the proteasome inhibitor MG132 prevented DHA decreases in β-catenin. They found that other proteins regulated by β-catenin such as matrix metalloproteinase (MMP)-7 and membrane type 1(MT-1)-MMP and VEGF were significantly reduced. Narayanan et al. [182] studied the effects of DHA on CaCo-2 colon cancer cell lines. They found by immunofluorescence that DHA decreased cytoplasmic and nuclear levels of β-catenin ($p<0.01$) and total cell β-catenin by 50% by western blot ($p<0.001$). Furthermore products of the β-catenin pathway were reduced such as cyclin D1. Lim et al. [183] studied the effects of DHA and EPA on three human cholangiocarcinoma cell lines CCLP1, HuCCT1, SG231. They found that exposure of the cells to either EPA or DHA inhibited growth whereas exposure to AA had no effect. They found that DHA and EPA caused dephosphorylation of GSK-3B and therefore GSK-3B activation with a decrease in β-catenin and a decrease of proteins regulated by the β-catenin/T-cell factor pathway such as c-MET.

1.3.3 Effects of ω-3 PUFAs on Membrane-Generated Lipid Compounds

1.3.3.1 Eicosanoids

Eicosanoids are widely believed to play a role in carcinogenesis and aggressiveness of cancer especially in colorectal cancer. Jacoby et al. [184] showed in the Min mouse model of adenomatous polyposis that celecoxib, a COX-2 inhibitor, decreased the number and size of adenomatous polyps. The animal studies were confirmed in a large prospective study in patients with a prior history of colorectal adenomatous polyps. In that study 1591 patients were randomized to either celecoxib (n=933) or placebo (n=628). At 3 years 33.6% of celecoxib patients had polyps compared to 49.3% of control patients (RR 0.64, $p<0.001$). At 3 years 5.3% of celecoxib patients had advanced adenomatous polyps compared to 10.4% of control patients (RR 0.49, $p<0.001$) [185]. These results support COX-2 as an important mediator of tumour cell growth.

Habbel et al. [186] studied effects of DHA on AA-induced proliferation of LS-174T human colon cancer cell line. They showed that AA stimulated growth of LS174T cells and that DHA inhibited growth of cells in a dose-dependent manner. Effects of DHA and AA on Bcl-2 and $p21^{WAF1}$ were assessed by RT-PCR. AA induced a 3.5-fold increase in Bcl-2 whereas DHA decreased Bcl-2 by 50%

(P<0.05). In the case of p21^{WAF1} DHA increased expression by 3.5-fold whereas AA had no effect. When LS-174T cells were incubated with 60 μmol/L AA, DHA-suppressed AA stimulated growth in a dose-dependent manner. They went on to study DHA's effects on prostaglandin E_2 (PGE_2) and showed that DHA inhibited AA metabolism to PGE_2 in a dose-dependent manner (p<0.05). Denkins et al. [187] studied the effects of ω-3 PUFAs on the 70 W human melanoma cancer cell line that metastasizes to the brain in murine xenografts. When the 70 W cells were incubated in the presence of 50 μM AA, EPA, or DHA for 24 hours, AA significantly increased COX-2 mRNA levels whereas EPA and DHA decreased COX-2 mRNA levels. These changes in mRNA levels were reflected in production of PGE_2 during incubation of 70 W cells with AA, which resulted in a 50-fold increase in PGE_2 vs. control (p<0.007), 12-fold higher than EPA (p=0.009) and 22-fold higher than DHA (p=0.001). When COX-2 metabolizes ω-3 PUFAs the product is PGE_3 not PGE_2; they therefore studied effects of PGE_2 and PGE_3 on 70 W cell invasiveness and found that PGE_2 significantly increased invasion by 70 W cells where PGE_3 significantly decreased it. Additionally, the tumour-promoting effect of high ω-6 PUFA diets on breast cancer has been correlated with a greater production of eicosanoids [188]. There are many mechanisms that have been studied that explain how ω-3 PUFAs alter eicosanoid metabolism (see earlier section in this chapter for more details). Additionally, PGE_2 has been reported to increase tumour progression by promoting tumour angiogenesis and tumour cell adhesion to endothelial cells [2].

1.3.3.2 ROS and Products of Lipid Peroxidation

ω-3 PUFAs which have multiple double bonds are very susceptible to free radical attack and cause oxidative stress (see earlier section of this chapter). Lipid peroxidation products can inhibit DNA synthesis [189] and cell division [190–192]. Mazière et al. [193] studied the effects of ω-3 and ω-6 PUFAs on cultured human fibroblasts. Control fibroblasts were incubated with 5×10^{-5} M oleic or the same concentration of the respective PUFA. Both ω-3 and ω-6 PUFAs increased the reactive oxygen species and lipid oxidation products; all PUFA treatments increased the transcription factors AP-1 and NFkB as well. Administration of vitamin E or *n*-acetyl-cysteine prevented AA from increasing AP1 and NFkB but they did not test the effects of anti-oxidants against ω-3 PUFAs. Schonberg et al. [194] examined four cancer cell lines with respect to their differential sensitivity to ω-3 PUFAs: the non-small cell lung cancer cell lines A-427 and SK-LU-1 and the glioblastoma cell lines A-172 and U-87 MG. The A-427 cell line was much more sensitive to DHA and EPA than the other three cell lines. Both vitamin E and sodium selenite substantially protected the A-427 cells from DHA's growth inhibition effects, but they also found no difference in lipid peroxidation products. They did find that the A-427 had the lowest levels of glutathione peroxidase of the four cell lines and they postulated that this was a factor in its sensitivity to ω-3 PUFAs. However, not all studies have demonstrated that ROS are the key mechanism for ω-3 PUFAs. Charmras et al. [195] studied the effects of DHA, EPA, AA, and LA on growth of MCF-7 cells. They found that EPA, DHA, and AA inhibited growth but that vitamin E had minimal effects on this inhibition. Similarly Grammatikos et al. [196] also studied effects of ω-3 PUFAs and ω-6 PUFAs and found that ALA, EPA, DHA, and AA inhibited MCF-7 growth and again vitamin E had no effect on ω-3 or ω-6 PUFA inhibition of MCF-7 growth, consistent with our work in MCF-7 and MDA-MB-231 cells [13].

1.3.3.3 Other Membrane-Generated Lipids Involved in Signalling

There is some evidence to suggest that ω-3 PUFAs may alter the formation of phospholipid-derived second messengers such as diacylglycerol (DAG) and ceramide (see earlier sections). DAG, derived from hydrolysis of PIP_2 by PLC enzymes, is the principal activator of PKC, which is involved in a diverse array of cellular responses, including proliferation [197]. Ceramide is believed to mediate anti-proliferative responses such as growth inhibition/cell cycle arrest, differentiation, and senescence in response to certain cytokines or stress-causing agonists [46]. Ceramide modulates components of various signalling pathways (e.g. Akt, phospholipase D, PKC, and MAPKs) by regulating ceramide-activated protein phosphatases, kinases, and proteases [46].

1.3.4 Effects of ω-3 PUFAs on Transcription Factors

ω-3 PUFAs can also have an influence on tumour growth by modulating the expression of genes potentially involved in cell transformation and tumourigenesis. Activation of PPARs is reported to play an important role in the anti-growth effects of ω-3 PUFAs [74]. PPARα is a transcription factor that is associated with cell differentiation and anti-proliferative effects. There are three PPAR isoforms that vary in their affinities for ligands, tissue distribution, and developmental expression [198–201]. More particularly, PPARγ appears to have antineoplastic effects through induction of cell cycle arrest and apoptosis, reprogramming of cell differentiation, and inhibition of angiogenesis [74]. In colon cancer cell lines activating PPARγ decreased cell growth and induced differentiation [202] and had similar effects in breast cancer cell lines [203]. PPARγ's natural occurring ligands are PUFAs such as LA, ALA, AA, and EPA [204, 205]. Metabolism of PUFAs by LOX produces metabolites that are actually more potent activators of PPAR than the parent PUFA. DHA is metabolized by LOX to 17-hydroxy and 7-hydroxy-DHA which are more potent than DHA at activating PPAR (17-OH and 7-OH-DHA PPARγ ED_{50} 5 μM vs. DHA's PPARγ ED_{50} 10–30 μM) [206]. Studies examining the effects of PPAR ligands in cancer cells have generally shown positive effects. Bonofiglio et al. [207] studied the effects of rosiglitazone, a PPARγ agonist, on growth of MCF 7 cells and showed that it inhibited cell growth in addition to inducing apoptosis. They demonstrated that PPARγ bound to NFκB-binding sequence of the p53 gene promoter and increased expression of p53 mRNA and $p21^{WAF1}$. Han et al. [208] demonstrated in H1792 and H1838 non-small cell lung cancer cell lines that rosiglitazone increased phosphatase and tensin homologue expression and decreased phosphorylation of Akt, thereby potently inhibiting the MTOR/Akt pathway. Given that ω-3 PUFAs and their LOX products are potent ligands for PPARγ it is very likely that some effects of PUFAs are through activation of PPARγ.

Another transcription factor affected by ω-3 PUFAs is the sterol regulatory element-binding proteins (SREPBs), the most common form being SREBP1 which regulates fatty acid and cholesterol biosynthesis [209]. In normal vascular endothelial cells downregulation of SREBP1 either by siRNA or 25-hydroxycholesterol suppresses DNA synthesis and leads to G_0/G_1 arrest [210]. Field et al. [211] studied the effects of PUFAs on SREBP-1 in CaCo-2 colon adenocarcinoma cells and found that EPA and DHA suppressed SREBP-1, acetyl-CoA carboxylase, and fatty acid synthase expression. Similarly Schonberg et al. [212] studied the effects of AA, EPA, and DHA on the human colon cancer cell lines SW480 and SW620. The SW480 and SW620 cell lines were both from the same patient but SW480 was from the primary tumour and SW620 was from a metastasis. DHA had the greatest growth inhibitory effects on both cell lines and led to significant accumulation of DHA in cholesterol esters in SW620 cells and in triglycerides in the SW480 cells. Both SREBP-1 and acyl CoA:cholesterol acyltransferase-1 (ACAT1) expressions were decreased by exposure to DHA.

1.3.5 Effects of ω-3 PUFAs on Ras and Protein Kinase C

Ras is a family of genes encoding small GTPases that are involved in cellular signal transduction. Activation of Ras signalling causes cell growth, differentiation, and survival. Singh et al. [213] studied effects of fish oil on tumour formation and Ras in the azoxymethane (AOM)-induced model of colon cancer in male F344 rats. The control group of rats received saline injections and low-fat corn oil diet. The experimental groups were: (1) low-fat corn oil +AOM, (2) high-fat corn oil + AOM, and (3) high-fat fish oil + AOM. They found that the percentage of animals with colon cancer was 57, 76, and 40% for the low-fat corn oil, high-fat corn oil, and high-fat fish oil diets, respectively. They found that levels of total Ras (H, N, and K-Ras) were significantly lower in rats fed fish oil; the amount of membrane-bound ras was 24, 33, and 12 ng/mg of protein for low-fat corn oil, high-fat corn oil, and high-fat fish oil diets, respectively. Davidson et al. [214] performed a similar set of experiments in male Sprougue-Dawley rats, using the AOM-induced model of colon cancer. Rats were fed either a diet containing either corn oil or fish oil. Rats fed corn oil had a 13% higher expression of Ras in membranes than rats fed fish oil ($p < 0.05$). Collett et al. [215] studied effects of pure DHA and LA on Ras localization and activity based on their previous work which utilized fish oil and corn oil. They exposed YAMC colon cells to either 0–100 μM DHA or LA for 72 hours. They found that DHA significantly decreased the number of cells and that this appeared to be due to suppression of growth. Neither DHA nor LA decreased total Ras levels compared to the control but DHA decreased the ratio of membrane-bound Ras to cytosolic Ras compared to control ($p<0.05$), whereas LA increased the ratio of membrane-bound Ras to cytosolic Ras ($p<0.05$). Given that Ras is only activated when bound to GTP, DHA decreased the proportion of GTP-bound Ras to

GDP-bound Ras when compared to LA treated and untreated cells ($p<0.05$). Ma et al. [90] studied the effects of ω-3 PUFAs on Ras in YAMC cells, male Sprague-Dawley rats, and C57BL/6 mice. They showed that H-Ras localized to caveolae in YAMC cells and mice and rat colon cells. For K-Ras there was no localization to caveolae in either mice colon cells or YAMC cells, but there was localization to caveolae in rat colon cells. Caveolin-1 expression in caveolae was decreased by 53% in mice fed the ω-3 PUFA diet compared to mice fed the ω-6 PUFA diet. Caveolin-1 expression was not decreased in total cell lysates. H-Ras expression in caveolae was decreased in mice fed the ω-3 PUFA diet, likely as a result of reduced caveolin-1 expression in caveolae; H-Ras levels were 74 ± 15 ng/mg protein in ω-3 PUFA mice compared to 134 ± 29 ng/mg protein in ω-6 PUFA mice. Furthermore, decreased levels of H-Ras decreased activation of Raf, the next protein in the MAPK pathway ($p<0.05$). Recently it was demonstrated that Ras plasma membrane targeting was inhibited only when DHA was sufficiently enriched in membranes to substantially increase membrane unsaturation [216], suggesting the mechanism by which ω-3 PUFAs inhibit Ras.

PKC has been implicated in carcinogenesis, especially in colorectal cancer [217–219]. ω-3 PUFAs have been found to suppress expression of PKC in both normal and cancerous tissues. Davidson et al. [219, 220] studied effects of PUFAs and fibre on colon cancer development in the AOM model of colon cancer in Sprague-Dawley rats in a $2 \times 2 \times 2$ factorial design. They found that ω-3 PUFAs decreased expression of PKC β_{II} in colon cancers compared to colon cancers induced in rats fed ω-6 PUFAs ($p<0.05$). Murray et al. [221] studied effects of ω-3 and ω-6 dietary PUFAs on carcinogenesis in transgenic mice expressing increased levels of PKC β_{II}. They found that animals fed the ω-6 PUFA diet had increased number of aberrant crypts ($p=0.009$) and increased proliferation index ($p=0.03$). ω-3 PUFAs decreased the amount of PKC found at the membrane compared to rats fed ω-6 PUFAs ($p<0.05$). They went on to develop rat intestinal epithelial cells that stably expressed PKC and analysed gene expression of the original cells and the transfected cells using microarray. They found that cells that over-expressed PKC had repressed levels of transforming growth factor-β.

1.3.6 Summary of the Effects of ω-3 PUFAs on Tumour Cell Growth

In summary, many possible mechanisms whereby ω-3 PUFAs may decrease tumour cell growth have been described in this section. These include decreasing key growth receptors and signals; altering the level of cyclins, cyclin-dependent kinases, and pRB; and modulating lipid-derived signals and transcription factors needed for cell growth. Further studies are needed to identify new mechanisms and to evaluate and verify these mechanisms in humans to gain more understanding of the effects of and many mechanisms by which ω-3 PUFAs alter tumour growth and death in human cancers.

1.4 Conclusions

In summary, the biologically favourable effects of the ω-3 PUFAs, EPA and DHA, are likely mediated through effects on many different pathways within tumours. These include directly changing the composition of cell membranes and membrane function, activating or suppressing signalling molecules, interacting with DNA as well as with proteins that affect the processing of transcription factors, and altering the activity of key enzymes. Some of the effects of ω-3 PUFAs may not be directly related to the fatty acid molecule itself but rather to their metabolites, such as ROS, DAG, ceramide, and eicosanoids. It is likely that a combination of these effects is responsible for the anti-tumour effects of ω-3 PUFAs. However, despite the accumulating evidence in human tumour cells and animal cancer models that ω-3 PUFAs modulate cell death and growth, at present, the cellular and molecular mechanisms are unclear. The challenge for the future researcher is to obtain a better understanding of the basic actions of ω-3 PUFAs so these lipids can be employed in the clinic as chemopreventive agents or adjuvants to current cancer therapies.

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References

1. Calviello G, Serini S, Piccioni E. N-3 polyunsaturated fatty acids and the prevention of colorectal cancer: molecular mechanisms involved. *Curr Med Chem*. 2007; 14:3059–3069.
2. Larsson SC, Kumlin M, Ingelman-Sundberg M, Wolk A. Dietary long-chain n-3 fatty acids for the prevention of cancer: a review of potential mechanisms. *Am J Clin Nutr*. 2004; 79:935–945.
3. De CR, Massaro M. Omega-3 fatty acids and the regulation of expression of endothelial pro-atherogenic and pro-inflammatory genes. *J Membr Biol*. 2005; 206:103–116.
4. Berquin IM, Edwards IJ, Chen YQ. Multi-targeted therapy of cancer by omega-3 fatty acids. *Cancer Lett*. 2008; 269:363–377.
5. Chapkin RS, Seo J, McMurray DN, Lupton JR. Mechanisms by which docosahexaenoic acid and related fatty acids reduce colon cancer risk and inflammatory disorders of the intestine. *Chem Phys Lipids*. 2008; 153:14–23.
6. Okada H, Mak TW. Pathways of apoptotic and non-apoptotic death in tumour cells. *Nat Rev Cancer*. 2004; 4:592–603.
7. Hengartner MO. The biochemistry of apoptosis. *Nature*. 2000; 407:770–776.
8. Lavrik I, Golks A, Krammer PH. Death receptor signaling. *J Cell Sci*. 2005; 118:265–267.
9. Simstein R, Burow M, Parker A, Weldon C, Beckman B. Apoptosis, chemoresistance, and breast cancer: insights from the MCF-7 cell model system. *Exp Biol Med*. 2003; 228:995–1003.
10. Green DR. Apoptotic pathways: ten minutes to dead. *Cell*. 2005; 121:671–674.
11. Yamamoto D, Kiyozuka Y, Adachi Y, Takada H, Hioki K, Tsubura A. Synergistic action of apoptosis induced by eicosapentaenoic acid and TNP-470 on human breast cancer cells. *Bst Cancer Res Treat*. 1999; 55:149–160.
12. Tsujita-Kyutoku M, Yuri T, Danbara N, Senzaki H, Kiyozuka Y, Uehara N et al. Conjugated docosahexaenoic acid suppresses KPL-1 human breast cancer cell growth in vitro and in vivo: potential mechanisms of action. *Breast Cancer Res*. 2004; 6:R291–R299.

13. Schley PD, Jijon HB, Robinson LE, Field CJ. Mechanisms of omega-3 fatty acid-induced growth inhibition in MDA-MB-231 human breast cancer cells. *Breast Cancer Res Treat*. 2005; 92:187–195.
14. Hilakivi-Clarke L, Olivo SE, Shajahan A, Khan G, Zhu Y, Zwart A et al. Mechanisms mediating the effects of prepubertal (n-3) polyunsaturated fatty acid diet on breast cancer risk in rats. *J Nutr*. 2005; 135:2946S–2952S.
15. Calviello G, Resci F, Serini S, Piccioni E, Toesca A, Boninsegna A et al. Docosahexaenoic acid induces proteasome-dependent degradation of beta-catenin, down-regulation of survivin and apoptosis in human colorectal cancer cells not expressing COX-2. *Carcinogenesis*. 2007; 28:1202–1209.
16. Narayanan BA, Narayanan NK, Reddy BS. Docosahexaenoic acid regulated genes and transcription factors inducing apoptosis in human colon cancer cells. *Int J Oncol*. 2001; 19:1255–1262.
17. Latham P, Lund EK, Brown JC, Johnson IT. Effects of cellular redox balance on induction of apoptosis by eicosapentaenoic acid in HT29 colorectal adenocarcinoma cells and rat colon in vivo. *Gut*. 2001; 49:97–105.
18. Chen ZY, Istfan NW. Docosahexaenoic acid is a potent inducer of apoptosis in HT-29 colon cancer cells. *Prost Leuk Ess Fatty Acids*. 2000; 63:301–308.
19. Hong MY, Lupton JR, Morris JS, Wang N, Carroll RJ, Davidson LA et al. Dietary fish oil reduces O6-methylguanine DNA adduct levels in rat colon in part by increasing apoptosis during tumor initiation. *Cancer Epidemiol Biomarkers Prev*. 2000; 9:819–826.
20. Hong MY, Chapkin RS, Barhoumi R, Burghardt RC, Turner ND, Henderson CE et al. Fish oil increases mitochondrial phospholipid unsaturation, upregulating reactive oxygen species and apoptosis in rat colonocytes. *Carcinogenesis*. 2002; 23:1919–1925.
21. Serini S, Trombino S, Oliva F, Piccioni E, Monego G, Resci F et al. Docosahexaenoic acid induces apoptosis in lung cancer cells by increasing MKP-1 and down-regulating p-ERK1/2 and p-p38 expression. *Apoptosis*. 2008; 13:1172–1183.
22. Mannini A, Kerstin N, Calorini L, Mugnai G, Ruggieri S. An enhanced apoptosis and a reduced angiogenesis are associated with the inhibition of lung colonisation in animals fed an n-3 polyunsaturated fatty acid-rich diet injected with a highly metastatic murine melanoma line. *Br J Nutr*. 2009; 101:688–693.
23. Edwards IJ, Sun H, Hu Y, Berquin IM, O'Flaherty JT, Cline JM et al. In vivo and in vitro regulation of syndecan 1 in prostate cells by n-3 polyunsaturated fatty acids. *J Biol Chem*. 2008; 283:18441–18449.
24. Narayanan NK, Narayanan BA, Reddy BS. A combination of docosahexaenoic acid and celecoxib prevents prostate cancer cell growth in vitro and is associated with modulation of nuclear factor-kappaB, and steroid hormone receptors. *Int J Oncol*. 2005; 26:785–792.
25. Heimli H, Finstad HS, Drevon CA. Necrosis and apoptosis in lymphoma cell lines exposed to eicosapentaenoic acid and antioxidants. *Lipids*. 2001; 36:613–621.
26. Chiu LC, Wan JM. Induction of apoptosis in HL-60 cells by eicosapentaenoic acid (EPA) is associated with downregulation of bcl-2 expression. *Cancer Lett*. 1999; 145:17–27.
27. Chiu LC, Wan JM, Ooi VE. Induction of apoptosis by dietary polyunsaturated fatty acids in human leukemic cells is not associated with DNA fragmentation. *Int J Oncol*. 2000; 17:789–796.
28. Calviello G, Palozza P, Piccioni E, Maggiano N, Frattucci A, Franceschelli P et al. Dietary supplementation with eicosapentaenoic and docosahexaenoic acid inhibits growth of Morris hepatocarcinoma 3924A in rats: effects on proliferation and apoptosis. *Int J Cancer*. 1998; 75:699–705.
29. Zhang W, Long Y, Zhang J, Wang C. Modulatory effects of EPA and DHA on proliferation and apoptosis of pancreatic cancer cells. *J Huazhong Univ Sci Technolog Med Sci*. 2007; 27:547–550.
30. Hawkins RA, Sangster K, Arends MJ. Apoptotic death of pancreatic cancer cells induced by polyunsaturated fatty acids varies with double bond number and involves an oxidative mechanism. *J Pathol*. 1998; 185:61–70.

31. Lai PBS, Ross JA, Fearon KCH, Anderson JD, Carter DC. Cell cycle arrest and induction of apoptosis in pancreatic cancer cells exposed to eicosapentaenoic acid in vitro. *Br J Cancer*. 1996; 74:1375–1383.
32. Merendino N, Loppi B, D'Aquino M, Molinari R, Pessina G, Romano C et al. Docosahexaenoic acid induces apoptosis in the human PaCa-44 pancreatic cancer cell line by active reduced glutathione extrusion and lipid peroxidation. *Nutr Cancer*. 2005; 52:225–233.
33. Colquhoun A. Induction of apoptosis by polyunsaturated fatty acids and its relationship to fatty acid inhibition of carnitine palmitoyltransferase I activity in Hep2 cells. *Biochem Mol Biol Int*. 1998; 45:331–336.
34. Serini S, Piccioni E, Merendino N, Calviello G. Dietary polyunsaturated fatty acids as inducers of apoptosis: implications for cancer. *Apoptosis*. 2009; 14:135–152.
35. Cheng J, Ogawa K, Kuriki K, Yokoyama Y, Kamiya T, Seno K et al. Increased intake of n-3 polyunsaturated fatty acids elevates the level of apoptosis in the normal sigmoid colon of patients polypectomized for adenomas/tumors. *Cancer Lett*. 2003; 193:17–24.
36. Courtney ED, Matthews S, Finlayson C, Di PD, Belluzzi A, Roda E et al. Eicosapentaenoic acid (EPA) reduces crypt cell proliferation and increases apoptosis in normal colonic mucosa in subjects with a history of colorectal adenomas. *Int J Colorectal Dis*. 2007; 22:765–776.
37. Pfrommer CA, Erl W, Weber PC. Docosahexaenoic acid induces ciap1 mRNA and protects human endothelial cells from stress-induced apoptosis. *Am J Physiol Heart Circ Physiol*. 2006; 290:H2178–H2186.
38. Kim HJ, Vosseler CA, Weber PC, Erl W. Docosahexaenoic acid induces apoptosis in proliferating human endothelial cells. *J Cell Physiol*. 2005; 204:881–888.
39. Siddiqui RA, Shaikh SR, Sech LA, Yount HR, Stillwell W, Zaloga GP. Omega 3-fatty acids: health benefits and cellular mechanisms of action. *Mini Rev Med Chem*. 2004; 4:859–871.
40. Fulda S, Debatin KM. Extrinsic versus intrinsic apoptosis pathways in anticancer chemotherapy. *Oncogene*. 2006; 25:4798–4811.
41. Carpinteiro A, Dumitru C, Schenck M, Gulbins E. Ceramide-induced cell death in malignant cells. *Cancer Lett*. 2008; 264:1–10.
42. Saelens X, Festjens N, Vande WL, van GM, Van LG, Vandenabeele P. Toxic proteins released from mitochondria in cell death. *Oncogene*. 2004; 23:2861–2874.
43. Degterev A, Boyce M, Yuan J. A decade of caspases. *Oncogene*. 2003; 22:8543–8567.
44. Fulda S. Tumor resistance to apoptosis. *Int J Cancer*. 2009; 124:511–515.
45. Reed JC. Apoptosis-targeted therapies for cancer. *Cancer Cell*. 2003; 3:17–22.
46. Ogretmen B, Hannun YA. Biologically active sphingolipids in cancer pathogenesis and treatment. *Nat Rev Cancer*. 2004; 4:604–616.
47. Debatin KM, Stahnke K, Fulda S. Apoptosis in hematological disorders. *Semin Cancer Biol*. 2003; 13:149–158.
48. Roth W, Isenmann S, Nakamura M, Platten M, Wick W, Kleihues P et al. Soluble decoy receptor 3 is expressed by malignant gliomas and suppresses CD95 ligand-induced apoptosis and chemotaxis. *Cancer Res*. 2001; 61:2759–2765.
49. Friesen C, Fulda S, Debatin KM. Deficient activation of the CD95 (APO-1/Fas) system in drug-resistant cells. *Leukemia*. 1997; 11:1833–1841.
50. Fulda S, Scaffidi C, Susin SA, Krammer PH, Kroemer G, Peter ME et al. Activation of mitochondria and release of mitochondrial apoptogenic factors by betulinic acid. *J Biol Chem*. 1998; 273:33942–33948.
51. Biondo PD, Brindley DN, Sawyer MB, Field CJ. The potential for treatment with dietary long-chain polyunsaturated n-3 fatty acids during chemotherapy. *J Nutr Biochem*. 2008; 19:787–796.
52. Sauer LA, Blask DE, Dauchy RT. Dietary factors and growth and metabolism in experimental tumors. *J Nutr Biochem*. 2007; 18:637–649.
53. Clandinin MT, Cheema S, Field CJ, Garg ML, Venkatraman J, Clandinin TR. Dietary fat: exogenous determination of membrane structure and cell function. *FASEB J*. 1991; 5:2761–2769.

54. Robinson LE, Clandinin MT, Field CJ. The role of dietary long-chain n-3 fatty acids in anticancer immune defense and R3230AC mammary tumor growth in rats: influence of diet fat composition. Breast Cancer Res Treat. 2002; 73:145–160.
55. Rose DP, Connolly JM, Rayburn J, Coleman M. Influence of diets containing eicosapentaenoic or docosahexaenoic acid on growth and metastasis of breast cancer cell in nude mice. *J Natl Cancer Inst*. 1995; 87:587–592.
56. Chapkin RS, Hong MY, Fan YY, Davidson LA, Sanders LM, Henderson CE et al. Dietary n-3 PUFA alter colonocyte mitochondrial membrane composition and function. *Lipids*. 2002; 37:193–199.
57. Ng Y, Barhoumi R, Tjalkens RB, Fan YY, Kolar S, Wang N et al. The role of docosahexaenoic acid in mediating mitochondrial membrane lipid oxidation and apoptosis in colonocytes. *Carcinogenesis*. 2005; 26:1914–1921.
58. Schley PD, Brindley DN, Field CJ. (n-3) PUFA alter raft lipid composition and decrease epidermal growth factor receptor levels in lipid rafts of human breast cancer cells. *J Nutr*. 2007; 137:548–553.
59. Finstad HS, Dyrendal H, Myhrstad MC, Heimli H, Drevon CA. Uptake and activation of eicosapentaenoic acid are related to accumulation of triacylglycerol in Ramos cells dying from apoptosis. *J Lipid Res*. 2000; 41:554–563.
60. Stillwell W, Jenski LJ, Crump FT, Ehringer W. Effect of docosahexaenoic acid on mouse mitochondrial membrane properties. *Lipids*. 1997; 32:497–506.
61. Ehringer W, Belcher D, Wassall SR, Stillwell W. A comparison of the effects of linolenic (18:3 omega 3) and docosahexaenoic (22:6 omega 3) acids on phospholipid bilayers. *Chem Phys Lipids*. 1990; 54:79–88.
62. Ma DW, Seo J, Switzer KC, Fan YY, McMurray DN, Lupton JR et al. N-3 PUFA and membrane microdomains: a new frontier in bioactive lipid research. *J Nutr Biochem*. 2004; 15:700–706.
63. Shaikh SR, Dumaual AC, Jenski LJ, Stillwell W. Lipid phase separation in phospholipid bilayers and monolayers modeling the plasma membrane. *Biochim Biophys Acta*. 2001; 1512:317–328.
64. Halliwell B, Chirico S. Lipid peroxidation: its mechanism, measurement, and significance. *Am J Clin Nutr*. 1993; 57(suppl):715S–725S.
65. Das UN. Essential fatty acids, lipid peroxidation and apoptosis. *Prost Leuk Ess Fatty Acids*. 1999; 61:157–163.
66. Nomura K, Imai H, Koumura T, Arai M, Nakagawa Y. Mitochondrial phospholipid hydroperoxide glutathione peroxidase suppresses apoptosis mediated by a mitochondrial death pathway. *J Biol Chem*. 1999; 274:29294–29302.
67. Wang TG, Gotoh Y, Jennings MH, Rhoads CA, Aw TY. Lipid hydroperoxide-induced apoptosis in human colonic CaCo-2 cells is associated with an early loss of cellular redox balance. *FASEB J*. 2000; 14:1567–1576.
68. Davis RJ. Signal transduction by the JNK group of MAP kinases. *Cell*. 2000; 103:239–252.
69. Karin M, Cao Y, Greten FR, Li ZW. NF-kappaB in cancer: from innocent bystander to major culprit. *Nat Rev Cancer*. 2002; 2:301–310.
70. Gonzalez MJ, Schemmel RA, Dugan LJr, Gray JI, Welsch CW. Dietary fish oil inhibits human breast carcinoma growth: a function of increased lipid peroxidation. *Lipids*. 1993; 28:827–832.
71. Masotti L, Casali E, Galeotti T. Lipid peroxidation in tumour cells. *Free Rad Biol Med*. 1988; 4:377–386.
72. Maziere C, Conte MA, Degonville J, Ali D, Maziere JC. Cellular enrichment with polyunsaturated fatty acids induces an oxidative stress and activates the transcription factors AP1 and NFkappaB. *Biochem Biophys Res Commun*. 1999; 265:116–122.
73. Dommels YE, Haring MM, Keestra NG, Alink GM, van Bladeren PJ, van OB. The role of cyclooxygenase in n-6 and n-3 polyunsaturated fatty acid mediated effects on cell proliferation, PGE(2) synthesis and cytotoxicity in human colorectal carcinoma cell lines. *Carcinogenesis*. 2003; 24:385–392.

74. Dupertuis YM, Meguid MM, Pichard C. Colon cancer therapy: new perspectives of nutritional manipulations using polyunsaturated fatty acids. *Curr Opin Clin Nutr Metab Care*. 2007; 10:427–432.
75. Ding WQ, Lind SE. Phospholipid hydroperoxide glutathione peroxidase plays a role in protecting cancer cells from docosahexaenoic acid-induced cytotoxicity. *Mol Cancer Ther*. 2007; 6:1467–1474.
76. Begin ME, Ells G, Horrobin DF. Polyunsaturated fatty acid-induced cytotoxicity against tumor cells and its relationship to lipid peroxidation. *J Natl Cancer Inst*. 1988; 80:188–194.
77. Chajes V, Sattler W, Stranzl A, Kostner GM. Influence of n-3 fatty acids on the growth of human breast cancer cells in vitro: relationship to peroxides and vitamin-E. *Bst Cancer Res Treat*. 1995; 34:199–212.
78. Cognault S, Jourdan ML, Germain E, Pitavy R, Morel E, Durand G et al. Effect of an αalinolenic acid-rich diet on rat mammary tumor growth depends on the dietary oxidative status. *Nutr Cancer*. 2000; 36:33–41.
79. Hofmanova J, Vaculova A, Kozubik A. Polyunsaturated fatty acids sensitize human colon adenocarcinoma HT-29 cells to death receptor-mediated apoptosis. *Cancer Lett*. 2005; 218:33–41.
80. Boudreau MD, Sohn KH, Rhee SH, Lee SW, Hunt JD, Hwang DH. Suppression of tumor cell growth both in nude mice and in culture by n-3 polyunsaturated fatty acids: mediation through cyclooxygenase-independent pathways. *Cancer Res*. 2001; 61:1386–1391.
81. Watkins SM, Carter LC, German JB. Docosahexaenoic acid accumulates in cardiolipin and enhances HT-29 cell oxidant production. *J Lipid Res*. 1998; 39:1583–1588.
82. Schlame M, Ren M. The role of cardiolipin in the structural organization of mitochondrial membranes. *Biochim Biophys Acta*. 2009; 1788:2080–2083.
83. Jemmerson R, Liu J, Hausauer D, Lam KP, Mondino A, Nelson RD. A conformational change in cytochrome c of apoptotic and necrotic cells is detected by monoclonal antibody binding and mimicked by association of the native antigen with synthetic phospholipid vesicles. *Biochemistry*. 1999; 38:3599–3609.
84. Bayir H, Fadeel B, Palladino MJ, Witasp E, Kurnikov IV, Tyurina YY et al. Apoptotic interactions of cytochrome c: redox flirting with anionic phospholipids within and outside of mitochondria. *Biochim Biophys Acta*. 2006; 1757:648–659.
85. Fisher M, Levine PH, Weiner BH, Johnson MH, Doyle EM, Ellis PA et al. Dietary n-3 fatty acid supplementation reduces superoxide production and chemiluminescence in a monocyte-enriched preparation of leukocytes. *Am J Clin Nutr*. 1990; 51:804–808.
86. Serhan CN, Hong S, Gronert K, Colgan SP, Devchand PR, Mirick G et al. Resolvins: a family of bioactive products of omega-3 fatty acid transformation circuits initiated by aspirin treatment that counter proinflammation signals. *J Exp Med*. 2002; 196:1025–1037.
87. Brown DA, London E. Functions of lipid rafts in biological membranes. *Annu Rev Cell Dev Biol*. 1998; 14:111–136.
88. Simons K, Toomre D. Lipid rafts and signal transduction. *Nat Rev Mol Cell Biol*. 2000; 1:31–39.
89. Fan YY, McMurray DN, Ly LH, Chapkin RS. Dietary (n-3) Polyunsaturated Fatty Acids Remodel Mouse T-Cell Lipid Rafts. *J Nutr*. 2003; 133:1913.
90. Ma DW, Seo J, Davidson LA, Callaway ES, Fan YY, Lupton JR et al. N-3 PUFA alter caveolae lipid composition and resident protein localization in mouse colon. *FASEB J*. 2004; 18:1040–1042.
91. Stulnig TM, Berger M, Sigmund T, Raederstorff D, Stockinger H, Waldhausl W. Polyunsaturated fatty acids inhibit T cell signal transduction by modification of detergent-insoluble membrane domains. *J Cell Biol*. 1998; 143:637–644.
92. Stulnig TM, Huber J, Leitinger N, Imre EM, Angelisova P, Nowotny P et al. Polyunsaturated eicosapentaenoic acid displaces proteins from membrane rafts by altering raft lipid composition. *J Biol Chem*. 2001; 276:37335–37340.

93. Shaikh SR, Edidin M. Polyunsaturated fatty acids, membrane organization, T cells, and antigen presentation. *Am J Clin Nutr*. 2006; 84:1277–1289.
94. Hulbert AJ, Turner N, Storlien LH, Else PL. Dietary fats and membrane function: implications for metabolism and disease. *Biol Rev Camb Philos Soc*. 2005; 80:155–169.
95. Sampath H, Ntambi JM. Polyunsaturated fatty acid regulation of gene expression. *Nutr Rev*. 2004; 62:333–339.
96. Deckelbaum RJ, Worgall TS, Seo T. N-3 fatty acids and gene expression. *Am J Clin Nutr*. 2006; 83:1520S–1525S.
97. Hong MY, Chapkin RS, Davidson LA, Turner ND, Morris JS, Carroll RJ et al. Fish oil enhances targeted apoptosis during colon tumor initiation in part by downregulating Bcl-2. *Nutr Cancer*. 2003; 46:44–51.
98. Berquin IM, Min Y, Wu R, Wu J, Perry D, Cline JM et al. Modulation of prostate cancer genetic risk by omega-3 and omega-6 fatty acids. *J Clin Invest*. 2007; 117:1866–1875.
99. Collett ED, Davidson LA, Fan YY, Lupton JR, Chapkin RS. n-6 and n-3 polyunsaturated fatty acids differentially modulate oncogenic Ras activation in colonocytes. *Am J Physiol Cell Physiol*. 2001; 280:C1066–C1075.
100. Davidson LA, Lupton JR, Jiang YH, Chapkin RS. Carcinogen and dietary lipid regulate ras expression and localization in rat colon without affecting farnesylation kinetics. *Carcinogenesis*. 1999; 20:785–791.
101. Heimli H, Giske C, Naderi S, Drevon CA, Hollung K. Eicosapentaenoic acid promotes apoptosis in Ramos cells via activation of caspase-3 and -9. *Lipids*. 2002; 37:797–802.
102. Arita K, Kobuchi H, Utsumi T, Takehara Y, Akiyama J, Horton AA et al. Mechanism of apoptosis in HL-60 cells induced by n-3 and n-6 polyunsaturated fatty acids. *Biochem Pharmacol*. 2001; 62:821–828.
103. Fuchs SY, Ougolkov AV, Spiegelman VS, Minamoto T. Oncogenic beta-catenin signaling networks in colorectal cancer. *Cell Cycle*. 2005; 4:1522–1539.
104. Gulbins E, Kolesnick R. Raft ceramide in molecular medicine. *Oncogene*. 2003; 22:7070–7077.
105. Kim WH, Kang KH, Kim MY, Choi KH. Induction of p53-independent p21 during ceramide-induced G1 arrest in human hepatocarcinoma cells. *Biochem Cell Biol*. 2000; 78:127–135.
106. Hellin AC, Bentires-Alj M, Verlaet M, Benoit V, Gielen J, Bours V et al. Roles of nuclear factor-kappaB, p53, and p21/WAF1 in daunomycin-induced cell cycle arrest and apoptosis. *J Pharmacol Exp Ther*. 2000; 295:870–878.
107. Wu M, Harvey KA, Ruzmetov N, Welch ZR, Sech L, Jackson K et al. Omega-3 polyunsaturated fatty acids attenuate breast cancer growth through activation of a neutral sphingomyelinase-mediated pathway. *Int J Cancer*. 2005; 117:340–348.
108. Siddiqui RA, Jenski LJ, Harvey KA, Wiesehan JD, Stillwell W, Zaloga GP. Cell-cycle arrest in Jurkat leukaemic cells: a possible role for docosahexaenoic acid. *Biochem J*. 2003; 371:621–629.
109. Fowler KH, McMurray DN, Fan Y-Y, Aukema HM, Chapkin RS. Purified dietary n-3 polyunsaturated fatty acids alter diacylglycerol mass and molecular species composition in concanavalin A-stimulated murine splenocytes. *Biochim Biophys Acta*. 1993; 1210:89–96.
110. Jolly CA, Jiang Y-H, Chapkin RS, McMurray DN. Dietary (n-3) polyunsaturated fatty acids suppress murine lymphoproliferation, interleukin-2 secretion, and the formation of diacylglycerol and ceramide. *J Nutr*. 1997; 127:37–43.
111. McMurray DN, Jolly CA, Chapkin RS. Effects of dietary n-3 fatty acids on T cell activation and T cell receptor-mediated signaling in a murine model. *J Infect Dis*. 2000; 182 Suppl 1:S103–S107.
112. Funk CD. Prostaglandins and leukotrienes: advances in eicosanoid biology. *Science*. 2001; 294:1871–1875.
113. Samuelsson B, Dahlen SE, Lindgren JA, Rouzer CA, Serhan CN. Leukotrienes and lipoxins: structures, biosynthesis, and biological effects. *Science*. 1987; 237:1171–1176.

114. Sheng H, Shao J, Morrow JD, Beauchamp RD, Dubois RN. Modulation of apoptosis and Bcl-2 expression by prostaglandin E2 in human colon cancer cells. *Cancer Res*. 1998; 58:362–366.
115. Chapkin RS, Akoh CC, Miller CC. Influence of dietary n-3 fatty acids on macrophage glycerophospholipid molecular species and peptidoleukotriene synthesis. *J Lipid Res*. 1991; 32:1205–1213.
116. Stillwell W, Wassall SR. Docosahexaenoic acid: membrane properties of a unique fatty acid. *Chem Phys Lipids*. 2003; 126:1–27.
117. Lands WE. Biochemistry and physiology of n-3 fatty acids. *FASEB J*. 1992; 6:2530–2536.
118. Trifan OC, Hla T. Cyclooxygenase-2 modulates cellular growth and promotes tumorigenesis. *J Cell Mol Med*. 2003; 7:207–222.
119. Hamid R, Singh J, Reddy BS, Cohen LA. Inhibition by dietary menhaden oil of cyclooxygenase-1 and -2 in N-nitrosomethylurea-induced rat mammary tumors. *Int J Oncol*. 1999; 14:523–528.
120. Narayanan BA, Narayanan NK, Desai D, Pittman B, Reddy BS. Effects of a combination of docosahexaenoic acid and 1,4-phenylene bis(methylene) selenocyanate on cyclooxygenase 2, inducible nitric oxide synthase and beta-catenin pathways in colon cancer cells. *Carcinogenesis*. 2004; 25:2443–2449.
121. Kliewer SA, Lehmann JM, Willson TM. Orphan nuclear receptors: shifting endocrinology into reverse. *Science*. 1999; 284:757–760.
122. Yaqoob P. Lipids and the immune response: from molecular mechanisms to clinical applications. *Curr Opin Clin Nutr Metab Care*. 2003; 6:133–150.
123. Kliewer SA, Willson TM. The nuclear receptor PPARγ – bigger than fat. *Curr Opin Genet Dev*. 1998; 8:576–581.
124. Hostetler HA, Petrescu AD, Kier AB, Schroeder F. Peroxisome proliferator-activated receptor alpha interacts with high affinity and is conformationally responsive to endogenous ligands. *J Biol Chem*. 2005; 280:18667–18682.
125. Fan YY, Spencer TE, Wang N, Moyer MP, Chapkin RS. Chemopreventive n-3 fatty acids activate RXRalpha in colonocytes. *Carcinogenesis*. 2003; 24:1541–1548.
126. Nicholson KM, Anderson NG. The protein kinase B/Akt signalling pathway in human malignancy. *Cell Signal*. 2002; 14:381–395.
127. Narayanan BA, Narayanan NK, Simi B, Reddy BS. Modulation of inducible nitric oxide synthase and related proinflammatory genes by the omega-3 fatty acid docosahexaenoic acid in human colon cancer cells. *Cancer Res*. 2003; 63:972–979.
128. Lee JY, Ye J, Gao Z, Youn HS, Lee WH, Zhao L et al. Reciprocal modulation of Toll-like receptor-4 signaling pathways involving MyD88 and phosphatidylinositol 3-kinase/AKT by saturated and polyunsaturated fatty acids. *J Biol Chem*. 2003; 278:37041–37051.
129. Shaikh IA, Brown I, Schofield AC, Wahle KW, Heys SD. Docosahexaenoic acid enhances the efficacy of docetaxel in prostate cancer cells by modulation of apoptosis: the role of genes associated with the NF-kappaB pathway. *Prostate*. 2008; 68:1635–1646.
130. Lo CJ, Chiu KC, Fu M, Lo R, Helton S. Fish oil decreases macrophage tumor necrosis factor gene transcription by altering the NF kappa B activity. *J Surg Res*. 1999; 82:216–221.
131. Mendelsohn J, Baselga J. Status of epidermal growth factor receptor antagonists in the biology and treatment of cancer. *J Clin Oncol*. 2003; 21:2787–2799.
132. Olayioye MA, Neve RM, Lane HA, Hynes NE. The ErbB signaling network: receptor heterodimerization in development and cancer. *Embo J*. 2000; 19:3159–3167.
133. Yarden Y, Sliwkowski MX. Untangling the ErbB signalling network. *Nat Rev Mol Cell Biol*. 2001; 2:127–137.
134. Kawamoto T, Sato JD, Le A, Polikoff J, Sato GH, Mendelsohn J. Growth stimulation of A431 cells by epidermal growth factor: identification of high-affinity receptors for epidermal growth factor by an anti-receptor monoclonal antibody. *Proc Natl Acad Sci USA*. 1983; 80:1337–1341.

135. Iihara K, Shiozaki H, Tahara H, Kobayashi K, Inoue M, Tamura S et al. Prognostic significance of transforming growth factor-alpha in human esophageal carcinoma. Implication for the autocrine proliferation. *Cancer*. 1993; 71:2902–2909.
136. Yasui W, Sumiyoshi H, Hata J, Kameda T, Ochiai A, Ito H et al. Expression of epidermal growth factor receptor in human gastric and colonic carcinomas. *Cancer Res*. 1988; 48:137–141.
137. Magnusson I, Rosen AV, Nilsson R, Macias A, Perez R, Skoog L. Receptors for epidermal growth factor and sex steroid hormones in human colorectal carcinomas. *Anticancer Res*. 1989; 9:299–301.
138. Barton CM, Hall PA, Hughes CM, Gullick WJ, Lemoine NR. Transforming growth factor alpha and epidermal growth factor in human pancreatic cancer. *J Pathol*. 1991; 163:111–116.
139. Selvaggi G, Novello S, Torri V, Leonardo E, De Giuli P, Borasio P et al. Epidermal growth factor receptor overexpression correlates with a poor prognosis in completely resected non-small-cell lung cancer. *Ann Oncol*. 2004; 15:28–32.
140. Couet J, Sargiacomo M, Lisanti MP. Interaction of a receptor tyrosine kinase, EGF-R, with caveolins. Caveolin binding negatively regulates tyrosine and serine/threonine kinase activities. *J Biol Chem*. 1997; 272:30429–30438.
141. Yamabhai M, Anderson RG. Second cysteine-rich region of epidermal growth factor receptor contains targeting information for caveolae/rafts. *J Biol Chem*. 2002; 277:24843–24846.
142. Roy S, Luetterforst R, Harding A, Apolloni A, Etheridge M, Stang E et al. Dominant-negative caveolin inhibits H-Ras function by disrupting cholesterol-rich plasma membrane domains. *Nat Cell Biol*. 1999; 1:98–105.
143. Puri C, Tosoni D, Comai R, Rabellino A, Segat D, Caneva F et al. Relationships between EGFR signaling-competent and endocytosis-competent membrane microdomains. *Mol Biol Cell*. 2005; 16:2704–2718.
144. Wary KK, Mariotti A, Zurzolo C, Giancotti FG. A requirement for caveolin-1 and associated kinase Fyn in integrin signaling and anchorage-dependent cell growth. *Cell*. 1998; 94:625–634.
145. Shih C, Padhy LC, Murray M, Weinberg RA. Transforming genes of carcinomas and neuroblastomas introduced into mouse fibroblasts. *Nature*. 1981; 290:261–264.
146. Natali PG, Nicotra MR, Bigotti A, Venturo I, Slamon DJ, Fendly BM et al. Expression of the p185 encoded by HER2 oncogene in normal and transformed human tissues. *Int J Cancer*. 1990; 45:457–461.
147. Slamon DJ, Godolphin W, Jones LA, Holt JA, Wong SG, Keith DE et al. Studies of the HER-2/neu proto-oncogene in human breast and ovarian cancer. *Science*. 1989; 244:707–712.
148. Yonemura Y, Ninomiya I, Yamaguchi A, Fushida S, Kimura H, Ohoyama S et al. Evaluation of immunoreactivity for erbB-2 protein as a marker of poor short term prognosis in gastric cancer. *Cancer Res*. 1991; 51:1034–1038.
149. Venter DJ, Tuzi NL, Kumar S, Gullick WJ. Overexpression of the c-erbB-2 oncoprotein in human breast carcinomas: immunohistological assessment correlates with gene amplification. *Lancet*. 1987; 2:69–72.
150. Hynes NE, Lane HA. ERBB receptors and cancer: the complexity of targeted inhibitors. *Nat Rev Cancer*. 2005; 5:341–354.
151. Slamon DJ, Clark GM, Wong SG, Levin WJ, Ullrich A, McGuire WL. Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. *Science*. 1987; 235:177–182.
152. Schley PD, Brindley DN, Field CJ. (n-3) PUFA alter raft lipid composition and decrease epidermal growth factor receptor levels in lipid rafts of human breast cancer cells. *J Nutr*. 2007; 137:548–553.
153. Menendez JA, Ropero S, Lupu R, Colomer R. Dietary fatty acids regulate the activation status of Her-2/neu (c-erbB-2) oncogene in breast cancer cells. *Ann Oncol*. 2004; 15:1719–1721.

154. Menendez JA, Lupu R, Colomer R. Exogenous supplementation with omega-3 polyunsaturated fatty acid docosahexaenoic acid (DHA; 22:6n-3) synergistically enhances taxane cytotoxicity and downregulates Her-2/neu (c-erbB-2) oncogene expression in human breast cancer cells. *Eur J Cancer Prev*. 2005; 14:263–270.
155. Myal Y, Shiu RP, Bhaumick B, Bala M. Receptor binding and growth-promoting activity of insulin-like growth factors in human breast cancer cells (T-47D) in culture. *Cancer Res*. 1984; 44:5486–5490.
156. Law JH, Habibi G, Hu K, Masoudi H, Wang MY, Stratford AL et al. Phosphorylated insulin-like growth factor-i/insulin receptor is present in all breast cancer subtypes and is related to poor survival. *Cancer Res*. 2008; 68:10238–10246.
157. Fuchs CS, Goldberg RM, Sargent DJ, Meyerhardt JA, Wolpin BM, Green EM et al. Plasma insulin-like growth factors, insulin-like binding protein-3, and outcome in metastatic colorectal cancer: results from intergroup trial N9741. *Clin Cancer Res*. 2008; 14:8263–8269.
158. Ebina Y, Ellis L, Jarnagin K, Edery M, Graf L, Clauser E et al. The human insulin receptor cDNA: the structural basis for hormone-activated transmembrane signalling. *Cell*. 1985; 40:747–758.
159. Ullrich A, Bell JR, Chen EY, Herrera R, Petruzzelli LM, Dull TJ et al. Human insulin receptor and its relationship to the tyrosine kinase family of oncogenes. *Nature*. 1985; 313:756–761.
160. Ullrich A, Gray A, Tam AW, Yang-Feng T, Tsubokawa M, Collins C et al. Insulin-like growth factor I receptor primary structure: comparison with insulin receptor suggests structural determinants that define functional specificity. *Embo J*. 1986; 5:2503–2512.
161. Belfiore A. The role of insulin receptor isoforms and hybrid insulin/IGF-I receptors in human cancer. *Curr Pharm Des*. 2007; 13:671–686.
162. Benyoucef S, Surinya KH, Hadaschik D, Siddle K. Characterization of insulin/IGF hybrid receptors: contributions of the insulin receptor L2 and Fn1 domains and the alternatively spliced exon 11 sequence to ligand binding and receptor activation. *Biochem J*. 2007; 403:603–613.
163. De Meyts P. Insulin and its receptor: structure, function and evolution. *Bioessays*. 2004; 26:1351–1362.
164. De Meyts P, Palsgaard J, Sajid W, Theede AM, Aladdin H. Structural biology of insulin and IGF-1 receptors. *Novartis Found Symp*. 2004; 262:160–171.
165. Hernandez-Sanchez C, Blakesley V, Kalebic T, Helman L, LeRoith D. The role of the tyrosine kinase domain of the insulin-like growth factor-I receptor in intracellular signaling, cellular proliferation, and tumorigenesis. *J Biol Chem*. 1995; 270:29176–29181.
166. Probst-Hensch NM, Wang H, Goh VH, Seow A, Lee HP, Yu MC. Determinants of circulating insulin-like growth factor I and insulin-like growth factor binding protein 3 concentrations in a cohort of Singapore men and women. *Cancer Epidemiol Biomarkers Prev*. 2003; 12:739–746.
167. Bhathena SJ, Berlin E, Judd JT, Kim YC, Law JS, Bhagavan HN et al. Effects of omega 3 fatty acids and vitamin E on hormones involved in carbohydrate and lipid metabolism in men. *Am J Clin Nutr*. 1991; 54:684–688.
168. Ghoshal AK, Xu Z, Wood GA, Archer MC. Induction of hepatic insulin-like growth factor binding protein-1 (IGFBP-1) in rats by dietary n-6 polyunsaturated fatty acids. *Proc Soc Exp Biol Med*. 2000; 225:128–135.
169. Kondoh N, Wakatsuki T, Ryo A, Hada A, Aihara T, Horiuchi S et al. Identification and characterization of genes associated with human hepatocellular carcinogenesis. *Cancer Res*. 1999; 59:4990–4996.
170. Chen J, Stavro PM, Thompson LU. Dietary flaxseed inhibits human breast cancer growth and metastasis and downregulates expression of insulin-like growth factor and epidermal growth factor receptor. *Nutr Cancer*. 2002; 43:187–192.

171. Kun Z, Haiyun Z, Meng W, Li N, Li Y, Li J. Dietary omega-3 polyunsaturated fatty acids can inhibit expression of granzyme B, perforin, and cation-independent mannose 6-phosphate/insulin-like growth factor receptor in rat model of small bowel transplant chronic rejection. *JPEN J Parenter Enteral Nutr*. 2008; 32:12–17.
172. Leake R. The cell cycle and regulation of cancer cell growth. *Ann NY Acad Sci*. 1996; 784:252–262.
173. Dictor M, Ehinger M, Mertens F, Akervall J, Wennerberg J. Abnormal cell cycle regulation in malignancy. *Am J Clin Path*. 1999; 112:S40–S52.
174. Albino AP, Juan G, Traganos F, Reinhart L, Connolly J, Rose DP et al. Cell cycle arrest and apoptosis of melanoma cells by docosahexaenoic acid: association with decreased pRb phosphorylation. *Cancer Res*. 2000; 60:4139–4145.
175. Khan NA, Nishimura K, Aires V, Yamashita T, Oaxaca-Castillo D, Kashiwagi K et al. Docosahexaenoic acid inhibits cancer cell growth via p27Kip1, CDK2, ERK1/ERK2, and retinoblastoma phosphorylation. *J Lipid Res*. 2006; 47:2306–2313.
176. Barascu A, Besson P, Le Floch O, Bougnoux P, Jourdan ML. CDK1-cyclin B1 mediates the inhibition of proliferation induced by omega-3 fatty acids in MDA-MB-231 breast cancer cells. *Int J Biochem Cell Biol*. 2006; 38:196–208.
177. Zhang W, Long Y, Zhang J, Wang C. Modulatory effects of EPA and DHA on proliferation and apoptosis of pancreatic cancer cells. *J Huazhong Univ Sci Technolog Med Sci*. 2007; 27:547–550.
178. Mosimann C, Hausmann G, Basler K. Beta-catenin hits chromatin: regulation of Wnt target gene activation. *Nat Rev Mol Cell Biol*. 2009; 10:276–286.
179. Weinberg RA. The biology of cancer. New York, NY: Garland Science, Taylor & Francis Group, LLC, 2007.
180. Takahashi-Yanaga F, Sasaguri T. Drug development targeting the glycogen synthase kinase-3beta (GSK-3beta)-mediated signal transduction pathway: inhibitors of the Wnt/beta-catenin signaling pathway as novel anticancer drugs. *J Pharmacol Sci*. 2009; 109:179–183.
181. Calviello G, Resci F, Serini S, Piccioni E, Toesca A, Boninsegna A et al. Docosahexaenoic acid induces proteasome-dependent degradation of beta-catenin, down-regulation of survivin and apoptosis in human colorectal cancer cells not expressing COX-2. *Carcinogenesis*. 2007; 28:1202–1209.
182. Narayanan BA, Narayanan NK, Desai D, Pittman B, Reddy BS. Effects of a combination of docosahexaenoic acid and 1,4-phenylene bis(methylene) selenocyanate on cyclooxygenase 2, inducible nitric oxide synthase and beta-catenin pathways in colon cancer cells. *Carcinogenesis*. 2004; 25:2443–2449.
183. Lim K, Han C, Xu L, Isse K, Demetris AJ, Wu T. Cyclooxygenase-2-derived prostaglandin E2 activates beta-catenin in human cholangiocarcinoma cells: evidence for inhibition of these signaling pathways by omega 3 polyunsaturated fatty acids. *Cancer Res*. 2008; 68:553–560.
184. Jacoby RF, Seibert K, Cole CE, Kelloff G, Lubet RA. The cyclooxygenase-2 inhibitor celecoxib is a potent preventive and therapeutic agent in the min mouse model of adenomatous polyposis. *Cancer Res*. 2000; 60:5040–5044.
185. Arber N, Eagle CJ, Spicak J, Racz I, Dite P, Hajer J et al. Celecoxib for the prevention of colorectal adenomatous polyps. *N Engl J Med*. 2006; 355:885–895.
186. Habbel P, Weylandt KH, Lichopoj K, Nowak J, Purschke M, Wang JD et al. Docosahexaenoic acid suppresses arachidonic acid-induced proliferation of LS-174T human colon carcinoma cells. *World J Gastroenterol*. 2009; 15:1079–1084.
187. Denkins Y, Kempf D, Ferniz M, Nileshwar S, Marchetti D. Role of omega-3 polyunsaturated fatty acids on cyclooxygenase-2 metabolism in brain-metastatic melanoma. *J Lipid Res*. 2005; 46:1278–1284.
188. Escrich E, Moral R, Grau L, Costa I, Solanas M. Molecular mechanisms of the effects of olive oil and other dietary lipids on cancer. *Mol Nutr Food Res*. 2007; 51:1279–1292.
189. Slater TF, Cheeseman KH, Proudfoot K. Free radicals, lipid peroxidation and cancer. 1984; 27:293.

190. Cornwell DG, Huttner JJ, Milo GE, Panganamala RV, Sharma HM, Geer JC. Polyunsaturated fatty acids, vitamin E, and the proliferation of aortic smooth muscle cells. *Lipids*. 1979; 14:194–207.
191. Huttner JJ, Gwebu ET, Panganamala RV, Milo GE, Cornwell DC, Sharma HM et al. Fatty acids and their prostaglandin derivatives: inhibitors of proliferation in aortic smooth muscle cells. *Science*. 1977; 197:289–291.
192. Huttner JJ, Milo GE, Panganamala RV, Cornwell DG. Fatty acids and the selective alteration of in vitro proliferation in human fibroblast and guinea-pig smooth-muscle cells. *In Vitro*. 1978; 14:854–859.
193. Maziere C, Conte MA, Degonville J, Ali D, Maziere JC. Cellular enrichment with polyunsaturated fatty acids induces an oxidative stress and activates the transcription factors AP1 and NFkappaB. *Biochem Biophys Res Commun*. 1999; 265:116–122.
194. Schonberg SA, Rudra PK, Noding R, Skorpen F, Bjerve KS, Krokan HE. Evidence that changes in Se-glutathione peroxidase levels affect the sensitivity of human tumour cell lines to n-3 fatty acids. *Carcinogenesis*. 1997; 18:1897–1904.
195. Chamras H, Ardashian A, Heber D, Glaspy JA. Fatty acid modulation of MCF-7 human breast cancer cell proliferation, apoptosis and differentiation. *J Nutr Biochem*. 2002; 13:711–716.
196. Grammatikos SI, Subbaiah PV, Victor TA, Miller WM. Diverse effects of essential (n-6 and n-3) fatty acids on cultured cells. *Cytotechnology*. 1994; 15:31–50.
197. Galli C, Marangoni F, Galella G. Modulation of lipid derived mediators by polyunsaturated fatty acids. *Prostaglandins Leukot Essent Fatty Acids*. 1993; 48:51–55.
198. Edwards IJ, O'Flaherty JT. Omega-3 Fatty Acids and PPARgamma in Cancer. *PPAR Res*. 2008; 2008:358052.
199. Feige JN, Gelman L, Michalik L, Desvergne B, Wahli W. From molecular action to physiological outputs: peroxisome proliferator-activated receptors are nuclear receptors at the crossroads of key cellular functions. *Prog Lipid Res*. 2006; 45:120–159.
200. Kersten S, Desvergne B, Wahli W. Roles of PPARs in health and disease. *Nature*. 2000; 405:421–424.
201. Lee CH, Olson P, Evans RM. Minireview: lipid metabolism, metabolic diseases, and peroxisome proliferator-activated receptors. *Endocrinology*. 2003; 144:2201–2207.
202. Sarraf P, Mueller E, Jones D, King FJ, DeAngelo DJ, Partridge JB et al. Differentiation and reversal of malignant changes in colon cancer through PPARgamma. *Nat Med*. 1998; 4:1046–1052.
203. Mueller E, Sarraf P, Tontonoz P, Evans RM, Martin KJ, Zhang M et al. Terminal differentiation of human breast cancer through PPAR gamma. *Mol Cell*. 1998; 1:465–470.
204. Berger J, Moller DE. The mechanisms of action of PPARs. *Annu Rev Med*. 2002; 53:409–435.
205. Houseknecht KL, Cole BM, Steele PJ. Peroxisome proliferator-activated receptor gamma (PPARgamma) and its ligands: a review. *Domest Anim Endocrinol*. 2002; 22:1–23.
206. Gonzalez-Periz A, Planaguma A, Gronert K, Miquel R, Lopez-Parra M, Titos E et al. Docosahexaenoic acid (DHA) blunts liver injury by conversion to protective lipid mediators: protectin D1 and 17S-hydroxy-DHA. *Faseb J*. 2006; 20:2537–2539.
207. Bonofiglio D, Aquila S, Catalano S, Gabriele S, Belmonte M, Middea E et al. Peroxisome proliferator-activated receptor-gamma activates p53 gene promoter binding to the nuclear factor-kappaB sequence in human MCF7 breast cancer cells. *Mol Endocrinol*. 2006; 20:3083–3092.
208. Han S, Roman J. Rosiglitazone suppresses human lung carcinoma cell growth through PPARgamma-dependent and PPARgamma-independent signal pathways. *Mol Cancer Ther*. 2006; 5:430–437.
209. Horton JD, Goldstein JL, Brown MS. SREBPs: activators of the complete program of cholesterol and fatty acid synthesis in the liver. *J Clin Invest*. 2002; 109:1125–1131.

210. Zhou RH, Yao M, Lee TS, Zhu Y, Martins-Green M, Shyy JY. Vascular endothelial growth factor activation of sterol regulatory element binding protein: a potential role in angiogenesis. *Circ Res*. 2004; 95:471–478.
211. Field FJ, Born E, Murthy S, Mathur SN. Polyunsaturated fatty acids decrease the expression of sterol regulatory element-binding protein-1 in CaCo-2 cells: effect on fatty acid synthesis and triacylglycerol transport. *Biochem J*. 2002; 368:855–864.
212. Schonberg SA, Lundemo AG, Fladvad T, Holmgren K, Bremseth H, Nilsen A et al. Closely related colon cancer cell lines display different sensitivity to polyunsaturated fatty acids, accumulate different lipid classes and downregulate sterol regulatory element-binding protein 1. *Febs J*. 2006; 273:2749–2765.
213. Singh J, Hamid R, Reddy BS. Dietary fat and colon cancer: modulating effect of types and amount of dietary fat on ras-p21 function during promotion and progression stages of colon cancer. *Cancer Res*. 1997; 57:253–258.
214. Davidson LA, Lupton JR, Jiang YH, Chapkin RS. Carcinogen and dietary lipid regulate ras expression and localization in rat colon without affecting farnesylation kinetics. *Carcinogenesis*. 1999; 20:785–791.
215. Collett ED, Davidson LA, Fan YY, Lupton JR, Chapkin RS. n-6 and n-3 polyunsaturated fatty acids differentially modulate oncogenic Ras activation in colonocytes. *Am J Physiol Cell Physiol*. 2001; 280:C1066–C1075.
216. Seo J, Barhoumi R, Johnson AE, Lupton JR, Chapkin RS. Docosahexaenoic acid selectively inhibits plasma membrane targeting of lipidated proteins. *FASEB J*. 2006; 20:770–772.
217. Craven PA, DeRubertis FR. Alterations in protein kinase C in 1,2-dimethylhydrazine induced colonic carcinogenesis. *Cancer Res*. 1992; 52:2216–2221.
218. Kahl-Rainer P, Sedivy R, Marian B. Protein kinase C tissue localization in human colonic tumors suggests a role for adenoma growth control. *Gastroenterology*. 1996; 110:1753–1759.
219. Ma DW, Seo J, Davidson LA, Callaway ES, Fan YY, Lupton JR et al. n-3 PUFA alter caveolae lipid composition and resident protein localization in mouse colon. *Faseb J*. 2004; 18:1040–1042.
220. Davidson LA, Brown RE, Chang WC, Morris JS, Wang N, Carroll RJ et al. Morphodensitometric analysis of protein kinase C beta(II) expression in rat colon: modulation by diet and relation to in situ cell proliferation and apoptosis. *Carcinogenesis*. 2000; 21:1513–1519.
221. Murray NR, Weems C, Chen L, Leon J, Yu W, Davidson LA et al. Protein kinase C betaII and TGFbetaRII in omega-3 fatty acid-mediated inhibition of colon carcinogenesis. *J Cell Biol*. 2002; 157:915–920.

Part II
ω-3 PUFAs and Colon Cancer

Chapter 2
ω-3 PUFAs and Colon Cancer: Epidemiological Studies

Yasumi Kimura

Abstract Colorectal cancer is the second most common cancer in developed countries according to its incidence and mortality rate. The incidence of colorectal cancer is particularly high in Oceania, North America, Western Europe, and Japan, and preventing this disease is therefore a high priority in developed countries. There is currently much interest in fish and omega-3 polyunsaturated fatty acids (ω-3 PUFAs) because of their preventive effect against colorectal cancer.

Epidemiological studies have shown that consumption of fish/ω-3 PUFAs can reduce the risk of colorectal cancer. However, the results are inconsistent: some studies have suggested a preventive effect, while others reported no association. In future epidemiological studies on the association between fish/ω-3 PUFAs and colorectal cancer the analysis methods should be improved to accurately measure the intake of fish/ω-3 PUFAs using dietary surveys and/or biomarkers. Furthermore, randomized controlled trials and/or subsite-specific analyses should yield more consistent results.

Keywords Colorectal cancer · Epidemiological studies · Fish · Omega-3 polyunsaturated fatty acids · Prevention

Abbreviation

OR	odds ratio
RR	relative risk
CI	confidence interval
FFQ	food frequency questionnaire
SQFFQ	semiquantitative food frequency questionnaire
RCT	randomized controlled trials
AI	apoptosis index

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EPA	eicosapentaenoic acid
DHA	docosahexaenoic acid
DPA	docosapentaenoic acid
HCA	heterocyclic amines
PAH	polycyclic aromatic hydrocarbons

2.1 Introduction

The worldwide incidence of colorectal cancer is approximately 945,000 per year, and the number of deaths is approximately 492,000. In developed countries, colorectal cancer is the second most common cancer in incidence and mortality rate. Regionally, the incidence rate of colorectal cancer is high in Oceania, North America, and Western Europe and has recently become so in Japan [1]. Thus, the prevention of colorectal cancer is a high priority in developed countries. There is currently much interest in fish/omega-3 polyunsaturated fatty acids (ω-3 PUFAs) because of their preventive effect against colorectal cancer.

This section describes epidemiological studies analyzing the relationship between fish intake and the prevention of colorectal cancer.

2.1.1 Definitions and Sources of Fish

Estimates of the number of fish species range between 15,000 and 40,000 [2]. Among all fish species, marine fish comprise 58.2% and freshwater fish comprise 41.2%. These species include edible fish, inedible fish, and shellfish. In this chapter, we focus on edible fish and shellfish. Some studies might include processed fish such as dried fish, salted fish, or smoked fish.

2.1.2 Composition of Fish

Fish is rich in proteins and fatty acids, and the composition of fatty acids varies between different fish species [3]. Fish is the source of fish oil rich in ω-3 PUFAs (EPA: eicosapentaenoic acid, DHA: docosahexaenoic acid). Recently, hatchery fish have become popular, and hatchery fish contains more fat than wild fish [3].

2.1.3 Consumption of Fish

The percentage of total energy derived from fish oil is 1.7% in Japan, which is the highest in the world. Similarly, it is 0.28% in France, 0.24% in Italy, 0.22% in the United Kingdom, and 0.20% in the United States [4]. In 2003, the amount of fish or seafood consumed (g/capita/day) was 181 g in Japan, 85 g in France, 71 g in Italy, and 63 g in the United Kingdom, according to Food Balance Sheet of the Food and Agriculture Organization [5].

2.2 Evidence from Epidemiological Studies

2.2.1 Ecological Studies

A study by Caygill et al. in 24 European countries revealed a significant negative association ($P = 0.036$) between current fish intake and the mortality rate from colorectal cancer. It also demonstrated a weak negative association ($P = 0.042$) between fish intake 10 years earlier and same mortality rate [6]. Another study reported a reduction in the risk of death from colorectal cancer by consuming fish and fish oil, if the intake of animal fat was at least 85 g per day [7].

2.2.2 Case–Control Studies

Case–control studies and cohort studies are summarized in Tables 2.1 and 2.2, respectively. The tables show first author, year of publication, country (no. of reference), study, site, the number of objects, gender, exposure assessment (no. of items), exposure definition, age, range of exposure, odds ratio (OR) (case–control study) or relative risk (RR) (cohort study), 95% confidence interval (CI) of risk estimation, adjustment for confounding effects. The tables also show a subsite specific analysis for colorectal cancer and colon (proximal colon, distal colon) or rectal cancer alone.

Table 2.1 shows a summary of case–control studies on colorectal cancer and fish intake. A decreased risk of colon cancer with increased fish intake can be found in 7 out of 18 studies. If we focus on the studies in Japan, where the fish intake is the highest in the world, a preventive association can be found in two out of six studies.

Bjelke et al. reported a low consumption of fish in subjects with colorectal cancer [8]. In Australia, the reduction in the risk of colorectal cancer for the highest vs the lowest quintiles of fish intake gave an odds ratio of 0.58 and a 95% CI of 0.43–0.78 [9]. In Northern Italy, the odds ratio for risk of rectal cancer in the highest vs. the lowest tertile of fish intake was 0.58 [10]. A study in Argentina showed that the odds ratio for colorectal cancer risk in the highest fish/seafood intake group relative to the lowest quartile group was extremely low at 0.27 (95% CI 0.13–0.60) [11]. Two studies in Italy reported that the OR of colorectal cancer in the highest vs. the lowest quintile of fish intake was 0.72 (95% CI 0.59–0.88) [12] and reported that the OR of colon cancer became 0.6 (95% CI 0.5–0.7) and that of rectal cancer became 0.5 (95% CI 0.3–0.6) by taking more than two servings of fish per week compared with less than one serving [13]. In Japan, Kimura et al. reported a significant reduction in the risk of distal colon cancer in the highest vs. the lowest quintiles of ω-3 PUFA

Table 2.1 Case–control studies on fish or omega-3 fatty acids intake and colorectal cancer

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Tajima et al. (1985) Japan [17] Japan Nagoya Case-control study	Colon Rectum	42:42 51:51 All	Food habits Self-administered FFQ	Fish	40–70	$\geq$ 4 times /wk vs. $\leq$ 1 time /wk	1.09 (nr) 1.44 (nr)	Age, sex
Macquart-Moulin et al. (1986) France [22] France Marseilles Case-control study	Colorectal	399:798 All	Semiquantitative FFQ	Fish	Average M: 65.4 F: 65.9	Highest vs. lowest quartile	0.86 (nr)	Age, sex, energy intake, and weight
Kune et al. (1987) Australia [9] The Melbourne Colorectal Cancer Study	Colorectal Colon Rectum	715:727 392 323 All	Diet history (>300) interviewed	Fish	Average 65	Continuous variable from highest to lowest quintiles	0.58* (0.43–0.78) 0.67* (nr) 0.53* (nr)	Age, sex
La Vecchia et al. (1988) Italy [10] Northern Italy Case-control Study	Colon Rectum	339:778 236 All	FFQ (29)	Fish	< 40–74	Highest vs. lowest tertile	0.89 (nr) 0.58* (nr)	Age, sex

Table 2.1 (continued)

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Lee et al. (1989) Singapore [21] Singapore Chinese Case-control study	Colorectal Colon Rectum	203:425 132 71 All	FFQ (116) interviewed	Fish	44–75	Highest vs. lowest tertile	1.03 (0.66–1.61) 1.15 (0.69–1.90) 0.81 (0.38–1.68)	N/A
Kato et al. (1990) Japan [18] Japan Nagoya Case-control study	Colon Rectum	132:578 91 All	Self-administered FFQ (25)	Fish & shellfish	39–75	Daily intake vs. less than daily intake	0.95 (0.63–1.43) 0.88 (0.55–1.43)	Age, sex, residence
Peters et al. (1992) USA [16] USA Los Angeles County, California Case–control Study	Colon	746:746 All	Semiquantitative FFQ (116)	Seafood	Average 61.4 y	Continuous variable (/ 10 servings/month)	1.00 (0.83–1.21)	Fat, protein, carbohydrates, alcohol, calcium, family history of colorectal cancer, weight, physical activity, (if female, pregnancy history)

Table 2.1 (continued)

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Iscovich et al. (1992) Argentina [11] Argentina La Plata Case-control Study	Colon	110:220 All	FFQ (140) interviewed	Fish & seafood	Average 65	Highest vs. lowest quartile	0.27* (0.13–0.60)	N/A
Bidoli et al. (1992) Italy [25] North-Eastern Italy Case-control study	Colon Rectum	123:699 125 All	FFQ	Fish	50–70	Highest vs. lowest tertile	1.3 (nr) 1.6 (nr)	Age, sex, and social status
Centonze et al. (1994) Italy [23] Southern Italy Case-control Study	Colorectal	119:119 All	FFQ (70)	Fish	34–90	Highest vs. lowest tertile	1.07 (0.56–2.05)	Age, sex, level of education, smoking status, modifications in diet in the previous 10 years

Table 2.1 (continued)

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Inoue et al. (1995) Japan [19]	Proximal	94:31,782	Food habits FFQ	Cooked or raw fish	Average Cases M 60.6	≥ 3–4 times/wk vs. less	0.9 (0.5–1.7)[b] 1.4 (0.8–2.5)[c]	Age
Japan Nagoya Case-control study	Distal	137			F 58.6 Controls M 50.3		1.2 (0.8–1.9)[b] 1.0 (0.6–1.7)[c]	
	Rectum	201 All			F 45.7		1.0 (0.7–1.4)[b] 1.4 (0.9–2.2)[c]	
Kampman et al. (1995) Netherlands [24] Dutch Case–control Study	Colon	232:259 All	Semiquantitative FFQ (289) interviewed	Fish	Average Cases 62 Controls 62	>24 g/day vs. < 5 g/day	1.13 (0.68–1.87)	Age, sex, urbanization level, total energy intake, alcohol use, cholecystectomy, family history of colon cancer

Table 2.1 (continued)

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Kotake et al. (1995) Japan [20] Japan 10 Medical Institutions Case–control study	Colon Rectum	187:363 176 All	Self-administered FFQ	Fish	Average Cases colon 63.3 rectum 59.6 Controls colon 62.2 rectum 59.4	Daily intakes vs. <1–2/wk	0.5 (0.21–1.20) 0.5 (0.21–1.41)	Age, sex
Franceschi et al. (1997) Italy [12] Italy Case–control Study	Colorectal	1,953:4,154 All	FFQ (79)	Fish	Median 62	Highest vs. lowest quintile	0.72** (0.59–0.88)	Age, sex, center, education, physical activity, total energy intake

Table 2.1 (continued)

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Le Marchand et al. (1997). USA [26] USA Hawaii Multiethnic Case–control Study	Colorectal	698:698[b] 494:494[c] All	Semiquantitative FFQ (>280) Interviewed	Fish	≤ 84	Highest vs. lowest quartile	1.1(0.7–1.6)[b] 1.1 (0.6–1.7)[c]	Age, family history of colorectal cancer, alcoholic drinks/week, pack-years, lifetime recreational activity, body mass index 5 years ago, energy intake, dietary fiber intake, calcium intake
Fernandez et al. (1999) Italy [13] Northern Italy Case–control Study	Colon Rectum	828:7990 498 All	FFQ (14–37)	Fish	< 75	≥ 2 servings/ wk vs. < 1 serving/ wk	0.6* (0.5–0.7) 0.5* (0.3–0.6)	Age, sex, area of residence, education, smoking, alcohol consumption, body mass index

Table 2.1 (continued)

Author (year) Country (No. of reference) Study	Site	Cases/controls[a], gender	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Odds ratio (95% CI)	Adjustment for confounding
Yang et al. (2003) Japan [15] Japan Nagoya Case–control Study	Colon Rectum	928:46,886 622 All	Food habits FFQ	Raw/cooked fish	40–79 y	> 4 times/wk vs. < 1 time/wk	0.68* (0.47–0.99)[b] 0.80 (0.52–1.24)[c] 1.13 (0.76–1.68)[b] 0.62 (0.33–1.16)[c]	Age
Kimura et al. (2007) Japan [14] Fukuoka Colorectal Cancer Study	Colorectal Proximal Distal Rectum	782:793 177 262 327 All	Semiquantitative Food Frequency Personal-computer software (148) interviewed	Fish & fish products Omega-3 PUFAs	Average Cases 61 Controls 59	Highest vs. lowest quintile	Proximal 0.68 (0.38–1.20) Distal 0.64 (0.39–1.06) Rectum 0.91 (0.57–1.43) Proximal 0.84 (0.45–1.55) Distal 0.56* (0.34–0.92) Rectum 0.88 (0.56–1.41)	Age, sex, residential area, body mass index 10 years before, parental colorectal cancer, smoking, alcohol use, type of job, leisure-time physical activity, dietary calcium and dietary fiber

* $P < 0.05$ ** $P<0.01$

[a]Number of cases and controls. nr: not reported N/A: not applicable

[b]Data are shown for men only.

[c]Data are shown for women only.

Table 2.2 Cohort studies on fish or omega-3 fatty acids intake and colorectal cancer

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Willett et al. (1990) USA [32] Nurses' Health Study	Colon	88,751[b]:150 (6 y) Female nurses	Semiquantitative FFQ (61)	Fish	34–59	≥ 1/day vs. < 1/month	1.06 (0.36–3.12)[b]	Age
Giovannucci et al (1994). USA [34] Health Professionals Follow-up Study	Colon	47,949[c]:205 (6 y) Male health professionals	Semiquantitative FFQ (131)	Fish	40–75	83.4 median g/day vs. 8.4 median g/day	1.06 (0.70–1.60)[c]	Age, total energy intake
Bostick et al. (1994) USA [33] Iowa Women's Health Study	Colon	35,215[b]:212 (4 y)	Semiquantitative FFQ (127)	Seafood	55–69	> 2.5 servings/wk vs. < 1 serving/wk	0.76 (0.49–1.19)[b]	Age, total energy intake, height, parity, total vitamin E intake, a total vitamin E by age interaction term, and vitamin A supplement intake.
				Omega-3 fatty acids		> 0.18 g/day vs. < 0.03 g/day	0.70 (0.45–1.09)[b]	

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Goldbohm et al. (1994) Netherlands [38] Netherlands Cohort Study	Colon	120,852 (M: 58,279, F: 62,573): 215 (M: 105, F: 110) (3.3 y)	Semiquantitative FFQ (150)	Fish	55–69	>20 g/day vs. 0 g/day	0.81 (0.56–1.17) 0.73 (0.44–1.21)[c] 0.87 (0.52–1.45)[b]	Age, sex, dietary fiber intake
Gaard et al. (1996) Norway [41] Norwegian National Health Screening Service Study	Colon	50,535 (M: 25,638, F: 24,897): 143 (M: 83, F: 60) (average 11.4 y)	Semiquantitative FFQ (80)	Fish meals	20–54	≥ 5 /wk vs. ≤ 2/wk	0.46 (0.19–1.11)[c] 0.81 (0.30–1.94)[b]	Age
Kato et al. (1997) USA [28] New York University Women's Health Study	Colorectal	14,727[b]:100 (average 7.1 y)	Semiquantitative FFQ (70)	Fish, shellfish	34–65	Highest vs. lowest quartile	0.49** (0.27–0.89)[b]	Age, total energy intake, place at enrollment, educational level

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Hsing et al. (1998) USA [35] Lutheran Brotherhood Study	Colorectal Colon Rectum	17,633[c]:145 120 25 (20 y) (Mortality)	FFQ (35)	Fish	$\geq$ 35 y	highest (> 4 times/month) vs. lowest (< 0.8 times/month) quartile	1.5 (0.9–2.6)[c] 1.4 (0.8–2.5)[c]	Age, smoking, alcohol intake and total energy
Pietinen et al. (1999) Finland [46] Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study	Colorectal	27,111[c]:185 (average 8 y) Male smokers	Semiquantitative FFQ (276)	Fish	50–69 at entry	Highest (medians 68 g/day) vs. lowest quartile (medians 13 g/day)	0.9 (0.6–1.4)[c]	Age, supplement group, smoking years, body mass index, alcohol, educational level, physical activity at work and calcium intake
				Omega-3 fatty acids (g)		highest (medians 0.7 g/day) vs. lowest quartile (medians 0.2 g/day)	1.2 (0.8–1.9)[c]	
Knekt et al. (1999) Finland [27] Finnish Follow-up Study	Colorectal	9,985:73 (24 y) All	Dietary history interview	Fish (smoked and salted)	15–99	Highest vs. lowest quartile	2.58 (1.21–5.51) (p value: nr)	Age, sex, municipality, smoking, and energy intake
				(other)			1.11 (0.55–2.28)	

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Ma et al. (2001) USA [36] Physicians' Health Study	Colorectal	14,916[c]:193 (13 y) Male physicians	FFQ (19)	Fish	40–84	Highest (0.35–2.03 servings/day, median 0.57) vs. lowest (≤ 0.14 servings/day, median 0.14) tertile	0.92(0.56–1.51)[c]	Age, cigarette smoking, body mass index, alcohol intake, multivitamin use, aspirin use, exercise, and molar ratio of IGF-I to IGFBP-3
Tiemersma et al. (2002) Netherlands [39] Dutch Prospective Monitoring Project on Cardiovascular Disease Risk Factors	Colorectal	> 36,000: nested case–control study Case 102 Control 537 (8.5 y) All	Semiquantitative FFQ	Fish	20–59	> 4 times/month vs. 0–1 times/month	0.5 (0.2–1.0)[c] 1.2 (0.6–2.4)[b]	Age, sex, center, total energy intake, alcohol consumption, and body height.

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Kobayashi et al. (2004) Japan [42] Japan Public Health Center-based prospective study	Colorectal Colon Rectum	88,658 (M: 42,525, F: 46,133): 705 (M: 454, F: 251) 456 (M: 300, F: 156) 249 (M: 154, F: 95) All	Self-administered FFQ (44: Cohort I) (52: Cohort II)	Fish	40–69	Highest vs. lowest quartile	Colon 1.07 (0.77–1.48) 1.07 (0.72–1.58)[c] 1.05 (0.61–1.82)[b] Rectal 0.95 (0.63–1.43) 1.31 (0.78–2.22)[c] 0.69 (0.35–1.36)[b]	Age, sex (All), area, family history of colorectal cancer, body mass index, physical activity, smoking status, alcohol intake, use of vitamin supplement, total energy intake, cereal, vegetable and meat intake
English et al. (2004) Australia [45] The Melbourne Collaborative Cohort Study	Colorectal Colon Rectum	37,112: 451 283 169 All (average 9 y)	FFQ (121)	Fish	27–75	Highest vs. lowest quartile	0.9 (0.7–1.2) 1.0 (0.7–1.4) 0.9 (0.6–1.4)	Age, total energy intake, sex, country of birth, fat, and cereal products

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Khan et al. (2004) Japan [44] Japan Hokkaido Cohort study	Colorectal	3,158 (M: 1524, F: 1,634) 29 (M: 15, F: 14) All	FFQ (37)	Fish	≥ 40	Comparison category (several times/week, everyday) vs. reference category (never, several times/year, several times/month)		Age and smoking
				Raw fish			1.1 (0.4–3.1)[c] 1.2 (0.3–4.5)[b]	
				Boiled fish			0.5 (0.2–1.4)[c] 1.2 (0.4–3.7)[b]	
				Salty fish			0.7 (0.2–2.0)[c] 1.6 (0.5–5.0)[b]	
Kojima et al. (2004) Japan [43] Japan Collaborative Cohort Study	Colon Rectum	107,824 (M: 45,181, F: 62,643): 284 (M: 138, F: 146) 173 (M: 116, F: 57) (Mortality) (average 9.9 y) All	FFQ (33)	Fish	40–79	Highest vs. lowest tertile	Colon 1.04 (0.65–1.66)[c] 0.97 (0.62–1.50)[b] Rectal 0.95 (0.60–1.51)[c] 0.90 (0.44–1.84)[b]	Age, family history of colorectal cancer, body mass index, frequency of alcohol intake, current smoking status, walking time per day, educational level, stratified by regions of enrollment

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Sanjoaquin et al. (2004) UK [37] Oxford Vegetarian Study		10,998 (M: 4,162, F: 6,836):	FFQ	Fish	16–89 at entry	≥ 1/wk vs. not eaten/wk		Age, sex, alcohol, smoking
	Colorectal	95 All (42.3% vegetarians) (17 y)					1.17 (0.71–1.92)	
Chao et al. (2005) USA [47] Cancer Prevention Study II Nutrition Cohort		148,610 (M: 69,664, F: 78,946) :1667	FFQ (68)	Poultry & fish	50–74	≥ 560 g/wk vs. ≤ 160 g/wk		Age, sex, total energy intake, education, body mass index, cigarette smoking, recreational physical activity, use of hormone therapy (women), multivitamin use, aspirin use, beer, wine, liquor, fruits, vegetables, high-fiber grain foods.
	Colorectal	667						
	Proximal	408					0.89 (0.69–1.15)	
	Distal	470					0.73 (0.53–1.01)	
	Recto-sigmoid & rectum	(1992/1993 through August 31, 2001) All					0.93 (0.68–1.26)	

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Larsson et al. (2005) Sweden [48] Swedish Mammography Cohort	Colorectal	61,433[b]:733	Self-administered FFQ (67)	Fish	40–75	≥ 2 servings/wk (median 3.0) vs. < 0.5 servings/wk (median 0.5)	1.08 (0.81–1.43)[b]	Age, body mass index, educational level, total energy intake, alcohol, saturated fat, calcium, folate, fruits, vegetables, whole-grain foods.
	Proximal	234					1.03 (0.63-1.67)[b]	
	Distal	155					0.83 (0.45–1.51)[b]	
	Rectum	230					1.08 (0.63–1.86)[b]	
		F(13.9 y)						
Norat et al. (2005) France [29] (10 European countries)[d] European Prospective Investigation into Cancer and Nutrition	Colorectal	478,040:1329	Most centers: Self-administered FFQ (88–266) Greece, Spain, Ragusa, Italy: Administered at a personal interview Malmö, Sweden: A questionnaire method combined with a food record	Fish	35–70	≥ 80 g/day vs. <10 g/day	0.69*** (0.54–0.88)	Age, sex, energy from nonfat sources, energy from fat sources, height, weight, occupational physical activity, smoking status, dietary fiber, alcohol intake, stratified for center.
	Colon	855					0.82** (0.60–1.11)	
	Proximal	351					0.85 (0.53–1.37)	
	Distal	391					0.70* (0.44–1.11)	
	Rectum	474 (4.8 y) All					0.49*** (0.32–0.76)	

Table 2.2 (continued)

Author (year) Country (No. of reference) Study	Site	Cohort size/cases[a] (years follow-up), population	Exposure assessment (no. of items)	Exposure definition	Age (y)	Range of exposure	Relative risk (95% CI)	Adjustment for confounding
Lüchtenborg et al. (2005) Netherlands [40] Netherlands Cohort Study		120,852 (M: 58,279, F: 62, 573): Subcohort 2,948:	Self-administered Semiquantitative FFQ (150)	Fish	55–69	Highest (median intake 30.5 g/day in men, 28.2 g/day in women) vs. lowest (0 g/day) quartile		Age, sex, family history of colorectal cancer, smoking status, body mass index, energy intake
	Colon	434					1.03 (0.76–1.40)	
	Rectum	154					0.94 (0.59–1.52)	
		All						
Hall et al. (2008) USA [30] Physicians' Health Study	Colorectal	21,406[c]:500	Semiquantitative FFQ (61)	Fish		≥5 times/week vs. < 1 time/week	Colorectal 0.63* (0.42–0.95)[c]	Age, smoking status, body mass index, multivitamin use, history of diabetes, random assignment to aspirin or placebo, vigorous exercise, alcohol intake, quartile of red meat intake.
	Colon	388						
	Rectum	112						
		(22 y) Male physicians		Omega-3 fatty acids		highest vs. lowest quartile	Colorectal 0.76* (0.59–0.98)[c]	

*$P<0.05$ **$P<0.01$ ***$P<0.001$

M, males; F, females; nr: not reported

[a]Number of baseline population and incident cases.

[b]Data are shown for women only.

[c]Data are shown for men only.

[d]Denmark, France, Germany, Greece, Italy, the Netherlands, Norway, Spain, Sweden, UK.

intake, with an OR of 0.56 (95% CI 0.34–0.92). The same study also reported a tendency toward decreased distal colon cancer risk in the highest vs the lowest quintile of fish/fish products intake, with an OR of 0.64 (95% CI 0.39–1.06) [14]. Yang et al. reported a reduction in colon cancer risk if fish were consumed more than four times per week compared with less than once per week (OR 0.68, 95% CI 0.47–0.99) [15]. In other studies, there was no significant association between fish intake and colorectal cancer [16–26].

2.2.3 Cohort Studies

The results of cohort studies on the association between fish intake and colorectal cancer are summarized in Table 2.2. A reduction in the risk of colorectal cancer was found in 3 out of 21 studies. In contrast, an increased risk of colorectal cancer with the consumption of smoked and salted fish was observed in Finland (RR 2.58, 95% CI 1.21–5.51) [27]. A cohort study of women at New York University showed that the relative risk of colorectal cancer was significantly reduced in the highest vs. the lowest quartile of fish intake (RR 0.49, 95% CI 0.27–0.89) [28]. In the European Prospective Investigation into Cancer and Nutrition (EPIC) cohort studies in ten European countries, the relative risk of colorectal cancer was smaller in those who consumed more than 80 g of fish per day than in those with less than 10 g fish intake (RR 0.69, 95% CI 0.54–0.88); the relative risk of colon cancer was 0.82 (95% CI 0.60–1.11) and for rectal cancer it was 0.49 (95% CI 0.32–0.76) [29]. A study of males in the United States showed a significant inverse association between fish intake and colorectal cancer risk, where the multivariate relative risk for those consuming fish more than five times per week compared with less than once per week was 0.63 (95% CI 0.42–0.95). The inverse association was similar for ω-3 PUFA intake, where the multivariate relative risk of colorectal cancer for the highest vs. the lowest quartile was 0.76 (95% CI 0.59–0.98) [30]. In a nested case–control study in Japan, the relative risk of colorectal cancer for the highest vs. the lowest quartile of serum ω-3 PUFA levels in men was 0.24 (95% CI 0.08–0.76) [31].

In contrast, many studies found no association between fish intake and the risk of colorectal cancer. In studies of female subjects [32, 33] and male subjects [34–36] in the United States, there was no significant association between the risk of colorectal or colon cancer and the frequency of fish or seafood intake. In a study of vegetarians in the United Kingdom [37], studies in the Netherlands [38–40], Norway [41], three studies in Japan [42–44], and a study in Australia [45], no association was detected between the risk of colorectal or colon cancer and fish intake. In a study of male smokers in Finland, there was no association between colorectal cancer and fish or ω-3 PUFA intake [46]. In a study in the United States, no association was found between the total amount of fish and chicken intake and distal colon cancer [47]. In a study in Sweden, fish intake had no association with cancers of the proximal colon, distal colon, and rectum [48].

2.2.4 Randomized Controlled Trials (RCT)

A randomized controlled trial by Anti et al. showed a significant reduction in the proliferation index of adenoma and in the level of arachidonic acid in mucosa caused by fish oil supplementation [49]. Chang et al. performed a 2-year dietary intervention study and showed that the apoptosis index (AI), the expression of Bax, or the ratio of Bax/Bcl-2 expression was significantly increased in the intervention group who consumed less fat (20–30% of total fat intake), decreased ω-6 PUFAs, and increased ω-3 PUFAs; the control group only decreased their intake of fat. This suggests that ω-3 PUFAs promote apoptosis in the normal colonic mucosa in humans [50].

2.3 Comprehensive Assessment

Both cohort studies and case–control studies have demonstrated an association between fish/ω-3 PUFA intake and a preventive effect on colorectal cancer, but the association is limited and inconsistent. In the meta-analysis by Geelen et al. [51], pooled relative risks for the highest compared with the lowest fish consumption category showed a slightly reduced colorectal cancer risk (RR 0.88, 95% CI 0.78–1.00). The World Cancer Research Fund reported that fish intake shows a limited suggestion of a decreased risk of colorectal cancer [52]. In cohort studies, an increased risk of colorectal cancer was found with smoked and salted fish [27]. In summary, the risk of colorectal cancer varies between studies. The risk also varies depending on cooking styles or ingestion of fresh, salted, or dried fish.

2.4 Discussion

The results of some epidemiological studies have shown that fish/ω-3 PUFA intake is effective in preventing colorectal cancer. However, the results are inconsistent. There have been a few interventional studies, but many more studies are expected.

2.4.1 Reasons for Inconsistent Results in Epidemiological Studies

2.4.1.1 The Relationship Between Serum ω-3 PUFA Levels and Fish Intake by FFQ

In several epidemiological studies, fish intake, as reported by food frequency questionnaire (FFQ), was used to estimate the intake of ω-3 PUFAs [53, 54]. Fish is the primary source of ω-3 PUFAs [55]. The results of the Japan Collaborative Cohort Study (JACC Study), a cross-sectional study of 1,257 controls in a nested case–control study [56], showed that the geometric mean of the percentage of total fatty acids in serum that comprised ω-3 PUFAs (EPA, docosapentaenoic acid (DPA) (ω-3), DHA) was related to the frequency of fish intake. However, the Spearman correlation coefficient between fish intake frequency and serum ω-3 PUFAs, after age adjustment, was rather low: 0.11–0.18. This result suggests that the relationship between the frequency of fish intake reported by FFQ and serum ω-3 PUFAs is not clear.

2.4.1.2 Portion Size and Fish Type in Dietary Assessment

Almost all studies used an FFQ or semiquantitative food frequency questionnaire (SQFFQ) for the assessment of fish intake. In the FFQ, it is difficult to calculate intake accurately because there is no question about portion size. This problem can be overcome by using an SQFFQ, in which a question about portion size is included [57]. Moreover, ω-3 PUFAs are more highly enriched in fatty fish [58], but fish type might not be included in the questionnaire. Accordingly, there might be misclassification in both the amount of fish consumed and the ω-3 PUFA intake.

2.4.1.3 Generation of Carcinogenic Substances by Cooking Fish at a High Temperature

The known carcinogens heterocyclic amines (HCA) and polycyclic aromatic hydrocarbons (PAH) are generated by cooking fish or meat at a high temperature. The risk of colorectal cancer might depend on the cooking methods used [59, 60]. Studies on fish intake and risk of colorectal cancer should therefore take the cooking method into account.

2.4.1.4 Definition of Fish

The definition of fish varies between studies, some including shellfish and crustaceans and others working on fish and chicken [47]. This could explain some of the heterogeneity in the results.

2.4.2 Subsite-Specific Analysis

Dietary factors may be differentially related to cancer risk at different subsites of the colorectum [45–48]. Few studies have analyzed each subsite separately; more are required.

2.5 Conclusions

Epidemiologic studies have suggested that fish consumption might prevent colorectal cancer, but the results are inconsistent. In the future, we need to obtain more accurate epidemiological results based on valid estimations of exposure, by devising analyses to correctly estimate fish/ω-3 PUFA intake by dietary surveys or using biomarkers, also considering the definition of fish and the cooking methods used.

References

1. Bernard W. and Stewart P. K. World Health Organization. *World Cancer Report*. International Agency for Research on Cancer. Lyon: IARC Press. 2003.
2. Cohen D. M. How many recent fishes are there? *Published by the Academy*. 1970; 38: 341–6.
3. Standard tables of food composition in Japan, Fifth Revised and Enlarged Edition. *Kagawa Education Institute of Nutrition*. 2008.
4. Sasaki S., Horacsek M. and Kesteloot H. An ecological study of the relationship between dietary fat intake and breast cancer mortality. *Prev Med*. 1993; 22(2): 187–202.
5. Food Balance Sheet, FAOSTAT *FAO*. 2003; http://faostat.fao.org/site/610/default.aspx#ancor
6. Caygill C. P. and Hill M. J. Fish, n-3 fatty acids and human colorectal and breast cancer mortality. *Eur J Cancer Prev*. 1995; 4(4): 329–32.
7. Caygill C. P., Charlett A. and Hill M. J. Fat, fish, fish oil and cancer. *Br J Cancer*. 1996; 74(1): 159–64.
8. Bjelke E. Epidemiologic Studies of Cancer of the Stomack, Colon, and Rectum; vol. iii Case-control study in Norway. Scand J Gastroenterol. 1974; 9 (Suppl. 31): 42–8.
9. Kune S., Kune G. A. and Watson L. F. Case-control study of dietary etiological factors: the Melbourne Colorectal Cancer Study. *Nutr Cancer*. 1987; 9(1): 21–42.
10. La Vecchia C., Negri E., Decarli A., D'Avanzo B., Gallotti L., Gentile A. and Franceschi S. A case-control study of diet and colo-rectal cancer in northern Italy. *Int J Cancer*. 1988; 41(4): 492–8.
11. Iscovich J. M., L'Abbe K. A., Castelleto R., Calzona A., Bernedo A., Chopita N. A., Jmelnitzsky A. C. and Kaldor J. Colon cancer in Argentina. I: Risk from intake of dietary items. *Int J Cancer*. 1992; 51(6): 851–7.
12. Franceschi S., Favero A., La Vecchia C., Negri E., Conti E., Montella M., Giacosa A., Nanni O. and Decarli A. Food groups and risk of colorectal cancer in Italy. *Int J Cancer*. 1997; 72(1): 56–61.
13. Fernandez E., Chatenoud L., La Vecchia C., Negri E. and Franceschi S. Fish consumption and cancer risk. *Am J Clin Nutr*. 1999; 70(1): 85–90.
14. Kimura Y., Kono S., Toyomura K., Nagano J., Mizoue T., Moore M. A., Mibu R., Tanaka M., Kakeji Y., Maehara Y., Okamura T., Ikejiri K., Futami K., Yasunami Y., Maekawa T., Takenaka K., Ichimiya H. and Imaizumi N. Meat, fish and fat intake in relation to subsite-specific risk of colorectal cancer: The Fukuoka Colorectal Cancer Study. *Cancer Sci*. 2007; 98(4): 590–7.
15. Yang C. X., Takezaki T., Hirose K., Inoue M., Huang X. E. and Tajima K. Fish consumption and colorectal cancer: a case-reference study in Japan. *Eur J Cancer Prev*. 2003; 12(2): 109–15.
16. Peters R. K., Pike M. C., Garabrant D. and Mack T. M. Diet and colon cancer in Los Angeles County, California. *Cancer Causes Control*. 1992; 3(5): 457–73.
17. Tajima K. and Tominaga S. Dietary habits and gastro-intestinal cancers: a comparative case-control study of stomach and large intestinal cancers in Nagoya, Japan. *Jpn J Cancer Res*. 1985; 76(8): 705–16.
18. Kato I., Tominaga S., Matsuura A., Yoshii Y., Shirai M. and Kobayashi S. A comparative case-control study of colorectal cancer and adenoma. *Jpn J Cancer Res*. 1990; 81(11): 1101–8.
19. Inoue M., Tajima K., Hirose K., Hamajima N., Takezaki T., Hirai T., Kato T. and Ohno Y. Subsite-specific risk factors for colorectal cancer: a hospital-based case-control study in Japan. *Cancer Causes Control*. 1995; 6(1): 14–22.

20. Kotake K., Koyama Y., Nasu J., Fukutomi T. and Yamaguchi N. Relation of family history of cancer and environmental factors to the risk of colorectal cancer: a case-control study. *Jpn J Clin Oncol*. 1995; 25(5): 195–202.
21. Lee H. P., Gourley L., Duffy S. W., Esteve J., Lee J. and Day N. E. Colorectal cancer and diet in an Asian population–a case-control study among Singapore Chinese. *Int J Cancer*. 1989; 43(6): 1007–16.
22. Macquart-Moulin G., Riboli E., Cornee J., Charnay B., Berthezene P. and Day N. Case-control study on colorectal cancer and diet in Marseilles. *Int J Cancer*. 1986; 38(2): 183–91.
23. Centonze S., Boeing H., Leoci C., Guerra V. and Misciagna G. Dietary habits and colorectal cancer in a low-risk area. Results from a population-based case-control study in southern Italy. *Nutr Cancer*. 1994; 21(3): 233–46.
24. Kampman E., Verhoeven D., Sloots L. and van 't Veer P. Vegetable and animal products as determinants of colon cancer risk in Dutch men and women. *Cancer Causes Control*. 1995; 6(3): 225–34.
25. Bidoli E., Franceschi S., Talamini R., Barra S. and La Vecchia C. Food consumption and cancer of the colon and rectum in north-eastern Italy. *Int J Cancer*. 1992; 50(2): 223–9.
26. Le Marchand L., Wilkens L. R., Hankin J. H., Kolonel L. N. and Lyu L. C. A case-control study of diet and colorectal cancer in a multiethnic population in Hawaii (United States): lipids and foods of animal origin. *Cancer Causes Control*. 1997; 8(4): 637–48.
27. Knekt P., Jarvinen R., Dich J. and Hakulinen T. Risk of colorectal and other gastro-intestinal cancers after exposure to nitrate, nitrite and N-nitroso compounds: a follow-up study. *Int J Cancer*. 1999; 80(6): 852–6.
28. Kato I., Akhmedkhanov A., Koenig K., Toniolo P. G., Shore R. E. and Riboli E. Prospective study of diet and female colorectal cancer: the New York University Women's Health Study. *Nutr Cancer*. 1997; 28(3): 276–81.
29. Norat T., Bingham S., Ferrari P., Slimani N., Jenab M., Mazuir M., Overvad K., Olsen A., Tjonneland A., Clavel F., Boutron-Ruault M. C., Kesse E., Boeing H., Bergmann M. M., Nieters A., Linseisen J., Trichopoulou A., Trichopoulos D., Tountas Y., Berrino F., Palli D., Panico S., Tumino R., Vineis P., Bueno-de-Mesquita H. B., Peeters P. H., Engeset D., Lund E., Skeie G., Ardanaz E., Gonzalez C., Navarro C., Quiros J. R., Sanchez M. J., Berglund G., Mattisson I., Hallmans G., Palmqvist R., Day N. E., Khaw K. T., Key T. J., San Joaquin M., Hemon B., Saracci R., Kaaks R. and Riboli E. Meat, fish, and colorectal cancer risk: the European Prospective Investigation into cancer and nutrition. *J Natl Cancer Inst*. 2005; 97(12): 906–16.
30. Hall M. N., Chavarro J. E., Lee I. M., Willett W. C. and Ma J. A 22-year prospective study of fish, n-3 fatty acid intake, and colorectal cancer risk in men. *Cancer Epidemiol Biomarkers Prev*. 2008; 17(5): 1136–43.
31. Kojima M., Wakai K., Tokudome S., Suzuki K., Tamakoshi K., Watanabe Y., Kawado M., Hashimoto S., Hayakawa N., Ozasa K., Toyoshima H., Suzuki S., Ito Y. and Tamakoshi A. Serum levels of polyunsaturated fatty acids and risk of colorectal cancer: a prospective study. *Am J Epidemiol*. 2005; 161(5): 462–71.
32. Willett W. C., Stampfer M. J., Colditz G. A., Rosner B. A. and Speizer F. E. Relation of meat, fat, and fiber intake to the risk of colon cancer in a prospective study among women. *N Engl J Med*. 1990; 323(24): 1664–72.
33. Bostick R. M., Potter J. D., Kushi L. H., Sellers T. A., Steinmetz K. A., McKenzie D. R., Gapstur S. M. and Folsom A. R. Sugar, meat, and fat intake, and non-dietary risk factors for colon cancer incidence in Iowa women (United States). *Cancer Causes Control*. 1994; 5(1): 38–52.
34. Giovannucci E., Rimm E. B., Stampfer M. J., Colditz G. A., Ascherio A. and Willett W. C. Intake of fat, meat, and fiber in relation to risk of colon cancer in men. *Cancer Res*. 1994; 54(9): 2390–7.
35. Hsing A. W., McLaughlin J. K., Chow W. H., Schuman L. M., Co Chien H. T., Gridley G., Bjelke E., Wacholder S. and Blot W. J. Risk factors for colorectal cancer in a prospective study among U.S. white men. *Int J Cancer*. 1998; 77(4): 549–53.

36. Ma J., Giovannucci E., Pollak M., Chan J. M., Gaziano J. M., Willett W. and Stampfer M. J. Milk intake, circulating levels of insulin-like growth factor-I, and risk of colorectal cancer in men. *J Natl Cancer Inst*. 2001; 93(17): 1330–6.
37. Sanjoaquin M. A., Appleby P. N., Thorogood M., Mann J. I. and Key T. J. Nutrition, lifestyle and colorectal cancer incidence: a prospective investigation of 10998 vegetarians and non-vegetarians in the United Kingdom. *Br J Cancer*. 2004; 90(1): 118–21.
38. Goldbohm R. A., van den Brandt P. A., van ′t Veer P. , Brants H. A. , Dorant E. , Sturmans F. and Hermus R. J. A prospective cohort study on the relation between meat consumption and the risk of colon cancer. *Cancer Res*. 1994; 54(3): 718–23.
39. Tiemersma E. W., Kampman E., Bueno de Mesquita H. B., Bunschoten A., van Schothorst E. M., Kok F. J. and Kromhout D. Meat consumption, cigarette smoking, and genetic susceptibility in the etiology of colorectal cancer: results from a Dutch prospective study. *Cancer Causes Control*. 2002; 13(4): 383–93.
40. Luchtenborg M., Weijenberg M. P., de Goeij A. F., Wark P. A., Brink M., Roemen G. M., Lentjes M. H., de Bruine A. P., Goldbohm R. A. , van ′t Veer P. and van den Brandt P. A. Meat and fish consumption, APC gene mutations and hMLH1 expression in colon and rectal cancer: a prospective cohort study (The Netherlands). *Cancer Causes Control*. 2005; 16(9): 1041–54.
41. Gaard M., Tretli S. and Loken E. B. Dietary factors and risk of colon cancer: a prospective study of 50,535 young Norwegian men and women. *Eur J Cancer Prev*. 1996; 5(6): 445–54.
42. Kobayashi M., Tsubono Y., Otani T., Hanaoka T., Sobue T. and Tsugane S. Fish, long-chain n-3 polyunsaturated fatty acids, and risk of colorectal cancer in middle-aged Japanese: the JPHC study. *Nutr Cancer*. 2004; 49(1): 32–40.
43. Kojima M., Wakai K., Tamakoshi K., Tokudome S., Toyoshima H., Watanabe Y., Hayakawa N., Suzuki K., Hashimoto S., Ito Y. and Tamakoshi A. Diet and colorectal cancer mortality: results from the Japan Collaborative Cohort Study. *Nutr Cancer*. 2004; 50(1): 23–32.
44. Khan M. M., Goto R., Kobayashi K., Suzumura S., Nagata Y., Sonoda T., Sakauchi F., Washio M. and Mori M. Dietary habits and cancer mortality among middle aged and older Japanese living in hokkaido, Japan by cancer site and sex. *Asian Pac J Cancer Prev*. 2004; 5(1): 58–65.
45. English D. R., MacInnis R. J., Hodge A. M., Hopper J. L., Haydon A. M. and Giles G. G. Red meat, chicken, and fish consumption and risk of colorectal cancer. *Cancer Epidemiol Biomarkers Prev*. 2004; 13(9): 1509–14.
46. Pietinen P., Malila N., Virtanen M., Hartman T. J., Tangrea J. A., Albanes D. and Virtamo J. Diet and risk of colorectal cancer in a cohort of Finnish men. *Cancer Causes Control*. 1999; 10(5): 387–96.
47. Chao A., Thun M. J., Connell C. J., McCullough M. L., Jacobs E. J., Flanders W. D., Rodriguez C., Sinha R. and Calle E. E. Meat consumption and risk of colorectal cancer. *Jama*. 2005; 293(2): 172–82.
48. Larsson S. C., Rafter J., Holmberg L., Bergkvist L. and Wolk A. Red meat consumption and risk of cancers of the proximal colon, distal colon and rectum: the Swedish Mammography Cohort. *Int J Cancer*. 2005; 113(5): 829–34.
49. Anti M., Armelao F., Marra G., Percesepe A., Bartoli G. M., Palozza P., Parrella P., Canetta C., Gentiloni N., De Vitis I. and et al. Effects of different doses of fish oil on rectal cell proliferation in patients with sporadic colonic adenomas. *Gastroenterology*. 1994; 107(6): 1709–18.
50. Cheng J., Ogawa K., Kuriki K., Yokoyama Y., Kamiya T., Seno K., Okuyama H., Wang J., Luo C., Fujii T., Ichikawa H., Shirai T. and Tokudome S. Increased intake of n-3 polyunsaturated fatty acids elevates the level of apoptosis in the normal sigmoid colon of patients polypectomized for adenomas/tumors. *Cancer Lett*. 2003; 193(1): 17–24.
51. Geelen A., Schouten J. M., Kamphuis C., Stam B. E., Burema J., Renkema J. M., Bakker E. J., van’t Veer P. and Kampman E. Fish consumption, n-3 fatty acids, and colorectal cancer: a meta-analysis of prospective cohort studies. *Am J Epidemiol*. 2007; 166(10): 1116–25.

52. World Cancer Research Fund and American Institute for Cancer Research. *Food, Nutrition, Physical Activity, and the Prevention of Cancer: A Global Perspective*. Washington, DC: American Institute for Cancer Research. 2007: 116–26.
53. de Deckere E. A. Possible beneficial effect of fish and fish n-3 polyunsaturated fatty acids in breast and colorectal cancer. *Eur J Cancer Prev*. 1999; 8(3): 213–21.
54. Larsson S. C., Kumlin M., Ingelman-Sundberg M. and Wolk A. Dietary long-chain n-3 fatty acids for the prevention of cancer: a review of potential mechanisms. *Am J Clin Nutr*. 2004; 79(6): 935–45.
55. Tokudome Y., Imaeda N., Ikeda M., Kitagawa I., Fujiwara N. and Tokudome S. Foods contributing to absolute intake and variance in intake of fat, fatty acids and cholesterol in middle-aged Japanese. *J Epidemiol*. 1999; 9(2): 78–90.
56. Wakai K., Ito Y., Kojima M., Tokudome S., Ozasa K., Inaba Y., Yagyu K. and Tamakoshi A. Intake frequency of fish and serum levels of long-chain n-3 fatty acids: a cross-sectional study within the Japan Collaborative Cohort Study. *J Epidemiol*. 2005; 15(6): 211–8.
57. Willett W. *Nutritional Epidemiology Second Edition*. Oxford, NY: Oxford University Press. 1998: 74–87.
58. Hjartaker A., Lund E. and Bjerve K. S. Serum phospholipid fatty acid composition and habitual intake of marine foods registered by a semi-quantitative food frequency questionnaire. *Eur J Clin Nutr*. 1997; 51(11): 736–42.
59. Sinha R., Peters U., Cross A. J., Kulldorff M., Weissfeld J. L., Pinsky P. F., Rothman N. and Hayes R. B. Meat, meat cooking methods and preservation, and risk for colorectal adenoma. *Cancer Res*. 2005; 65(17): 8034–41.
60. de Kok T. M. and van Maanen J. M. Evaluation of fecal mutagenicity and colorectal cancer risk. *Mutat Res*. 2000; 463(1): 53–101.

Chapter 3
ω-3 PUFAs and Colon Cancer: Experimental Studies and Human Interventional Trials

Simona Serini, Elisabetta Piccioni, and Gabriella Calviello

Abstract Colon cancer represents one of the most frequent forms of cancer worldwide. There is plenty of evidence to support the notion that chemoprevention is a major component of colon cancer control, and that dietary fats influence the rate of incidence of this kind of cancer. In particular, a variety of experimental studies conducted on animals subject to chemical carcinogenesis of colon, transplanted with colon cancer cells, or representing a genetic model of colon cancer have demonstrated the ability of diets at high content of ω-3 polyunsaturated fatty acids (PUFAs) to reduce the risk, growth, and progression of colon cancer. Several of these studies have also emphasized the importance of a reduced dietary ω-6 PUFA/ω-3 PUFA ratio to diminish the risk. The beneficial anti-tumor effects of ω-3 PUFAs have been also largely demonstrated in colon cancer cells treated in vitro with these fatty acids. The synergic action of combinations of ω-3 PUFAs and other anti-neoplastic agents has also been demonstrated both in vivo and in vitro. On the contrary, just a few human interventional trials have been conducted so far, but there is complete agreement among them and with the experimental studies on the anti-tumor efficacy of increased dietary intakes of ω-3 PUFAs.

Keywords ω-3 PUFAs · Growth-inhibiting effect · Colon cancer · Experimental studies · Interventional trials

3.1 Introduction

Colorectal cancer represents one of the most frequent forms of cancer worldwide and especially among Western populations [1]. According to a recent estimate [2], it represents the second prevalent cancer in Europe for incidence (2.9% of all cases

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of cancer) and mortality (12.2% of all deaths for cancer). Its incidence has been strictly related to the dietary habits of populations, and, accordingly, it has recently decreased as a result of lifestyle changes in the West [3]. On the other hand, its incidence among Japanese, previously much lower than that of Western populations, is now increasing in relation to the change of dietary habits observed in Japan in the last few decades [4]. Interestingly, in China and in other developing countries, economic development and the adoption of earlier Western lifestyles have led to an increased incidence, similar to that present in developed countries more than two decades ago [5].

The high incidence of colon cancer among Western populations was ascribed to the high intake of some classes of dietary fat. At first the finger was pointed at the high level of saturated fatty acids in the Western diet [6], particularly in meat and dairy products, and subsequently also at the ω-6 polyunsaturated fatty acids (PUFAs) present at high levels in fat of both terrestrial animals and vegetables [7]. Conversely, it was suggested that the ω-3 PUFAs, contained at high levels in fish tissues, could exert a beneficial role against colon cancer [8, 9]. Accordingly, the results of the "European Prospective Investigation into Cancer and Nutrition" published in 2005 [10] confirmed the association between colon cancer risk and a high consumption of red meat and supported the inverse association with intake of fish reported by previous studies [11–13].

It was suggested that coming back to a dietary regimen similar to that of our prehistoric ancestors, presumably able to provide higher levels of ω-3 PUFAs and much lower levels of ω-6 PUFAs than those daily ingested at the present time, we could restore a dietary ω-3 PUFA/ω-6 PUFA ratio much higher (about ten times and more higher) and healthier for us. In particular, if this hypothesis is correct, a dietary switch to an ω-3 PUFA richer diet would allow to come back to a much lower degree of incidence of colorectal cancer.

3.2 Experimental Studies

3.2.1 Animal Studies

All the findings obtained through preclinical studies with animals have been conducted utilizing three different types of animal models: (a) chemically induced colon carcinogenesis rat models; (b) colon tumor cells transplanted into animals; (c) genetic models for the induction of colon cancer. We will analyze separately the results obtained using each type of model in the following three sections.

3.2.1.1 Chemically Induced Colon Carcinogenesis in Rat Models

These represent very useful models since the morphology of the tumors obtained and their histochemical properties and biological behavior appear very similar to those of human colon cancers [14]. For this reason most of the experiments performed to investigate the anti-tumor effect of ω-3 PUFAs in animals have been carried out using this approach. In particular, since the late 1980s and the beginning of 1990s, the effects of ω-3 PUFAs on colon carcinogenesis induced by a colon-specific carcinogen, azoxymethane (AOM), have been extensively studied. Since then, a clear inhibitory effect of dietary fish oils (FO), as well as of purified EPA or DHA on the incidence and multiplicity (number of tumors/rat) of preneoplastic foci or tumors [15–18] was observed. On the contrary, the same reports showed the tumor-promoting action of diets containing high levels of corn oil (CO), a vegetable oil rich in ω-6 PUFAs. Such beneficial effects of ω-3 PUFA-enriched diets were initially related to the diminished production of prostaglandin E_2 (PGE_2) [15] or decreased activity of ornithine decarboxylase [17] in the tumors. Similar beneficial anti-cancer effects were also observed in rats subject to colon carcinogenesis with 1,2-dimethyl-hydrazine (DMH) [19–21] and fed with diets at high levels of FO. It is worth noticing that from the beginning great efforts have been made to find ω-3 PUFA sources alternative to FO, given the possible contamination of fish with dangerous carcinogenic compounds [22]. Hence, different studies were also performed at that initial stage to evaluate the effect of perilla oil in rats subject to chemical carcinogenesis with 7,12-dimethylbenz[a]anthracene (DMBA) and DMH [23] as well as *N*-methyl-*N*-nitrosourea (MNU) [24, 25] or AOM [26]. Perilla oil is a vegetable oil with high content of α-linolenic acid (18:3 ω-3, ALA), and diets enriched in this oil decreased both the number of colonic aberrant crypt foci (ACF) and the incidence and number of tumors/rats. The protective effects of ω-3 PUFAs against AOM-induced colon carcinogenesis were later confirmed also using another vegetable oil at high content of ALA, flaxseed oil [27, 28], and also very recently with the use of a microalgal oil, even though considerable differences in fatty acid composition and total amount of ω-3 PUFAs exist between the microalgal oil and fish oils [29]. Moreover, in an attempt to identify which of the ω-3 PUFAs contained in fish oil may be considered the most active chemopreventive component, until the beginning the effect of supplementation of purified fatty acids such as EPA (4.7%) [15] or DHA (0.7–1.0 ml, given by gavage for periods ranging from 4 to 12 weeks) [16, 30, 31] given to animals subject to chemical carcinogenesis was investigated by different authors. It was observed that either EPA or DHA, if supplemented alone with the diet, had the potential to inhibit the incidence and multiplicity of AOM-induced colon tumors with the same efficiency of FO [15, 31]. Moreover, DHA was also shown to hamper the development of 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP)-induced aberrant colon crypt foci in rats [16, 30]. Reddy et al. [32] showed that colon tumor incidence and multiplicity were significantly reduced in groups of rats fed a diet enriched with FO either at initiation or at post-initiation phases of carcinogenesis with AOM, as compared to groups fed CO. This effect was related by the same authors to the enhanced activities of phospholipase A_2 (PLA_2) and phosphatidylinositol-specific phospholipase C (PI-PLC), the formation of prostaglandins and thromboxane A_2 [33], as well as the levels of diacylglycerol kinase (DGK) and PKC activities, all produced in the colonic mucosa by a diet at high levels of ω-6 PUFAs [34]. However, they did not exclude also that dietary ω-6 PUFAs may enhance colon tumor promotion by increasing the levels of cecal bacterial 7 alpha-dehydroxylase and of secondary bile acids, such as deoxycholic acid and lithocholic acid, all involved in colon carcinogenesis [34]. The same authors also observed that the carcinogenic treatment with AOM caused increased expression of cyclooxygenase-2 (COX-2) in colonic mucosa, and that dietary ω-6 PUFAs and ω-3 PUFAs influenced in opposite ways the AOM-induced COX-2-enhanced expression. The induction of COX-2 expression by ω-6 PUFAs and, conversely, the inhibition of COX-2 expression by ω-3 PUFAs were related to the opposite effects of these fatty acids on tumor incidence and multiplicity in colon [35]. Also the expression of Ras-p21, the gene product implicated in the etiology of many human malignancies, was induced by AOM carcinogenesis in colonic mucosa and was influenced by the levels and quality of PUFAs ingested. Diets containing high levels of CO, and hence rich in ω-6 PUFAs, increased the expression of both Ras-p21 and the enzyme that catalyzes the farnesylation of ras precursors (farnesyl protein transferase, FPTase). This enables the anchorage of Ras to plasma membrane, a crucial step in colon tumorigenesis [35, 36]. Conversely, diets containing high levels of FO, and hence rich in ω-3 PUFAs, exerted their anti-tumor activity by interfering with posttranslational modification and membrane localization of Ras-p21 [35, 36]. Considering all these results, however, the controversy still remained if the chemopreventive effects of diets at high levels of ω-3 PUFAs were due to the direct action of ω-3 PUFAs or to a reduction in the content of ω-6 PUFAs. The findings of Davidson et al. [37] were decisive in this sense, since they showed that the protective effects exerted by the increased intake of ω-3 PUFAs at both the initiation (DNA adduct formation) and promotional (aberrant crypt foci) stages of rat AOM-induced carcinogenesis were related to the unique ability of ω-3 PUFAs to directly alter colonic gene expression profiles in vivo, and in particular to the capacity to up-regulate genes involved in colonic cell apoptosis and differentiation. For instance, in this respect it is worth noting the ω-3 PUFA-induced overexpression of retinoid X receptor-α (RXR-α), a component of retinoid X receptors [38], a family of nuclear receptors involved in cancer chemoprevention, and for which DHA represents a possible ligand. Similarly, the adipocyte lipid binding protein (ALBP) was up-regulated by ω-3 PUFAs at colon level. This is a binding protein shown to influence cell signaling by peroxisome proliferator-activated receptor-γ (PPAR-γ), a nuclear receptor and transcription factor which functions to induce terminal differentiation in the colon [39, 40], thereby suppressing colon carcinogenesis [41].

The protective effect of dietary fish oils during AOM-induced carcinogenesis has been related to their ability to induce differentiation and to increase the percentage of apoptotic cells in crypts of both proximal and distal colon [42]. It was observed that a long period (38 weeks) of treatment with a diet enriched in ω-3 PUFAs increased the percentage of apoptotic cells in the colonic mucosa of rats subject to AOM-induced carcinogenesis [43], as compared to a diet rich in saturated fatty acids or ω-6 PUFAs. This finding is worth noting since suppression of apoptosis is considered a predictive intermediate biomarker of tumor development. Interestingly, it was also observed that diets enriched in ω-6 PUFAs induced a decreased expression of Polo-like kinase-3 (PLK3) in colon tumors from AOM-treated rats [44]. PLK3, known to exert pro-apoptotic effects, was down-regulated in AOM-induced colon tumors. In this case, however, dietary ω-3 PUFAs did not revert but simply did not exacerbate the down-regulation of PLK3 present in tumor cells, nevertheless demonstrating that only ω-6 PUFAs exert pro-carcinogenic action. It was also suggested that induction of apoptosis may represent an important mechanism through which fish oils may protect against tumor development in AOM-treated rats during initiation, resulting in the reduction of *O*6-methylguanine DNA adduct levels in rat colon [45]. The incremental stimulation of apoptosis was observed in the top one-third of the crypts, as adduct level increased. The ability of dietary fish oil to decrease oxidative DNA damage in colon mucosa by enhancing apoptosis was observed also in colon cells of dextran sodium sulfate (DSS, an inducer of oxidative DNA damage)-treated rats [46]. The ability of fish oils to down-regulate the anti-apoptotic protein Bcl-2 has also been invoked to explain their pro-apoptotic effect during colon tumor initiation [47]. Another possibility put forward to explain the ω-3 PUFA chemopreventive action during chemical carcinogenesis is based on their ability to regulate the colonic levels of the so-called I-compounds, which are bulky covalent DNA modifications that are derived from metabolic intermediates of nutrients [48]. Some I-compounds may play protective roles against cancer and many carcinogens and promoters decreased their tissue levels. It was observed that AOM treatment significantly reduced I-compound levels in colon of rats, whereas fish oil diet protected colon cells from the reduction.

Finally, a mention should be done of the suppression of AOM-induced carcinogenesis in transgenic mice carrying the *fat-1* gene and showing endogenously increased tissue levels of ω-3 PUFAs [49]. This gene encodes an ω-3 fatty acid desaturase which has been recently cloned from *Caenorhabditis elegans* and expressed in mammalian cells [50]. When the mice were subject to pro-inflammatory treatment with dextran sodium sulfate combined with AOM carcinogenesis a lower incidence of colon tumors and a reduced tumor growth rate was observed [51, 52] with respect to wild mice. All the results obtained using this model concur to suggest that the anti-tumorigenic effect of high endogenous levels of ω-3 PUFAs could be mediated, at least in part, via its anti-inflammatory properties. It was observed that in *fat-1* animals the colonic inflammation which followed the dextran treatment and a 2-week recovery period was inhibited as compared to the wild-type counterparts [52]. In combination, the reversion to normal of a series of inflammatory parameters was observed, including the decreased activity of nuclear factor-kappa B (NF-κB), the increased expression of transforming growth factor beta in the colons, and the lower expression of inducible nitric oxide synthase in the tumors of *fat-1* animals [51], as well as the decreased CD3(+), CD4(+) T helper, and macrophage cell numbers per colon [52].

Figure 3.1 summarizes the results of the experimental studies performed using animals subject to colon carcinogenesis and treated with ω-3 PUFAs. The anti-tumoral effects of ω-3 PUFAs and the molecular mechanisms proposed are shown in the figure.

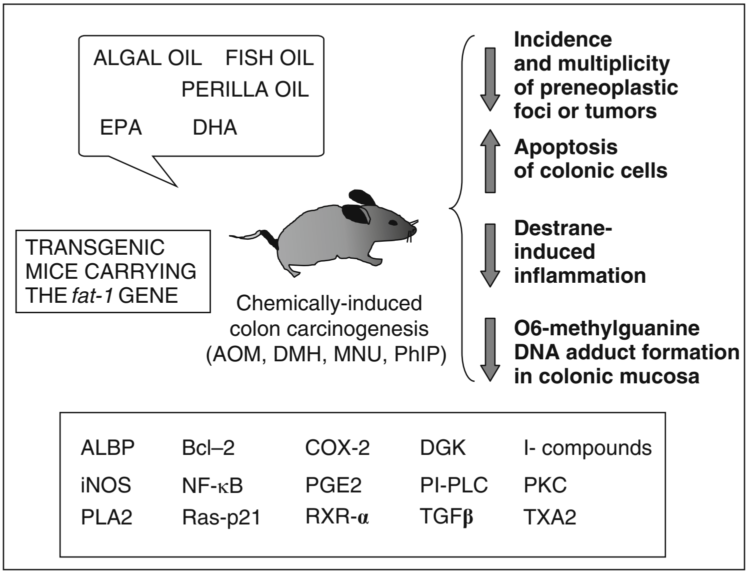

Fig. 3.1 Experimental studies performed on animals subject to chemical carcinogenesis of colon. The animals subject to chemical carcinogenesis were either dietary treated with ω-3 PUFAs or already endogenously enriched with ω-3 PUFAs (transgenic *fat-1* mouse model). On the *right*: the observed anti-tumoral effects are shown. The *down* and *up arrows* indicate the inhibition or stimulation of the indicated processes. *Bottom box*: the molecular factors known to be involved in the regulation of tumor growth, cell cycle, and apoptosis and reported to be affected by ω-3 PUFA treatments in these studies are shown. ALBP, adipocyte lipid binding protein; AOM, azoxymethane; Bcl-2, B-cell lymphoma-2; COX-2: cyclooxygenase-2; DGK, diacylglycerol kinase; DMH, 1, 2-dimethyl-hydrazine; iNOS, inducible nitric oxide synthase; NF-κB, nuclear factor kappa B; MNU, *N*-methyl-*N*-nitrosourea; PGE_2, prostaglandin E_2; PhIP, 2-amino-1-methyl-6-phenylimidazo [4,5-b]pyridine; PI-PLC, phosphatidylinositol-specific phospholipase C; PKC, protein kinase C; PLA_2, phospholipase A_2; TGF-β, transforming growth factor-beta; RXR-α, retinoid X receptor-alpha; TXA_2, thromboxane A_2. See text for further details

3.2.1.2 Tumor Cells Transplanted into Animals

The first model of transplanted colonic cancer cells to be used was the murine MAC16 adenocarcinoma colon cell line implanted in NMRI syngenic mice [53, 54]. MAC16 are cachexia-inducing cells, and oral administration of purified EPA (1.25–2.50 g/kg) in rats bearing MAC16 implants inhibited both host weight loss and tumor growth rate in a dose-dependent manner, even though the EPA inhibitory effect on tumor growth was visible only during the first week of treatment and not later. This effect was not related to an increased incorporation of this fatty acid in tumor cells and involved an increase in the rate of cell loss. In agreement, EPA and DHA exerted also inhibitory effects on the growth of the colon carcinoma 26 (CC26) cell line implanted s.c. in CDF1 mice [55]. This tumor cell line shows high metastatic potential when injected i.v., and DHA was shown to be particularly efficient in inhibiting the number of metastatic nodules in lung [56].

This effect was related to the ability of DHA to decrease the expression of membrane metalloproteinase-9 (MMP-9), whose activity is crucial for tumor invasion. In agreement, Iwamoto et al. [57] found that dietary EPA inhibited the formation of liver metastatic foci of ACL-15 colon cancer cells injected in F344 rats, as compared to dietary LA or standard chow. On the basis of in vitro experiments the authors suggest that the effect was related to EPA ability to inhibit the expression of vascular cell adhesion molecules and tumor cell proliferation at the secondary site. However, it should be remembered that, in complete disagreement with all the above reported results, it was also observed [58] that syngeneic Wag-Rij rats fed diets enriched in fish oil (20% v/w of a fish oil containing 70% ω-3 PUFAs) and i.v. injected with the cell line CC531 showed a much increased number of liver metastases as compared to rats fed low-fat diets. On the other hand, more recently, other authors [59], injecting the same cell line CC531 in the spleen of Wag-Rij rats fed a ω-3 PUFA-enriched diet (15% v/w, EPA plus DHA), found that this treatment did decrease the incidence of malignant metastatic tumor growth in the liver. Several methodological differences may be invoked to explain the discrepancies in the results, including the via used for cell injection, the type of ω-3 PUFAs added to the diets (more purified in the more recent work), and the length of the pre-treatment with ω-3 PUFAs before cell injection (3 weeks and 3 days for the first and second work, respectively).

Moreover, several authors reported the in vivo ω-3 PUFA-induced growth inhibition of tumors originated from human colon cancer cell lines injected in athymic nude mice [60–66]. Among the human colon cell lines subcutaneously implanted there are SW620, LS174, Colo 320 and Colo 205, HT-29, HCT-116, WiDr, DDL1. It is interesting to note that the growth in nude mice of both the p53 wild-type human colon cancer cells Colo205 and the p53 mutant WiDr was inhibited by dietary FO, demonstrating that the effect of ω-3 PUFAs was irrespective of the cell p53 status [63, 66]. Moreover, a diet enriched with golden algal oil, which furnishes almost exclusively DHA, was shown to inhibit the growth of these cells in vivo more efficiently than the diet enriched with FO, suggesting for DHA a primary suppressive role against colon tumor growth. On the other hand, Calder et al. [67] reported that the growth of HT-29 cells in nude mice was not influenced by the dietary source of fat (200 g/kg of diet) and was not different if the animals were dietary treated with either safflower oil (rich in ω-6 PUFAs) or Menhaden oil (a fish oil rich in ω-3 PUFAs). However, more recently, inoculating HT-29 cells in nude mice supplemented either with EPA- or DHA-ethyl esters (1 g/kg/die) [65], we observed a marked inhibition of tumor growth. Moreover, HT-29 cells growing in nude mice fed EPA or DHA showed a reduced expression of COX-2 and VEGF, lower PGE_2 levels, and reduced neoangiogenesis, as demonstrated by the decreased formation of neovessels in the tumors, thus suggesting that the inhibition of tumor growth was strictly related to the inhibition of angiogenesis, with a COX-2-dependent mechanism [65]. However, dietary ω-3 PUFAs were shown to inhibit the in vivo growth of HCT-116 cells, which are constitutionally COX-2 negative cells, suggesting that the mechanism underlying the growth suppressive effect may also be COX-2 unrelated [62]. Furthermore, Tzuzuki et al. [64] found that dietary conjugated EPA (CEPA) inhibited more efficiently than EPA itself the growth of colon cancer DDL1 cells in vivo,

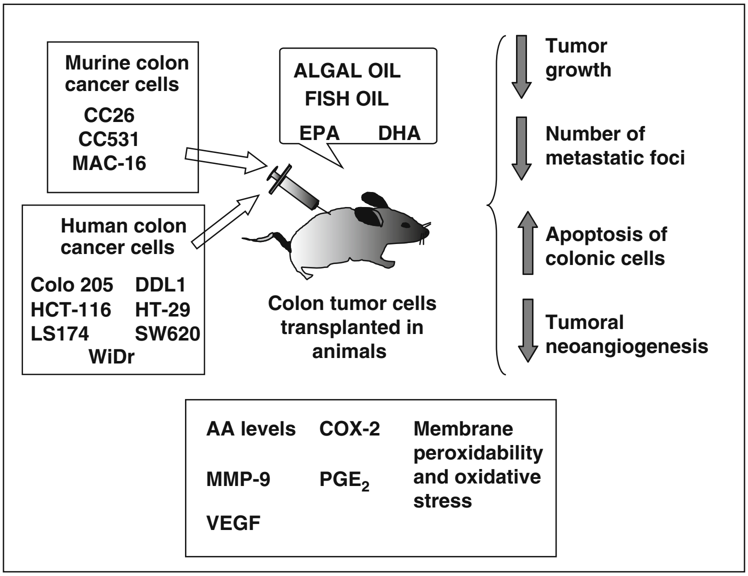

Fig. 3.2 Experimental studies performed transplanting colon cancer cells in animals. The animals were dietary treated with ω-3 PUFAs. The cells of murine origin were implanted in syngeneic animals. The cells of human origin were implanted in immunodeficient animals. On the *right*: the observed anti-tumoral effects are shown. The *down* and *up arrows* indicate the inhibition or stimulation of the indicated processes. *Bottom box*: the molecular factors known to be involved in the regulation of tumor growth, cell cycle, apoptosis, and angiogenesis and reported to be affected by ω-3 PUFA treatments in these studies are shown. AA, arachidonic acid; COX-2, cyclooxygenase-2; MMP-9, matrix metalloproteinase-9; PGE_2, prostaglandin E_2; VEGF, vascular endothelial growth factor. See text for further details

and this effect was related to the higher peroxidability of CEPA and the consequent increased level of oxidative stress, DNA fragmentation, and apoptosis [64].

Figure 3.2 summarizes the results of the experimental studies performed using colon cancer cells transplanted in animals treated with ω-3 PUFAs. The observed ω-3 PUFA anti-tumoral effects, and the molecular mechanisms proposed are shown in the figure.

3.2.1.3 Genetic Models for the Induction of Colon Cancer

Several reports have investigated the effects of ω-3 PUFAs in a murine model of familial adenomatous polyposis, the multiple intestinal neoplasia (Min) mouse model (APC min/+). This is a genetic model carrying a germ-line mutation of adenomatous polyposis colon (APC) gene at codon 850, resulting in the formation of a nonfunctional truncated gene product [68]. The somatic mutation of the remaining wild-type allele causes the spontaneous development of tumors throughout the intestinal tract of mice. In the APC min/+ mouse, however, the tumors which arise are multiple adenomas localized primary in the small intestine and seldom in the colon [69]; thus, for many aspects, it cannot be completely compared to human colon cancers. The first report [70] showed that dietary fish oil given to APC min/+ mice reduced significantly the number and diameter of small intestinal tumors. The same effect was noticed also in the large intestine, but in that tract it did not reach statistical significance. These effects were noticed in both sexes, even though not always at the same degree. Fish oil treatment also reduced the formation of aberrant crypt foci (ACF) in these animals, but only in females. Afterward, in females the effect of EPA, alone or in combination with arachidonic acid (AA) was investigated. EPA was able to reduce the number of tumors, but the concomitant ingestion of AA, which led to increased AA tissue content and AA-derived prostaglandin production, inhibited this effect. These findings suggested that the tumor inhibitory effect of EPA was strictly related to its ability to reduce AA tissue content and inhibit its metabolism. More recently, it was proven that also the dietary treatment with flaxseed, containing high levels of ALA, was able to negatively affect the number and size of adenomas developed in small intestine [71]. Interestingly, the authors found that also the flaxseed-derived oil, with the same fatty acid composition of flaxseed, but lacking its lignans, retained the same anti-tumor effect. Since lignans contained in flaxseed are themselves chemopreventive compounds, these findings showed that ALA was mainly responsible for the anti-tumor effect observed for flaxeed oil.

A different genetic model of colon carcinogenesis was developed in mice [72] on the basis of the observation that protein kinase C βII (PKCβII) promotes colon carcinogenesis and that PKCβII was induced during colon carcinogenesis in rodents and humans [73]. Transgenic mice overexpressing PKCβII exhibited hyperproliferation, colon carcinogenesis, and marked repression of transforming growth factor β receptor type II (TGFβRII) and COX-2 expression at colonic level [74, 75]. Dietary ω-3 PUFAs inhibited colonic PKCβII activity in vivo, blocked PKCβII-mediated hyperproliferation and enhanced carcinogenesis, and repressed TGFβRII and COX-2 expression in the colonic epithelium of transgenic PKCβII mice. These findings suggested that prevention of colon cancer by dietary ω-3 PUFAs involved inhibition of colonic PKCβII and COX-2 signaling, as well as restoration of TGF-β responsiveness.

3.2.2 In Vitro Studies with Normal and Tumor Colonic Cells

No sooner the first in vivo studies had demonstrated the efficacy of dietary ω-3 PUFAs against the growth of colon tumors in animals subject to colon carcinogenesis than the growth-inhibiting effect was confirmed also in colon cancer cell lines cultured in vitro [76]. Crucial tasks for the first in vitro studies were the demonstration of the ω-3 PUFA inhibiting effect on colon cancer cell growth and the identification of the possible molecular mechanisms underlying that effect. Nowadays, that the inhibiting effect has been well recognized, the identification of the molecular mechanisms underlying their action has become the major aim of this research field. The detailed analysis of all the reports investigating the molecular mechanisms of ω-3 PUFA action in colon cancer cells goes beyond the scope of this chapter; however, a brief summary will be given for the hypothesized pathways of ω-3 PUFA action, identified through the in vitro studies on colon cancer cells. Several works in vitro have shown that the induction of apoptosis is the main biological mechanism through which ω-3 PUFAs, and DHA in particular, decrease colon cancer cell growth and may prevent colon carcinogenesis [77, 78]. The increased lipid peroxidation in the cells following ω-3 PUFA treatment and the consequent induction of oxidative stress and oxidative adducts have been largely invoked as a possible mechanism underlying the decreased cell growth [79–81]. In particular, increase in phospholipid oxidation and Ca^{2+} accumulation in mitochondria has been recently proposed as a possible mechanism of ω-3 PUFA-induced apoptosis in colonocytes [82, 83]. Moreover, studies performed on colon cancer cells in vitro have added strength to the hypothesis that ω-3 PUFA-induced alteration in the intracellular levels of AA and AA-oxygenated metabolites with pro-inflammatory and pro-carcinogen activities (especially prostaglandin E_2) may represent a major pathway for their anti-cancer action [65, 84]. A number of works in vitro have also pointed out that the anti-neoplastic action of these compounds may be mediated by the alteration in the expression of many molecular factors which are involved in colon carcinogenesis [85]. Among these factors we can include several transcription factors and some of their components (β-catenin; hypoxia inducible factor 1a, HIF-1α; peroxisome proliferator-activated receptors, PPARs; nuclear sterol regulatory element-binding protein 1, nSREBP1; NF-κB isoforms) [86–92]; kinases and other molecules which function as key regulators of signal transduction (AKT/protein kinase B, AKT/PKB; p38, PI3-kinase, ERK1/2, PKC, Ras) [65, 93–95]; metabolic enzymes involved in the production of bioactive molecules (inducible nitric oxide synthase, iNOS; COX-2) [65, 84, 96] or in the modification of the extracellular matrix (matrix metalloproteinases, MMPs) [90]; factors involved in the regulation of cell cycle progression (p21, cyclin D1 and E) [88, 97], apoptosis (caspases, Bcl-2 family of proteins, c-myc, the anti-apoptotic protein myeloid cell leukemia sequence-1, Mcl-1; survivin) [78, 88, 90, 98], and angiogenesis (VEGF, HIF-1α) [65, 90]; factors involved in ER stress-induced response (eukaryotic translation initiation factor 2, eIF2α) [99].

It was recently shown that whereas the normal colonic mucosa cells were not affected by the treatment in vitro with DHA, colon cancer cells specifically reduce their viability and undergo apoptosis [93, 94], suggesting a specific growth-inhibiting effect in neoplastic cells. This recent finding confirmed the earlier observations of Tsai et al. [100] who, however, used a more complicated cellular model and investigated only the effect on cell growth and not on apoptosis. They showed the selective growth inhibitory effects of EPA or DHA treatments (10–20 μg/mL) only on the sigmoid colon cancer oncogene transformant NIH3T3 cell line (transfected with a DNA fragment from sigmoid colon cancer containing the ret-II oncogene), whereas no effects were observed in the parental NIH3T3 line. However, this matter awaits further confirmation since, by comparing cells transformed or not with oncogenic ras, it was also shown that only non-transformed cells were sensitive to the apoptogenic effects of DHA, indicating that ras transformation alters sensitivity to this dietary chemopreventive agent [101]. This subject, however, is quite controversial, since suppression of apoptosis is generally considered a predictive intermediate biomarker of tumor development during colon carcinogenesis [102], and thus a desirable outcome for normal colon cells following chemoprevention.

Figure 3.3 summarizes the results of experimental studies performed using colon cancer cells cultured in vitro in the presence of ω-3 PUFAs. The observed effects and the molecular mechanisms proposed are shown in the figure.

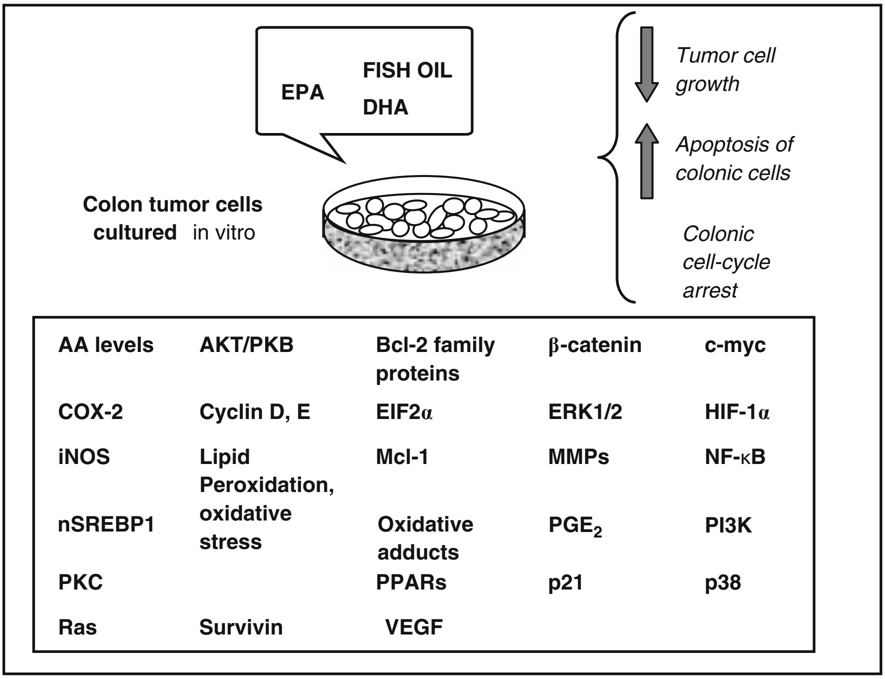

Fig. 3.3 Experimental studies performed on colon cancer cells cultured in vitro. The cells were treated with ω-3 PUFAs added in the culture media. On the *right*: the observed anti-tumoral effects are shown. The *down* and *up arrows* indicate the inhibition or stimulation of the indicated processes. *Bottom box*: the molecular factors known to be involved in the regulation of cell cycle, apoptosis, and angiogenesis and reported to be affected by ω-3 PUFA treatments in these studies are shown. AA, arachidonic acid; Bcl-2, B-cell lymphoma-2; COX-2, cyclooxygenase-2; EIF2α, eukaryotic translation initiation factor 2; ERK1/2, extracellular signal regulated kinase1/2; HIF, hypoxia inducible factor; iNOS, inducible nitric oxide synthase; MMPs, matrix metalloproteinases; Mcl-1, myeloid cell leukemia sequence-1; NF-κB, nuclear factor kappa B; nSREBP1, nuclear sterol regulatory element-binding protein 1; PPARs, peroxisome proliferator-activated receptors; PGE_2, prostaglandin E_2; PI3K, phosphoinositide-3 kinase; PKB, protein kinase B; PKC, protein kinase C; VEGF, vascular endothelial growth factor. See text for further details

3.2.3 *In Vitro and In Vivo Combined Treatments with ω-3 PUFAs and Other Anti-neoplastic Agents*

To conclude this section regarding the experimental works performed to study the efficacy of ω-3 PUFAs against colon cancer, we will analyze the studies conducted to evaluate the efficacy of combinations of ω-3 PUFAs and other anti-neoplastic agents against this kind of cancer, both in vivo on animal models and in cells cultured in vitro. This is a field that in the past few years has received great attention. Many works have been conducted to investigate the possibility that ω-3 PUFA treatment could enhance the efficacy of conventional chemotherapic agents and radiotherapy, as well as that of mono-target drugs. This approach is very interesting since ω-3 PUFAs are natural components of our diet, considered safe within the range of dosages known to be beneficial against many chronic disorders. It was hypothesized that the possible ω-3 PUFA potentiating action toward conventional or novel anti-neoplastic drugs, whose beneficial action is often accompanied by detrimental side effects, would have allowed the use of these drugs at lower and safer dosages. As far as colon cancer is concerned, many works have been published in the past few years demonstrating the effectiveness of these fatty acids to enhance the effect of a number of anti-neoplastic drugs and treatments [103, 104].

The growth of MAC16 colon adenocarcinoma cells in syngeneic mice was synergistically inhibited by a combination of 5-fluorouracil (5-FU) and EPA or DHA, a combination of epothilone and DHA, and a combination of EPA and cyclophosphamide [105]. Moreover, in combination with 5-FU, EPA prevented also the development of cachexia [105], and DHA the lesions produced by 5-FU in the intestinal mucosa [106]. The potentiating effects of ω-3 PUFAs on 5-FU efficacy were observed also in vitro using different colon cancer cell lines [78, 107]. Increased efficacy was also observed in vitro for the following two drugs when accompanied by dietary treatment with ω-3 PUFAs: tumor necrosis factor (TNF)-related apoptosis-inducing ligand (TRAIL) [108], which is a member of the TNF family and a promising anti-cancer agent [109, 110]; the nucleoside analogue arabinosylcytosine (araC) [95], which is particularly used in the treatment of leukemias [111], but for which there are also indications against human colon carcinoma [112].

Combinations of ω-3 PUFAs with chemopreventive agents considered useful for the secondary prevention of colon cancer in high-risk patients have been also largely investigated. To this regard the experimental approach used by Reddy et al. [113] who treated rats subject to AOM-induced carcinogenesis with a diet at high levels of fish oil and containing also low doses of celecoxib, a specific inhibitor of COX-2, is interesting. Even though COX-2 inhibitors such as celecoxib had been previously considered highly promising chemopreventive agents against colon carcinogenesis, subsequently it was observed that administration of high doses of these compounds over time could induce considerable side effects. Interestingly, however, the efficiency of low doses of celecoxib in inhibiting tumor incidence and multiplicity in this rat model of carcinogenesis was highly improved by high doses of dietary ω-3 PUFAs, suggesting that this combination of dietary ω-3 PUFAs with a prolonged treatment with lower and safer doses of celecoxib may be considered a promising approach for chemoprevention. Moreover, it has been shown that the treatment in vitro of colon cancer Caco-2 cells with a combination of low doses of a synthetic organoselenium compound, 1,4-phenylene bis(methylene) selenocyanate (p-XSC) and DHA induced synergistically cell growth inhibition and apoptosis and altered the expression of several molecular factors involved in colon carcinogenesis (COX-2; β-catenin; NF-κB, cyclin D1) [89]. The effect of this combination is interesting, since p-XSC was shown to be less toxic and even more efficacious against colon carcinogenesis than inorganic selenium and selenomethionine [114], compounds contained at high levels in natural selenium yeast and largely known not only for their chemopreventive action against colon cancer but also for possessing strong toxicity [115, 116]. Furthermore, the preventive action of a combination of ω-3 PUFAs with butyrate [82, 117], a short chain fatty acid generated by fermentable fibers in the colon lumen, has been shown. This finding is worth noting since fibers are widely recognized as chemopreventive agents against colon cancer. It has been ascertained that ω-3 PUFAs prime colonocytes such that butyrate can initiate apoptosis [118, 119]. One possible mechanism underlying this effect was identified in rats subject to AOM-induced carcinogenesis. The expression of the inhibitor of cell cycle progression p21(Waf1/Cip1) was repressed during carcinogenesis, and butyrate ingestion reverted its level to normal. However, the butyrate-induced enhancement in p21 resulted in an increase in apoptosis and a decreased number of aberrant crypt foci (ACF) only when fish oil was ingested concomitantly to butyrate [120]. Further findings obtained in vitro or ex vivo on normal and neoplastic colonocytes have allowed the identification of other possible molecular mechanisms involved in the ω-3 PUFA potentiation of butyrate pro-apoptotic action [82, 98, 118, 119], including enhanced lipid peroxidation, mitochondrial calcium loading, decrease of MMP, activation of caspase-3 and -9, and decrease of anti-apoptotic Mcl-1 protein. In agreement with these results, it was also reported that the combined dietary administration of pectin and fish oil exerted strong protective effect against AOM- and radiation-induced carcinogenesis in rats, by synergistically enhancing apoptosis of colonocytes, probably acting through the inhibition of two molecular pathways involved in colon carcinogenesis, the COX and Wnt/beta-catenin pathways, and by decreasing PPARδ expression and PGE_2 levels [121].

In this context the recent observation [122] that ALA subject in vitro for 42 h to the action of bifidobacteria is converted mainly to its conjugated isomer C18:3 c9,t11,c15 (CALA), and that CALA exhibits more powerful growth-inhibiting activity than ALA toward SW480 colon cancer cells, should also be mentioned. This is a very interesting finding since bifidobacteria are specific beneficial inhabitants of our colon, where, through their fermentation activity, cooperate to maintain a healthy micro-environment. For this reason their probiotic supplementation is often used in different pathological conditions with the aim to restore the normal colonic flora. In our opinion, the bifidobacteria-induced potentiation of the anti-neoplastic activity of ω-3 PUFAs, if confirmed by further work also in vivo, could open the way to combined treatments of bifidobacteria/ω-3 PUFAs designed to improve the anti-neoplastic efficacy of these fatty acids at colonic levels.

3.3 Interventional Trials

Even though the majority of experimental studies performed to investigate the efficacy of ω-3 PUFAs against cancer have been on the whole extremely encouraging, the human interventional trials performed so far are, conversely, extremely scarce. That in our opinion was related, as we already discussed in Chapter 1, to the discrepancies existing among the outcomes of the epidemiological studies which might have discouraged from designing more interventional trials.

As far as colon cancer is concerned, pioneeristic human interventional trials using supplementation of EPA and DHA were performed in our laboratory during the early 1990s [123, 124], just after the first preclinical studies with animals had been published. In these trials the enrolled subjects were at high risk for colon cancer for sporadic polyposis of colon. In the first study [123] it was found that quite high levels of ω-3 PUFAs (4.1 g EPA/day and 3.6 g DHA/day) supplemented for periods ranging from 2 weeks to 3 months deeply modified the pattern of fatty acids of colonic mucosa and plasma in these subjects, with a marked enrichment in both EPA and DHA and a considerable decrease in AA and LA. The atypical pattern of proliferation of the rectal mucosal cells initially present in these patients, which is considered a marker of risk for colon cancer, reverted to normal following the dietary supplementation. Two-week treatment was enough to cause all the effects described. In the second study [124] it was observed that lower levels of EPA and DHA (2.5 g EPA + DHA/day) were able to elicit the same beneficial effects either after 30 days or 6 months and only in patients with abnormal baseline proliferation pattern. This observation is particularly worth noting, since recently it has been established that doses of ω-3 PUFAs lower than 3.0 g/day do not induce harmful effects and can be considered absolutely safe [125]. The vitamin E content of colonic mucosal and erythrocyte membranes of these patients decreased after a 2-week treatment; however, prolonging the treatment, it reverted to normal levels. On the other hand, vitamin E levels in plasma never changed [124, 126]. It was hypothesized that during the first 2 weeks the dietary ω-3 PUFAs, incorporated at higher levels in cell membranes, elevated the degree of membrane peroxidability and caused an enhanced requirement and consumption of vitamin E. However, subsequently, the normal membrane vitamin E content was restored. Since rectal proliferation has been considered an intermediate biomarker of colon cancer risk also in healthy people, a double-blind, crossover trial was carried out on healthy volunteers supplemented with fish oil (4.4 g ω-3 PUFAs/day) or corn oil for 4 weeks, and a clear anti-proliferative effect of ω-3 PUFAs was observed also in this case [127]. Reduction of rectal cell proliferation observed after fish oil supplementation was accompanied by decreased PGE_2 release from rectal biopsies, suggesting a role for PGE_2 in the ω-3 PUFA effect. The effect was not detectable when the healthy volunteers supplemented with fish oil ate a diet at high level of fat (50% of energy) and with a low ω-6/ω-3 PUFA ratio [128], suggesting that the diet to be beneficial should introduce also a decreased level of ω-6 PUFAs besides furnishing a higher level of ω-3 PUFAs. A protective effect of a supplementation at high levels of ω-3

PUFAs (9 g/day) given for 3 and 6 months to patients at risk for colon cancer having had stage 1 or stage 2 colon carcinoma or adenomatous polyps was confirmed also by Huang and colleagues [129]. They observed a strongly positive correlation between the ω-6/ω-3 PUFA ratio in plasma phospholipids and the high proliferation of colon mucosa cells at baseline, as well as the decrease of both the ω-6/ω-3 plasmatic PUFA ratio and the high level of proliferation in colonic mucosa cells with the ω-3 PUFA supplementation. More recently, patients polypectomized for colorectal adenomas/tumors were advised to reduce their dietary intake of fat and of ω-6 PUFAs and to increase that of ω-3 PUFAs. After 2 years an increased apoptosis index and Bax/Bcl-2 ratio was observed in the sigmoid colon mucosa biopsies from these patients [130] as compared to biopsies from control patients advised to reduce just their total dietary intake of fat. This finding suggested that ω-3 PUFAs promote apoptosis of normal mucosa cells, and that this could represent another pathway through which ω-3 PUFAs may exert preventive effect against colon cancer in humans. In agreement with all the above reported results, recently it was observed that supplementation with highly purified EPA (2 g/day, for 3 months) was also able to inhibit crypt cell proliferation and to induce apoptosis in colon mucosa of patients with colorectal adenomas [131].

On the whole, the outcomes of the published human trials were all in agreement and demonstrated the efficacy of ω-3 PUFAs in reducing the histological markers related to colon carcinogenesis. However, given the scarcity of these human interventional studies, a higher number of them, enrolling a larger sample of patients, would be required to definitely ascertain the effectiveness of ω-3 PUFA treatments against colon cancer and to establish the doses necessary to provide protection.

3.4 Conclusion

The efficacy of ω-3 PUFAs against colon cancer has been demonstrated by plenty of experimental in vivo studies on animals and in vitro studies on cultured colon cancer cells. A number of these studies demonstrated the efficacy of the therapeutic strategy of using combinations of these dietary fatty acids and other anti-neoplastic agents active against colon cancer and had the aim to improve the effectiveness of the anti-neoplastic treatments and to lower their dosage. Several human interventional trials have demonstrated the anti-tumoral efficacy of ω-3 PUFAs. However, on the whole, it would be crucial that more and larger human interventional trials will be designed in the next future with the aim of demonstrating the efficacy of treatment with ω-3 PUFAs against colon cancer, both alone and in combination with other anti-neoplastic agents, at increasing dosages and for periods of variable length.

References

1. Cunningham D, Findlay M. The chemotherapy of colon can no long be ignored. *Eur J Cancer*. 1993; 29A(15):2077–9.
2. Ferlay J, Autier P, Boniol M, Heanue M, Colombet M, Boyle P. Estimates of the cancer incidence and mortality in Europe in 2006. *Ann Oncol*. 2007; 18(3):581–9.

3. Slattery ML. Diet, lifestyle, and colon cancer. *Semin Gastrointest Dis*. 2000; 11(3):142–6.
4. Minami Y, Nishino Y, Tsubono Y, Tsuji I, Hisamichi S. Increase of colon and rectal cancer incidence rates in Japan: trends in incidence rates in Miyagi Prefecture, 1959–1997. *J Epidemiol*. 2006; 16(6):240–8.
5. Li FY, Lai MD. Colorectal cancer, one entity or three. *J Zhejiang Univ Sci B*. 2009; 10(3):219–29.
6. World Cancer Research Fund and American Institute for Cancer Research . Food, Nutrition and the Prevention of Cancer: A Global Perspective. American Institute for Cancer Research, Washington, DC, 1997.
7. Whelan J, McEntee MF. Dietary (n-6) PUFA and intestinal tumorigenesis. *J Nutr*. 2004; 134(12 Suppl):3421S–26S.
8. Caygill CP, Hill MJ. Fish, n-3 fatty acids and human colorectal and breast cancer mortality. *Eur J Cancer Prev*. 1995; 4(4):329–32.
9. Caygill CP, Charlett A, Hill MJ. Fat, fish, fish oil and cancer. *Br J Cancer*. 1996; 74(1): 159–64.
10. Norat T, Bingham S, Ferrari P, Slimani N, Jenab M, Mazuir M, et al. Meat, fish, and colorectal cancer risk: the European Prospective Investigation into cancer and nutrition. *J Natl Cancer Inst*. 2005; 97(12):906–16.
11. Willett WC, Stampfer MJ, Colditz GA, Rosner BA, Speizer FE. Relation of meat, fat, and fiber intake to the risk of colon cancer in a prospective study among women. *N Engl J Med*. 1990; 323(24):1664–72.
12. Norat T, Lukanova A, Ferrari P, Riboli E. Meat consumption and colorectal cancer risk: dose-response meta-analysis of epidemiological studies. *Int J Cancer*. 2002; 98(2):241–56.
13. Sandhu MS, White IR, McPherson K. Systematic review of the prospective cohort studies on meat consumption and colorectal cancer risk: a meta-analytical approach. *Cancer Epidemiol Biomarkers Prev*. 2001; 10(5):439–46.
14. Reddy BS. Studies with the azoxymethane-rat preclinical model for assessing colon tumor development and chemoprevention. *Environ Mol Mutagen*. 2004; 44(1):26–35.
15. Minoura T, Takata T, Sakaguchi M, Takada H, Yamamura M, Hioki K, et al. Effect of dietary eicosapentaenoic acid on azoxymethane-induced colon carcinogenesis in rats. *Cancer Res*. 1988; 48(17):4790–4.
16. Takahashi M, Minamoto T, Yamashita N, Kato T, Yazawa K, Esumi H. Effect of docosahexaenoic acid on azoxymethane-induced colon carcinogenesis in rats. *Cancer Lett*. 1994; 83(1–2):177–84.
17. Reddy BS, Sugie S. Effect of different levels of omega-3 and omega-6 fatty acids on azoxymethane-induced colon carcinogenesis in F344 rats. *Cancer Res*. 1988; 48(23): 6642–7.
18. Deschner EE, Lytle JS, Wong G, Ruperto JF, Newmark HL. The effect of dietary omega-3 fatty acids (fish oil) on azoxymethanol-induced focal areas of dysplasia and colon tumor incidence. *Cancer*. 1990; 66(11):2350–6.
19. Lindner MA. A fish oil diet inhibits colon cancer in mice. *Nutr Cancer*. 1991; 15(1):1–11.
20. Paulsen JE, Stamm T, Alexander J. A fish oil-derived concentrate enriched in eicosapentaenoic and docosahexaenoic acid as ethyl esters inhibits the formation and growth of aberrant crypt foci in rat colon. *Pharmacol Toxicol*. 1998; 82(1):28–33.
21. Latham P, Lund EK, Johnson IT. Dietary n-3 PUFA increases the apoptotic response to 1,2-dimethylhydrazine,reduces mitosis and suppresses the induction of carcinogenesis in the rat colon. *Carcinogenesis*. 1999; 20(4):645–50.
22. Hamilton MC, Hites RA, Schwager SJ, Foran JA, Knuth BA, Carpenter DO. Lipid composition and contaminants in farmed and wild salmon. *Environ Sci Technol*. 2005; 39(22), 8622–9.
23. Hirose M, Masuda A, Ito N, Kamano K, Okuyama H. Effects of dietary perilla oil, soybean oil and safflower oil on 7,12-dimethylbenz[a]anthracene (DMBA) and 1,2-dimethylhydrazine (DMH)-induced mammary gland and colon carcinogenesis in female SD rats. *Carcinogenesis*. 1990; 11(5):731–5.

24. Narisawa T, Takahashi M, Kotanagi H, Kusaka H, Yamazaki Y, Koyama H, et al. Inhibitory effect of dietary perilla oil rich in the n-3 polyunsaturated fatty acid alpha-linolenic acid on colon carcinogenesis in rats. *Jpn J Cancer Res*. 1991; 82(10):1089–96.
25. Narisawa T, Fukaura Y, Yazawa K, Ishikawa C, Isoda Y, Nishizawa Y. Colon cancer prevention with a small amount of dietary perilla oil high in alpha-linolenic acid in an animal model. *Cancer*. 1994; 73(8):2069–75.
26. Onogi N, Okuno M, Komaki C, Moriwaki H, Kawamori T, Tanaka T, et al. Suppressing effect of perilla oil on azoxymethane-induced foci of colonic aberrant crypts in rats. *Carcinogenesis*. 1996; 17(6):1291–6.
27. Dwivedi C, Natarajan K, Matthees DP. Chemopreventive effects of dietary flaxseed oil on colon tumor development. *Nutr Cancer*. 2005; 51(1):52–8.
28. Williams D, Verghese M, Walker LT, Boateng J, Shackelford L, Chawan CB. Flax seed oil and flax seed meal reduce the formation of aberrant crypt foci (ACF) in azoxymethane-induced colon cancer in Fisher 344 male rats. *Food Chem Toxicol*. 2007; 45(1):153–9.
29. van Beelen VA, Spenkelink B, Mooibroek H, Sijtsma L, Bosch D, Rietjens IM, et al. An n-3 PUFA-rich microalgal oil diet protects to a similar extent as a fish oil-rich diet against AOM-induced colonic aberrant crypt foci in F344 rats. *Food Chem Toxicol*. 2009; 47(2):316–20.
30. Takahashi M, Totsuka Y, Masuda M, Fukuda K, Oguri A, Yazawa K, et al. Reduction in formation of 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP)-induced aberrant crypt foci in the rat colon by docosahexaenoic acid(DHA). *Carcinogenesis*. 1997; 18(10):1937–41.
31. Takahashi M, Fukutake M, Isoi T, Fukuda K, Sato H, Yazawa K, et al. Suppression of azoxymethane-induced rat colon carcinoma development by a fish oil component, docosahexaenoic acid (DHA). *Carcinogenesis*. 1997; 18(7):1337–42.
32. Reddy BS, Burill C, Rigotty J. Effect of diets high in omega-3 and omega-6 fatty acids on initiation and postinitiation stages of colon carcinogenesis. *Cancer Res*. 1991; 51(2):487–91.
33. Rao CV, Simi B, Wynn TT, Garr K, Reddy BS. Modulating effect of amount and types of dietary fat on colonic mucosal phospholipase A2, phosphatidylinositol-specific phospholipase C activities, and cyclooxygenase metabolite formation during different stages of colon tumor promotion in male F344 rats. *Cancer Res*. 1996; 56(3):532–7.
34. Reddy BS, Simi B, Patel N, Aliaga C, Rao CV. Effect of amount and types of dietary fat on intestinal bacterial 7 alpha-dehydroxylase and phosphatidylinositol-specific phospholipase C and colonic mucosal diacylglycerol kinase and PKC activities during stages of colon tumor promotion. *Cancer Res*. 1996; 56(10):2314–20.
35. Singh J, Hamid R, Reddy BS. Dietary fat and colon cancer: modulation of cyclooxygenase-2 by types and amount of dietary fat during the postinitiation stage of colon carcinogenesis. *Cancer Res*. 1997; 57(16):3465–70.
36. Singh J, Hamid R, Reddy BS. Dietary fish oil inhibits the expression of farnesyl protein transferase and colon tumor development in rodents. *Carcinogenesis*. 1998; 19(6):985–9.
37. Davidson LA, Nguyen DV, Hokanson RM, Callaway ES, Isett RB, Turner ND, et al. Chemopreventive n-3 polyunsaturated fatty acids reprogram genetic signatures during colon cancer initiation and progression in the rat. *Cancer Res*. 2004; 64(18):6797–804.
38. Fan YY, Spencer TE, Wang N, Moyer MP, Chapkin RS. Chemopreventive n-3 fatty acids activate RXRalpha in colonocytes. *Carcinogenesis*. 2003; 24(9):1541–8.
39. Gupta RA, Sarraf P, Mueller E, Brockman JA, Prusakiewicz JJ, Eng C, et al. Peroxisome proliferator-activated receptor gamma-mediated differentiation: a mutation in colon cancer cells reveals divergent and cell type-specific mechanisms. *J Biol Chem*. 2003; 278(25):22669–77.
40. Lefebvre M, Paulweber B, Fajas L, Woods J, McCrary C, Colombel JF, et al. Peroxisome proliferator-activated receptor gamma is induced during differentiation of colon epithelium cells. *J Endocrinol*. 1999; 162(3):331–40.
41. Girnun GD, Smith WM, Drori S, Sarraf P, Mueller E, Eng C, et al. APC-dependent suppression of colon carcinogenesis by PPARgamma. *Proc Natl Acad Sci U S A*. 2002; 99(21):13771–6.

42. Chang WL, Chapkin RS, Lupton JR. Fish oil blocks azoxymethane-induced rat colon tumorigenesis by increasing cell differentiation and apoptosis rather than decreasing cell proliferation. *J Nutr*. 1998; 128(3):491–7.
43. Rao CV, Hirose Y, Indranie C, Reddy BS. Modulation of experimental colon tumorigenesis by types and amounts of dietary fatty acids. *Cancer Res*. 2001; 61(5):1927–33.
44. Dai W, Liu T, Wang Q, Rao CV, Reddy BS. Down-regulation of PLK3 gene expression by types and amount of dietary fat in rat colon tumors. *Int J Oncol*. 2002; 20(1):121–6.
45. Hong MY, Lupton JR, Morris JS, Wang N, Carroll RJ, Davidson LA, et al. Dietary fish oil reduces O6-methylguanine DNA adduct levels in rat colon in part by increasing apoptosis during tumor initiation. *Cancer Epidemiol Biomarkers Prev*. 2000; 9(8):819–26.
46. Hong MY, Bancroft LK, Turner ND, Davidson LA, Murphy ME, Carroll RJ, et al. Fish oil decreases oxidative DNA damage by enhancing apoptosis in rat colon. *Nutr Cancer*. 2005; 52(2):166–75.
47. Hong MY, Chapkin RS, Davidson LA, Turner ND, Morris JS, Carroll RJ, et al. Fish oil enhances targeted apoptosis during colon tumor initiation in part by downregulating Bcl-2. *Nutr Cancer*. 2003; 46(1):44–51.
48. Zhou GD, Popovic N, Lupton JR, Turner ND, Chapkin RS, Donnelly KC. Tissue-specific attenuation of endogenous DNA I-compounds in rats by carcinogen azoxymethane: possible role of dietary fish oil in colon cancer prevention. *Cancer Epidemiol Biomarkers Prev*. 2005; 14(5):1230–5.
49. Kang JX, Wang J, Wu L, Kang ZB. Transgenic mice: fat-1 mice convert n-6 to n-3 fatty acids. *Nature*. 2004; 427(6974):504.
50. Kang ZB, Ge Y, Chen Z, Cluette-Brown J, Laposata M, Leaf A, et al. Adenoviral gene transfer of Caenorhabditis elegans n-3 fatty acid desaturase optimizes fatty acid composition in mammalian cells. *Proc Natl Acad Sci USA*. 2001; 98(7):4050–4.
51. Nowak J, Weylandt KH, Habbel P, Wang J, Dignass A, Glickman JN, et al. Colitis-associated colon tumorigenesis is suppressed in transgenic mice rich in endogenous n-3 fatty acids. *Carcinogenesis*. 2007; 28(9):1991–5.
52. Jia Q, Lupton JR, Smith R, Weeks BR, Callaway E, Davidson LA, et al. Reduced colitis-associated colon cancer in Fat-1 (n-3 fatty acid desaturase) transgenic mice. *Cancer Res*. 2008; 68(10):3985–91.
53. Beck SA, Smith KL, Tisdale MJ. Anticachectic and antitumor effect of eicosapentaenoic acid and its effect on protein turnover. *Cancer Res*. 1991; 51(22):6089–93.
54. Hudson EA, Beck SA, Tisdale MJ. Kinetics of the inhibition of tumour growth in mice by eicosapentaenoic acid-reversal by linoleic acid. *Biochem Pharmacol*. 1993; 45(11):2189–94.
55. Iigo M, Nakagawa T, Ishikawa C, Iwahori Y, Asamoto M, Yazawa K, et al. Inhibitory effects of docosahexaenoic acid on colon carcinoma 26 metastasis to the lung. *Br J Cancer*. 1997; 75(5):650–5.
56. Suzuki I, Iigo M, Ishikawa C, Kuhara T, Asamoto M, Kunimoto T, et al. Inhibitory effects of oleic and docosahexaenoic acids on lung metastasis by colon-carcinoma-26 cells are associated with reduced matrix metalloproteinase-2 and -9 activities. *Int J Cancer*. 1997; 73(4):607–12.
57. Iwamoto S, Senzaki H, Kiyozuka Y, Ogura E, Takada H, Hioki K, et al. Effects of fatty acids on liver metastasis of ACL-15 rat colon cancer cells. *Nutr Cancer*. 1998; 31(2):143–50.
58. Griffini P, Fehres O, Klieverik L, Vogels IM, Tigchelaar W, Smorenburg SM, et al. Dietary omega-3 polyunsaturated fatty acids promote colon carcinoma metastasis in rat liver. *Cancer Res*. 1998; 58(15):3312–9.
59. Gutt CN, Brinkmann L, Mehrabi A, Fonouni H, Müller-Stich BP, Vetter G, et al. Dietary omega-3-polyunsaturated fatty acids prevent the development of metastases of colon carcinoma in rat liver. *Eur J Nutr*. 2007; 46(5):279–85.
60. Sakaguchi M, Imray C, Davis A, Rowley S, Jones C, Lawson N, et al. Effects of dietary N-3 and saturated fats on growth rates of the human colonic cancer cell lines SW-620 and LS 174T in vivo in relation to tissue and plasma lipids. *Anticancer Res*. 1990; 10(6):1763–8.

61. Sakaguchi M, Rowley S, Kane N, Imray C, Davies A, Jones C, et al. Reduced tumour growth of the human colonic cancer cell lines COLO-320 and HT-29 in vivo by dietary n-3 lipids. *Br J Cancer*. 1990; 62(5):742–7.
62. Boudreau MD, Sohn KH, Rhee SH, Lee SW, Hunt JD, Hwang DH. Suppression of tumor cell growth both in nude mice and in culture by n-3 polyunsaturated fatty acids: mediation through cyclooxygenase-independent pathways. *Cancer Res*. 2001; 61(4):1386–91.
63. Kato T, Hancock RL, Mohammadpour H, McGregor B, Manalo P, Khaiboullina S, et al. Influence of omega-3 fatty acids on the growth of human colon carcinoma in nude mice. *Cancer Lett*. 2002; 187(1–2):169–77.
64. Tsuzuki T, Igarashi M, Miyazawa T. Conjugated eicosapentaenoic acid (EPA) inhibits transplanted tumor growth via membrane lipid peroxidation in nude mice. *J Nutr*. 2004; 134(5):1162–6.
65. Calviello G, Di Nicuolo F, Gragnoli S, Piccioni E, Serini S, Maggiano N, et al. n-3 PUFAs reduce VEGF expression in human colon cancer cells modulating the COX-2/PGE2 induced ERK-1 and -2 and HIF-1alpha induction pathway. *Carcinogenesis*. 2004; 25(12):2303–10.
66. Kato T, Kolenic N, Pardini RS. Docosahexaenoic acid (DHA), a primary tumor suppressive omega-3 fatty acid, inhibits growth of colorectal cancer independent of p53 mutational status. *Nutr Cancer*. 2007; 58(2):178–87.
67. Calder PC, Davis J, Yaqoob P, Pala H, Thies F, Newsholme EA. Dietary fish oil suppresses human colon tumour growth in athymic mice. *Clin Sci* (*Lond*). 1998; 94(3):303–11.
68. Petrik MB, McEntee MF, Johnson BT, Obukowicz MG, Whelan J. Highly unsaturated (n-3) fatty acids, but not alpha-linolenic, conjugated linoleic or gamma-linolenic acids, reduce tumorigenesis in Apc(Min/+) mice. *J Nutr*. 2000; 130(10):2434–43.
69. Boolbol SK, Dannenberg AJ, Chadburn A, Martucci C, Guo XJ, Ramonetti JT, et al. Cyclooxygenase-2 overexpression and tumor formation are blocked by sulindac in a murine model of familial adenomatous polyposis. *Cancer Res*. 1996; 56(11):2556–60.
70. Paulsen JE, Elvsaas IK, Steffensen IL, Alexander J. A fish oil derived concentrate enriched in eicosapentaenoic and docosahexaenoic acid as ethyl ester suppresses the formation and growth of intestinal polyps in the Min mouse. *Carcinogenesis*. 1997; 18(10):1905–10.
71. Oikarinen SI, Pajari AM, Salminen I, Heinonen SM, Adlercreutz H, Mutanen M. Effects of a flaxseed mixture and plant oils rich in alpha-linolenic acid on the adenoma formation in multiple intestinal neoplasia (Min) mice. *Br J Nutr*. 2005; 94(4):510–8.
72. Murray NR, Davidson LA, Chapkin RS, Clay Gustafson W, Schattenberg DG, Fields AP. Overexpression of protein kinase C betaII induces colonic hyperproliferation and increased sensitivity to colon carcinogenesis. *J Cell Biol*. 1999; 145(4):699–711.
73. Gökmen-Polar Y, Murray NR, Velasco MA, Gatalica Z, Fields AP. Elevated protein kinase C betaII is an early promotive event in colon carcinogenesis. *Cancer Res*. 2001; 61(4):1375–81.
74. Murray NR, Weems C, Chen L, Leon J, Yu W, Davidson LA, et al. Protein kinase C betaII and TGFbetaRII in omega-3 fatty acid-mediated inhibition of colon carcinogenesis. *J Cell Biol*. 2002; 157(6):915–20.
75. Yu W, Murray NR, Weems C, Chen L, Guo H, Ethridge R, et al. Role of cyclooxygenase 2 in protein kinase C beta II-mediated colon carcinogenesis. *Biol Chem*. 2003; 278(13):11167–74.
76. Mengeaud V, Nano JL, Fournel S, Rampal P. Effects of eicosapentaenoic acid, gamma-linolenic acid and prostaglandin E1 on three human colon carcinoma cell lines. *Prostaglandins Leukot Essent Fatty Acids*. 1992; 47(4):313–9.
77. Tanaka Y, Goto K, Matsumoto Y, Ueoka R. Remarkably high inhibitory effects of docosahexaenoic acid incorporated into hybrid liposomes on the growth of tumor cells along with apoptosis. *Int J Pharm*. 2008; 359(1-2):264–71.
78. Calviello G, Di Nicuolo F, Serini S, Piccioni E, Boninsegna A, Maggiano N, et al. Docosahexaenoic acid enhances the susceptibility of human colorectal cancer cells to 5-fluorouracil. *Cancer Chemother Pharmacol*. 2005; 55(1):12–20.

79. Latham P, Lund EK, Brown JC, Johnson IT. Effects of cellular redox balance on induction of apoptosis by eicosapentaenoic acid in HT29 colorectal adenocarcinoma cells and rat colon in vivo. *Gut*. 2001; 49(1):97–105.
80. Pan J, Keffer J, Emami A, Ma X, Lan R, Goldman R, et al. Acrolein-derived DNA adduct formation in human colon cancer cells: its role in apoptosis induction by docosahexaenoic acid. *Chem Res Toxicol*. 2009; 22(5):798–806.
81. van Beelen VA, Roeleveld J, Mooibroek H, Sijtsma L, Bino RJ, Bosch D, et al. A comparative study on the effect of algal and fish oil on viability and cell proliferation of Caco-2 cells. *Food Chem Toxicol*. 2007; 45(5):716–24.
82. Kolar SS, Barhoumi R, Lupton JR, Chapkin RS: Docosahexaenoic acid and butyrate synergistically induce colonocyte apoptosis by enhancing mitochondrial Ca2+ accumulation. *Cancer Res*. 2007; 67:5561–8.
83. Ng Y, Barhoumi R, Tjalkens RB, Fan YY, Kolar S, Wang N, et al. The role of docosahexaenoic acid in mediating mitochondrial membrane lipid oxidation and apoptosis in colonocytes. *Carcinogenesis*. 2005; 26(11):1914–21.
84. Swamy MV, Cooma I, Patlolla JM, Simi B, Reddy BS, Rao CV. Modulation of cyclooxygenase-2 activities by the combined action of celecoxib and docosahexaenoic acid: novel strategies for colon cancer prevention and treatment. *Mol Cancer Ther*. 2004; 3(2):215–21.
85. Habermann N, Lund EK, Pool-Zobel BL, Glei M. Modulation of gene expression in eicosapentaenoic acid and docosahexaenoic acid treated human colon adenoma cells. *Genes Nutr*. 2009; 4(1):73–6.
86. Collett ED, Davidson LA, Fan YY, Lupton JR, Chapkin RS. n-6 and n-3 polyunsaturated fatty acids differentially modulate oncogenic Ras activation in colonocytes. *Am J Physiol Cell Physiol*. 2001; 280(5):C1066–75.
87. Allred CD, Talbert DR, Southard RC, Wang X, Kilgore MW. PPARgamma1 as a molecular target of eicosapentaenoic acid in human colon cancer (HT-29) cells. *J Nutr*. 2008; 138(2):250–6.
88. Narayanan BA, Narayanan NK, Reddy BS. Docosahexaenoic acid regulated genes and transcription factors inducing apoptosis in human colon cancer cells. *Int J Oncol*. 2001; 19(6):1255–62.
89. Narayanan BA, Narayanan NK, Desai D, Pittman B, Reddy BS. Effects of a combination of docosahexaenoic acid and 1, 4-phenylene bis(methylene) selenocyanate on cyclooxygenase 2, inducible nitric oxide synthase and beta-catenin pathways in colon cancer cells. *Carcinogenesis*. 2004; 25:2443–9.
90. Calviello G, Resci F, Serini S, Piccioni E, Toesca A, Boninsegna A, et al. Docosahexaenoic acid induces proteasome-dependent degradation of beta-catenin, down-regulation of survivin and apoptosis in human colorectal cancer cells not expressing COX-2. *Carcinogenesis*. 2007; 28(6):1202–9.
91. Lee JY, Hwang DH. Docosahexaenoic acid suppresses the activity of peroxisome proliferator-activated receptors in a colon tumor cell line. *Biochem Biophys Res Commun*. 2002; 298(5):667–74.
92. Schønberg SA, Lundemo AG, Fladvad T, Holmgren K, Bremseth H, Nilsen A, et al. Closely related colon cancer cell lines display different sensitivity to polyunsaturated fatty acids, accumulate different lipid classes and downregulate sterol regulatory element-binding protein 1. *FEBS J*. 2006; 273(12):2749–65.
93. Toit-Kohn JL, Louw L, Engelbrecht AM. Docosahexaenoic acid induces apoptosis in colorectal carcinoma cells by modulating the PI3 kinase and p38 MAPK pathways. *J Nutr Biochem*. 2009; 20(2):106–14.
94. Engelbrecht AM, Toit-Kohn JL, Ellis B, Thomas M, Nell T, Smith R. Differential induction of apoptosis and inhibition of the PI3-kinase pathway by saturated, monounsaturated and polyunsaturated fatty acids in a colon cancer cell model. *Apoptosis*. 2008; 13(11):1368–77.

95. Cha MC, Lin A, Meckling KA. Low dose docosahexaenoic acid protects normal colonic epithelial cells from araC toxicity. *BMC Pharmacol* 2005; 5:7.
96. Narayanan BA, Narayanan NK, Simi B, Reddy BS. Modulation of inducible nitric oxide synthase and related proinflammatory genes by the omega-3 fatty acid docosahexaenoic acid in human colon cancer cells. *Cancer Res*. 2003; 63(5):972–9.
97. Danbara N, Yuri T, Tsujita-Kyutoku M, Sato M, Senzaki H, Takada H, et al. Conjugated docosahexaenoic acid is a potent inducer of cell cycle arrest and apoptosis and inhibits growth of colo 201 human colon cancer cells. *Nutr Cancer*. 2004; 50(1):71–9.
98. Hofmanová J, Vaculová A, Lojek A, Kozubík A. Interaction of polyunsaturated fatty acids and sodium butyrate during apoptosis in HT-29 human colon adenocarcinoma cells. *Eur J Nutr*. 2005; 44(1):40–51.
99. Jakobsen CH, Størvold GL, Bremseth H, Follestad T, Sand K, Mack M, et al. DHA induces ER stress and growth arrest in human colon cancer cells: associations with cholesterol and calcium homeostasis. *J Lipid Res*. 2008; 49(10):2089–100.
100. Tsai WS, Nagawa H, Kaizaki S, Tsuruo T, Muto T. Inhibitory effects of n-3 polyunsaturated fatty acids on sigmoid colon cancer transformants. *J Gastroenterol*. 1998; 33(2):206–12.
101. Turner ND, Zhang J, Davidson LA, Lupton JR, Chapkin RS. Oncogenic ras alters sensitivity of mouse colonocytes to butyrate and fatty acid mediated growth arrest and apoptosis. *Cancer Lett*. 2002; 186(1):29–35.
102. Davidson LA, Brown RE, Chang WC, Morris JS, Wang N, Carroll RJ, et al. Morphodensitometric analysis of protein kinase C beta(II) expression in rat colon: modulation by diet and relation to in situ cell proliferation and apoptosis. *Carcinogenesis*. 2000; 21(8):1513–9.
103. Biondo PD, Brindley DN, Sawyer MB, Field CJ. The potential for treatment with dietary long-chain polyunsaturated n-3 fatty acids during chemotherapy. *J Nutr Biochem*. 2008; 19(12):787–96.
104. Calviello G, Serini S, Piccioni E, Pessina G. Antineoplastic effects of n-3 polyunsaturated fatty acids in combination with drugs and radiotherapy: preventive and therapeutic strategies. *Nutr Cancer*. 2009; 61(3):287–301.
105. Wynter MP, Russell ST, Tisdale MJ. Effect of n-3 fatty acids on the antitumour effects of cytotoxic drugs. *In Vivo*. 2004; 18(5):543–7.
106. Gómez de Segura IA, Valderrábano S, Vázquez I, Vallejo-Cremades MT, Gómez-García L, et al. Protective effects of dietary enrichment with docosahexaenoic acid plus protein in 5-fluorouracil-induced intestinal injury in the rat. *Eur J Gastroenterol Hepatol*. 2004; 16(5):479–85.
107. Jordan A, Stein J. Effect of an omega-3 fatty acid containing lipid emulsion alone and in combination with 5-fluorouracil (5-FU) on growth of the colon cancer cell line Caco-2. *Eur J Nutr*. 2003; 42(6):324–31.
108. Vaculová A, Hofmanová J, Andera L, Kozubík A. TRAIL and docosahexaenoic acid cooperate to induce HT-29 colon cancer cell death. *Cancer Lett*. 2005; 229(1):43–8.
109. Nagane M, Huang HJ, Cavenee WK. The potential of TRAIL for cancer chemotherapy. *Apoptosis*. 2001; 6(3):191–7.
110. Kim K, Fisher MJ, Xu SQ, el-Deiry WS. Molecular determinants of response to TRAIL in killing of normal and cancer cells. *Clin Cancer Res*. 2000; 6(2):335–46.
111. Gandhi V, Estey E, Plunkett W. Modulation of arabinosylcytosine metabolism during leukemia therapy. *Adv Exp Med Biol*. 1994; 370: 119–24.
112. Grem JL, Geoffroy F, Politi PM, Cuddy DP, Ross DD, Nguyen D, et al. Determinants of sensitivity to 1-beta-D-arabinofuranosylcytosine in HCT 116 and NCI-H630 human colon carcinoma cells. *Mol Pharmacol.* 1995; 48(2):305–15.
113. Reddy BS, Patlolla JM, Simi B, Wang SH, Rao CV. Prevention of colon cancer by low doses of celecoxib, a cyclooxygenase inhibitor, administered in diet rich in omega-3 polyunsaturated fatty acids. *Cancer Res*. 2005; 65(17):8022–7.

114. Reddy BS, Rivenson A, Kulkarni N, Upadhyaya P, el-Bayoumy K. Chemoprevention of colon carcinogenesis by the synthetic organoselenium compound 1,4-phenylenebis (methylene)selenocyanate. *Cancer Res*. 1992; 52(20):5635–40.
115. Clark LC, Combs GF Jr, Turnbull BW, Slate EH, Chalker DK, Chow J, et al. Effects of selenium supplementation for cancer prevention in patients with carcinoma of the skin. A randomized controlled trial. Nutritional Prevention of Cancer Study Group. *JAMA*. 1996; 276(24):1957–63.
116. el-Bayoumy K, Rao CV, Reddy BS. Multiorgan sensitivity to anticarcinogenesis by the organoselenium 1,4-phenylenebis(methylene)selenocyanate. *Nutr Cancer*. 2001; 40(1): 18–27.
117. Kolar SS, Barhoumi R, Callaway ES, Fan YY, Wang N, Lupton JR, et al. Synergy between docosahexaenoic acid and butyrate elicits p53-independent apoptosis via mitochondrial Ca(2+) accumulation in colonocytes. *Am J Physiol Gastrointest Liver Physiol*. 2007; 293(5):G935–43.
118. Chapkin RS, Hong MY, Fan YY, Davidson LA, Sanders LM, Henderson CE, et al. Dietary n-3 PUFA alter colonocyte mitochondrial membrane composition and function. *Lipids*. 2002; 37(2):193–9.
119. Hong MY, Chapkin RS, Barhoumi R, Burghardt RC, Turner ND, Henderson CE, et al. Fish oil increases mitochondrial phospholipid unsaturation, upregulating reactive oxygen species and apoptosis in rat colonocytes. *Carcinogenesis*. 2002; 23(11):1919–25.
120. Crim KC, Sanders LM, Hong MY, Taddeo SS, Turner ND, Chapkin RS, et al. Upregulation of p21Waf1/Cip1 expression in vivo by butyrate administration can be chemoprotective or chemopromotive depending on the lipid component of the diet. *Carcinogenesis*. 2008; 29(7):1415–20.
121. Vanamala J, Glagolenko A, Yang P, Carroll RJ, Murphy ME, Newman RA, et al. Dietary fish oil and pectin enhance colonocyte apoptosis in part through suppression of PPARdelta/PGE2 and elevation of PGE3. *Carcinogenesis*. 2008; 29(4):790–6.
122. Coakley M, Banni S, Johnson MC, Mills S, Devery R, Fitzgerald G, et al. Inhibitory effect of conjugated alpha-linolenic acid from bifidobacteria of intestinal origin on SW480 cancer cells. *Lipids*. 2009; 44(3):249–56.
123. Anti M, Marra G, Armelao F, Bartoli GM, Ficarelli R, Percesepe A, et al. Effect of omega-3 fatty acids on rectal mucosal cell proliferation in subjects at risk for colon cancer. *Gastroenterology*. 1992; 103(3):883–91.
124. Anti M, Armelao F, Marra G, Percesepe A, Bartoli GM, Palozza P, et al. Effects of different doses of fish oil on rectal cell proliferation in patients with sporadic colonic adenomas. *Gastroenterology*. 1994; 107(6):1709–18.
125. De Maria G. Omega-3. An overview. Focus on Omega-3. *Agro Food Industry hi-tech*. 2008; 19(suppl):4–9.
126. Palozza P, Anti M, Franceschelli P, Marra G, Calviello G, Armelao F, et al. In Oxygen paradox; KJA Davies Ed.; The University of Padova Press, Padova, 1995, pp. 653–664.
127. Bartram HP, Gostner A, Scheppach W, Reddy BS, Rao CV, Dusel G, et al. Effects of fish oil on rectal cell proliferation, mucosal fatty acids, and prostaglandin E2 release in healthy subjects. *Gastroenterology*. 1993; 105(5):1317–22.
128. Bartram HP, Gostner A, Reddy BS, Rao CV, Scheppach W, Dusel G, et al. Missing antiproliferative effect of fish oil on rectal epithelium in healthy volunteers consuming a high-fat diet: potential role of the n-3:n-6 fatty acid ratio. *Eur J Cancer Prev*. 1995; 4(3):231–7.
129. Huang YC, Jessup JM, Forse RA, Flickner S, Pleskow D, Anastopoulos HT, et al. n-3 fatty acids decrease colonic epithelial cell proliferation in high-risk bowel mucosa. *Lipids*. 1996; 31(Suppl.):S313–7.
130. Cheng J, Ogawa K, Kuriki K, Yokoyama Y, Kamiya T, Seno K, et al. Increased intake of n-3 polyunsaturated fatty acids elevates the level of apoptosis in the normal sigmoid colon of patients polypectomized for adenomas/tumors. *Cancer Lett*. 2003; 193(1):17–24.

131. Courtney ED, Matthews S, Finlayson C, Di Pierro D, Belluzzi A, Roda E, et al. Eicosapentaenoic acid (EPA) reduces crypt cell proliferation and increases apoptosis in normal colonic mucosa in subjects with a history of colorectal adenomas. *Int J Colorectal Dis*. 2007; 22(7):765–76.

Part III
ω-3 PUFAs and Hormone-Related Cancers (Breast and Prostate)

Chapter 4
ω-3 PUFAs and Breast Cancer: Epidemiological Studies

Paul D. Terry and Pamela J. Mink

Abstract ω-3 polyunsaturated fatty acids, particularly the long-chain eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), have been shown to inhibit the proliferation of breast cancer cell lines in vitro and to reduce the risk and progression of this tumor in animal experiments. However, the evidence from epidemiological studies remains unclear, despite a large number of studies published to date. The majority of the studies have not supported a consistent inverse association between fish consumption or omega-3 fatty acid intake and risk of breast cancer. Nevertheless, the results of several studies do suggest potentially important beneficial effects. Future epidemiological studies likely will benefit from the assessment of specific fatty acids in the diet, including EPA and DHA, and the ratio of these to omega-6 fatty acids, the consideration of host factors such as menopausal status, tumor and prognostic factors such as tumor hormone receptor status, tumor size, stage of breast cancer at diagnosis, and factors that might modify the association between diet and breast cancer.

Keywords Fatty acids · ω-3 · Fatty acids · ω-6 · Breast neoplasms · Neoplasms · Hormone dependent · Epidemiological studies

4.1 Introduction

Dietary fat intake is among the most widely studied dietary risk factors for breast cancer, yet its roles in influencing endogenous sex hormone levels [1–5] and cancer risk [6–8] remain unclear. In recent years, increasing attention has been paid to the intake of specific fatty acids [9], rather than total fat intake, and notable among these have been "marine fatty acids" [10, 11]. Long-chain eicosapentaenoic

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acid (EPA, 20:5n–3) and docosahexaenoic acid (DHA, 22:6n–3), omega-3 (ω-3) polyunsaturated fatty acids (PUFAs) contained primarily in fatty fish, have been shown consistently to inhibit the proliferation of breast cancer cell lines in vitro and to reduce the risk and progression of this disease in animal experiments [12, 13]. However, whether high-intake long- or short-chain ω-3 PUFAs can actually lower breast cancer risk in human populations remains to be clarified. Our primary aim is to review the current epidemiological literature on long-chain ω-3 PUFAs and breast cancer risk. We also summarize evidence regarding the short-chain ω-3 PUFA α-linolenic acid (ALA) contained primarily in seed oils [e.g., canola (rapeseed) oil, flaxseed oil] and in the chloroblasts of green leafy vegetables. We review published studies identified through searches of Medline, Web of Science, and EMBASE databases and cross-matching the references of relevant articles. Virtually all published reports are in the English language, and we have restricted our review to those.

4.2 Breast Cancer

Breast cancer is the most commonly diagnosed cancer among women and is the leading cause of female cancer mortality in the world [14]. The estimated annual incidence is more than 1 million cases worldwide, more than 200,000 cases in the United States, and more than 400,000 cases in Europe [15–17]. A comparison of breast cancer data worldwide shows tremendous variation in both incidence and mortality rates, with greater than fivefold differences observed between low-risk and high-risk areas [14, 18, 19]. In the United States, breast cancer incidence rates have been rising slowly for the past few decades and may now be leveling off [20].

Differences in eating patterns across countries suggest several possible dietary components that could affect breast cancer risk. The focus of this review is on ω-3 fatty acids, although other dietary factors have been investigated in relation to breast cancer, including intakes of total fat and saturated fat, and consumption of red meat, fruit and vegetables, and soy products, among others [21–24]. The results of migrant studies suggest that the observed geographic differences in breast cancer incidence are due, at least in part, to differences in environmental (i.e., non-genetic) exposures. For example, dietary factors are thought to be most responsible for the change in incidence rates among migrants [25], although correlated changes in other lifestyle factors, such as obesity and physical activity, may also play important roles.

4.3 EPA and DHA

EPA and DHA are polyunsaturated fatty acids found mainly in fatty fish; hence, they are often referred to as marine fatty acids. EPA and DHA can also be biosynthesized in humans from ALA, sources of which include flaxseed, canola oil, soybeans, walnuts, and green leafy vegetables. However, the conversion of ALA by the body to the more active longer chain metabolites is inefficient: <5–10% for EPA and 2–5% for DHA [26]. A common feature of most of the proposed mechanisms by which long-chain ω-3 PUFA fatty acids might lower cancer risk (see below) is the inhibition of eicosanoid production from ω-6 fatty acid precursors [12], which include linoleic acid (LA), found primarily in vegetable oils, and arachidonic acid (AA), found mainly in (grain-fed) animal products. Therefore, epidemiological studies increasingly have examined the interrelated effects of ω-3 and ω-6 fatty acids on cancer risk.

4.4 Epidemiological Studies

4.4.1 Ecological Studies of Fish Consumption

Cross-national studies have shown inverse associations between per capita consumption of fish and breast cancer incidence and mortality rates [27–31]. Within populations, such as those living in Japan [32], Iceland [33], Alaska [34], and Greenland [35], reductions over time in the relative contribution of fish to total fat intake have coincided with increased incidence rates of breast. Although not without merit, ecological studies, which are based on comparisons between or within populations, suffer from important limitations, including the fact that variations in exposure on a population level do not always correspond to variations among individuals within any given population, and the limited ability to adjust for potentially confounding factors [36]. Furthermore, because the relative amounts of EPA and DHA in fish oil vary between fish species [37], the interpretation of "total fish consumption" in epidemiological studies can be problematical, because the absolute and relative amounts of fatty acids reflected in this measure vary greatly among populations.

4.4.2 Case–Control Studies

The majority of epidemiological studies on fish consumption or marine fatty acid intake and breast cancer risk published to date have used the case–control design. Case–control studies have several limitations, including their vulnerability to selection and recall biases [38]. It is also well established that the collection of dietary information from individuals, especially long-term diet, involves varying degrees of measurement error [39], which, if random with respect to disease status, can drive associations toward (and sometimes away from) the null association. Biomarkers of fatty acid intake, such as measures from serum, adipose tissue, and erythrocyte membranes, may be more objective than dietary recall and have been used in epidemiological studies of ω-3 PUFAs and breast cancer reviewed here (Figs. 4.1 and 4.2). However, none of these sources necessarily capture long-term intake better than dietary recall instruments [39].

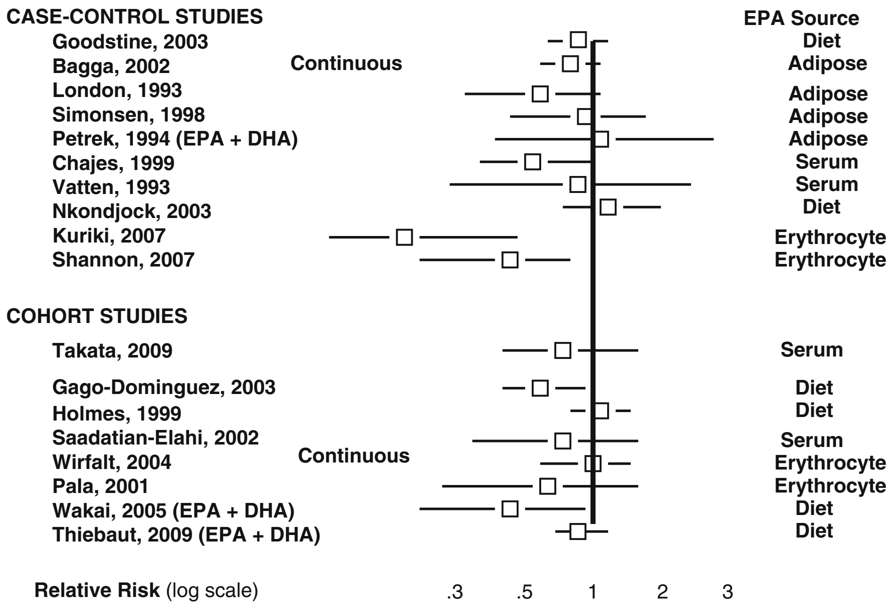

Fig. 4.1 Measures of EPA and breast cancer (highest vs. lowest quantile)

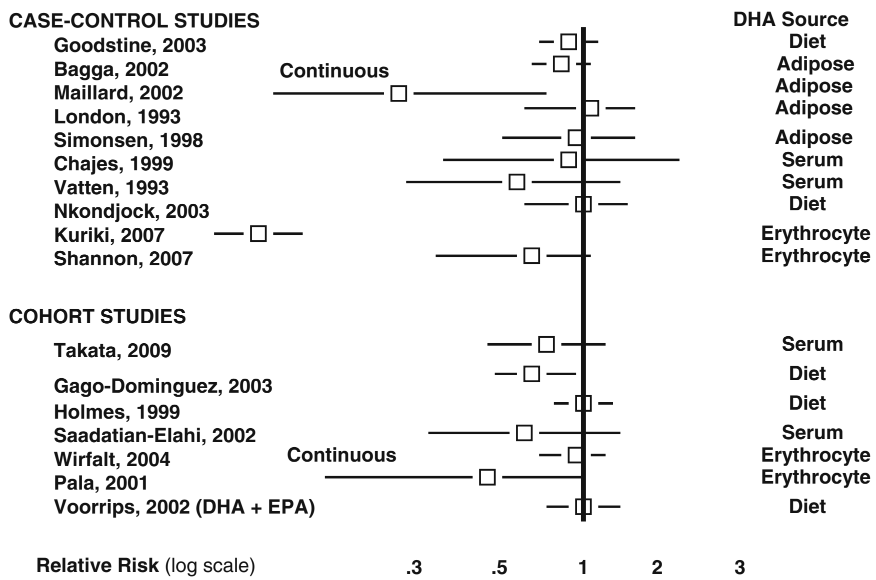

Fig. 4.2 Measures of DHA and breast cancer (highest vs. lowest quantile)

4.4.2.1 Fish Consumption

At least 19 case–control studies [40–58] have examined the association between total fish consumption and breast cancer risk and generally have not shown a clear association with breast cancer risk. Some of these studies [40–44, 46–49, 52, 53, 56–58] have examined total fish or seafood consumption without accounting for the type of fish consumed. Of these, no clear association between total fish consumption and breast cancer risk was observed in one study in the United States [42], two studies in Italy [43, 46], one study in Uruguay [53], one study in China [58], and two studies in Singapore [56, 57]. The null studies have several features in common, including mostly (with the possible exception of China) a low per capita intake of ω-3 fatty acids in the studied populations and narrow or unclear ranges of exposure. Two studies in Canada, a country with a relatively low per capita intake of ω-3 fatty acids [59], showed mixed results with total fish consumption [40, 41], whereby statistically significant inverse associations were evident only for premenopausal women overall [40] or for premenopausal women with tumors that were negative for estrogen receptors [41]. Mixed results were also observed in two studies in Argentina [50] (both based on the same case series), suggesting an inverse association with fish, but not "seafood" per se. Of the four remaining case–control studies that examined only total fish consumption, one small study in Switzerland [44] found a 30% reduced risk among women in the highest tertile of consumption compared to those in the lowest, another small study in Spain [48] found a 70% reduced risk, also based on tertiles (neither study included confidence intervals), and two studies in Japan [49, 52] showed a weak inverse [52] and a null association [49], respectively. Both Spain and Japan are countries with relatively high per capita intakes of ω-3 fatty acids [59].

4.4.2.2 Long-Chain ω-3 PUFAs EPA and DHA

Three case–control studies [60–62] have examined dietary intake of ω-3 fatty acids in relation to breast cancer risk. Two of these studies showed no clear association [60, 61] (Fig. 4.3), whereas both dietary DHA and EPA were associated inversely with risk in a small study in Finland [62]; that study also observed an inverse association with adipose DHA (significant differences were in terms of mean fatty acid levels; data not shown).

Ten case–control studies have examined the association between adipose tissue [62–66], erythrocyte membrane [67, 68], or serum phospholipid [69–71] long-chain ω-3 PUFAs and breast cancer risk (Figs. 4.1 and 4.2). In a study in Sweden [69], a relatively high level of EPA in serum phospholipids was associated with a halving of breast cancer risk (although the confidence intervals included unity), whereas DHA and the ratio of EPA to LA both showed weaker associations. Three relatively recent studies in Japan and China [67, 68, 71], countries with relatively high per capita long-chain ω-3 PUFA intake, showed moderate to strong inverse associations with risk, two [67, 68] reaching statistical significance. In contrast, two studies in the United States [63, 65] found essentially no association between adipose tissue

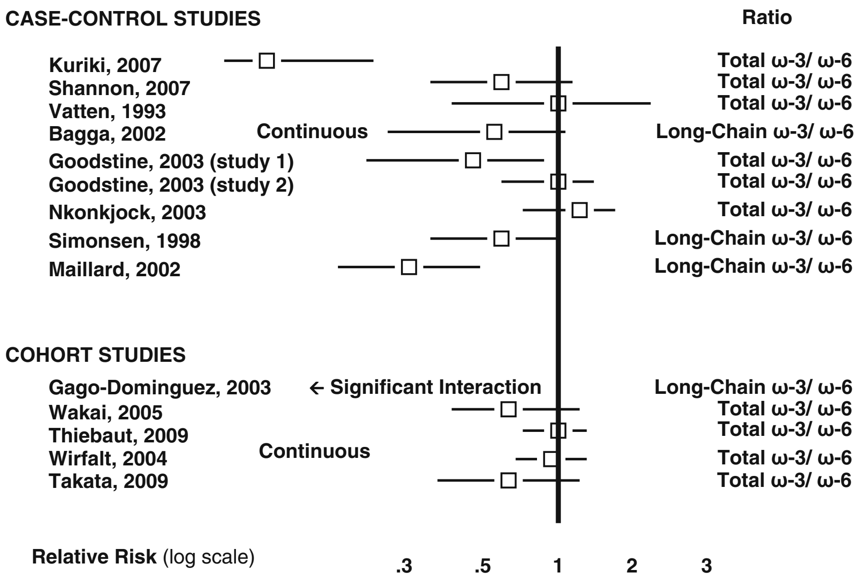

Fig. 4.3 Omega-3/omega-6 ratios and breast cancer (highest vs. lowest quantile)

long-chain ω-3 PUFAs and breast cancer risk. In a study in France [64], both adipose DHA and the ratio of total ω-3 to ω-6 fatty acids in adipose tissue were strongly and inversely associated with risk. A study in Finland [62] reported a statistically significant inverse association with adipose tissue DHA, but not EPA, although the sample size of this study was very small. These findings are consistent with those from a small study in Norway [70], which found a statistically non-significant moderate reduction in breast cancer risk among women with the highest serum phospholipid levels of DHA compared to those with the lowest, but only a weak association with EPA. In a multi-center study [66], total adipose ω-3 PUFAs were not associated with breast cancer risk, and adipose EPA and DHA levels were associated inversely, but weakly, with risk. In this study, women in the highest tertile of the ratio of ω-3 to ω-6 fatty acids had a 30% lower risk than women in the lowest tertile, suggesting that the intake of ω-3 PUFAs relative to that of ω-6 might be a more relevant measure of exposure with respect to breast cancer risk than either group of fatty acids examined independently. Indeed, evidence from in vivo studies suggests that the modulation of eicosanoid biosynthesis depends more on the ratio than the absolute levels of these fatty acid groups [12].

There was no clear pattern in the findings according to the source of the ω-3 measures, including diet, adipose tissue, serum, plasma, or erythrocyte membranes. In general, adipose tissue may provide the best measure of long-term intake of dietary fatty acids, while blood lipid fractions and erythrocyte membranes may provide a better measure of short-term dietary changes [72]. Although adipose tissue measures can discriminate between long-term fish eaters and non-fish eaters, the exact nature of the dose–response relationship between marine fatty acids intake and tissue marker levels remains unclear [72].

4.4.3 Prospective Cohort Studies

Cohort studies have several advantages over those of case–control design [38], although they are not without limitations. For example, changes in diet during follow-up can lead to the misclassification of long-term exposure if, as has generally been the case, exposure is not updated after the baseline assessment. Because non-differential misclassification can, under many circumstances, attenuate an association that might exist, random changes in diet over time would tend to mask a true association between fish consumption and cancer risk. Cohort studies can also be limited by losses to follow-up, where assumptions regarding potential bias due to systematic losses must be made.

4.4.3.1 Fish Consumption

The results of eight prospective cohort studies [73–80] that have examined the association between fish consumption and breast cancer risk have generally shown no clear association. More than half of these studies were conducted in the United States [73, 74, 78–80], a country with relatively low per capita intake of ω-3 PUFAs [59]. As we have argued previously [10, 11], the range of dietary intake of long-chain ω-3 PUFAs in populations with generally low intake may be too narrow to observe an association with risk in these epidemiological studies. Studies conducted in countries with relatively high per capita intake of ω-3 PUFAs [59], such as Japan [77], Norway [75, 76], and Sweden [81], although not always showing statistical significance, tend to suggest inverse associations with high fish consumption. Overall, the association with fish consumption in prospective cohort studies to date remains unclear, but generally provides no clear, consistent evidence of beneficial effects at present.

4.4.3.2 Long-Chain ω-3 PUFAs EPA and DHA

The results of prospective cohort studies of EPA and DHA levels and breast cancer risk are mixed, but show some tendency toward a small to moderate inverse association (Figs. 4.1 and 4.2); however, results from only two [82, 83] in nine prospective cohort studies [71, 82–89] were statistically significant. As was observed in the case–control studies overall, there was no clear pattern in results according to the source of PUFA measurement. However, the number of studies using specific PUFA data from any one source remains small.

4.4.4 Fatty Acid Ratios and Breast Cancer

Evidence from in vivo studies suggests that the modulation of eicosanoid biosynthesis, such as reduced PGE_2 biosynthesis, depends more on the ratio than the absolute levels of ω-3 and ω-6 fatty acids [12, 90]. Therefore, one might expect that the epidemiological evidence for an inverse association with breast cancer would be more compelling for the ratio of long-chain ω-3 to ω-6.

Three case–control studies that examined this measure observed statistically significant inverse associations with risk [64, 66, 91], whereas one case–control study [69] and one prospective cohort study [71] found an inverse association (albeit weaker) with increasing ratio of EPA to arachidonic acid (AA). In addition, Gago-Dominguez and colleagues [82] found a positive association between total ω-6 intake and breast cancer risk among women in the lowest quartile levels of long-chain ω-3 PUFA intake in a prospective cohort study. Most studies, however, examined the ratio of total ω-3 to total ω-6 PUFAs [60, 61, 67, 68, 70, 71, 83, 87, 88], showing an overall tendency toward a weak to moderate inverse association with breast cancer risk (Fig. 4.3), perhaps slightly stronger than what was observed for the individual long-chain ω-3 PUFAs (Figs. 4.1 and 4.2).

4.4.5 Short-Chain ω-3 PUFA α-Linolenic Acid (ALA)

ALA is an essential fatty acid which cannot be produced in the body and, therefore, it must be taken in through the diet. A relatively small percent of ALA is desaturated and elongated into EPA and DHA [92]. Results from epidemiologic studies of dietary intake of ALA and breast cancer are mixed (Fig. 4.4). Case–control studies from Finland and Canada reported no clear association with breast cancer risk [61, 62], whereas a study from Uruguay reported a significant positive association [93] and an Italian study of linolenic acid observed a significant inverse association [94]. In contrast, a Dutch cohort study [89] reported a 70% reduced risk of breast cancer (highest quintile vs. lowest; p-trend $= 0.006$). A recent prospective study from France [88] observed no significant overall association between ALA and breast cancer, but found significant associations in opposite directions depending on the food source. Specifically, ALA intake levels from fruits and vegetables and from vegetable oils were associated inversely with breast cancer incidence, whereas ALA intakes from nut mixes and from processed foods were associated positively [88]. These latter results do not necessarily indicate a direct effect of dietary ALA and may be explained, at least in part, by uncontrolled or residual confounding. The authors suggest that the diversity of food sources for ALA may explain the inconsistent findings across previous studies of dietary ALA and breast cancer cited above [88].

Additional epidemiologic studies have examined biomarkers of ALA intake and breast cancer risk. A US case–control observed no clear association across increasing quintiles of ALA levels in subcutaneous adipose tissue [63]. Two case–control studies, one conducted in the United States and one in Europe, of ALA

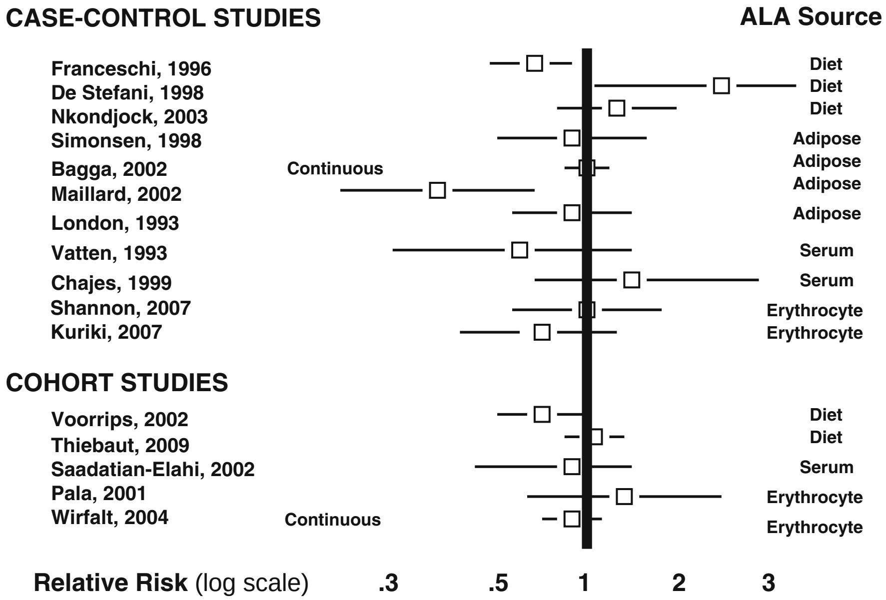

Fig. 4.4 Alpha-linolenic acid and breast cancer (highest vs. lowest quantile)

concentrations in breast adipose tissue reported non-significant odds ratios close to the null [66, 91], and a Finnish study reported no significant differences between ALA composition of phospholipids in breast adipose tissue in pre- or postmenopausal cases and controls [62]. In contrast, a French case–control study reported a significant 60% reduced risk among women in the highest vs. lowest tertile of adipose ALA (p-value for trend = 0.01) [64]. Three case–control studies of ALA in serum phospholipids and breast cancer [69, 70, 86] reported non-significant odds ratios above and below the null; confidence intervals in those studies were generally wide and p-values for tests for trend were not statistically significant. Finally, there were no material or significant associations observed in two case–control studies [67, 68] conducted in China and Japan, respectively, one Italian nested case–control study [85], or one Swedish cohort study [87] of breast cancer risk and concentrations of ALA in erythrocytes.

A meta-analysis of studies of biomarkers of ALA and breast cancer risk published in 2004 [95] reported results that differed by study design with a summary risk ratio for cohort studies greater than 1.0 (albeit not statistically significant) and a summary risk ratio for case–control studies significantly lower than 1.0. The authors noted that interpretation of their results is limited by heterogeneity across studies in adjustment for potential confounding factors and publication bias, which also influence interpretation of non-quantitative reviews such as this. Thus, taken together, studies of ALA biomarkers to date have not indicated a consistent pattern of association, either inverse or positive.

4.5 Discussion

The evidence from epidemiological studies of long-chain ω-3 PUFA intake and breast cancer risk remains unclear, despite a large number of studies published to date. Nevertheless, although limited, the available evidence does suggest that there might be an inverse association between the long-chain ω-3 PUFAs EPA and DHA, one that is perhaps strongest for the ratio of long-chain ω-3 to ω-6 PUFAs. There remains too few of these latter studies, however. Studies of short-chain ω-3 PUFA ALA have not indicated a clear pattern of associations. The recent study by Thiebaut and colleagues [88] suggests that further research on dietary source of ALA may prove fruitful, although it remains to be clarified whether observed associations in their study are causal.

Studies of both cross-national and intra-national secular trends have shown inverse associations between per capita consumption of long-chain ω-3 PUFAs and breast cancer incidence and mortality rates [27–31]. Moreover, the shift toward a Western diet usually involves a concurrent decline in ω-3 and a rise in ω-6 PUFA intake, such as that observed in Japan over the past several decades (with a concurrent rise in breast cancer incidence) [32]. Whereas the intakes of these two classes of fatty acids were, for most of human history, similar in quantity (i.e., an intake ratio near unity), modern diets now heavily favor ω-6 intake; for example, one cross-national study of food disappearance data [59] estimated the ratio of per capita ω-3 to ω-6 fatty acid intake in the United States to be approximately 0.003, a ratio that is consistent with what has been observed in adipose tissue levels [96]. Indeed, the results of several human and animal studies suggest that reductions in epithelial cell proliferation rates, mammary tumorigenesis, and PGE_2 biosynthesis can best be achieved with a relatively high ratio of ω-3 (particularly long-chain ω-3) to ω-6 intake, for example, 0.5 and higher [97–101]. These latter findings have some, albeit limited, support from the relatively few epidemiology studies that have considered the ratio of these PUFAs; nonetheless, they do suggest directions for future studies.

Overall, the relatively convincing protective effects of dietary long-chain ω-3 PUFAs on breast cancer observed in various animal models and mechanistic experiments in vitro have not been supported clearly by the results of epidemiological studies. Several reasons for this discrepancy have been suggested, including the higher levels of exposure used in the animal and in vitro studies, the occurrence of measurement error in the epidemiological studies, together with other methodological issues [10], and the more general difficulty of extrapolating the results of animal experiments and in vitro studies to free-living humans [102]. Nonetheless, an increasing number of epidemiological studies of ω-3 PUFA intake and cancer have been published in recent years and these studies have, to varying degrees, attempted to address at least some of these conceptual and methodological issues. Additional studies along these lines are needed to clarify whether, and if so, to what extent dietary short-chain and long-chain ω-3 PUFAs may influence breast cancer risk, the importance of relative vs. absolute levels of specific fatty acids in the diet, and individual factors (including genetic) that might play a role in any association.

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References

1. Wu AH, Pike MC, Stram DO. Meta-analysis: Dietary fat Intake, serum estrogen levels, and the risk of breast cancer. J Natl Cancer Inst 1999; 91:529–34.
2. Ballard-Barbash R, Forman MR, Kipnis V. Dietary fat, serum estrogen levels, and breast cancer risk: A multifaceted story. J Natl Cancer Inst 1999; 91:492–4.
3. Holmes MD, Schisterman EF, Spiegelman D, Hunter DJ, Willett WC. Re: Meta-analysis: dietary fat intake, serum estrogen levels, and the risk of breast cancer. J Natl Cancer Inst 1999; 91:1511–2.
4. Nagata C, Takatsuka N, Kawakami N, Shimizu H. Total and monounsaturated fat intake and serum estrogen concentrations in premenopausal Japanese women. Nutr Cancer 2000; 38:37–9.
5. Nagata C, Takatsuka N, Kawakami N, Shimizu H. Relationships between types of fat consumed and serum estrogen and androgen concentrations in Japanese men. Nutr Cancer 2000; 38:163–7.
6. Thiebaut AC, Kipnis V, Chang SC, et al. Dietary fat and postmenopausal invasive breast cancer in the national institutes of health-AARP diet and health study cohort. J Natl Cancer Inst 2007; 99:451–62.
7. Hunter DJ, Spiegelman D, Adami HO, et al. Cohort studies of fat intake and the risk of breast cancer–a pooled analysis. N Engl J Med 1996; 334:356–61.
8. Smith-Warner SA, Spiegelman D, Adami HO, et al. Types of dietary fat and breast cancer: A pooled analysis of cohort studies. Int J Cancer 2001; 92:767–74.
9. Willett WC. Specific fatty acids and risks of breast and prostate cancer: Dietary intake. Am J Clin Nutr 1997; 66:1557S–63S.
10. Terry PD, Rohan TE, Wolk A. Intakes of fish and marine fatty acids and the risks of cancers of the breast and prostate and of other hormone-related cancers: A review of the epidemiologic evidence. Am J Clin Nutr 2003; 77:532–43.
11. Terry PD, Terry JB, Rohan TE. Long-chain (n-3) fatty acid intake and risk of cancers of the breast and the prostate: Recent epidemiological studies, biological mechanisms, and directions for future research. J Nutr 2004; 134: 3412S–20S.
12. Rose DP, Connolly JM. Omega-3 fatty acids as cancer chemopreventive agents. Pharmacol Ther 1999; 83:217–44.
13. Ip C. Review of the effects of trans fatty acids, oleic acid, n-3 polyunsaturated fatty acids, and conjugated linoleic acid on mammary carcinogenesis in animals. Am J Clin Nutr 1997; 66:1523S–29S.
14. Parkin DM, Bray F, Ferlay J, Pisani P. Global cancer statistics, 2002. CA Cancer J Clin 2005; 55:74–108.
15. Ferlay J, Autier P, Boniol M, Heanue M, Colombet M, Boyle P. Estimates of the cancer incidence and mortality in Europe in 2006. Ann Oncol 2007; 18:581–92.
16. Jemal A, Siegel R, Ward E, et al. Cancer statistics, 2006. CA Cancer J Clin 2006; 56:106–30.
17. Smigal C, Jemal A, Ward E, et al. Trends in breast cancer by race and ethnicity: Update 2006. CA Cancer J Clin 2006; 56:168–83.
18. Parkin DM, Pisani P, Ferlay J. Global cancer statistics. CA Cancer J Clin 1999; 49:33–64, 1.
19. Whelan SL, Parkin DM, Masuyer E. Patterns of Cancer in Five Continents. Lyon: IARC Scientific Publications n. 102, 1990.
20. Horner MJ RL, Krapcho M, Neyman N, Aminou R, Howlader N, Altekruse SF, Feuer EJ, Huang L, Mariotto A, Miller BA, Lewis DR, Eisner MP, Stinchcomb DG, Edwards BK (eds.). SEER Cancer Statistics Review, 1975–2006, National Cancer Institute. Bethesda, MD, http://seer.cancer.gov/csr/1975_2006/, based on November 2008 SEER data submission, posted to the SEER web site, 2009.

21. Willett WC. Diet and breast cancer. J Intern Med 2001; 249:395–411.
22. Key TJ, Allen NE, Spencer EA, Travis RC. The effect of diet on risk of cancer. Lancet 2002; 360:861–8.
23. Lof M, Weiderpass E. Impact of diet on breast cancer risk. Curr Opin Obstet Gynecol 2009; 21:80–5.
24. Michels KB, Mohllajee AP, Roset-Bahmanyar E, Beehler GP, Moysich KB. Diet and breast cancer: A review of the prospective observational studies. Cancer 2007; 109:2712–49.
25. Armstrong B, Doll R. Environmental factors and cancer incidence and mortality in different countries, with special reference to dietary practices. Int J Cancer 1975; 15:617–31.
26. Davis BC, Kris-Etherton PM. Achieving optimal essential fatty acid status in vegetarians: Current knowledge and practical implications. Am J Clin Nutr 2003; 78:640S–46S.
27. Hebert JR, Rosen A. Nutritional, socioeconomic, and reproductive factors in relation to female breast cancer mortality: Findings from a cross-national study. Cancer Detect Prev 1996; 20:234–44.
28. Kaizer L, Boyd NF, Kriukov V, Tritchler D. Fish consumption and breast cancer risk: An ecological study. Nutr Cancer 1989; 12:61–8.
29. Sasaki S, Horacsek M, Kesteloot H. An ecological study of the relationship between dietary fat intake and breast cancer mortality. Prev Med 1993; 22:187–202.
30. Guo WD, Chow WH, Zheng W, Li JY, Blot WJ. Diet, serum markers and breast cancer mortality in China. Jpn J Cancer Res 1994; 85:572–7.
31. Caygill CP, Charlett A, Hill MJ. Fat, fish, fish oil and cancer. Br J Cancer 1996; 74:159–64.
32. Karmali RA. Omega-3 fatty acids and cancer: A review. In: Lands WE, (ed.). Proceedings of the AOCS Short Course on Polyunsaturated Fatty Acids and Eicosanoids, Mississippi, May 1987:13–16. Champaign, IL: American Oil Chemists' Society, 1987:222–32.
33. Bjarnason O, Day N, Snaedal G, Tulinius H. The effect of year of birth on the breast cancer age-incidence curve in Iceland. Int J Cancer 1974; 13:689–96.
34. Lanier AP, Kelly JJ, Smith B, et al. Alaska native cancer update: Incidence rates 1989–1993. Cancer Epidemiol Biomarkers Prev 1996; 5:749–51.
35. Nielsen NH, Hansen JP. Breast cancer in Greenland–selected epidemiological, clinical, and histological features. J Cancer Res Clin Oncol 1980; 98:287–99.
36. Poole C. Ecologic analysis as outlook and method. Am J Public Health 1994; 84:715–6.
37. Childs MT, King IB, Knopp RH. Divergent lipoprotein responses to fish oils with various ratios of eicosapentaenoic acid and docosahexaenoic acid. Am J Clin Nutr 1990; 52: 632–9.
38. Rothman KJ, Greenland S. Modern Epidemiology, 2nd ed. Philadelphia, PA: Lipincott-Raven, 1998: p. 319.
39. Willett WC. Nutritional Epidemiology, 2nd ed. New York: Oxford University Press, 1998.
40. Hislop TG, Coldman AJ, Elwood JM, Brauer G, Kan L. Childhood and recent eating patterns and risk of breast cancer. Cancer Detect Prev 1986; 9:47–58.
41. Hislop TG, Kan L, Coldman AJ, Band PR, Brauer G. Influence of estrogen receptor status on dietary risk factors for breast cancer. CMAJ 1988; 138:424–30.
42. Ambrosone CB, Freudenheim JL, Sinha R, et al. Breast cancer risk, meat consumption and N-acetyltransferase (NAT2) genetic polymorphisms. Int J Cancer 1998; 75:825–30.
43. Fernandez E, Chatenoud L, La Vecchia C, Negri E, Franceschi S. Fish consumption and cancer risk. Am J Clin Nutr 1999; 70:85–90.
44. Levi F, La Vecchia C, Gulie C, Negri E. Dietary factors and breast cancer risk in Vaud, Switzerland. Nutr Cancer 1993; 19:327–35.
45. Shannon J, Cook LS, Stanford JL. Dietary intake and risk of postmenopausal breast cancer (United States). Cancer Causes Control 2003; 14:19–27.
46. Toniolo P, Riboli E, Protta F, Charrel M, Cappa AP. Calorie-providing nutrients and risk of breast cancer. J Natl Cancer Inst 1989; 81:278–86.
47. Mannisto S, Pietinen P, Virtanen M, Kataja V, Uusitupa M. Diet and the risk of breast cancer in a case-control study: Does the threat of disease have an influence on recall bias? J Clin Epidemiol 1999; 52:429–39.

48. Landa MC, Frago N, Tres A. Diet and the risk of breast cancer in Spain. Eur J Cancer Prev 1994; 3:313–20.
49. Kato I, Miura S, Kasumi F, et al. A case-control study of breast cancer among Japanese women: With special reference to family history and reproductive and dietary factors. Breast Cancer Res Treat 1992; 24:51–9.
50. Iscovich JM, Iscovich RB, Howe G, Shiboski S, Kaldor JM. A case-control study of diet and breast cancer in Argentina. Int J Cancer 1989; 44:770–6.
51. Hirose K, Takezaki T, Hamajima N, Miura S, Tajima K. Dietary factors protective against breast cancer in Japanese premenopausal and postmenopausal women. Int J Cancer 2003; 107:276–82.
52. Hirose K, Tajima K, Hamajima N, et al. A large-scale, hospital-based case-control study of risk factors of breast cancer according to menopausal status. Jpn J Cancer Res 1995; 86:146–54.
53. De Stefani E, Ronco A, Mendilaharsu M, Guidobono M, Deneo-Pellegrini H. Meat intake, heterocyclic amines, and risk of breast cancer: A case-control study in Uruguay. Cancer Epidemiol Biomarkers Prev 1997; 6:573–81.
54. Dai Q, Shu XO, Jin F, Gao YT, Ruan ZX, Zheng W. Consumption of animal foods, cooking methods, and risk of breast cancer. Cancer Epidemiol Biomarkers Prev 2002; 11:801–8.
55. Lubin JH, Burns PE, Blot WJ, Ziegler RG, Lees AW, Fraumeni JF, Jr. Dietary factors and breast cancer risk. Int J Cancer 1981; 28:685–9.
56. Lee HP, Gourley L, Duffy SW, Esteve J, Lee J, Day NE. Dietary effects on breast-cancer risk in Singapore. Lancet 1991; 337:1197–200.
57. Lee HP, Gourley L, Duffy SW, Esteve J, Lee J, Day NE. Risk factors for breast cancer by age and menopausal status: A case-control study in Singapore. Cancer Causes Control 1992; 3:313–22.
58. Shu XO, Jin F, Dai Q, et al. Soyfood intake during adolescence and subsequent risk of breast cancer among Chinese women. Cancer Epidemiol Biomarkers Prev 2001; 10:483–8.
59. Hursting SD, Thornquist M, Henderson MM. Types of dietary fat and the incidence of cancer at five sites. Prev Med 1990; 19:242–53.
60. Goodstine SL, Zheng T, Holford TR, et al. Dietary (n-3)/(n-6) fatty acid ratio: Possible relationship to premenopausal but not postmenopausal breast cancer risk in U.S. women. J Nutr 2003; 133:1409–14.
61. Nkondjock A, Shatenstein B, Ghadirian P. A case-control study of breast cancer and dietary intake of individual fatty acids and antioxidants in Montreal, Canada. Breast 2003; 12: 128–35.
62. Zhu ZR, Agren J, Mannisto S, et al. Fatty acid composition of breast adipose tissue in breast cancer patients and in patients with benign breast disease. Nutr Cancer 1995; 24:151–60.
63. London SJ, Sacks FM, Stampfer MJ, et al. Fatty acid composition of the subcutaneous adipose tissue and risk of proliferative benign breast disease and breast cancer. J Natl Cancer Inst 1993; 85:785–93.
64. Maillard V, Bougnoux P, Ferrari P, et al. N-3 and N-6 fatty acids in breast adipose tissue and relative risk of breast cancer in a case-control study in Tours, France. Int J Cancer 2002; 98:78–83.
65. Petrek JA, Hudgins LC, Levine B, Ho M, Hirsch J. Breast cancer risk and fatty acids in the breast and abdominal adipose tissues. J Natl Cancer Inst 1994; 86:53–6.
66. Simonsen N, van't Veer P, Strain JJ, et al. Adipose tissue omega-3 and omega-6 fatty acid content and breast cancer in the EURAMIC study. European community multicenter study on antioxidants, myocardial infarction, and breast cancer. Am J Epidemiol 1998; 147: 342–52.
67. Kuriki K, Hirose K, Wakai K, et al. Breast cancer risk and erythrocyte compositions of n-3 highly unsaturated fatty acids in Japanese. Int J Cancer 2007; 121:377–85.
68. Shannon J, King IB, Moshofsky R, et al. Erythrocyte fatty acids and breast cancer risk: A case-control study in Shanghai, China. Am J Clin Nutr 2007; 85:1090–7.

69. Chajes V, Hulten K, Van Kappel AL, et al. Fatty-acid composition in serum phospholipids and risk of breast cancer: An incident case-control study in Sweden. Int J Cancer 1999; 83:585–90.
70. Vatten LJ, Bjerve KS, Andersen A, Jellum E. Polyunsaturated fatty acids in serum phospholipids and risk of breast cancer: A case-control study from the Janus serum bank in Norway. Eur J Cancer 1993; 29A:532–8.
71. Takata Y, King IB, Neuhouser ML, et al. Association of serum phospholipid fatty acids with breast cancer risk among postmenopausal cigarette smokers. Cancer Causes Control 2009; 20:497–504.
72. Hunter D. Biochemical indicators of dietary intake. In: Willett WC, (ed.). Nutritional Epidemiology, 2nd ed. New York: Oxford University Press, 1998.
73. Stampfer MJ, Willett WC, Colditz GA, Speizer FE. Intake of cholesterol, fish and specific types of fat in relation to risk of breast cancer. In: Lands WE, (ed.). Proceedings of the AOCS Short Course on Polyunsaturated Fatty Acids and Eicosanoids, Mississippi, May 1987: 13–16. Champaign, IL: American Oil Chemists' Society, 1987:248–52.
74. Toniolo P, Riboli E, Shore RE, Pasternack BS. Consumption of meat, animal products, protein, and fat and risk of breast cancer: a prospective cohort study in New York. Epidemiology 1994; 5:391–7.
75. Vatten LJ, Solvoll K, Loken EB. Frequency of meat and fish intake and risk of breast cancer in a prospective study of 14,500 Norwegian women. Int J Cancer 1990; 46:12–5.
76. Lund E, Bonaa KH. Reduced breast cancer mortality among fishermen's wives in Norway. Cancer Causes Control 1993; 4:283–7.
77. Key TJ, Sharp GB, Appleby PN, et al. Soya foods and breast cancer risk: A prospective study in Hiroshima and Nagasaki, Japan. Br J Cancer 1999; 81:1248–56.
78. Holmes MD, Colditz GA, Hunter DJ, et al. Meat, fish and egg intake and risk of breast cancer. Int J Cancer 2003; 104:221–7.
79. Gertig DM, Hankinson SE, Hough H, et al. N-acetyl transferase 2 genotypes, meat intake and breast cancer risk. Int J Cancer 1999; 80:13–7.
80. Folsom AR, Demissie Z. Fish intake, marine omega-3 fatty acids, and mortality in a cohort of postmenopausal women. Am J Epidemiol 2004; 160:1005–10.
81. Terry P, Rohan TE, Wolk A, Maehle-Schmidt M, Magnusson C. Fish consumption and breast cancer risk. Nutr Cancer 2002; 44:1–6.
82. Gago-Dominguez M, Yuan JM, Sun CL, Lee HP, Yu MC. Opposing effects of dietary n-3 and n-6 fatty acids on mammary carcinogenesis: The Singapore Chinese health study. Br J Cancer 2003; 89:1686–92.
83. Wakai K, Tamakoshi K, Date C, et al. Dietary intakes of fat and fatty acids and risk of breast cancer: A prospective study in Japan. Cancer Sci 2005; 96:590–9.
84. Holmes MD, Hunter DJ, Colditz GA, et al. Association of dietary intake of fat and fatty acids with risk of breast cancer. JAMA 1999; 281:914–20.
85. Pala V, Krogh V, Muti P, et al. Erythrocyte membrane fatty acids and subsequent breast cancer: A prospective Italian study. J Natl Cancer Inst 2001; 93:1088–95.
86. Saadatian-Elahi M, Toniolo P, Ferrari P, et al. Serum fatty acids and risk of breast cancer in a nested case-control study of the New York university women's health study. Cancer Epidemiol Biomarkers Prev 2002; 11:1353–60.
87. Wirfalt E, Vessby B, Mattisson I, Gullberg B, Olsson H, Berglund G. No relations between breast cancer risk and fatty acids of erythrocyte membranes in postmenopausal women of the Malmo Diet Cancer cohort (Sweden). Eur J Clin Nutr 2004; 58:761–70.
88. Thiebaut AC, Chajes V, Gerber M, et al. Dietary intakes of omega-6 and omega-3 polyunsaturated fatty acids and the risk of breast cancer. Int J Cancer 2009; 124: 924–31.
89. Voorrips LE, Brants HA, Kardinaal AF, Hiddink GJ, van den Brandt PA, Goldbohm RA. Intake of conjugated linoleic acid, fat, and other fatty acids in relation to postmenopausal breast cancer: The Netherlands cohort study on diet and cancer. Am J Clin Nutr 2002; 76:873–82.

90. Tapiero H, Ba GN, Couvreur P, Tew KD. Polyunsaturated fatty acids (PUFA) and eicosanoids in human health and pathologies. Biomed Pharmacother 2002; 56:215–22.
91. Bagga D, Anders KH, Wang HJ, Glaspy JA. Long-chain n-3-to-n-6 polyunsaturated fatty acid ratios in breast adipose tissue from women with and without breast cancer. Nutr Cancer 2002; 42:180–5.
92. Brenna JT. Efficiency of conversion of alpha-linolenic acid to long chain n-3 fatty acids in man. Curr Opin Clin Nutr Metab Care 2002; 5:127–32.
93. De Stefani E, Deneo-Pellegrini H, Mendilaharsu M, Ronco A. Essential fatty acids and breast cancer: A case-control study in Uruguay. Int J Cancer 1998; 76:491–4.
94. Franceschi S, Favero A, Decarli A, et al. Intake of macronutrients and risk of breast cancer. Lancet 1996; 347:1351–6.
95. Saadatian-Elahi M, Norat T, Goudable J, Riboli E. Biomarkers of dietary fatty acid intake and the risk of breast cancer: A meta-analysis. Int J Cancer 2004; 111:584–91.
96. Godley PA, Campbell MK, Gallagher P, Martinson FE, Mohler JL, Sandler RS. Biomarkers of essential fatty acid consumption and risk of prostatic carcinoma. Cancer Epidemiol Biomarkers Prev 1996; 5:889–95.
97. Abou-el-Ela SH, Prasse KW, Farrell RL, Carroll RW, Wade AE, Bunce OR. Effects of D, L-2-difluoromethylornithine and indomethacin on mammary tumor promotion in rats fed high n-3 and/or n-6 fat diets. Cancer Res 1989; 49:1434–40.
98. Bartram HP, Gostner A, Reddy BS, et al. Missing anti-proliferative effect of fish oil on rectal epithelium in healthy volunteers consuming a high-fat diet: Potential role of the n-3:n-6 fatty acid ratio. Eur J Cancer Prev 1995; 4:231–7.
99. Bartram HP, Gostner A, Scheppach W, et al. Effects of fish oil on rectal cell proliferation, mucosal fatty acids, and prostaglandin E2 release in healthy subjects. Gastroenterology 1993; 105:1317–22.
100. Deschner EE, Lytle JS, Wong G, Ruperto JF, Newmark HL. The effect of dietary omega-3 fatty acids (fish oil) on azoxymethanol-induced focal areas of dysplasia and colon tumor incidence. Cancer 1990; 66:2350–6.
101. Noguchi M, Minami M, Yagasaki R, et al. Chemoprevention of DMBA-induced mammary carcinogenesis in rats by low-dose EPA and DHA. Br J Cancer 1997; 75:348–53.
102. Bukowski JA, Schnatter AR, Korn L. Using epidemiological studies to check the consistency of the cancer risks predicted by high-dose animal experiments: A methodological review. Risk Anal 2001; 21:601–11.

Chapter 5
ω-3 PUFAs and Prostate Cancer: Epidemiological Studies

Pierre Astorg

Abstract This chapter reviews all studies published before 31 May 2009 and having searched associations between the intakes or the blood levels of ω-3 polyunsaturated fatty acids (PUFAs) and prostate cancer risk, including studies on fish or seafood intake. The association between higher α-linolenic acid intakes or blood levels and an increased risk of advanced prostate cancer, which was found in several earlier studies and had raised some concern, was not confirmed in most later studies, especially in cohort studies. However, the heterogeneity of the results calls for further research. Among the numerous studies having looked for the association between fish or long-chain ω-3 PUFA intakes and prostate cancer risk, only a few have found a decreased risk with higher intakes, and the association appears to depend on COX-2 genetic variants. A strong decrease in the risk of metastatic or fatal prostate cancer in higher fish consumers was observed in some cohort studies, suggesting a possible protective effect of long-chain ω-3 PUFAs on the later stages of cancer progression. Further studies based on improved methodology, i.e. in particular, accurate measure of exposure, assessment of cancer stage and grade, and controlling for confounding variables, especially for PSA screening, are strongly awaited.

Keywords α-Linolenic acid · Long-chain omega-3 PUFA · Fish · Prostate · Cancer

5.1 Introduction

5.1.1 Prostate Cancer: Epidemiological Characteristics

Prostate cancer is the second most frequent cancer in men in the world and the most frequent in developed countries [1]. Prostate cancer incidence varies considerably between countries, and rates of the disease differ by 100-fold or more between

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countries with the lowest rates such as in China or Bangladesh, and the highest rates such as in the USA [1–3]. Still more than for other cancers, age is a prominent risk factor of prostate cancer: The rate is very low until 50 years, but increases exponentially thereafter [2, 4]. Prostate cancer has an important genetic component. Its heritability has been estimated to 42% on the basis of twin studies [5], which is higher than that for the other cancers. Genetic factors are present at the population level, with large differences of incidence between races, the black men from the USA having the highest rate in the world [2, 4]. Genetics has a likely influence at the individual level, since a family history of prostate cancer is also a risk factor [2, 4]. Age, race/ethnicity, and family history are the only established risk factors of prostate cancer. However, the large differences of rates between countries suggest a strong influence of environmental factors. In particular, dietary factors favouring or preventing the development of prostate cancer have been the subject of much epidemiological research in the two past decades [reviews [6–8]]. Among them, as for other cancers, ω-3 fatty acids have raised much interest and continue to do so.

Prostate cancer incidence has been increasing in many countries, in part because of an increasing rate of case detection [3]. The screening of prostate cancer through prostate-specific antigen (PSA) testing – coupled to prostate biopsy histological examination – was first introduced into clinical practice in the USA in the mid- to late 1980s, then in Canada, United Kingdom, Australia around 1990 [3], and later on in other countries. The widespread use of PSA testing in these countries has caused a sharp rise in the incidence of asymptomatic/localized cases, which otherwise had remained undiagnosed, while the rate of advanced cases tended at the same time to decrease [3]. Introduction of PSA screening has thus changed the characteristics of diagnosed prostate cancers and interferes with the research of its risk or protection factors, which are not expected to be the same at the early (small localized tumour), middle (regional tumour), or final (distant metastases) stages of the disease [9]. Thus, a protection or risk factor of prostate cancer, which had been observed in studies preceding the PSA era, may no longer be detected in more recent studies because of the enrichment of the case pool with preclinical cases, which, in earlier studies, would have been classified among the non-cases [9]. To try to overcome this problem, many recent epidemiological studies have distinguished cases categories, according to the stage of the disease (localized, regional/advanced, metastatic) or to its pathological score (Gleason score, measuring tumour aggressiveness), or both. In the absence of such information, the country where the study was realized and the period of case assessment are of importance to estimate the type of cases involved.

Case finding through PSA testing, instead of through consulting for symptoms, has another consequence in epidemiological studies: consulting for PSA testing is a voluntary practice and is typically a component of a health-conscious behaviour. There is thus a greater probability of finding more localized cancers in health-conscious men, PSA screening propensity being confounded with other components of this behaviour, such as avoiding overweight, making physical exercise or eating a healthy diet (containing, for example, fish and ω-3 fatty acids). This risk of confounding will be suspected in studies where some dietary factor, especially among those expected to be "healthy", is associated with an increase in the risk of localized cancer, but not of advanced cancer.

Finally, not all localized cancers evolve to a clinical disease. At variance, many small localized, non-aggressive cancers remain latent (without clinical symptoms) throughout the subject's life. Autopsy studies have shown that most aged men have histologically defined prostate cancer: about 50% of men in their fifties, more than 75% for men older than 85 years [2]. The high frequency of latent prostate cancer in middle age and aged men supports the view that the important process for dietary prevention is the modulation of cancer progression to a clinical disease, rather than the blockade of cancer initiation. Thus, in addition to the possible detection bias introduced by PSA screening, this will have for consequence that a greater attention will be brought to the risks associated with advanced or aggressive cases in studies performed in the PSA era.

5.1.2 ω-3 Polyunsaturated Fatty Acids (PUFAs): Biochemical and Nutritional Considerations

α-Linolenic acid (ALA), the precursor of the ω-3 family, although of exclusive vegetable origin, is present in the human diet in both animal and vegetable products. There are few concentrated dietary sources of ALA (more than 1g/100 g edible part): some vegetable oils (mainly rapeseed and soybean oils), margarines and dressings containing these oils, and walnuts. However, many animal or vegetable foods, including oils and fats, meats, dairy products, green vegetables, nuts, contain smaller but significant amounts of ALA (0.1–1g/100 g edible part). In populations with a relatively low consumption of rapeseed/soybean oils and margarines, such as Mediterranean countries, including France, ALA intake is relatively low and animal foods are important contributors [10, 11]. At variance with linoleic acid, which readily incorporates in adipose tissue triglycerides and in all blood lipid fractions, ALA is only present at low concentrations in blood or tissue lipids, especially in plasma or erythrocyte phospholipids, where its concentrations are very low (0.1–0.2% of total fatty acids) [12–14]. As a consequence, ALA blood level is not always correlated with its intake level [12, 14]. ALA in total blood, total plasma or serum, plasma cholesteryl esters, or adipose tissue appear as more suitable biomarkers of ALA intake than the small amounts of ALA found in erythrocytes or plasma PL [13, 15, 16].

Dietary long-chain ω-3 PUFAs mainly consist of eicosapentaenoic acid (EPA, 20:5 *n*–3), docosapentaenoic acid (DPA, 22:5 *n*–3), and docosahexaenoic acid (DHA, 22:6 *n*–3) and are brought in the human diet exclusively by animal products. Among them, fish and other seafoods, and especially fatty fish, are by far the main contributors to long-chain ω-3 PUFA intake: in a French population sample taking no or little fish oil supplements (in 1994–1996), 60% of the long-chain ω-3 PUFA intake and about two-thirds of EPA and DHA intakes were provided by fish and seafood, whereas meats, poultry, and eggs were the main contributors of ω-3

DPA intake [11]. The bioconversion of ALA to long-chain ω-3 PUFAs occurs only at a low rate in adult man: less than 5% of ingested ALA is converted to long-chain PUFAs and less than 0.5% is converted to DHA [17]. Thus, in omnivores, blood and tissue long-chain ω-3 PUFAs are provided mainly, if not exclusively, from exogenous sources (diet and supplements). As a consequence, the blood levels of long-chain ω-3 PUFAs in the general population are correlated to their own levels of intake, not to the intake of ALA [14]. At variance with ALA, the percentages of long-chain ω-3 PUFAs in all blood fractions are good biomarkers of their intake levels [15, 16].

5.1.3 Studies Selection

We have included in this review all the studies that have looked for the association of prostate cancer with the intake levels, or with the blood or tissue levels of ALA or of long-chain ω-3 PUFAs, satisfying these conditions: (1) case–control studies based on dietary questionnaires, having included at least 100 cases; (2) case–control studies based on biomarkers, having included at least 50 cases; (3) all cohort or nested case–control studies, based either on questionnaires or on biomarkers. Many studies have distinguished case subgroups, according either to the grade (histological score) or to the stage of the tumour: high-grade (aggressive) tumours had a Gleason score ≥ 7, low-grade (non-aggressive) tumours < 7; localized tumours were confined within the prostate; advanced tumours had progressed through the prostate capsule, spread to adjacent tissues, or metastasized. The main results of these studies are presented in Tables 5.1, 5.2, 5.3, 5.4, and 5.5. A few ecological studies and some biomarker-based case–control studies of small size will also be mentioned in the text. Cohort studies and case–control studies nested within cohorts have been performed in a predefined population and have a prospective design: dietary intakes or biomarker fatty acid levels have been measured at or near baseline, and incident cancer cases have been recorded during the follow-up. These prospective studies are thus free of the selection and recall biases that usually flaw many case–control studies, and their design is better suited to detect a causal relation. For these main reasons, these studies are a priori of greater weight in this review.

In the case–control and cohort studies based on dietary questionnaires that we report below, adjustments for potentially confounding dietary variables have been made, at least for total energy intake, and often for other dietary variables such as intakes of total lipid, animal fat. In contrast, dietary variables have rarely been included as adjustment variables in studies based on biomarkers.

5.2 ALA and Prostate Cancer

Some concern has arisen in the past years about the possible promoter effect of ALA on prostate cancer [18–20]. As a matter of fact, in the first two studies which have looked for the association between ALA intake and prostate cancer risk, a US cohort

Table 5.1 Intake of α-linolenic acid and prostate cancer: case–control studies

Study, country, period	Dietary assessment	No. of cases and controls	Total fat	18:2 ω–6 LA	18:3 ω–3 ALA	Comments	References
Sweden, 1989–1994	FFQ, 68 items	526 cases 536 population controls 296 advanced cases	1.08 (0.77–1.52) 1.15 (0.77–1.72)	1.19 (0.84–1.68) 1.19 (0.79–1.77)	0.93 (0.65–1.32) 0.82 (0.54–1.23)	ORs are only adjusted for age and total energy intake	Andersson et al. [26]
Canada, 1990–1993	Diet history interview, 143 items	215 cases, most localized 593 controls, most with BPH	1.02 (0.57–1.85)	1.57 (0.85–2.93)	0.98 (0.54–1.78)	All men have been either surgically treated for BPH or screened for prostate cancer	Meyer et al. [27]
Canada, 1990–1992	Diet history interview, 143 items	142 advanced cases 242 localized cases	1.59 (0.87–2.91)	0.66 (0.36–1.20)	1.04 (0.58–1.89)	ORs: advanced cases vs. localized cases	Bairati et al. [28]
Uruguay, 1994–1998	FFQ, 64 items	217 cases 431 hospital controls	1.33 (0.75–2.34)	0.71 (0.42–1.20) p for trend 0.07	**1.91** (1.12–3.25) p for trend 0.009	After adjustment for intake of other fatty acids, fruits and vegetables: ALA **3.91** (1.50–10.1) p for trend 0.001	De Stéfani et al. [29]
Spain, 1994–1998	FFQ, 89 items	217 cases 434 controls (hospital + population)	1.2 (0.9–1.6)	1.0 (0.7–1.4) (AGPI n–6)	**2.5** (1.8–3.4) p for trend 0.001	After adjustment for other fatty acids, animal fats, and cholesterol: ALA **3.1** (1.2–4.7) p for trend 0.001	Ramon et al. [30]
Australia, 1994–1997	FFQ, 121 items	858 cases 905 population controls	1.0 (0.8–1.4)	1.0 (0.7–1.3)	**0.8** (0.6–1.0)	Cases: Gleason score ≥ 5. No significant trend for ALA	Hodge et al. [32]

Table 5.1 (continued)

Study, country, period	Dietary assessment	No. of cases and controls	Total fat	18:2 ω–6 LA	18:3 ω–3 ALA	Comments	References
Italy, 1991–2002	FFQ, 78 items	1294 cases 1451 hospital controls	1.0 (0.8–1.3)	**0.8** (0.6–1.0) p for trend 0.02	**0.7** (0.6–0.9) p for trend 0.003	ALA, continuous OR (for 1 g/day): **0.84** (0.75–0.94)	Bidoli et al. [33]
Sweden, 2001–2002	FFQ, 261 items	1499 cases 1130 hospital controls	n.d.	**1.36** (1.01–1.84) p for trend 0.03 (n–6 PUFA)	1.35 (0.99–1.84) p for trend 0.03	Cases: 55% localized, 41% advanced, 4% unknown. ORs are adjusted for intakes of many nutrients and foods	Hedelin et al. [31]
USA, 2001–2004	FFQ	466 advanced or aggressive cases 478 hospital controls	n.d.	0.97 (0.56–1.67)	0.81 (0.48–1.35) p for trend 0.11	Cases: Gleason score ≥7 (aggressive), or stage ≥T2c (advanced), or PSA > 10 ng/ml. Further adjustment for PSA testing did not change the results	Fradet et al. [34]

Values are odds ratios (OR) for the highest tertile, quartile, or quintile relative to the lowest one, with their 95% confidence interval, as well as the probability of the trend test if <0.10. ORs in bold are significant for p = 0.05. All ORs are at least adjusted for age and total energy intake and in most cases for other variables such as family history of prostate cancer, BMI, smoking, ethnicity, socio-economic status, education, residence, dietary variables.
Tumour stage: localized = confined to the prostate, advanced = regional or metastatic; tumour score: low grade = Gleason score <7, high grade = Gleason score ≥7.
Abbreviations: ALA: α-linolenic acid; BPH: benign prostatic hyperplasia; DHA: docosahexaenoic acid; DPA: docosapentaenoic acid; EPA: eicosapentaenoic acid; FFQ: food frequency questionnaire; LA: linoleic acid.

Table 5.2 Intake of α-linolenic acid and prostate cancer: cohort studies

Study, country, period	Cohort size	Dietary assessment	Follow-up, years	No. of cases and controls	Total fat	18:2 ω–6 LA	18:3 ω–3 ALA	Comments	Reference
Health Professionals Follow-Up Study, USA, 1986–1990	48,855 men	FFQ, 131 items	3.5	279 cases	1.32 (0.91–1.92)	0.88 (0.55–1.43)	1.25 (0.82–1.92)	ORs adjusted for intakes of other fatty acids; association of ALA with advanced and fatal cases after adjustment for meat fat: **2.67** (1.48–4.81) p for trend=0.02	Giovannucci et al. [21]
				126 advanced and fatal cases	1.68 (0.97–2.90)	0.64 (0.32–1.32)	**3.43** (1.67–7.04) p for trend 0.002		
Netherlands Cohort Study, Netherlands 1986–1992	58,279 men	FFQ, 150 items	6.3	642 cases	1.10 (0.80–1.52)	0.78 (0.56–1.09)	0.76 (0.66–1.04) p for trend 0.09	No association of LA or ALA with localized or advanced cases	Schuurman et al. a [37]
ATBC Study, Finland, 1985–1993	29,133 men	FFQ, 276 items	5–8	198 cases 198 controls	n.d.	0.92 (0.54–1.59)	1.16 (0.64–2.13)	Case–control study nested within a cohort of male smokers; unadjusted ORs	Männistö et al. [38]
Kuopio Ischaemic Heart Disease Risk Factor Study, Finland, 1984–1999	2002 men	4-day food record	4–16 mean 12.6	46 cases	n.d.	0.57 (0.27–1.19)	No significant association	ORs are adjusted for many dietary items	Laaksonen et al. [39]

Table 5.2 (continued)

Study, country, period	Cohort size	Dietary assessment	Follow-up, years	No. of cases and controls	Total fat	18:2 ω–6 LA	18:3 ω–3 ALA	Comments	Reference
Health Professionals Follow-Up study, USA, 1986–2000	47,866 men	FFQ, 131 items	14	2965 cases	n.d.	1.06 (0.89–1.26)	1.09 (0.93–1.26)	Association of ALA with advanced cancer persists after adjustment for red meat intake; ALA from vegetable sources **2.02** (1.35–3.03) p for trend < 0.001	Leitzmann et al. [35]
				448 advanced cases	n.d.	0.80 (0.52–1.24)	**1.98** (1.34–2.93) p for trend 0.001		
PLCO Cancer Screening Trial, USA, 1993–2001	29,592 men	FFQ, 137 items	5.1	1898 cases	n.d.	n.d.	0.94 (0.81–1.09)	All men have been screened annually for 5 years by PSA test. No association with ALA from different sources; no association of ALA with low- or high-grade cases	Koralek et al. [41]
				285 advanced cases	n.d.	n.d.	0.83 (0.58–1.19)		

Table 5.2 (continued)

Study, country, period	Cohort size	Dietary assessment	Follow-up, years	No. of cases and controls	Total fat	18:2 ω–6 LA	18:3 ω–3 ALA	Comments	Reference
Health Professionals Follow-Up Study, USA, 1986–2002	47,750 men	FFQ, 131 items	15	3544 cases	n.d.	n.d.	**1.12** (1.01–1.25) p for trend 0.03	ALA was associated with low-grade cancer: **1.22** (1.04–1.43) p for trend 0.02, not with high-grade cancer; the associations of ALA with advanced and fatal cancer were similar before or during the PSA era. ORs adjusted for intakes of processed meat, fish, and tomato sauce	Giovanucci et al. [36]
				523 advanced and fatal cases	n.d.	n.d.	**1.57** (1.19–2.07) p for trend 0.05		
				312 fatal cases	n.d.	n.d.	**1.53** (1.07–2.20) p for trend 0.01		
Multiethnic Cohort Study, USA (Hawaii and Los Angeles) 1993–2002	82,483 men	FFQ, 180 items	8	4404 cases	0.99 (0.89–1.09)	1.03 (0.93–1.14)	0.92 (0.84–1.02)	ALA inversely associated with all-case cancer in white men: **0.79** (0.64–0.99) p for trend 0.06, and in latino men 0.83 (0.67–1.01) p for trend 0.03	Park et al. [42]
				1278 advanced or high-grade cases	0.90 (0.75–1.09)	1.04 (0.86–1.27) (AGPI n–6)	0.89 (0.74–1.06)		

Table 5.2 (continued)

Study, country, period	Cohort size	Dietary assessment	Follow-up, years	No. of cases and controls	Total fat	18:2 ω–6 LA	18:3 ω–3 ALA	Comments	Reference
Malmö Diet and Cancer Cohort, Sweden, 1991–2005	10,564 men	FFQ, 168 items	11	817 cases	0.99 (0.79–1.24)	0.98 (0.79–1.22)	0.92 (0.73–1.15)	Population with high intakes of ALA: first quintile 0.5–1.6 g/day, fifth quintile 2.4–10.7 g/day; ORs adjusted for intakes of calcium, meat, fruit, and vegetables	Wallström et al. [40]
				281 advanced cases	1.11 (0.75–1.66)	0.95 (0.67–1.36)	0.93 (0.64–1.36)		

Values are odds ratios (OR) for the highest tertile, quartile, or quintile relative to the lowest one, with their 95% confidence interval, as well as the probability of the trend test if <0.10. ORs in bold are significant for $p = 0.05$. All ORs (except in Mannistö et al.) were at least adjusted for age and total energy intake, and in most cases for other variables such as family history of prostate cancer, BMI, smoking, ethnicity, socio-economic status, education, residence, dietary variables.

Tumour stage: localized = confined to the prostate, advanced = regional or metastatic; tumour score: low grade = Gleason score < 7, high grade = Gleason score ≥ 7.

Abbreviations: ATBC study: Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study; ALA: α-linolenic acid; BPH: benign prostatic hyperplasia; DHA: docosahexaenoic acid; DPA: docosapentaenoic acid; EPA: eicosapentaenoic acid; FFQ: food frequency questionnaire; LA: linoleic acid; PLCO: Prostate, Lung, Colorectal, and Ovarian.

Table 5.3 Blood or tissue polyunsaturated fatty acids and prostate cancer: case–control, cohort, and nested case–control studies

Country, period	Study type	Follow-up, year	No. of cases and controls	Blood or tissue fraction	18:2 ω–6 LA	18:3 ω–3 ALA	20:5 ω–3 EPA	22:6 ω–3 DHA	Comments	Reference
Physician's Health Study, USA, 1982–1990	Nested case–control	6	120 cases and 120 population controls	Plasma CE	0.71 (0.32–1.56)	2.22 (0.93–5.29) p for trend 0.04	0.87 (0.41–1.82)	n.d.	ORs ALA adjusted for linoleic acid and meat; low mean ALA level: 0.15%	Gann et al., [22]
USA, 1989–1991	Case–control	–	89 cases and 38 hospital controls, all with prostate biopsy	erythrocytes	**3.54** (1.0–12.5) p for trend 0.04	1.69 (0.54–5.26)	0.74 (0.23–2.33)	0.36 (0.10–1.27)	No change for adjustment for other fatty acids; mean ALA levels: 0.18% in erythrocytes, 0.82% in adipose tissue	Godley et al., [44]
				adipose tissue	2.47 (0.66–9.26)	2.73 (0.70–10.6)	0.54 (0.18–1.62)	1.11 (0.30–4.14)		
Norway 1973–1994	Prospective case–control	0.2–19 mean 11.6	141 cases and 282 population controls, all blood donors	serum	0.9 (0.5–1.7)	**2.0** (1.1–3.6) p for trend 0.03	1.2 (0.6–2.1)	1.0 (0.5–1.8)	ORs unadjusted; ALA, second and third quartiles: 1.4 (0.8–2.5), 1.5 (0.9–2.7); low mean ALA level: 0.15%	Harvei et al. [52]

Table 5.3 (continued)

Country, period	Study type	Follow-up, year	No. of cases and controls	Blood or tissue fraction	18:2 ω–6 LA	18:3 ω–3 ALA	20:5 ω–3 EPA	22:6 ω–3 DHA	Comments	Reference
New-Zealand, 1996–1997	Case–control	–	285 cases and 427 population controls	erythrocyte PC	n.d.	n.d.	**0.59** (0.37–0.95) p for trend 0.03	**0.62** (0.39–0.98) p for trend 0.01	ORs adjusted for total PUFA intake; advanced cancers: EPA **0.54** (0.31–0.96) p for trend 0.03; DHA 0.66 (0.39–1.13) p for trend 0.04	Norrish et al., [59]
USA, 1995–1997	Case–control	–	67 cases and 156 population controls	erythrocytes	2.1 (0.9–4.8) p for trend 0.05	**2.6** (1.1–5.8) p for trend 0.01	1.3 (0.6–3.0)	1.0 (0.4–2.3)	Age-adjusted ORs; low mean ALA level: 0.20%	Newcomer et al., [43]
ATBC study, Finland, 1985–1994	Nested case–control	5–8	198 cases and 198 controls	serum CE	0.77 (0.43–1.39)	0.97 (0.54–1.75)	1.12 (0.61–2.04)	0.71 (0.40–1.26)	ORs unadjusted; vitamin E-supplemented group: 18:2n–6 **0.17** (0.04–0.68) p for trend 0.02; mean ALA level: 0.70%	Männistö et al., [38]

Table 5.3 (continued)

Country, period	Study type	Follow-up, year	No. of cases and controls	Blood or tissue fraction	18:2 ω–6 LA	18:3 ω–3 ALA	20:5 ω–3 EPA	22:6 ω–3 DHA	Comments	Reference
Kuopio Ischaemic Heart Disease Risk Factor Study, Finland, 1984–1999	Cohort, 2002 men	4–16 mean 12.6	46 cases	serum	**0.37** (0.16–0.86) p for trend 0.02	No association	No association	No association	ORs are adjusted for many dietary items	Laaksonen et al., [39]
USA, men with prostate cancer treated by radical prostatec-tomy, 1996–2000	Cases sample	–	52 locally advanced and 144 organ-confined cancers	Non-tumour prostate tissue	–	No association	–	0.72 (0.52–1.01) (EPA + DHA) 0.37 (0.09–1.56) (EPA + DHA)	Continuous OR for advanced cancer vs. organ-confined cancer	Freeman et al., [46]
		2–6	14 biochemi-cal recurrences after 2–6 year		–	1.17 (0.72–1.90)	–		continuous OR for biochemical recurrence; mean ALA level: about 1.3%	Freeman et al., [92]

Table 5.3 (continued)

Country, period	Study type	Follow-up, year	No. of cases and controls	Blood or tissue fraction	18:2 ω–6 LA	18:3 ω–3 ALA	20:5 ω–3 EPA	22:6 ω–3 DHA	Comments	Reference
Physician's Health Study, USA, 1982–1995	Nested case–control	13	476 cases 476 controls	Total blood	**0.62** (0.41–0.95) p for trend 0.03	1.31 (0.89–1.95)	**0.57** (0.36–0.92) p for trend 0.02	**0.60** (0.39–0.93) p for trend 0.07	mean ALA level: 0.35%; Association between ALA and localized cases no longer significant after adjustment for trans-fatty acids; associations with ALA, EPA, or DHA were not different in low- and high-grade tumours	Chavarro et al., [53]
			289 localized cases		**0.55** (0.32–0.94) p for trend 0.04	**1.66** (1.02–2.71) p for trend 0.05	**0.46** (0.24–0.86) p for trend 0.02	**0.53** (0.30–0.94) p for trend 0.02		
			108 advanced cases		0.67 (0.28–1.58)	1.04 (0.45–2.38)	1.27 (0.49–3.29)	0.98 (039–2.50)		
The Multiethnic Cohort Study, USA	Nested case–control	1.9	376 cases and 729 controls	erythrocytes	0.91 (0.62–1.33)	0.94 (0.50–1.75)	1.11 (0.73–1.67)	1.11 (0.73–1.69)	In white men (50 cases), positive association of cancer risk with ALA, DPA, and DHA, significant only for DHA; mean ALA level: 0.57%	Park et al., [55]
			102 advanced or high-grade cases		1.02 (0.52–1.99)	0.60 (0.17–2.14)	1.61 (0.79–3.25)	1.05 (0.51–2.16)		

Table 5.3 (continued)

Country, period	Study type	Follow-up, year	No. of cases and controls	Blood or tissue fraction	18:2 ω–6 LA	18:3 ω–3 ALA	20:5 ω–3 EPA	22:6 ω–3 DHA	Comments	Reference
EPIC cohort study, Europe	Nested case–control	4.2	962 cases and 1061 controls	Plasma PL	0.88 (0.64–1.19)	1.06 (0.75–1.50)	1.31 (0.96–1.81) p for trend 0.09	**1.39** (1.02–1.90)	DPA is not associated with prostate cancer: 0.95 (0.65–1.39); no difference between localized and advanced cases; mean ALA level: 0.24%	Crowe et al., [56]
			286 high-grade cases		0.83 (0.46–1.50)	1.79 (0.91–3.53) p for trend 0.02	**2.00** (1.07–3.76) p for trend 0.03	1.41 (0.76–2.62)		

Values are odds ratios (OR) for the highest tertile, quartile, or quintile relative to the lowest one, with their 95% confidence interval, as well as the probability of the linear trend test if <0.10. ORs in bold are significant for $p = 0.05$. ORs are adjusted for age and in many studies for other variables such as family history of prostate cancer, BMI, education, smoking. Tumour stage: localized = confined to the prostate, advanced = regional or metastatic; tumour score: low grade = Gleason score < 7 (non-aggressive), high grade = Gleason score ≥7 (aggressive).

Abbreviations: ALA: α-linolenic acid; ATBC: Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study; CE: cholesteryl esters; DHA: docosahexaenoic acid; DPA: docosapentaenoic acid; EPA: eicosapentaenoic acid

EPIC: European Prospective Investigation Into Cancer and Nutrition; LA: linoleic acid; PC: phosphatidylcholine; PL: phospholipids.

Table 5.4 Intake of ω-3 polyunsaturated fatty acids or of fish and seafood and prostate cancer: case–control studies

Country, period	No. of cases and controls	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
Japan, before 1981	100 cases 100 population controls	FFQ, 6 items	n.d.	**0.43** $p<0.05$	Fish/seafood intake: regularly vs. rarely or never	Mishina et al. [74]
Italy, 1986–1990	271 cases 685 controls	FFQ, 14 items	n.d.	0.79 (0.53–1.17)	Fish intake: twice a week or more vs. less than once a week	Talamini et al. [67]
United Kingdom 1989–1992	328 cases 328 population controls	FFQ, 83 items	n.d.	0.72 (0.44–1.18) (fatty fish)	No association with energy or total fat intake	Key et al. [72]
Italy, 1983–1996	127 cases 3293 hospital controls	FFQ	n.d.	0.7 (0.4–1.1)	Fish intake: twice a week or more vs. less than once a week	Fernandez et al. [73]
New Zealand, 1996–1997	285 cases 427 population controls	FFQ, 107 items	0.96 (0.63–1.48) (EPA)	n.d.	DHA: 1.10 (0.71–1.70). EPA and DHA in erythrocytes were associated with a decrease in prostate cancer risk (see Table 5.3)	Norrish et al. [59]
Uruguay, 1994–1997	175 cases 233 hospital controls	FFQ, 64 items	n.d.	0.9 (0.5–1.8)	Fish intake: >52 g/day vs. no intake	Deneo-Pellegrini et al. [68]
USA, 1993–1996	443 localized cases 162 advanced cases 592 population controls	FFQ, 99 items	1.05 (0.68–1.63) 0.84 (0.44–1.58) (EPA+DHA)	n.d.	EPA+DHA intake: > 0.24 g/day vs. < 0.03 g/day; ORs adjusted for PSA testing	Kristal et al. [69]
Australia, 1994–1997	858 cases 905 population controls	FFQ, 121 items	EPA: 0.8 (0.6–1.1)	n.d.	DHA: 1.0 (0.7–1.4) DHA intake: > 0.24 g/day vs. < 0.10 g/day	Hodge et al. [32]

Table 5.4 (continued)

Country, period	No. of cases and controls	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
Japan, 1996–2002	140 cases 140 hospital controls	FFQ, 102 items	n.d.	0.45 (0.20–1.02) p for trend 0.04	Fish intake: > 130 g/day vs. < 47 g/day	Sonoda et al. [75]
Taiwan, 1996–1998	237 cases 481 hospital controls	FFQ	n.d.	1.12 (0.80–1.56)	Fish intake: regularly vs. rarely or never; in men with BMI > 25: fish **2.45** (1.14–5.24)	Chen et al. [70]
Sweden, 2001–2002	1499 cases 1130 hospital controls	FFQ, 261 items	**0.70** (0.51–0.97) p for trend 0.05 (EPA+DHA)	**0.36** (0.18–0.72) p for trend <0.01 (fatty fish)	Lean fish **1.45** (1.12–1.88) p for trend <0.01; shellfish **1.81** (1.28–2.56) p for trend <0.01; fatty fish intake: ≥ 5 times/week vs. never; interaction of salmon-type fish intake with a COX-2 polymorphism (see text)	Hedelin et al. [31]
Canada, NECSS study, 1994–1997	1534 cases 1607 population controls	FFQ, 60 items	1.07 (0.87–1.31) (fish fat)	1.10 (0.84–1.42)	Preserved fish 0.79 (0.61–1.02); fish intake (fresh and canned fish): twice a week or more vs. never	Mina et al. [71]
Canada, 2003–2006	386 cases 268 controls	FFQ, 13 items	n.d.	**0.54** (0.32–0.89) p for trend < 0.02	Cases and controls were assessed in men who underwent prostate biopsy; no association of fish with high-grade cancer (188 cases); fish intake: > 4 times/week vs. once a week or less	Amin et al. [76]

Table 5.4 (continued)

Country, period	No. of cases and controls	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
USA, 2001–2004	466 aggressive cases 478 hospital controls	FFQ	**0.35** (0.24–0.52) p for trend <0.0001 (EPA)	**0.43** (0.29–0.63) p for trend <0.0001 (dark fish)	Cases: Gleason score ≥7 (aggressive), or stage ≥ T2c (advanced), or PSA > 10 ng/ml; DHA **0.36** (0.25–0.53) p for trend < 0.0001; white fish **0.66** (0.45–0.96); shellfish **0.51** (0.35–0.74) p for trend <0.0001; further adjustment for PSA testing did not change the results; fish intake: once a week or more vs. never; interaction of ω-3 LC-PUFA intake with a COX-2 polymorphism (see text)	Fradet et al. [34]

Values are odds ratios (OR) or hazard ratios (HR) for the highest tertile, quartile, or quintile relative to the lowest one, with their 95% confidence interval, as well as the probability of the trend test if < 0.10. ORs or HRs in bold are significant for $p = 0.05$. All ORs are at least adjusted for age, and in many studies for other variables such as family history of prostate cancer, BMI, smoking, ethnicity, socio-economic status, education, residence, dietary variables. Tumour stage: localized = confined to the prostate; advanced = regional or metastatic; tumour score: low grade = Gleason score < 7; high grade = Gleason score ≥7. Abbreviations: DHA: docosahexaenoic acid; DPA: docosapentaenoic acid; EPA: eicosapentaenoic acid; NECSS: National Enhanced Cancer Surveillance System; PUFA: polyunsaturated fatty acids

Table 5.5 Intake of ω-3 polyunsaturated fatty acids or of fish and seafood and prostate cancer: cohort studies

Country, year	Cohort size	No. of cases	Follow-up, year	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
Hawaii, USA, 1965–1986	7999 men	174 cases	17.5	FFQ, 20 items	n.d.	1.22 (0.74–2.01)	population of Japanese ancestry; fish intake: ≥5 times/week vs. ≤once a week	Severson et al. [77]
Seventh-Day Adventists, USA, 1974–1982	14,000 men	180 cases	6	FFQ	n.d.	1.57 (0.88–2.78)	Mostly vegetarian population; fish intake: ≥once a week vs. never	Mills et al. [83]
Health Professionals Follow-Up Study, USA, 1986–1990	47,855 men	126 advanced or fatal cases	3.5	FFQ, 131 items	0.90 (0.51–1.61) (ω-3 PUFA from fish)	n.d.	ω-3 PUFA from fish: 0.55 g/day vs. 0.05 g/day; no data on all-case cancer (279 cases)	Giovannucci et al. [21]
Health Survey in Hawaii, USA, 1975–1989	20,316 men	198 cases	9–14	FFQ, 13 items	n.d.	1.2 (0.8–1.9)	Fish intake: >37 g/day (fourth quartile) vs. <13 g/day (first quartile)	LeMarchand et al. [78]
Norway, 1977–1992	25,708 men	72 cases	12.4	FFQ	n.d.	No association	No association with the intake of fish or of cod liver oil supplements	Veierod et al. [79]

Table 5.5 (continued)

Country, year	Cohort size	No. of cases	Follow-up, year	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
Netherlands Cohort Study, 1986–1993	58,279 men	642 cases	6.3	FFQ, 150 items	1.03 (0.75–1.40) (DHA)	1.03 (0.80–1.34)	EPA 1.00 (0.73–1.35); no association of fish, EPA, or DHA with tumour subgroups (latent/nonlatent, localized/advanced); fish intake: 32 g/day vs. none; DHA intake: 0.18 vs. 0.03 g/day; EPA intake 0.10 vs. 0.05 g/day	Schuurman et al a,b [37, 80]
Swedish Twin Registry, 1967–1997	6272 men	466 cases	21.4	FFQ	n.d.	**0.43** (0.30–0.70) p for trend 0.05	Fish intake: very often (a large part of the diet) vs. rarely or never. HR adjusted for meat, fruit and vegetables, and milk intakes	Terry et al. [88]
		340 fatal cases				**0.27** (0.18–0.51) p for trend 0.01		
Health Professionals Follow-Up Study, USA, 1986–1998	47,883 men	2482 cases	12	FFQ, 131 items	n.d.	0.93 (0.80–1.08)	No association with intake of seafood; fish intake: >3 times/week vs. <twice/month; further adjustment for meat, vitamin D, and retinol intakes did not change the association with fish	Augustsson et al. [86]
		617 advanced cases			n.d.	0.83 (0.61–1.13)		
		278 metastatic cases			**0.76** (0.58–0.98) (continuous for 0.5 g/day)	**0.56** (0.37–0.86)		

Table 5.5 (continued)

Country, year	Cohort size	No. of cases	Follow-up, year	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
ATBC Study, Finland, 1985–1994	23,133 men	198 cases and 198 controls, from the cohort	5–8	FFQ, 276 items	1.31 (0.74–2.32) (DHA)	n.d.	EPA 1.22 (0.68–2.20); DHA intake 0.43 vs. 0.11 g/day; EPA intake 0.20 vs. 0.05 g/day	Männistö et al. [38]
Health Professionals Follow-Up Study, USA, 1986–1998	47,866 men	2965 cases	14	FFQ, 131 items	0.89 (0.77–1.04) p for trend 0.04	n.d.	EPA+DHA intake: >0.21% of TEI vs. <0.06% of TEI; HRs adjusted for many dietary variables	Leitzmann et al. [35]
		448 advanced cases			0.74 (0.49–1.08) p for trend 0.08 (EPA+DHA)	n.d.		
Life Time Study Cohort, Japan, 1963–1996	18,115 men	196 cases	16.9	FFQ	n.d.	**1.54** (1.03–2.31) p for trend 0.03	Men living in Hiroshima or Nagasaki, most of them survivors from atomic bombing; Fish intake: daily vs. <2 times/week	Allen et al. [84]
Health Professionals Follow-Up Study, USA, 1986–1998	51,529 men	392 progression outcomes from 1202 cases	10	FFQ, 131 items	n.d.	0.73 (0.52–1.02)	Men diagnosed with local/regional cancer; continuous HR for fish intake, 1 serving/day, adjusted for pre- and post-diagnostic diet: **0.52** p = 0. 006	Chan et al. [87]

Table 5.5 (continued)

Country, year	Cohort size	No. of cases	Follow-up, year	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
CLUE II Study, USA 1989–2004	3892 men	199 cases	15	FFQ	n.d.	0.86 (0.44–1.67)	Fish intake: ≥5 times/week vs. ≤1 time/week	Rohrmann et al. [81]
		54 advanced cases				0.92 (0.27–3.21)		
Multiethnic Cohort Study, USA (Hawaii and Los Angeles) 1993–2002	82,483 men from 4 ethnic groups	4404 cases	8	FFQ, 180 items	0.99 (0.89–1.09)	1.04 (0.93–1.15)	EPA: no association. Fish intake: 17.8 vs. 1.2 g/1000 kcal; DHA intake 0.07 vs 0.01 g/1000 kcal; EPA intake: 0.037 vs. 0.004 g/ 1000 kcal	Park et al. [42]
		1278 advanced or high-grade cases			1.07 (0.88–1.30) (DHA)	1.01 (0.82–1.23)		
Malmö Diet and Cancer Cohort, Sweden, 1991–2005	10,564 men	817 cases	11	FFQ, 168 items	**1.28** (1.01–1.62) p for trend 0.06	n.d.	EPA+DHA intake: ≥0.94 vs. <0.22 g/day; HRs adjusted for energy calcium, vegetable, fruit, and meat intakes	Wallström et al. [40]
		281 advanced cases			0.86 (0.58–1.28) (EPA+DHA)	n.d.		
EPIC study, Europe, 1995–2004	142,520 men	2727 cases	8.7	FFQ	1.00 (0.93–1.07)	1.05 (0.91–1.20)	White fish 1.03 (0.90–1.18) Fatty fish 1.07 (0.95–1.21) No association of fish and shellfish lipid intake with the risk of low-grade, high-grade, advanced, or localized cancer	Allen et al. [82]; Crowe et al. [57]
		533 advanced cases			1.05 (0.91–1.21) (continuous for 1% of TEI)			
Japan, 1994–2001	24,895 men	95 cases	7	FFQ, 40 items	n.d.	0.72 (0.40–1.33)	Men older than 70:0.44 (0.18–1.11), p for trend 0.08	Sato et al. [89]

Table 5.5 (continued)

Country, year	Cohort size	No. of cases	Follow-up, year	Dietary assessment	Long-chain ω–3 PUFA	Fish and seafood	Comments	Reference
Miyako Study, Japan, 1986–2003	5589 men	21 fatal cases	13.4	FFQ	n.d.	**0.12** (0.05–0.32)	Fish intake: ≥2 times/week vs. less often. HR adjusted for vegetable, fruit, and meat intakes	Pham et al. [90]
Physician's Health Study, USA, 1983–2006	20,167 men	2161 cases 230 fatal cases	22	FFQ	1.09 (0.95–1.25) **0.65** (**0.42**–0.99) p for trend 0.02 (seafood ω-3 PUFA)	1.11 (0.95–1.30) **0.52** (0.30–0.91) p for trend 0.05	Canned tuna, dark meat fish: no association; other fish **1.33** (1.13–1.58) p for trend 0.01 Canned tuna **0.57** (0.33–0.99); dark meat fish 0.64 (0.39–1.04) p for trend 0.04; other fish **0.58** (0.35–0.97); HRs adjusted for tumour stage or grade at diagnosis, and dietary variables. Fish intake: ≥5 times/week vs. <once a week	Chavarro et al. [91]

Values are odds ratios (OR) or hazard ratios (HR) for the highest tertile, quartile, or quintile relative to the lowest one, with their 95% confidence interval, as well as the probability of the trend test if <0.10. ORs or HRs in bold are significant for $p = 0.05$. All ORs are at least adjusted for age, and in many studies for other variables such as family history of prostate cancer, BMI, smoking, ethnicity, socio-economic status, education, and dietary variables.

Tumour stage: localized = confined to the prostate; advanced = regional or metastatic; tumour score: low grade = Gleason score < 7; high grade = Gleason score ≥7.

Abbreviations: ATBC: α-Tocopherol, Beta-Carotene Cancer Prevention Study; DHA: docosahexaenoic acid; DPA: docosapentaenoic acid; EPA: eicosapentaenoic acid; EPIC: European Prospective Investigation Into Cancer and Nutrition; PUFA: polyunsaturated fatty acids study based on a dietary questionnaire [21] and a case–control study nested in another US cohort, based on blood biomarkers [22], an increased risk of advanced prostate cancer has been found to be associated with higher intakes or blood levels of ALA. This increase in risk has also been found in some subsequent studies performed in different countries during the following years (see below), confirming the somewhat alarming first results. More recently, however, many other studies, among which large cohort studies, have brought results which have very substantially modified the picture.

5.2.1 Ecological Studies

Very few ecological studies have examined the possibility of a relation between the intake of ALA and prostate cancer. On the basis of national US data from 1930 to 1992, a strong correlation was found each year between prostate cancer mortality and the apparent consumption of vegetable oil, margarine, and vegetable shortening, which could suggest that linoleic acid or ALA, of which these foods are main dietary sources, might be risk factors for prostate cancer [23]. However, in a previous study based on cancer rates and adipose tissue fatty acid composition from 11 centres in 8 countries in Europe and Israel, no significant correlations were observed between prostate cancer incidence and the concentrations of total ω-6 or ω-3 fatty acids (mainly linoleic acid and ALA, respectively) in adipose tissue [24]. Similarly, no correlation was found between serum concentrations of ALA determined in samples of Japanese men from six populations (five regions in Japan and São Paulo, Brazil) and prostate cancer mortality in these populations [25].

5.2.2 Case–Control Studies Based on Dietary Questionnaires

Nine case–control studies based on food questionnaires have been published and gave heterogeneous results (Table 5.1). Three of them did not find any association of ALA with prostate cancer risk [26–28]. Among them, one study compared mostly localized cases with controls which had been screened and found free of subclinical cancer [27], and another one compared advanced and localized prostate cancer cases [28]. The third study was a large study with population controls and a large proportion of advanced cases: no association with ALA was found, either with all-stage or with advanced prostate cancer [26]. At variance, a strongly increased risk of cancer was found with higher intakes of ALA in two other studies in Uruguay [29] and Spain [30]. In these studies, the ORs were, as in other studies, adjusted at least for age and energy intake, and they were not attenuated by further adjustment for other fatty acids or animal fats. In the Uruguayan study, ALA from both vegetable and animal sources was associated with an increased cancer risk [29]. A recent large case–control study in Sweden has also found an increased risk with higher ALA intake [31]. However, the association was only marginally significant

(OR 1.35, 95% CI 0.99–1.84, *p* for trend 0.03, for the highest quartile of intake vs. the lowest), after adjustment for many dietary variables, and it was not significant when only adjusted for age and energy intake (OR 1.06, 95% CI 0.85–1.32). A similar positive association was also found with linoleic acid intake in this study, appearing only when fully adjusted. In contrast, two other large case–control studies in Australia and Italy found a significantly *decreased* risk of all-stage prostate cancer with the higher intakes of ALA [32, 33]. A recent study based only on advanced or aggressive cases did not found any significant association of prostate cancer with ALA intake, but a non-significant tendency for a decreased risk [34]. Considering the studies where all-stage cases were included (all except the Canadian studies [27, 28] and the recent US study [34]), the heterogeneity of the results cannot be explained by differences between studies in the repartition of cancer stages, since these studies have been performed in countries where PSA screening was not of widespread use in the period considered (1989–2002). Finally, the range of ALA intake was similar in studies which have found an increased risk of prostate cancer and in those which have not, and thus cannot explain the heterogeneity of the results either (Table 5.1).

5.2.3 Cohort Studies Based on Dietary Questionnaires

As mentioned above, a strongly increased risk of advanced prostate cancer associated with higher ALA intakes was observed in a first survey of the US Health Professionals Follow-Up Study (HPFS) [21], which persisted after adjustment for meat fat intake. This increase in risk was observed from the second quintile of ALA intake, i.e. 0.93 g/day on average. This increased risk was confirmed in two subsequent studies from the same cohort, involving more cases [35, 36], although the increase in risk tended to be less marked with time, since the OR of 3.43 in 1993 was attenuated to 1.98 in 2004 and 1.57 in 2007. The association was not found between ALA intake and localized prostate cancer. Again, the association with advanced cancer persisted after adjustment for meat, saturated, monounsaturated, polyunsaturated, or trans-fat intake [35] and other dietary factors [36], and it was about the same for ALA from vegetable sources as for ALA from animal sources [35]. In addition, the association was the same for advanced cases detected in the period where PSA screening was not yet widely used (1986–1992) as in the PSA era (1992–2002) [36]. However, of the six other cohort or nested case–control studies based on a dietary questionnaire, performed either in North European countries [37–40] or in the USA [41, 42], none found any association of ALA intake with an increase in the risk of prostate cancer, whatever localized or advanced, low grade or high grade. In contrast, ALA intake was associated in two of these cohort studies with a *decrease* in the risk of all-case prostate cancer: non-significantly in the Dutch study [37], significantly in ethnical subgroups (whites and latinos) of the large Multiethnic Cohort Study in Hawaii and Los Angeles, USA [42]. In a study in the screening arm of the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer

Screening Trial (USA), all men have been screened annually by PSA test and digital rectal examination, and there was thus no possible confounding between dietary habits and screening propensity [41]: no association was found between all-case, organ-confined, advanced, or aggressive prostate cancer risk and ALA intake, either total ALA or ALA from animal or vegetable sources [41]. Again, the range of ALA intake was similar in the HPFS study, in which an increased risk was found, and in the other studies (Table 5.2).

5.2.4 Case–Control or Cross-Sectional Studies Based on Biomarkers

Two case–control studies based on biomarkers show an increased prostate cancer risk with the highest quartile of ALA in erythrocytes [43] or in erythrocytes and adipose tissue [44] relatively to the first one (Table 5.3): in one of them, the association is not significant, mainly because of the small number of controls [44]; in both of them, an increased risk was also observed for linoleic acid, suggesting a non-specific association with PUFAs [43, 44] (Table 5.3).

Two studies were performed in men undergoing prostatectomy for localized carcinoma. One study on 49 men found a lower level of ALA in men with locally advanced tumours than in men with organ-confined tumours [45], whereas the other one, of larger size (196 men), found no association of ALA level in prostate tissue with tumour extension [46]. Several small studies (35 men or less by group) have compared the ALA levels in blood, prostate, or adipose tissue of men with prostate cancer or with benign prostate hyperplasia (BPH) [47–50], or have looked at the relationship between the levels of ALA and PSA [49, 51], but their results were inconsistent. Here again, the results are heterogeneous and no conclusion can be drawn.

5.2.5 Case–Control Studies Nested Within Cohorts, Based on Biomarkers

Among the prospective studies based on biomarkers, one performed in Norway from a population of blood donors showed an increased risk of prostate cancer (all-case) with the highest serum levels of ALA [52], but there was no adjustment for other fatty acids or dietary variables. Two nested, biomarker-based case–control studies have been performed within the Physician's Health Study cohort, after 6 and 13 years of follow-up, respectively [22, 53]. In the first one, in which cancer stage or grade was not taken into account, plasma levels of ALA in the second, third, and fourth quartiles were associated with increased risk of prostate cancer relative to the first quartile, and this increase in risk persisted after adjustment for several factors, like blood linoleic acid and meat intake [22]. The OR for ALA seemed to depend on linoleic acid status: it was much greater in men in the lowest quartile of linoleic acid (8.56, 95% CI 2.05–35.8) than in men in the other quartiles (1.87, 95% CI 0.89–3.91). Plasma linoleic acid was not associated with prostate cancer [22]. In a second study nested in this cohort, no significant association between total blood ALA and all-case prostate cancer was found, whereas a significantly decreased risk was observed for linoleic acid [53]. No association was found between ALA level and the risk of advanced prostate cancer. However, an increased risk of *localized* prostate cancer was observed for the highest ALA levels (1.66, 95% CI 1.02–2.71 in the highest quintile), but it was attenuated and no longer significant after adjustment for trans-fatty acids levels [53]. In this nested case–control study, the risk of localized, low-grade prostate cancer was positively associated with blood levels of *trans* isomers of 18:1 and 18:2 [54]. In a case–control study nested within the Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study (ATBC Study), an interventional cohort of smokers in Finland, no association was found between all-case prostate cancer risk and ALA or linoleic acid levels in plasma cholesteryl esters (CE) [38], a result identical to that based on dietary questionnaires in the same study (see above). It must be noticed that the biomarker used in this study, plasma CE fatty acids, is considered a valuable marker of ALA intake. A nested case–control study based on baseline erythrocyte membrane fatty acids was performed within the Multiethnic Cohort Study in Hawaii [55]. No association of ALA level (and of linoleic acid level) with either all-case or advanced or high-grade prostate cancer risk was observed after a mean of 2 years of follow-up [55], a result consistent with that observed in a study from the same cohort, based on dietary questionnaires [42]. A large nested case–control study (962 cases and 1061 controls) was performed within the European multi-cohort EPIC, based on fatty acids from plasma phospholipids (PL) and incident cases during 4 years of follow-up. No association of ALA level with all-case prostate cancer risk was observed, as well as with localized or advanced prostate cancer risk [56]. A marginally significantly increased risk of high-grade cancer was found to be associated with the highest ALA levels (significant trend) in plasma PL. In this study, as in most other biomarker-based studies, odds ratios were adjusted neither for other fatty acids nor for dietary intake variables [56]. In another study from the EPIC multi-cohort, however, no association was found between prostate cancer risk and total fat intake, as well as intakes of saturated, monounsaturated, or polyunsaturated fatty acids [57]. High intakes of fat and of main fatty acid classes (saturated, monounsaturated, polyunsaturated) were, in contrast, associated with a decrease in high-grade prostate cancer risk [57] (Table 5.3).

5.2.6 Discussion

The results of the studies on ALA and prostate cancer present obvious heterogeneity, which can be due to several reasons. The first one may be a classification bias due to a lack of precision in the measure of ALA intake. Epidemiological studies looking for associations between food or nutrients and diseases are subject to classification biases due to the lack of precision in the measure of food/nutrient intake. In many countries, such as France or other European countries, ALA is provided by a large number of different foods, among which many of industrial products (margarines, dressings, biscuits, etc.) with varying ALA contents, but neither the food questionnaires nor the food composition tables are sufficiently detailed to allow a precise estimate of individual ALA intake [11, 58]. Unfortunately, the use of blood or tissue biomarkers does not overcome this problem: contrary to linoleic acid, ALA is present in blood or adipose tissue only at low levels, and its levels in blood or blood fractions are less correlated to its intake levels than those of linoleic acid or of long-chain ω-3 PUFAs [14]. This is particularly true for the low ALA levels found in plasma or erythrocyte phospholipids, which cannot be considered as good biomarkers of ALA intake.

Nutritional epidemiology is often challenged by confounding biases, due to the fact that many nutrients vary together in the diet and that it is thus often difficult to distinguish the effects of each one from those of the others. Studies dealing with ALA are particularly exposed to these biases. Still more than other fatty acids and especially more than linoleic acid, ALA intake is strongly correlated with total energy and total lipid intake [11, 58], and with intake of foods with a high fat content. Studies in which no adjustment have been made for these variables are especially subject to confounding biases: this is the case for most studies based on biomarkers, since only a few of them have made adjustment for dietary variables [22, 39, 59].

The possibility of a selection bias due to confounding of PSA screening with a health-conscious food behaviour is to be considered in countries and periods where PSA screening was of significant use, i.e. in US and northwestern Europe in the 1990s and later. Such a bias can be suspected when an increased risk with ALA was found with localized prostate cancer, but not for advanced cancer. This could be the case for a recent case–control study nested within the Physician's Health Study cohort [53], since the follow-up period (1982–1995) partly reaches the period of use of PSA screening in the USA (from the late 1980s).

Finally, another reason of heterogeneity of the results is the possibility of a publication bias, the results from smaller studies being published more often when they are significant than when they are not. The possibility of such a bias was explored in studies dealing with all-case prostate cancer by plotting the logarithm of the odds ratio vs. the sample size: the plot was asymmetrical, with a lack of small studies with an odds ratio near 1 (non-significant) or smaller than 1, indicating a likely publication bias. In fact, higher OR values were found mainly in smaller case–control or nested case–control studies [22, 29, 30, 43, 44, 52]. No such bias was suggested in the less numerous studies dealing with advanced cases, but only one of them, the HPFS cohort, found an increased risk of advanced cancer with higher ALA intake [21, 35, 36], the other studies found no association [26, 28, 34, 40–42, 53, 55, 56].

Thirteen studies can be considered as using the most solid methodology, i.e. large case–control studies (>450 cases) [26, 31–34] and cohort studies [35, 37, 40–42] based on a dietary questionnaire; large nested case–control studies, based on blood biomarkers of ALA intake (198 cases or more) [38, 53, 56]. In these 13 studies, the

US cohort studies (HPFS), Physician's Health Study, and Multiethnic Cohort Study have been surveyed two or three times, but are each counted only once. Four of these studies have found an increased risk or prostate cancer (all stage, or any stage or grade) [31, 35, 53, 56], five did not find any association [26, 34, 38, 40, 41], and four have found a decreased risk [32, 33, 37, 42]. Among these 13 studies, 9 [26, 34, 35, 37, 40–42, 53, 56] have looked for the risk of advanced or high-grade prostate cancer. Two of them found an increased risk of advanced [35] or high-grade [56] prostate cancer with the higher intakes of levels of ALA, the others did not find any association. A recent meta-analysis of 16 studies found an increased pooled OR for ALA and all-case prostate cancer, just reaching significance (1.20, 95% CI 1.01–1.43, for the highest vs. the lowest quantile of ALA intake or blood level), with significant heterogeneity and evidence for a publication bias [60]. However, it did not include the four most recent studies [34, 40, 55, 56], all of which found no association. A meta-analysis including these studies would very likely give a non-significant pooled OR.

Overall, no definitive general conclusion can be drawn for the studies having tested the association between ALA intake and prostate cancer, since there is still unexplained heterogeneity between results. However, most of the recent studies (2003 and later) did not confirm the positive association found in earlier studies, especially earlier US cohort studies [21, 22], so that the realty of this increased risk of advanced prostate cancer associated with higher ALA intake appears now much less likely than previously. More large well-designed prospective studies, controlling the main possible sources of bias (in particular, confounding with dietary variables and PSA screening) will be necessary to establish firm conclusions.

5.3 Long-Chain ω-3 PUFAs and Prostate Cancer

As fish and seafood are by far the main source of long-chain ω-3 PUFAs, especially fatty fish [61], their intakes in population are highly correlated [11]. For this reason, fish and seafood intakes, and especially fatty fish intake, are good proxies for long-chain ω-3 PUFA intake, and we have therefore included in this review the studies having searched the association of fish and/or seafood intake and prostate cancer. Only a few of them have distinguished between fish/seafood categories.

5.3.1 Ecological Studies

Two international comparison studies of dietary factors of prostate cancer mortality have been performed, based on IARC/WHO mortality data and FAO statistics on apparent food consumption. Fish consumption by 1980 was negatively related with prostate cancer mortality in 1985–1989 in 59 countries over the world [62]; however, this relation was no longer found in a later study involving 71 countries, based on fish consumption in 1990–1992 and prostate cancer mortality in

2000 [63]. This discrepancy is perhaps due to the fact that prostate cancer mortality decreased in the 1990s in many Western countries, but not in other countries, due to unequal progresses in screening and treatment [64]. Serum concentrations of ω-3 PUFAs (mainly EPA and DHA) determined in samples of Japanese men from six populations (five regions in Japan and São Paulo, Brazil) were found to be significantly negatively correlated to prostate cancer mortality in these populations [25]. Greenland Inuits, whose diet is rich in long-chain ω-3 PUFAs from fish and sea mammals, have a very low rate of prostate cancer [65]. An autopsy study on 61 Inuit men found only one case of prostate cancer, and this man had a lower ω-3 PUFA concentration in adipose tissue than non-case men [66].

5.3.2 Case–Control Studies Based on Dietary Questionnaires

Of the 14 case–control studies realized to date to seek an association between dietary intake of fish or long-chain ω-3 PUFAs and prostate cancer, 7 did not find any association [32, 59, 67–71]. In one of these studies where cancer stages were distinguished, no association between long-chain ω-3 PUFA intake and either localized or advanced cases was found [69]. In a large case–control study in Canada, a decreased risk of prostate cancer was associated with high intakes of preserved fish (smoked, dried, or salted), but not of fresh or canned fish [71]. Two studies [72, 73] found a non-significantly decreased risk with fish intake. In contrast, five case–control studies have found a strong and significant decrease in prostate cancer risk with increasing levels of fish intakes [31, 74–76]. Compared to the other-case control studies, four of these studies, which have been done in Japan, Sweden, and Canada, are characterized by a large range of fish consumption: the highest quartile or quintile was, for example, more than four times/week [76], more than 130 g/day [75], or more than four times/week of fatty fish [31]. In a recent study in Canada [76], all men were submitted to prostate cancer biopsy following high or raising PSA level or suspicious digital rectal examination; controls were those where no significant pathology was found. There was thus no possible detection bias due to PSA screening in this study. The large Swedish case–control study has examined the association of prostate cancer risk with the intake of different types of fish and seafood [31]: interestingly, the intake of fatty fish and of long-chain ω-3 PUFAs (EPA + DHA) was associated with a decrease in prostate cancer risk; a strong decrease was observed to be associated with higher intakes of fatty fish (>four times/week) [31]. In contrast, lean fish and shellfish intakes were found to be associated with a significantly *increased* risk of prostate cancer [31]. Since cancer stages were not distinguished in this study, no explanation can be proposed for this increase in risk. In a study based only on advanced or aggressive cases, a strong decrease in the risk was associated with higher intakes of long-chain ω-3 PUFAs, and this decreased risk was observed for all types of fish and shellfish [34]. In this US study, the mean intakes of fish or of long-chain ω-3 PUFAs were lower than that in the Swedish study, but the range of long-chain ω-3 PUFA intake was very large (67 mg in the first quartile, 588 mg in the fourth), and the decrease in risk was observed from the second quartile of intake (167 mg) (Table 5.4).

In brief, a few case–control studies suggest that a high fish intake, especially of fatty fish, or a high long-chain ω-3 PUFA intake, can be associated with a lower risk of prostate cancer, but the amount and the type of fish or seafood eaten are likely to be of importance for the existence of the association.

5.3.3 Cohort Studies Based on Dietary Questionnaires

The association between fish or long-chain ω-3 PUFA intake and prostate cancer risk has been examined in 15 cohorts, one of them, the US Health Professionals Follow-Up Study (HPFS), having been surveyed several times during the follow-up. Most of these studies did not find any significant association [37, 38, 42, 57, 77–82], including for advanced or high-grade cases [37, 42, 57, 80–82]. Among these studies, some are large cohorts in the USA or in Europe [37, 42, 57, 80, 82]. Three studies have observed an increased risk of prostate cancer associated with higher intakes of fish or long-chain ω-3 PUFAs [40, 83, 84]. Two of these studies have been performed among special populations, likely having risk factors somewhat different from those of the general population in the same countries: a mostly vegetarian population (US Seventh-Day Adventists), with a low mean fish intake [83], and a population of men living in Hiroshima and Nagasaki (Japan), most of them survivors of the 1945 nuclear bombing [84]. In the Malmö Diet and Cancer Cohort (Sweden), higher long-chain ω-3 PUFA intakes were found to be associated with an increased risk of all-case prostate cancer (OR 1.28, 95% CI 1.01–1.62), but not of advanced prostate cancer [40]. However, the follow-up of this cohort was contemporary of increasing cancer incidence in this country, due to increasing PSA screening practice [85]. Although the widespread fish consumption in Sweden is less characteristic of a health-conscious behaviour than in other western countries, it is possible that the increased risk of prostate cancer associated with fish intake was due to confounding with PSA testing practice, which tends to increase the incidence of asymptomatic, localized cases (Table 5.5).

A decreased risk of prostate cancer associated with higher fish or long-chain ω-3 PUFA intakes has been found in only five cohort studies. Whereas the first survey of the US HPFS cohort did not find any association of long-chain ω-3 PUFA intake with advanced/fatal prostate cancer [21], later surveys, including 2500–3000 cases, allowed to show a marginally significant association of long-chain ω-3 PUFA intake with a lower rate of all-case or advanced prostate cancer [35], and especially a marked decrease in the risk of metastatic cancer with the higher intakes of fish and long-chain ω-3 PUFAs [86]. Moreover, in the same cohort, a study of prostate cancer cases outcome in men diagnosed with local or regional prostate cancer found a significant reduction of the risk of progression, recurrence, or death in men with a higher fish intake after diagnosis (continuous OR 0.52, $p = 0.006$, for one serving/day, adjusted for pre- and post-diagnostic diet) [87]. In a Swedish cohort with a long follow-up (up to 30 years), cases, most of them fatal, were diagnosed in 1967–1997, i.e. mainly before the PSA era. Fish intake was associated with a strong decrease in cancer risk and with a still stronger decrease in cancer death (by 73%, *p* for trend 0.01) [88]. As a high consumption of fish, especially of fatty fish, is traditional in Sweden, these results suggest a protective effect of long-chain ω-3 fatty acids on prostate cancer progression. In a cohort of health insurance beneficiaries in Japan, where prostate cancer incidence is much lower than that in western countries, aged men ($\geq$70 years) in the highest quartile of fish intake had a non-significantly decreased risk of prostate cancer (by 56%, *p* for trend 0.08), a trend which was not found for younger men [89]. Another Japanese study found 21 cases of fatal prostate cancer after 10–17 years of follow-up of the participants to a regional health survey. Men eating fish two times/week or more had a strongly decreased risk (by 88%, p<0.05) of prostate cancer death compared to those who ate fish less often [90]. In the US Physician's Health Study [91], no association was found between fish or long-chain ω-3 PUFA intake and all-case prostate cancer risk after 22 years of follow-up. Cancer risk was not associated with consumption of seafood categories, except with an undefined "other fish" category which was associated with a significant increase in risk. However, this association with "other fish" intake was stronger in cases detected by PSA screening, indicating that it was likely due to confounding with health-conscious behaviour [91]. In this cohort, in contrast, fish and long-chain ω-3 PUFA intake, as well as all fish categories, were associated with a significant decrease in *fatal* prostate cancer risk (by 48 and 35%, respectively).

5.3.4 Case–Control Studies Based on Biomarkers

These studies were based on the percentages of long-chain ω-3 PUFAs in fatty acids from total blood, serum or plasma, erythrocytes, serum or plasma phospholipids, phosphatidylcholine, or cholesteryl esters, all of them being good biomarkers of long-chain ω-3 PUFA intake (Table 5.3).

Of the three case–control studies having tested the association of blood levels of long-chain ω-3 PUFAs with prostate cancer, one found no association [43], one found a non-significantly decreased risk [44], and one a significantly decreased risk of both all-case and advanced prostate cancer [59]. In 196 men undergoing prostatectomy for localized carcinoma, there were lower levels of EPA and of DHA in prostatic tissue of men with locally advanced tumours than in men with organ-confined tumours [46]. Several small studies (35 men or less by group) have compared the levels of long-chain ω-3 PUFAs in blood, prostate, or adipose tissue of men with prostate cancer or with benign prostate hyperplasia (BPH), but gave heterogeneous results, some studies finding lower serum levels of EPA, DPA, or DHA in men with prostate cancer than in men with BPH or in normal controls [47, 48], but not the others [49, 50]. Several of these studies looked at the relationship between the levels of long-chain ω-3 PUFAs and PSA, but with no consistent findings [49–51].

5.3.5 Case–Control Studies Nested in Cohorts Based on Biomarkers

Eight nested case–control studies having looked for the association of long-chain ω-3 PUFAs and prostate cancer. Among them, four, performed within the Physician's Health Study cohort in the USA [22], a cohort in Norway [52], and two cohorts in Finland [38, 39], did not find any association. A small prospective study found a non-significant decreased risk of biochemical recurrence (return of PSA raising) with higher EPA + DHA levels after 2–6 years of follow-up of 196 men having undergone prostatectomy for localized prostate cancer [92], but it was based on only 14 recurrent cases. The three most recent studies nested within cohorts in the USA (Physician's Health Study, Multiethnic Cohort Study) and Europe (EPIC multi-cohort), of greater power than the previous studies, did not give consistent results. In the second study nested within the Physician's Health Study, the highest levels of EPA, DPA, and DHA in total blood were associated with a significantly decreased risk of all-case and localized prostate cancer, but not of advanced prostate cancer [53]. In contrast, no significant association was found in the Multiethnic Cohort Study between long-chain ω-3 PUFAs and all-case or advanced prostate cancer [55]. A non-significantly increased risk of all-case cancer was even observed for DPA and DHA in the subgroup of white men [55]. In the large case–control study nested in the European EPIC cohort, an increased risk of all-case cancer was observed in men with higher EPA and DHA levels (significant in the fourth quintile of EPA and in the fifth quintile of DHA, compared to the first one) [56]. There was no difference between localized and advanced cases, but higher EPA levels were associated with an increased risk of high-grade cancer [56] (Table 5.3).

In these three cohorts, the association between intake of long-chain ω-3 PUFAs and prostate cancer risk has been also tested (see above, Section 5.3.2). In the Multiethnic Cohort Study, a null result was found for both intakes and erythrocyte levels of long-chain ω-3 PUFAs [42, 55]. In the EPIC cohort, in contrast, an increased risk of all-case or high-grade prostate cancer with higher levels of long-chain ω-3 PUFAs was observed in the biomarker study [56], whereas the study based on intakes found no association [57]. In the Physician's Health Study, a protective association of fish or long-chain ω-3 PUFAs was found both in the study based on questionnaires and in the study based on biomarkers, but not in a consistent manner: blood EPA and DHA were associated with a decreased risk or all-case and localized cases, but not of advanced cases, after 8 years of follow-up [53], whereas fish and long-chain ω-3 PUFA intakes were associated with a decreased risk of fatal cases, but not of all-case prostate cancer, after a 22 years of follow-up [91].

5.3.6 Interaction of Long-Chain ω-3 PUFAs with COX-2 Polymorphisms

Cyclooxygenase-2 (COX-2), an inducible enzyme catalysing the first step of prostaglandin biosynthesis from arachidonic acid, plays a central role in the carcinogenic process of many organs, in particular in prostate carcinogenesis [93]. Increased COX-2 expression is observed in prostate tumour cells [93, 94] and has been associated with prostate cancer recurrence or progression [95, 96]. COX-2 blockade by treatment with anti-inflammatory drugs decreases prostate cancer risk [93, 94]. Moreover, genetic variations of the COX-2 gene have been found to alter the risk of prostate cancer [34, 97–100].

Long-chain ω-3 PUFAs are COX-2 inhibitors, and this property is one of those which determine their anti-carcinogenic potency. In two recent case–control studies [31, 34], the effects of some COX-2 polymorphisms on the association between fish or long-chain ω-3 PUFA intake and prostate cancer risk were investigated. In the Swedish study [31], a strong interaction of one COX-2 single-nucleotide polymorphism (SNP) (rs5275: +6365 T/C) with salmon-type fish intake was found: intake of salmon-type fish was associated with a strongly decreased risk of prostate cancer in carriers of the variant allele (TC or CC) (OR 0.28, 95% CI 0.18–0.45), but there was no association in non-carriers (OR 1.10, 95% CI 0.64–1.89) (p for interaction <0.01). In the US study [34], based only on advanced or aggressive cases, a similar interaction was found for another COX-2 SNP (rs4648310: +8897 A/G): the decrease in prostate cancer risk with long-chain PUFA intake was stronger in carriers of the variant allele (AG or GG) (OR 0.07, 95% CI 0.01–0.41) than in non-carriers (OR 0.61, 95% CI 0.47–0.81) (p for interaction 0.02). The finding that the association between long-chain ω-3 PUFA intake and prostate cancer risk depends of COX-2 genetic variants in human populations is a strong argument in favour of a genuine causal role of long-chain ω-3 PUFAs in the modulation of prostate cancer risk: they may act through COX-2 inhibition in populations, not only in experimental systems. It also brings another piece of explanation to the fact that this association is not regularly observed, since it might depend on some genetic characteristics of the studied population.

5.3.7 Discussion

In summary, most of the case–control or cohort studies based either on dietary questionnaires or on blood biomarkers did not find any significant association between prostate cancer and the intake of fish or long-chain ω-3 PUFAs, or the long-chain ω-3 PUFA blood levels. Only a few studies have found a decreased risk of prostate cancer (any stage or grade) with higher fish/long-chain ω-3 PUFA intake or blood levels: ten studies (among 30) based on dietary questionnaires, either case–control [31, 34, 74–76] or cohort studies [86–88, 90, 91], and two studies (among 11) based on blood biomarkers [53, 59]. The same is true if one considers only the studies dealing with advanced or high-grade prostate cancer: only three studies (among nine, two case–control and seven cohorts) have found a protective association [34, 35, 88]. Some confounding of fish eating with cancer detection through PSA testing may have masked an association between long-chain ω-3 fatty acids and localized cancer in studies performed in the PSA era, but this is much less likely for advanced cases.

On the other hand, the observation of an interaction of COX-2 genetic variation and long-chain PUFA intake in their association with prostate cancer risk is encouraging further research. On the whole, however, there is to date only a weak epidemiological evidence to suggest a possible preventive effect of long-chain ω-3 fatty acids on prostate cancer in populations.

However, more convincing results have been produced by the few studies having considered the latest stages of prostate cancer: metastatic cancer and death. The risk of metastatic cancer has been examined in only one study, a survey of the HPFS cohort in the USA: it was markedly decreased by higher intakes of fish or long-chain ω-3 PUFAs, whereas there was no significant association with all-case and advanced cancer [86]. Moreover, the three studies which have considered the risk of fatal prostate cancer have found a strongly decreased risk of prostate cancer death in men with higher intakes of fish or long-chain ω-3 PUFAs [88, 90, 91]. One can wonder whether a higher survival rate in fish-eating men might be in part due to confounding with a higher care-seeking propensity. However, the results have been obtained either in countries where fish eating is traditional and widespread (Japan and Sweden) [88, 90], thus less likely to be associated with a specially health-conscious behaviour, or in cohorts of US health professionals [86, 91], which are expected to have equal and high access and propensity to care. In addition, the unusual strength of the associations and the fact that they were found in different populations suggest a genuine effect of fish or long-chain ω-3 PUFA intake on prostate cancer progression to its late stages and to death, rather than a result of confounding. Two prospective studies on prostate cancer progression after diagnosis [87] or on biochemical recurrence after radical prostatectomy for prostate cancer [92], suggesting a protective effect of eating fish or long-chain ω-3 PUFAs, are consistent with this conclusion. Thus, there is still limited but significant evidence to suggest a protective effect of fish or long-chain ω-3 PUFA intake on prostate cancer progression and metastasis and on prostate cancer mortality.

5.4 Concluding Remarks

Epidemiological research on ω-3 PUFAs and prostate cancer has been driven, as for other cancers, by the anti-carcinogenic potency that these fatty acids had shown in experimental animal and cellular models. In addition, the unexpected finding in the 1990s that ALA intake was associated with an increase in prostate cancer incidence in several studies has raised concern about the possible harmful effects of this fatty acid, effects which had – and have to date – no mechanistic or experimental basis. Epidemiological research on prostate cancer has been complicated, from the late 1980s in the USA and later on in many other countries, by the introduction and the rapid spreading of PSA testing in clinical practice, which has considerably modified the nature of diagnosed cases, as well as of the factors associated with them: more preclinical localized cases in more health-conscious men. Prostate cancer evolves – and can be diagnosed – at very different phases from a focus of cancer cells in the prostate epithelium to bone metastases, and there are some reasons to suspect that risk or protection factors may not be the same throughout the carcinogenic process. Thus, the more recent studies have looked for the association between ω-3 PUFAs and prostate cancer of different stages or grades. Concerning ALA, the finding of an increased risk of advanced cancer with higher ALA intakes has not been confirmed by recent studies. The intake or the status of long-chain ω-3 PUFAs has been associated with a lower prostate cancer risk in some studies, but not in most of them, so that no firm conclusion can be drawn to date. Encouraging results have been brought by the few studies which have looked at the risk of fatal prostate cancer, which appeared to be much lower in men with a high fish or long-chain ω-3 PUFA intake. Interaction of ω-3 PUFA intake with polymorphisms of genes linked to PUFA metabolism, such as COX-2, calls for further study. Overall, although the first works have been published in the 1980s, the results are far from being definitive, in part because of the changes in prostate cancer incidence resulting from PSA screening practice. There is a need for more well-designed observational studies, especially prospective studies as well as of interventional studies on diagnosed cases to investigate the phases of cancer progression which could be slowed or inhibited by long-chain ω-3 PUFAs.

References

1. Parkin DM, Bray F, Ferlay J, Pisani P. Global cancer statistics, 2002. CA Cancer J Clin 2005;55:74–108.
2. Grönberg H. Prostate cancer epidemiology. Lancet 2003;361:859–64.
3. Baade PD, Youlden DR, Krnjacki LJ. International epidemiology of prostate cancer: Geographical distribution and secular trends. Mol Nutr Food Res 2009;53:171–84.
4. Hsing AW, Chokkalingam AP. Prostate cancer epidemiology. Front Biosci. 2006;11: 1388–413.
5. Lichtenstein P, Holm NV, Verkasalo PK et al. Environmental and heritable factors in the causation of cancer–analyses of cohorts of twins from Sweden, Denmark, and Finland. N Engl J Med 2000;343:78–85.
6. Chan JM, Gann PH, Giovannucci EL. Role of diet in prostate cancer development and progression. J Clin Oncol. 2005;23:8152–60.
7. Sonn GA, Aronson W, Litwin MS. Impact of diet on prostate cancer: a review. Prostate Cancer Prostatic Dis. 2005;8:304–10.
8. Ma RW, Chapman K. A systematic review of the effect of diet on prostate cancer prevention and treatment. J Hum Nutr Diet 2009. Jun;22 (3):187–99
9. Platz EA, De Marzo AM, Giovannucci E. Prostate cancer association studies: pitfalls and solutions to cancer misclassification in the PSA era. J Cell Biochem. 2004;91:553–71.
10. Hulshof KF, Erp-Baart MA, Anttolainen M et al. Intake of fatty acids in western Europe with emphasis on trans fatty acids: the TRANSFAIR Study. Eur J Clin Nutr 1999;53:143–57.
11. Astorg P, Arnault N, Czernichow S, Noisette N, Galan P, Hercberg S. Dietary intakes and food sources of n–6 and n–3 PUFA in French adult men and women. Lipids 2004;39:527–35.
12. Fuhrman BJ, Barba M, Krogh V et al. Erythrocyte membrane phospholipid composition as a biomarker of dietary fat. Ann Nutr Metab 2006;50:95–102.
13. Sun Q, Ma J, Campos H, Hankinson SE, Hu FB. Comparison between plasma and erythrocyte fatty acid content as biomarkers of fatty acid intake in US women. Am J Clin Nutr 2007;86:74–81.

14. Astorg P, Bertrais S, Laporte F et al. Plasma n–6 and n–3 polyunsaturated fatty acids as biomarkers of their dietary intakes: a cross-sectional study within a cohort of middle-aged French men and women. Eur J Clin Nutr 2008;62:1155–61.
15. Baylin A, Campos H. The use of fatty acid biomarkers to reflect dietary intake. Curr Opin Lipidol 2006;17:22–7.
16. Hodson L, Skeaff CM, Fielding BA. Fatty acid composition of adipose tissue and blood in humans and its use as a biomarker of dietary intake. Prog Lipid Res 2008;47:348–80.
17. Plourde M, Cunnane SC. Extremely limited synthesis of long chain polyunsaturates in adults: implications for their dietary essentiality and use as supplements. Appl Physiol Nutr Metab 2007;32:619–34.
18. Astorg P. Dietary N-6 and N–3 polyunsaturated fatty acids and prostate cancer risk: a review of epidemiological and experimental evidence. Cancer Causes Control 2004;15:367–86.
19. Attar-Bashi NM, Frauman AG, Sinclair AJ. Alpha-linolenic acid and the risk of prostate cancer. What is the evidence? J Urol 2004;171:1402–7.
20. Brouwer IA, Katan MB, Zock PL. Dietary alpha-linolenic acid is associated with reduced risk of fatal coronary heart disease, but increased prostate cancer risk: a meta-analysis. J Nutr 2004;134:919–22.
21. Giovannucci E, Rimm EB, Colditz GA et al. A prospective study of dietary fat and risk of prostate cancer. J Natl Cancer Inst 1993;85:1571–9.
22. Gann PH, Hennekens CH, Sacks FM, Grodstein F, Giovannucci EL, Stampfer MJ. Prospective study of plasma fatty acids and risk of prostate cancer. J Natl Cancer Inst 1994;86:281–6.
23. Colli JL, Colli A. Comparisons of prostate cancer mortality rates with dietary practices in the United States. Urol Oncol 2005;23:390–8.
24. Bakker N, Van't Veer P, Zock PL. Adipose fatty acids and cancers of the breast, prostate and colon: an ecological study. EURAMIC Study Group. Int J Cancer 1997;72:587–91.
25. Kobayashi M, Sasaki S, Hamada GS, Tsugane S. Serum n–3 fatty acids, fish consumption and cancer mortality in six Japanese populations in Japan and Brazil. Jpn J Cancer Res 1999;90:914–21.
26. Andersson SO, Wolk A, Bergstrom R et al. Energy, nutrient intake and prostate cancer risk: a population-based case-control study in Sweden. Int J Cancer 1996;68:716–22.
27. Meyer F, Bairati I, Fradet Y, Moore L. Dietary energy and nutrients in relation to preclinical prostate cancer. Nutr Cancer 1997;29:120–6.
28. Bairati I, Meyer F, Fradet Y, Moore L. Dietary fat and advanced prostate cancer. J Urol 1998;159:1271–5.
29. De Stefani E, Deneo-Pellegrini H, Boffetta P, Ronco A, Mendilaharsu M. Alpha-linolenic acid and risk of prostate cancer: a case-control study in Uruguay. Cancer Epidemiol Biomarkers Prev 2000;9:335–8.
30. Ramon JM, Bou R, Romea S et al. Dietary fat intake and prostate cancer risk: a case-control study in Spain. Cancer Causes Control 2000;11:679–85.
31. Hedelin M, Chang ET, Wiklund F et al. Association of frequent consumption of fatty fish with prostate cancer risk is modified by COX-2 polymorphism. Int J Cancer 2007;120: 398–405.
32. Hodge AM, English DR, McCredie MR et al. Foods, nutrients and prostate cancer. Cancer Causes Control 2004;15:11–20.
33. Bidoli E, Talamini R, Bosetti C et al. Macronutrients, fatty acids, cholesterol and prostate cancer risk. Ann Oncol 2005;16:152–7.
34. Fradet V, Cheng I, Casey G, Witte JS. Dietary Omega-3 Fatty Acids, Cyclooxygenase-2 Genetic Variation, and Aggressive Prostate Cancer Risk. Clin Cancer Res. 2009;15:2559–66.
35. Leitzmann MF, Stampfer MJ, Michaud DS et al. Dietary intake of n–3 and n–6 fatty acids and the risk of prostate cancer. Am J Clin Nutr 2004;80:204–16.

36. Giovannucci E, Liu Y, Platz EA, Stampfer MJ, Willett WC. Risk factors for prostate cancer incidence and progression in the Health Professionals Follow-Up Study. Int J Cancer 2007;121:1571–8.
37. Schuurman AG, van den Brandt PA, Dorant E, Brants HA, Goldbohm RA. Association of energy and fat intake with prostate carcinoma risk: results from The Netherlands Cohort Study. Cancer 1999;86:1019–27.
38. Mannistö S, Pietinen P, Virtanen MJ et al. Fatty acids and risk of prostate cancer in a nested case-control study in male smokers. Cancer Epidemiol Biomarkers Prev 2003;12:1422–8.
39. Laaksonen DE, Laukkanen JA, Niskanen L et al. Serum linoleic and total polyunsaturated fatty acids in relation to prostate and other cancers: a population-based cohort study. Int J Cancer 2004;111:444–50.
40. Wallström P, Bjartell A, Gullberg B, Olsson H, Wirfält E. A prospective study on dietary fat and incidence of prostate cancer (Malmo, Sweden) . Cancer Causes Control 2007;18: 1107–21.
41. Koralek DO, Peters U, Andriole G et al. A prospective study of dietary alpha-linolenic acid and the risk of prostate cancer (United States). Cancer Causes Control 2006;17:783–91.
42. Park SY, Murphy SP, Wilkens LR, Henderson BE, Kolonel LN. Fat and meat intake and prostate cancer risk: The multiethnic cohort study. Int J Cancer 2007;121:1339–45.
43. Newcomer LM, King IB, Wicklund KG, Stanford JL. The association of fatty acids with prostate cancer risk. Prostate 2001;47:262–8.
44. Godley PA, Campbell MK, Gallagher P, Martinson FE, Mohler JL, Sandler RS. Biomarkers of essential fatty acid consumption and risk of prostatic carcinoma. Cancer Epidemiol Biomarkers Prev 1996;5:889–95.
45. Freeman VL, Meydani M, Yong S Freeman VLet al. Prostatic levels of fatty acids and the histopathology of localized prostate cancer. J Urol 2000;164:2168–72.
46. Freeman VL, Meydani M, Hur K, Flanigan RC. Inverse association between prostatic polyunsaturated fatty acid and risk of locally advanced prostate carcinoma. Cancer 2004;101:2744–54.
47. Yang YJ, Lee SH, Hong SJ, Chung BC. Comparison of fatty acid profiles in the serum of patients with prostate cancer and benign prostatic hyperplasia. Clin Biochem 1999;32:405–9.
48. Mamalakis G, Kafatos A, Kalogeropoulos N, Andrikopoulos N, Daskalopulos G, Kranidis A. Prostate cancer vs hyperplasia: relationships with prostatic and adipose tissue fatty acid composition. Prostaglandins Leukot Essent Fatty Acids 2002;66:467–77.
49. Christensen JH, Fabrin K, Borup K, Barber N, Poulsen J. Prostate tissue and leukocyte levels of n-3 polyunsaturated fatty acids in men with benign prostate hyperplasia or prostate cancer. BJU Int 2006;97:270–3.
50. Ritch CR, Wan RL, Stephens LB et al. Dietary fatty acids correlate with prostate cancer biopsy grade and volume in Jamaican men. J Urol 2007;177:97–101.
51. Ritch CR, Brendler CB, Wan RL, Pickett KE, Sokoloff MH. Relationship of erythrocyte membrane polyunsaturated fatty acids and prostate-specific antigen levels in Jamaican men. BJU Int 2004;93:1211–5.
52. Harvei S, Bjerve KS, Tretli S, Jellum E, Robsahm TE, Vatten L. Prediagnostic level of fatty acids in serum phospholipids: omega-3 and omega-6 fatty acids and the risk of prostate cancer. Int J Cancer 1997;71:545–51.
53. Chavarro JE, Stampfer MJ, Li H, Campos H, Kurth T, Ma J. A Prospective Study of Polyunsaturated Fatty Acid Levels in Blood and Prostate Cancer Risk. Cancer Epidemiol Biomarkers Prev 2007;16:1364–70.
54. Chavarro JE, Stampfer MJ, Campos H, Kurth T, Willett WC, Ma J. A prospective study of trans-Fatty Acid levels in blood and risk of prostate cancer. Cancer Epidemiol Biomarkers Prev 2008;17:95–101.
55. Park SY, Wilkens LR, Henning SM et al. Circulating fatty acids and prostate cancer risk in a nested case-control study: the Multiethnic Cohort. Cancer Causes Control 2009;20:211–23.

56. Crowe FL, Allen NE, Appleby PN et al. Fatty acid composition of plasma phospholipids and risk of prostate cancer in a case-control analysis nested within the European Prospective Investigation into Cancer and Nutrition. Am J Clin Nutr 2008;88:1353–63.
57. Crowe FL, Key TJ, Appleby PN et al. Dietary fat intake and risk of prostate cancer in the European Prospective Investigation into Cancer and Nutrition. Am J Clin Nutr 2008;87:1405–13.
58. Dennis LK, Snetselaar LG, Smith BJ, Stewart RE, Robbins ME. Problems with the assessment of dietary fat in prostate cancer studies. Am J Epidemiol 2004;160:436–44.
59. Norrish AE, Skeaff CM, Arribas GL, Sharpe SJ, Jackson RT. Prostate cancer risk and consumption of fish oils: a dietary biomarker-based case-control study. Br J Cancer 1999;81:1238–42.
60. Simon JA, Chen YH, Bent S. The relation of {alpha}-linolenic acid to the risk of prostate cancer: a systematic review and meta-analysis. Am J Clin Nutr 2009;89:1558S–64S.
61. Bemrah N, Sirot V, Leblanc JC, Volatier JL. Fish and seafood consumption and omega 3 intake in French coastal populations: CALIPSO survey. Public Health Nutr 2009;12: 599–608.
62. Hebert JR, Hurley TG, Olendzki BC, Teas J, Ma Y, Hampl JS. Nutritional and socioeconomic factors in relation to prostate cancer mortality: a cross-national study. J Natl Cancer Inst 1998;90:1637–47.
63. Colli JL, Colli A. International comparisons of prostate cancer mortality rates with dietary practices and sunlight levels. Urol Oncol 2006;24:184–94.
64. Bouchardy C, Fioretta G, Rapiti E et al. Recent trends in prostate cancer mortality show a continuous decrease in several countries. Int J Cancer 2008;123:421–9.
65. Prener A, Storm HH, Nielsen NH. Cancer of the male genital tract in Circumpolar Inuit. Acta Oncol 1996;35:589–93.
66. Dewailly E, Mulvad G, Sloth PH, Hansen JC, Behrendt N, Hart Hansen JP. Inuit are protected against prostate cancer. Cancer Epidemiol Biomarkers Prev 2003;12:926–7.
67. Talamini R, Franceschi S, La Vecchia C, Serraino D, Barra S, Negri E. Diet and prostatic cancer: a case-control study in northern Italy. Nutr Cancer 1992;18:277–86.
68. Deneo-Pellegrini H, De Stefani E, Ronco A, Mendilaharsu M. Foods, nutrients and prostate cancer: a case-control study in Uruguay. Br J Cancer 1999;80:591–7.
69. Kristal AR, Cohen JH, Qu P, Stanford JL. Associations of energy, fat, calcium, and vitamin D with prostate cancer risk. Cancer Epidemiol Biomarkers Prev 2002;11:719–25.
70. Chen YC, Chiang CI, Lin RS, Pu YS, Lai MK, Sung FC. Diet, vegetarian food and prostate carcinoma among men in Taiwan. Br J Cancer 2005;93:1057–61.
71. Mina K, Fritschi L, Johnson KC. An inverse association between preserved fish and prostate cancer: results from a population-based case-control study in Canada. Nutr Cancer 2008;60:222–6.
72. Key TJ, Silcocks PB, Davey GK, Appleby PN, Bishop DT. A case-control study of diet and prostate cancer. Br J Cancer 1997;76:678–87.
73. Fernandez E, Chatenoud L, La Vecchia C, Negri E, Franceschi S. Fish consumption and cancer risk. Am J Clin Nutr 1999;70:85–90.
74. Mishina T, Watanabe H, Araki H, Nakao M. Epidemiological study of prostatic cancer by matched-pair analysis. Prostate 1985;6:423–36.
75. Sonoda T, Nagata Y, Mori M et al. A case-control study of diet and prostate cancer in Japan: possible protective effect of traditional Japanese diet. Cancer Sci 2004;95:238–42.
76. Amin M, Jeyaganth S, Fahmy N et al. Dietary habits and prostate cancer detection: a case-control study. Can.Urol Assoc J 2008;2:510–5.
77. Severson RK, Nomura AM, Grove JS, Stemmermann GN. A prospective study of demographics, diet, and prostate cancer among men of Japanese ancestry in Hawaii. Cancer Res 1989;49:1857–60.
78. Le Marchand L, Kolonel LN, Wilkens LR, Myers BC, Hirohata T. Animal fat consumption and prostate cancer: a prospective study in Hawaii. Epidemiology 1994;5:276–82.

79. Veierod MB, Laake P, Thelle DS. Dietary fat intake and risk of lung cancer: a prospective study of 51,452 Norwegian men and women. Eur J Cancer Prev 1997;6:540–9.
80. Schuurman AG, van den Brandt PA, Dorant E, Goldbohm RA. Animal products, calcium and protein and prostate cancer risk in The Netherlands Cohort Study. Br J Cancer 1999;80:1107–13.
81. Rohrmann S, Platz EA, Kavanaugh CJ, Thuita L, Hoffman SC, Helzlsouer KJ. Meat and dairy consumption and subsequent risk of prostate cancer in a US cohort study. Cancer Causes Control 2007;18:41–50.
82. Allen NE, Key TJ, Appleby PN et al. Animal foods, protein, calcium and prostate cancer risk: the European Prospective Investigation into Cancer and Nutrition. Br J Cancer 2008;98:1574–81.
83. Mills PK, Beeson WL, Phillips RL, Fraser GE. Cohort study of diet, lifestyle, and prostate cancer in Adventist men. Cancer 1989;64:598–604.
84. Allen NE, Sauvaget C, Roddam AW et al. A prospective study of diet and prostate cancer in Japanese men. Cancer Causes Control 2004;15:911–20.
85. Kvåle R, Auvinen A, Adami HO et al. Interpreting trends in prostate cancer incidence and mortality in the five Nordic countries. J Natl Cancer Inst 2007;99:1881–7.
86. Augustsson K, Michaud DS, Rimm EB et al. A prospective study of intake of fish and marine fatty acids and prostate cancer. Cancer Epidemiol Biomarkers Prev 2003;12:64–7.
87. Chan JM, Holick CN, Leitzmann MF et al. Diet after diagnosis and the risk of prostate cancer progression, recurrence, and death (United States). Cancer Causes Control 2006;17: 199–208.
88. Terry P, Lichtenstein P, Feychting M, Ahlbom A, Wolk A. Fatty fish consumption and risk of prostate cancer. Lancet 2001;357:1764–6.
89. Sato F, Shimazu T, Kuriyama S et al. [Fish intake and the risk of prostate cancer in Japan: a prospective cohort study]. Nippon Hinyokika Gakkai Zasshi 2008;99:14–21.
90. Pham TM, Fujino Y, Kubo T et al. Fish intake and the risk of fatal prostate cancer: findings from a cohort study in Japan. Public Health Nutr 2009;12:609–13.
91. Chavarro JE, Stampfer MJ, Hall MN, Sesso HD, Ma J. A 22-y prospective study of fish intake in relation to prostate cancer incidence and mortality. Am J Clin Nutr 2008;88: 1297–303.
92. Freeman VL, Flanigan RC, Meydani M. Prostatic fatty acids and cancer recurrence after radical prostatectomy for early-stage prostate cancer. Cancer Causes Control 2007;18: 211–8.
93. Patel MI, Kurek C, Dong Q. The arachidonic acid pathway and its role in prostate cancer development and progression. J Urol 2008;179:1668–75.
94. Harris RE. Cyclooxygenase-2 (cox-2) blockade in the chemoprevention of cancers of the colon, breast, prostate, and lung. Inflammopharmacology 2009;17:55–67.
95. Cohen BL, Gomez P, Omori Y et al. Cyclooxygenase-2 (COX-2) expression is an independent predictor of prostate cancer recurrence. Int J Cancer 2006;119:1082–7.
96. Khor LY, Bae K , Pollack A et al. COX-2 expression predicts prostate-cancer outcome: analysis of data from the RTOG 92-02 trial. Lancet Oncol 2007;8:912–20.
97. Cheng I, Liu X, Plummer SJ, Krumroy LM, Casey G, Witte JS. COX2 genetic variation, NSAIDs, and advanced prostate cancer risk. Br J Cancer 2007;97:557–61.
98. Panguluri RC, Long LO, Chen W et al. COX-2 gene promoter haplotypes and prostate cancer risk. Carcinogenesis 2004;25:961–6.
99. Shahedi K, Lindstrom S, Zheng SL et al. Genetic variation in the COX-2 gene and the association with prostate cancer risk. Int J Cancer 2006;119:668–72.
100. Fernandez P, de Beer PM, van der ML, Heyns CF. COX-2 promoter polymorphisms and the association with prostate cancer risk in South African men. Carcinogenesis 2008;29: 2347–50.

Chapter 6
ω-3 PUFAs: Interventional Trials for the Prevention and Treatment of Breast and Prostate Cancer

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Abstract Due to global changes in dietary sources and habits, the human diet has shifted to include an unfavorable ratio of ω-6 to ω-3 PUFAs. This is believed to have drastic consequences on health, increasing the risk for cardiovascular and inflammation-related diseases as well as cancer. Epidemiology and experimental studies suggest that ω-3 PUFAs are protective against certain types of cancer, including colon, breast, and prostate cancer. The available data have motivated the recent implementation of interventional clinical trials where cancer patients or individuals at high risk of developing cancer receive dietary ω-3 PUFA supplementation. In this review, we summarize the objectives of ongoing and recently closed clinical trials of ω-3 PUFA supplements for breast and prostate cancer prevention and treatment that are listed in the ClinicalTrials.gov database, a registry of federally and privately supported clinical trials conducted in the United States and worldwide (http://clinicaltrials.gov/).

Keywords: ω-3 PUFA · Clinical trials · Breast cancer · Prostate cancer

Abbreviations

ADH	atypical ductal hyperplasia
ALA	α-linolenic acid
CAM	complementary and alternative medicines
COX-2	cyclooxygenase-2
DCIS	ductal carcinoma in situ
DHA	docosahexaenoic acid
EPA	eicosapentaenoic acid
eIF2α	eukaryotic initiation factor 2α
ER	estrogen receptor

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FASN	fatty acid synthase
HER2	human epidermal growth factor receptor 2
IGF-1	insulin-like growth factor-1
IGFIR	insulin-like growth factor I receptor
IGFBP-3	insulin-like growth factor-binding protein 3
IL-6	interleukin-6
Ki-67	proliferation marker
PAI-1	plasminogen activator inhibitor 1
PR	progesterone receptor
PSA	prostate-specific antigen
PUFA	poly-unsaturated fatty acid
RBC	red blood cells
SHBG	sex hormone-binding globulin
SREBP	sterol response element-binding protein
TUNEL	terminal deoxynucleotidyl transferase-mediated dUTP nick end labelling
tPA	tissue plasminogen activator
VEGF	vascular endothelial growth factor
vWF	von Willebrand factor.

6.1 Introduction

Consumption of dietary fat can have profound influences on human health. Polyunsaturated fatty acids (PUFAs) are generally believed to be overall healthier than saturated fats, but the type of PUFA is also important. Both ω-3 and ω-6 PUFAs are essential fatty acids that are required in the diet. However, an excess of ω-6 PUFAs relative to ω-3 PUFAs is believed to increase risk for a number of conditions, including cardiovascular disease, troubles related to inflammation, and cancer. The modern Western diet contains an increased ratio of ω-6 to ω-3 PUFAs as compared to the diet of our hunter–gatherer ancestors [1, 2]. Restoring a healthier balance of ω-3 to ω-6 PUFAs is an attractive approach for cancer chemoprevention. Indeed, unlike many other chemopreventive agents, ω-3 PUFAs are essential nutrients with additional beneficial effects on cardiovascular and inflammatory diseases.

Perhaps not surprisingly, dietary fat has particular implications for hormone-related cancers. In general, increased consumption of total fat as well as ω-6 PUFAs is associated with increased incidence of breast and prostate cancer. Other chapters in this book describe the epidemiological studies linking ω-3 PUFA consumption and risk of breast and prostate cancer, as well as experimental evidence for the effects of these fatty acids on cancer cells in culture and animal models. Collectively, these studies raised the possibility of using ω-3 PUFAs in the prevention and treatment of cancer. In the present chapter, we will focus on clinical intervention trials using fish oil, flaxseed, or purified ω-3 PUFAs in the prevention and treatment of breast and prostate cancer. Ongoing or recently completed clinical trials which include dietary interventions with ω-3 PUFA supplements for prostate and breast cancer are summarized in Tables 6.1 and 6.2, respectively. These trials were listed

Table 6.1 Clinical trials with dietary ω-3 intervention for prostate cancer*

Study ID	Dates	Title	Intervention	Measured outcomes	PI	Institution
NCT00049309	1/03–5/06	Low-Fat Diet and/or Flaxseed in Preventing Prostate Cancer	Dietary supplement: flaxseed, low-fat diet	Proliferation, apoptosis, PSA, and other biomarkers	Wendy Demark-Wahnefried, PhD	Duke University
NCT00253643	7/05–2/09	Fish Oil and Green Tea Extract in Preventing Prostate Cancer in Patients Who Are at Risk for Developing Prostate Cancer	Dietary supplement: fish oil, green tea extract	Proliferation, lipid metabolism: FASN, SREBP, apoptosis, bone loss	Jackilen Shannon, PhD	Knight Cancer Institute at Oregon Health and Science University
NCT00402285	4/03–4/10	Lycopene or ω-3 Fatty Acid Nutritional Supplements in Treating Patients With Stage I or Stage II Prostate Cancer	Dietary supplement: lycopene, ω-3 fatty acids (DHA/EPA)	Gene expression pattern, correlation with diet, cancer progression (2-year follow-up	William Aronson, MD	UCSF Helen Diller Family Comprehensive Cancer Center
NCT00433797	6/07–4/09	Dietary Intervention With Phytochemicals and Polyunsaturated Fatty Acids in Prostate Cancer Patients	Dietary supplement: phytochemicals and PUFA	PSA, biomarkers of oxidative stress/damage, antioxidant status, inflammation, apoptosis, DNA microarrays	Wolfgang Lilleby, MD Sigbjørn Smeland, MD, PhD	Rikshospitalet-Radiumhospitalet, Oslo

Table 6.1 (continued)

Study ID	Dates	Title	Intervention	Measured outcomes	PI	Institution
NCT00458549	7/06–NS	Polyun saturated Fatty Acids in Treating Patients With Prostate Cancer Undergoing Prostate Biopsy and/or Surgery	Dietary supplement: fish oil	eIF2α phosphorylation, correlation of eIF2α phosphorylation with relapse-free survival (outcome data from tissue repository)	Jose A. Halperin, MD	Dana-Farber Cancer Institute
NCT00798876	8/05–8/09	Low-Fat Fish Oil Diet for Prostate Cancer Prevention	Dietary supplement: fish oil, vitamin E, low-Fat Diet	IGF-1, IGFBP, serum, and tissue fatty acids	William Aronson, MD	UCLA and Western Los Angles VA
NCT00836615	1/06–12/09	Low-Fat Diet and ω-Fatty Acids for Prostate Cancer Prevention	Dietary supplement: low-fat diet + fish oil	Fatty acid ratio	William Aronson, MD	University of California, Los Angeles

*eIF2α: eukaryotic initiation factor 2α, FASN: fatty acid synthase, IGF-1: insulin-like growth factor-1, IGFBP: insulin-like growth factor-binding proteins, PSA: prostate-specific antigen, SREBP: sterol response element-binding protein. The two shaded studies appear to be one single study.

Table 6.2 Clinical trials with dietary ω-3 intervention for breast cancer*

Study ID	Dates	Title	Intervention	Measured outcomes	PI	Institution
NCT00010829	1/01–12/05	Macrobiotic Diet and Flax Seed: Effects on Estrogens, Phytoestrogens, & Fibrinolytic Factors	Dietary supplement: flax seed + low-fat macrobiotic diet	Estrogens and metabolites, antigens to PAI-1 and tPA, fibrin D-dimer, vWF, endothelial function, antioxidant vitamins, retinoids	Dr. Lawrence Kushi	Columbia University, Teachers College
NCT00114296	4/05–	ω-3 Fatty Acids in Preventing Breast Cancer in Women at High Risk of Developing Breast Cancer	Dietary supplement: ω-3 fatty acids (DHA/EPA)	Mammographic breast density, atypia in ductal lavage, hormones, lipid peroxidation	Agustin Garcia, MD	Cedars-Sinai Medical Center
NCT00612560	11/07–6/10	Flaxseed, Aromatase Inhibitors and Breast Tumor Characteristics	Drug: Anastrozole; dietary supplement: flaxseed	Tumor characteristics, proliferation and apoptosis, ER, PR, HER2 expression, steroid and growth hormone serum levels, Mammostrat recurrence score	Susan E. McCann, PhD	Roswell Park Cancer Institute

Table 6.2 (continued)

Study ID	Dates	Title	Intervention	Measured outcomes	PI	Institution
NCT00635908	12/07–4/11	Flaxseed and/or Anastrozole in Treating Postmenopausal Women Undergoing Surgery for Newly Diagnosed Stage I or Stage II Breast Cancer	Drug: anastrozole; dietary supplement: flaxseed	Changes from initial tumor biopsy to tumor resection in markers of proliferation, apoptosis, signaling factors, Mammostrat recurrence scores, steroid and growth hormone profiles	Swati Kulkarni, MD	Roswell Park Cancer Institute
NCT00627276	12/07–8/08	ω-3 Fatty Acids in Treating Women With Newly Diagnosed Ductal Carcinoma In Situ and/or Atypical Ductal Hyperplasia	Dietary supplement: ω-3 fatty acids (EPA, DHA)	Genetic markers for breast cancer risk/progression (microarray analysis), fatty acids in RBC and nipple aspirates, presence of DCIS, ADH, invasive cancer	Jackilen Shannon, PhD	Oregon Health and Science University Cancer Institute

Table 6.2 (continued)

Study ID	Dates	Title	Intervention	Measured outcomes	PI	Institution
NCT00723398	9/08–10/12	Nutritional Supplements and Hormonal Manipulations for Breast Cancer Prevention	Dietary supplement: lovaza (ω-3 PUFA); drug: raloxifene	Breast density, biomarkers of oxidative stress, estrogen metabolites, C-reactive protein, IL-6, IGF-1, IGFBP-3, lipid panel, and complete blood count	Andrea Manni, MD	Penn State University
NCT00794989	4/08–10/12	Flaxseed in Preventing Breast Cancer in Premenopausal Women at Risk of Developing Breast Cancer	Dietary supplement: flaxseed	Development of primary breast cancer, changes in biomarkers from baseline to 6 months (Ki-67, act. caspase-3, cyclin D1, survivin, VEGF, IGF-1, IGFBP-3)	Swati Kulkarni, MD	Roswell Park Cancer Institute

*ADH: atypical ductal hyperplasia, DCIS: ductal carcinoma in situ, ER: estrogen receptor, HER2: human epidermal growth factor receptor 2, IGF-1: insulin-like growth factor-1, IGFBP: insulin-like growth factor-binding protein 3, IL-6: interleukin-6, Ki-67: proliferation marker, PAI-1: plasminogen activator inhibitor 1, PR: progesterone receptor, RBC: red blood cells, tPA: tissue plasminogen activator, VEGF: vascular endothelial growth factor, vWF: von Willebrand factor. The two shaded studies appear to be one single study.

in the ClinicalTrials.gov database (http://clinicaltrials.gov/) as of May 2009. The ClinicalTrials.gov database is “a registry of federally and privately supported clinical trials conducted in the United States and around the world”. It currently lists a total of 73,166 trials with locations in 167 countries. All available trials using dietary ω-3 PUFA supplements in the prevention or treatment of these two cancer types were selected for this review. Of note, since our last review of this database in January 2008 [3], five new studies of this kind have been added to the database. Some of the trials focus exclusively on ω-3 PUFAs, while others include them as part of a regimen with other dietary supplements, chemotherapeutic agents, or behavioral interventions (e.g., counseling for maintaining a low-fat diet or prepared meals distributed to the study participants). As most of these trials have not yet been published, our discussion will be limited to the objectives and methodologies of the trials, with results included whenever possible.

6.2 Prevention and Treatment of Prostate Cancer with ω-3 PUFAs

Prostate cancer offers good promise for chemoprevention, since the progression of this type of cancer is typically slow. Thus, there is a wide window of opportunity for chemopreventive agents capable of delaying cancer progression. Intermediate biomarkers commonly used to determine the biological effects of dietary supplements on prostate cancer include serum prostate-specific antigen (PSA), proliferation and apoptosis markers in prostate biopsies or surgical specimen, steroid hormones and their binding proteins, as well as growth factors and other signaling molecules. Some studies also measure fatty acids in the blood, in red blood cells, or in prostate tissue to determine the impact of the dietary intervention on lipid profiles. Two of the ongoing studies listed in Table 6.1 additionally make use of DNA microarrays to investigate gene expression patterns.

A pilot study in prostate cancer patients suggested that a low-fat, flaxseed-supplemented diet decreased PSA and cholesterol levels as well as benign prostatic epithelial cell proliferation in biopsy specimens from patients that were scheduled for a repeated prostate biopsy [4]. The same group recently published the results of a phase II multisite randomized controlled trial (NCT00049309) [5]. This trial was designed to assess the effect of pre-surgical flaxseed supplementation and/or dietary fat restriction on the biology of prostate cancer. Tumor proliferation was assessed by Ki-67 staining of prostatectomy specimens, and apoptosis was measured using the terminal deoxynucleotidyl transferase-mediated dUTP nick end labeling (TUNEL) method. Biomarkers also included serum PSA, total testosterone, sex hormone-binding globulin (SHBG), insulin-like growth factor-I (IGF-I) and IGF-binding protein-3 (IGFBP-3), C-reactive protein, and serum lipids. The effect of diets was determined by investigating levels of lignans in the urine and seminal fluid as well as fatty acid profiles of erythrocytes and prostatic tissue. A total of 161 patients completed the study, randomized in four arms. Eligible patients had biopsy-confirmed prostatic carcinoma and elected prostatectomy as their primary treatment, with exclusion of patients reporting recent flaxseed use or consuming diets with less than 30% kilocalories from fat. Flaxseed supplementation was in the form of 30 g ground flaxseed daily for an average of 30 days before prostatectomy. In this study, flaxseed supplementation resulted in significantly lower proliferation rates ($P < 0.002$). Median Ki-67-positive cells/total nuclei ratios ($\times 100$) were 1.66 for the flaxseed-supplemented diet and 1.50 for the flaxseed-supplemented, low-fat diet versus 3.23 and 2.56 for the control and low-fat diet, respectively. Patients on low-fat diets experienced significant decreases in serum cholesterol ($P = 0.048$), but no other significant differences were observed between arms, including no differences in side effects. This study suggests that flaxseed supplementation is safe and may be protective for prostate cancer. Flaxseed is a rich source in ω-3 PUFAs, in the form of α-linolenic acid (ALA). Inconsistent data exist relative to the effect of ALA on cancer risk [6], which may be due in part to different dietary sources of ALA. Part of the biological effects of flaxseed, however, may be attributed to lignans, which possess antioxidant, antiproliferative, antiangiogenic, and phytoestrogenic effects and reduce testosterone and 5α-reductase, which converts testosterone to the more potent dihydrotestosterone [7–9].

A pilot study in nine men with untreated localized or regional adenocarcinoma of the prostate was designed to determine whether a 3-month intervention with a low-fat, fish oil-supplemented diet affects the ratio of ω-3 to ω-6 fatty acids in plasma and gluteal fat. The study was also designed to evaluate the feasibility of measuring cyclooxygenase-2 (COX-2) expression in prostate needle biopsy specimens taken at the same sites before and after the diet. This 3-month intervention was able to increase the ω-3:ω-6 PUFA ratio in plasma ($P = 0.002$) and gluteal adipose tissue ($P = 0.002$) [10]. Moreover, COX-2 expression as determined by RT-PCR decreased in four of seven patients. Additional studies will be needed to determine the effects of ω-3 PUFAs on COX-2 expression and prostate cancer progression in patients.

In clinical trial NCT00253643, the effects of fish oil on lipid metabolism in prostate tissue samples are being evaluated in 144 patients with prostatic intraepithelial neoplasia or who are otherwise at elevated risk for prostate cancer compared to the average population. The original study, initiated in 2005, had two arms, but was later modified to include green tea extracts and now has four arms (fish oil and green tea extracts, green tea extracts, fish oil, and placebo). Treatment in both arms continues for up to 20 weeks in the absence of disease progression or unacceptable toxicity, and patient's are scheduled for a prostate biopsy on the last day of study treatment. After completion of study treatment, patients are followed for 30 days. The measured outcomes include pre- and post-intervention measurements of fatty acid synthase (FASN) and sterol regulatory element-binding protein (SREBP) expression, cell proliferation and apoptosis, and bone formation and loss. In addition, phospholipid membrane composition will be investigated after the intervention. To our knowledge, the results of this study, which was scheduled to be completed in February 2009, have not yet been published.

An ongoing study initiated in 2003 (NCT00402285) was designed to compare the long-term effects of ω-3 fatty acids with those of another nutritional supplement, lycopene. This randomized, placebo-controlled study in 114 patients with stage I or II prostate adenocarcinoma includes three treatment arms: placebo only, oral lycopene, and oral ω-3 fatty acids from fish oil (eicosapentaenoic acid [EPA] and docosahexaenoic acid [DHA]). Patients are stratified according to dietary intake of tomato and fish before randomization. Treatment is scheduled for up to 90 days or until post-treatment biopsy (a maximum of 104 days) in the absence of disease progression, with follow-up every 3 months for 2 years. The primary objective of this study is to compare gene expression patterns in normal prostate tissue of the patients at baseline and after treatment in the three experimental arms. Secondary objectives include a correlation of gene expression patterns with progression at 12 months and a correlation of expression patterns with self-reported dietary intake. In addition, the effect of lycopene or ω-3 PUFA supplements on the incidence of tumor progression over the 2-year follow-up period will be determined. The study authors also aim to determine new candidate molecular targets for lycopene and ω-3 response pathways.

A Norwegian study initiated in 2007 (NCT00433797) is also comparing dietary tomato supplements with a multi-diet supplement consisting of grape juice, pomegranate juice, tomato, green tea, black tea, soy, selenium, and PUFAs, although the nature or quantity of the PUFAs is not discussed in the available study summary. In addition, as PUFAs are part of a multisupplement, the effects of these fatty acids in isolation cannot be determined by this study design. Nonetheless, this study could help determine if short-term consumption of dietary supplements can have a beneficial impact on the biology of the prostate. A total of 102 patients with localized prostate cancer will be included in the study. At the time of inclusion, the participants will be randomized to three groups. The intervention groups include control group, tomato group, and multi-diet group. The intervention period is 3 weeks and will be completed before prostatectomy or radiation therapy. The primary outcome is serum PSA at baseline, after intervention, and at follow-up. Secondary outcomes include biomarkers of inflammation (acute phase proteins, cytokines, chemokines, and other inflammatory mediators), biomarkers of antioxidant status (vitamin C, vitamin E, glutathione, carotenoids, total antioxidant capacity, and total phenolics), biomarkers of oxidative stress (malondialdehyde, isoprostanes, 8-hydroxy-deoxyguanosine, oxidized vitamin C, total lipid peroxides, and protein carbonyls), apoptosis markers in prostate tissue, and DNA microarrays in blood cells and prostate tissue.

A clinical trial initiated in 2006 (NCT00458549) is investigating the effects of ω-3 and ω-6 PUFAs on prostate markers in patients with prostate cancer undergoing prostate biopsy and/or surgery. In the first part of this trial, patients undergo prostate biopsy and blood sample collection at baseline. Samples are analyzed by gas–liquid chromatography to determine the concentrations of ω-3 and ω-6 PUFAs in red blood cell membranes. Biopsy samples are analyzed for eukaryotic initiation factor 2α (eIF2α) phosphorylation by immunohistochemistry. The stated objective of this first part is to compare fatty acid analysis of red blood cells, levels of eIF2α phosphorylation, and tumor differentiation grade, as measured by Gleason score, in patients undergoing prostate biopsy. In the second part of this trial, some patients who were subsequently diagnosed with prostate cancer and are pursuing a total prostatectomy are randomized to one of two treatment arms, receiving either ω-3 PUFAs (fish oil) or ω-6 PUFAs (corn oil) once daily for 28 days prior to tumor resection. The primary objective of this part of the trial is to determine if neoadjuvant ω-3 PUFAs induce phosphorylation of eIF2α in these patients. In addition, relapse-free survival is evaluated as a secondary objective, although the time of follow-up is unclear from the study description. A third part of the trial is a retrospective study using tumor samples and associated pathology data (Gleason score) and clinical data (PSA values and patient's outcome) obtained from a prostate tissue repository. eIF2α phosphorylation is analyzed in these samples. The objective of this part of the study is to correlate eIF2α phosphorylation status, Gleason scores, and time to PSA failure. A total of 600 patients and 1000 stored tissue samples will be accrued for this study.

NCT00798876 is a dietary intervention trial of a low-fat diet with ω-3 fatty acid supplements in men undergoing radical prostatectomy which was initiated in 2005. Patients are randomized to receive either a standard Western diet or a low-fat diet with fish oil and vitamin E supplements for 4–8 weeks, at the end of which patients will undergo the prostatectomy. All food and beverages are provided to the study participants by the study staff, which will also conduct dietary interviews and estimate the body fat, lean weight, and the rate at which the participants burn fat. Blood draws will be performed at the beginning of the study and again at the time of the surgery to measure changes in PSA, fatty acids, insulin, and various hormones. The primary objective of this study is to establish and validate intermediate biomarkers for prostate cancer prevention trials. Secondary objectives are to establish and validate insulin growth factor-1 (IGF-1) and insulin growth factor-binding proteins (IGFBP) as well as serum and tissue fatty acids as relevant intermediate biomarkers for prostate cancer prevention trials. Estimated enrollment is 70 patients and the estimated completion date is August 2009. A similar intervention trial for men scheduled for radical prostatectomy (NCT00836615) from the same group is listed in clinical trials.gov. The subjects will be randomized to one of two 4-week nutrition programs – either a balanced Western diet or a low-fat diet with fish oil capsules. The primary outcome is to determine fatty acid ratio and the enrollment is also 70 patients. The latter study appears to be an extension of the first and was submitted in February 2009, with an estimated study completion date of December 2009.

6.3 Prevention and Treatment of Breast Cancer with ω-3 PUFAs

Dietary interventions with ω-3 PUFAs have been assessed for prevention and/or treatment in women at high risk of developing breast cancer or diagnosed with this malignancy. As was the case for prostate cancer, intermediate biomarkers of the mammary gland biology and cancer progression are often used in the study design (Table 6.2). These include mammographic breast density, markers of proliferation and apoptosis, markers of oxidative stress, lipid profiles, steroid hormones and their metabolites, hormone receptors, and other signaling molecules. Two of the studies use genetic markers for breast cancer risk (DNA microarray analysis or Mammostrat, a prognostic immunohistochemistry test for postmenopausal, node negative, estrogen receptor-expressing breast cancer patients using five antibodies).

A phase II clinical trial initiated in 2001 (NCT00010829) investigates the use of dietary interventions with flaxseed supplement as a means to maintain health in women between 50 and 72 years of age [11]. This investigation is not only focused on breast cancer but also on cardiovascular diseases, osteoporosis, and endometrial cancer. Approximately 120 women were scheduled to be enrolled, randomized into three arms to receive an American Heart Association (AHA) Step 1 diet, an AHA Step 2 diet supplemented with 10 g/day flaxseed, or a macrobiotic dietary intervention. The study design included blood and urine sample collection at baseline, and at 3, 6, 9, and 12 months to examine differences in multiple biomarkers, including estrogens and metabolites, antigens to plasminogen activator inhibitor (PAI) and tissue plasminogen activator (tPA), fibrin D-dimer, von Willebrand factor (vWF), antioxidant vitamins, and retinoids. Endothelial function, as measured by high-resolution ultrasound, was also assessed. Flaxseed supplements were under consideration in this study due to the fact that lignans are phytoestrogens; however, the presence of the ω-3 PUFA ALA in flaxseed could also modulate some of the markers measured in the study. While the trial was scheduled to be completed by December 2005, no publications were reported in the trial database or found on PubMed. However, the same group has been active in investigating the use of complementary and alternative medicines (CAM), including ω-3 PUFAs, in women diagnosed with breast cancer [12]. Over 95% of participants reported use of some form of CAM, with ω-3 PUFAs used by 33.7% of the women in this study [13]. Other studies report a similar prevalence in use of CAM in cancer patients [14–18]. Thus, when evaluating clinical trials on the use of nutritional supplements, it is important to take into account the CAM strategies already used by patients.

An ongoing breast cancer prevention trial initiated in 2005 (NCT00114296) investigates the effect of marine ω-3 PUFAs in women at high risk of developing breast cancer. High-risk criteria included increased 5-year Gail risk, family history of breast cancer, BRCA1/BRCA2 mutations, atypical hyperplasia, lobular or ductal carcinoma in situ, history of invasive stage I breast cancer in remission, or history of ovarian cancer in remission for at least 5 years. The primary objective is to determine the effect of ω-3 PUFAs on mammographic breast density at 1 year, as measured by the Madena method. Secondary objectives are to investigate the effects of this supplement on breast epithelial cell atypia and proliferation in ductal lavage samples, hormone and growth factor levels in the blood, expression of estrogen-related proteins in ductal lavage, and lipid peroxidation. In addition, the correlation between lipid peroxidation-related genes and mammographic breast density will be determined in patients treated with ω-3 PUFAs. The estimated enrollment for this study is 80 women, randomized to one of two arms (ω-3 fatty acids or placebo administered three times daily for 12 months in the absence of the development of ductal carcinoma in situ or invasive carcinoma of the breast or unacceptable toxicity).

A multicenter randomized phase II study initiated in 2007 (NCT00627276) investigates the efficacy of marine ω-3 fatty acids in treating women with newly diagnosed ductal carcinoma in situ (DCIS) and/or atypical ductal hyperplasia

(ADH). Patients are randomized to receive either oral ω-3 fatty acids or placebo (olive oil) capsules three times daily for up to 8 weeks. Patients undergo blood, urine, nipple aspirate, and breast tissue sample collection at baseline and after completion of study treatment. Blood samples are analyzed for genetic markers for breast cancer risk and progression by microarray analysis and also serve to determine the levels of fatty acids in red blood cells. Nipple aspirate samples are used for fatty acid measurements to determine the extent to which ω-3 PUFA supplementation alters the fatty acid profile of breast tissue. Breast tissue samples are analyzed for the presence of DCIS, ADH, and/or invasive cancer. After completion of the 8-week study treatment, patients are followed at 30 days. Objectives are to determine the effect of ω-3 fatty acids on markers of breast cancer progression in women with newly diagnosed DCIS and/or ADH. In addition, this study also plans to determine the effect of ω-3 fatty acids on specific targets identified by microarray in breast cancer cells and in primary cultures from benign and malignant breast tissue biopsies. The estimated enrollment for this study is 40 patients, and the estimated primary completion date was August 2008, although, as of May 2009, the trial is still listed as currently recruiting patients.

Several recently initiated studies investigate flaxseed supplements, either as a chemopreventive agent for women at high risk of developing breast cancer or as a complementary therapy to surgery in women newly diagnosed with stage I or II breast cancer. Although the rationale of these studies is to investigate phytoestrogenic effects of flaxseed, the results may also be informative for the effects of ω-3 PUFAs on breast cancer. The prevention trial NCT00794989, initiated in 2008, is a two-part study to determine the effect of flaxseed in preventing breast cancer in premenopausal women at increased risk of developing primary breast cancer. The first part of the study is an open-label trial to develop consistent methods for collecting samples obtained by random periareolar fine needle aspiration (RPFNA) and for performing immunohistochemistry or quantitative real-time PCR. In the second part, patients will be randomized to one of two arms: an intervention arm with consumption of ground flaxseed daily with prepared food and an observation arm. One of the primary objectives is to determine if consumption of ground flaxseed for 6 months will prevent the development of primary breast cancer. Other primary objectives include investigations on the effects of flaxseed on intermediate biomarkers of proliferation (Ki-67) and apoptotic rates (activated caspase-3) as well as estrogen-regulated genes (cyclin D1, survivin, and vascular endothelial growth factor [VEGF]) from baseline to 6 months. In addition, changes in serum IGF-1 and IGFBP-3 levels from baseline to 6 months will be evaluated. A secondary objective of this study is to evaluate the feasibility and tolerance of long-term flaxseed consumption and determine factors that lead to decreased compliance. The estimated enrollment is 60 patients, and the estimated primary completion date is October 2012.

Another trial from the same institution (NCT00612560), initiated in 2007, is a placebo-controlled randomized study to examine the effect of flaxseed consumption compared to aromatase inhibitors (anastrozole), as well as a flaxseed and aromatase inhibitor combination as a complementary approach to treating estrogen receptor-positive breast cancer. This study, performed in a pre-surgical setting (with the dietary intervention performed from biopsy to tumor resection), has a planned enrollment of 100 patients. Four arms will be compared: flaxseed (25 mg per day), anastrozole, flaxseed plus anastrozole, and placebo. Patients eligible for the study must be postmenopausal and newly diagnosed with incident, primary, invasive, estrogen receptor-positive clinical stage I or II breast cancer, with at least 2 weeks between the pre-surgical visit and the surgery. The study will assess the effects of flaxseed and/or anastrozole on steroid hormone and tumor-related characteristics associated with long-term survival and to investigate the potential interaction between flaxseed and anastrozole on tumor expression of Ki-67, caspase, estrogen receptors (ER) α and β, progesterone receptor (PR), human epidermal growth factor receptor 2 (HER2), IGF-1, and IGFIR. Recurrence scores are as estimated by the Mammostrat antibody panel (Applied Genomics Incorporated). A third study from this institution listed on clinicaltrials.gov (NCT00635908) appears to have similar outcomes and inclusion criteria and may represent the same study.

The breast cancer preventive role of ω-3 PUFAs in combination with the antiestrogen raloxifene is under investigation in the NCT00723398 trial, initiated in 2008. This trial is based on the hypothesis that the combination of a low dose of raloxifene with ω-3 fatty acids will exert a synergistic chemopreventive effect on decreased proliferation and increased apoptosis of premalignant mammary cells. This study, with a proposed enrollment of 372 women, will have five arms (control, 60 mg raloxifene, 30 mg raloxifene, 4 g lovaza – an ω-3 dietary supplement, 4 g lovaza plus 30 mg raloxifene). Raloxifene and lovaza will be administered daily for 2 years. The primary objective of this study is to evaluate breast density at baseline and every 6 months for 2 years, with secondary outcomes including biomarkers of oxidative stress (urinary 8-(isoprostane) F-2α and 8OHdG, lymphocyte 8-OHdG, DNA etheno adducts), estrogen metabolites (urinary 2-OHE1, 4-OHE1, and 16α-OHE1), C-reactive protein, IL-6, IGF-1, IGFBP-3, lipid panel, and complete blood count. This study is not yet open for participant recruitment and has an estimated completion date of October 2012.

6.4 Nutritional Support and Immunomodulation of Cancer Patients with ω-3 PUFAs

Although not specifically directed toward prostate or breast cancer patients, several trials address the potential of nutritional supplements containing ω-3 PUFAs to reduce cancer-associated weight loss (cachexia) for patients with advanced malignancies. Initial clinical trials suggested that ω-3 PUFAs could stabilize weight loss or lead to weight gain in advanced cancer patients with cachexia [19–22]. However, the results of larger trials have been inconsistent, due in part to relatively poor tolerance of these dietary supplements [23–26]. A review of randomized controlled trials assessing oral EPA compared with placebo or control in patients with advanced cancer failed to find sufficient evidence to support the use of oral EPA in treatment of cancer cachexia [27]. However, another meta-analysis of clinical trials or prospective observational studies with enteral supplements of ω-3 PUFAs concluded that the prolonged administration of EPA and DHA to patients with advanced cancer in doses of at least 1.5 g/day was associated with an improvement in clinical, biological, and quality of life parameters [28]. The benefits of ω-3 PUFA supplementation may therefore be only realized when patients are able to consume high doses of the dietary supplement, which is limited by gastrointestinal side effects [24, 26]. Synthetic COX-2 inhibitors such as celecoxib may also be beneficial for cachexic cancer patients, alone or in combination with ω-3 PUFAs [29, 30]. Several ongoing and recent trials are listed in the clinical trial database, which for the most part focus on patients with colorectal cancer or hematologic malignancies.

Nutritional supplements enriched in ω-3 PUFAs have also been evaluated in clinical studies for their ability to improve the outcome of other cancer treatments. The current standard treatment of advanced prostate cancer is hormonal ablation of the gonadotropin–luteinizing hormone–testosterone axis. One prominent side effect of this approach is osteoporosis. Higher ω-3 to -6 PUFA ratios have been linked to increased bone mineral density in older adults [31]. This suggests that ω-3 PUFA supplementation may have a salutatory affect on bone density in patients treated with hormonal ablation. Breast cancer patients receiving DHA supplements during anthracyclin chemotherapy experienced improved time to progression, overall survival, and tolerance of side effects; this effect was most pronounced in patients with high incorporation of DHA in the plasma [32]. As ω-3 PUFAs are able to reduce infection and inflammation, their possible immunomodulatory effects have received particular attention [33–35]. Administration of ω-3 PUFAs either before or after major abdominal surgery for cancer was shown to reduce inflammatory cytokines [34, 36] and improve liver and pancreas function [37, 38]. In addition, patients receiving parenteral immunonutrition with a nutritional supplement containing glutamine or ω-3 PUFAs had reduced incidence of infectious complications and improved immune function compared to patients who received a nutritional supplement without glutamine or ω-3 PUFAs [33].

6.5 Conclusions

Our search for clinical trials on the prevention or treatment of breast and prostate cancer in the ClinicalTrials.gov database returned seven relevant trials for each of the two cancer types (Tables 6.1 and 6.2). In most studies, dietary PUFAs were administered as part of a reduced-fat diet. Taking the apparently duplicate studies into account, the prostate cancer studies included three studies using fish oil, one study using flaxseed, and two studies using PUFAs of unspecified origin. Of these studies, three focused on ω-3 PUFAs alone, whereas the other three included them with other dietary supplements (lycopene, phytochemicals, or green tea extracts), alone or in combination. Few reports have yet been published on the results of the above clinical studies. One study showed that flaxseed, administered to prostate cancer patients prior to surgery as a supplement to a low-fat diet, decreased the proliferation of prostate cancer cells as determined by Ki-67 staining. Most of the prostate cancer studies included a relatively short period of administration of the dietary supplements, ranging from 3 to 20 weeks, with only one study offering an extended follow-up of 2 years after treatment. Thus, the information that can be expected from these short-term studies relates primarily to acute changes in intermediate biomarkers of prostate cancer. For the breast cancer studies, three studies utilized ground flaxseed and the other three used purified marine ω-3 PUFA supplements. Again, three studies focused exclusively on flaxseed or specific ω-3 PUFAs, whereas the other three also included other treatments (macrobiotic or American Heart Association Step 1–2 diet, aromatase inhibitors, or antiestrogens). The average duration of the dietary interventions in breast cancer trials was longer than that for prostate cancer, with two short-term studies (2 and 8 weeks, respectively) and four longer studies ranging from 6 months to 2 years. Additional studies investigate the use of ω-3 PUFA supplements for the nutritional support of advanced cancer patients or to improve the outcome of other therapies, with some apparent benefits when high doses of ω-3 PUFAs are tolerated. Further studies will be required in this area to confirm the usefulness of this approach.

Many patients faced with a diagnosis of cancer turn to complementary and alternative medicines, and ω-3 PUFAs are among the most frequently used nutritional supplements on the market. The marine ω-3 PUFAs in particular are attractive dietary supplements, as they are beneficial for cardiovascular health, reduce inflammatory and immune diseases, and could help prevent or slow the progression of cancer. Therefore, determining if these fatty acids are safe and have demonstrable efficacy for cancer treatment in the clinical setting is of timely importance. One can only hope that the results of the ongoing and closed studies will quickly be made available to the scientific community for evaluation. Trial reports will be very informative to help in the design of future studies. In addition, greater visibility of clinical studies on dietary supplements, in part via registration to public databases, will improve communications between research groups and between the scientific community and the public.

References

1. Simopoulos AP. Essential fatty acids in health and chronic disease. Am J Clin Nutr 1999;70 (3 Suppl.):560S–9S.
2. Simopoulos AP. The importance of the ratio of omega-6/omega-3 essential fatty acids. Biomed Pharmacother 2002;56(8):365–79.
3. Berquin IM, Edwards IJ, Chen YQ. Multi-targeted therapy of cancer by omega-3 fatty acids. Cancer Lett 2008;269(2):363–77.
4. Demark-Wahnefried W, Robertson CN, Walther PJ, Polascik TJ, Paulson DF, Vollmer RT. Pilot study to explore effects of low-fat, flaxseed-supplemented diet on proliferation of benign prostatic epithelium and prostate-specific antigen. Urology 2004;63(5):900–4.

5. Demark-Wahnefried W, Polascik TJ, George SL, et al. Flaxseed supplementation (not dietary fat restriction) reduces prostate cancer proliferation rates in men presurgery. Cancer Epidemiol Biomarkers Prev 2008;17(12):3577–87.
6. Simon JA, Chen YH, Bent S. The relation of alpha-linolenic acid to the risk of prostate cancer: A systematic review and meta-analysis. Am J Clin Nutr 2009;89(5):1558S–64S.
7. Denis L, Morton MS, Griffiths K. Diet and its preventive role in prostatic disease. Eur Urol 1999;35(5–6):377–87.
8. Evans BA, Griffiths K, Morton MS. Inhibition of 5 alpha-reductase in genital skin fibroblasts and prostate tissue by dietary lignans and isoflavonoids. J Endocrinol 1995;147(2):295–302.
9. McCann MJ, Gill CI, McGlynn H, Rowland IR. Role of mammalian lignans in the prevention and treatment of prostate cancer. Nutr Cancer 2005;52(1):1–14.
10. Aronson WJ, Glaspy JA, Reddy ST, Reese D, Heber D, Bagga D. Modulation of omega-3/omega-6 polyunsaturated ratios with dietary fish oils in men with prostate cancer. Urology 2001;58(2):283–8.
11. Kushi LH, Cunningham JE, Hebert JR, Lerman RH, Bandera EV, Teas J. The macrobiotic diet in cancer. J Nutr 2001;131(11 Suppl.):3056S–64S.
12. Kwan ML, Ambrosone CB, Lee MM, et al. The Pathways Study: A prospective study of breast cancer survivorship within Kaiser Permanente Northern California. Cancer Causes Control 2008;19(10):1065–76.
13. Greenlee H, Kwan ML, Ergas IJ, et al. Complementary and alternative therapy use before and after breast cancer diagnosis: The pathways study. Breast Cancer Res Treat 2009 117(3): 653–65.
14. Boon H, Westlake K, Stewart M, et al. Use of complementary/alternative medicine by men diagnosed with prostate cancer: Prevalence and characteristics. Urology 2003;62(5):849–53.
15. Chan JM, Elkin EP, Silva SJ, Broering JM, Latini DM, Carroll PR. Total and specific complementary and alternative medicine use in a large cohort of men with prostate cancer. Urology 2005;66(6):1223–8.
16. Lengacher CA, Bennett MP, Kip KE, et al. Frequency of use of complementary and alternative medicine in women with breast cancer. Oncol Nurs Forum 2002;29(10):1445–52.
17. Politi MC, Rabin C, Pinto B. Biologically based complementary and alternative medicine use among breast cancer survivors: relationship to dietary fat consumption and exercise. Support Care Cancer 2006;14(10):1064–9.
18. Tagliaferri M, Cohen I, Tripathy D. Complementary and alternative medicine in early-stage breast cancer. Semin Oncol 2001;28(1):121–34.
19. Wigmore SJ, Barber MD, Ross JA, Tisdale MJ, Fearon KC. Effect of oral eicosapentaenoic acid on weight loss in patients with pancreatic cancer. Nutr Cancer 2000;36(2):177–84.
20. Gogos CA, Ginopoulos P, Salsa B, Apostolidou E, Zoumbos NC, Kalfarentzos F. Dietary omega-3 polyunsaturated fatty acids plus vitamin E restore immunodeficiency and prolong survival for severely ill patients with generalized malignancy: A randomized control trial. Cancer 1998;82(2):395–402.
21. Burns CP, Halabi S, Clamon GH, et al. Phase I clinical study of fish oil fatty acid capsules for patients with cancer cachexia: Cancer and leukemia group B study 9473. Clin Cancer Res 1999;5(12):3942–7.
22. Mantovani G, Maccio A, Madeddu C, et al. A phase II study with antioxidants, both in the diet and supplemented, pharmaconutritional support, progestagen, and anti-cyclooxygenase-2 showing efficacy and safety in patients with cancer-related anorexia/cachexia and oxidative stress. Cancer Epidemiol Biomarkers Prev 2006;15(5):1030–4.
23. Bruera E, Strasser F, Palmer JL, et al. Effect of fish oil on appetite and other symptoms in patients with advanced cancer and anorexia/cachexia: A double-blind, placebo-controlled study. J Clin Oncol 2003;21(1):129–34.
24. Fearon KC, Von Meyenfeldt MF, Moses AG, et al. Effect of a protein and energy dense N-3 fatty acid enriched oral supplement on loss of weight and lean tissue in cancer cachexia: A randomised double blind trial. Gut 2003;52(10):1479–86.

25. Jatoi A, Rowland K, Loprinzi CL, et al. An eicosapentaenoic acid supplement versus megestrol acetate versus both for patients with cancer-associated wasting: A North Central Cancer Treatment Group and National Cancer Institute of Canada collaborative effort. J Clin Oncol 2004;22(12):2469–76.
26. Burns CP, Halabi S, Clamon G, et al. Phase II study of high-dose fish oil capsules for patients with cancer-related cachexia. Cancer 2004;101(2):370–8.
27. Dewey A, Baughan C, Dean T, Higgins B, Johnson I. Eicosapentaenoic acid (EPA, an omega-3 fatty acid from fish oils) for the treatment of cancer cachexia. Cochrane Database Syst Rev 2007; Jan 24(1):CD004597.
28. Colomer R, Moreno-Nogueira JM, Garcia-Luna PP, et al. N-3 fatty acids, cancer and cachexia: a systematic review of the literature. Br J Nutr 2007;97(5):823–31.
29. Cerchietti LC, Navigante AH, Castro MA. Effects of eicosapentaenoic and docosahexaenoic n-3 fatty acids from fish oil and preferential Cox-2 inhibition on systemic syndromes in patients with advanced lung cancer. Nutr Cancer 2007;59(1):14–20.
30. Lai V, George J, Richey L, et al. Results of a pilot study of the effects of celecoxib on cancer cachexia in patients with cancer of the head, neck, and gastrointestinal tract. Head Neck 2008;30(1):67–74.
31. Weiss LA, Barrett-Connor E, von Muhlen D. Ratio of n-6 to n-3 fatty acids and bone mineral density in older adults: the Rancho Bernardo Study. Am J Clin Nutr 2005;81(4):934–8.
32. Bougnoux P, Hajjaju N, Baucher MA, et al. Docosahexaenoic acid (DHA) intake during first line chemotherapy improves survival in metastatic breast cancer. Proc Am Assoc Cancer Res 2006;47:1237 (abstract #5276).
33. Klek S, Kulig J, Szczepanik AM, Jedrys J, Kolodziejczyk P. The clinical value of parenteral immunonutrition in surgical patients. Acta Chir Belg 2005;105(2):175–9.
34. Nakamura K, Kariyazono H, Komokata T, Hamada N, Sakata R, Yamada K. Influence of preoperative administration of omega-3 fatty acid-enriched supplement on inflammatory and immune responses in patients undergoing major surgery for cancer. Nutrition 2005;21(6):639–49.
35. Zheng Y, Li F, Qi B, et al. Application of perioperative immunonutrition for gastrointestinal surgery: a meta-analysis of randomized controlled trials. Asia Pac J Clin Nutr 2007;16(1 Suppl.):253–7.
36. Aiko S, Yoshizumi Y, Tsuwano S, Shimanouchi M, Sugiura Y, Maehara T. The effects of immediate enteral feeding with a formula containing high levels of omega-3 fatty acids in patients after surgery for esophageal cancer. JPEN J Parenter Enteral Nutr 2005;29(3):141–7.
37. Heller AR, Rossel T, Gottschlich B, et al. Omega-3 fatty acids improve liver and pancreas function in postoperative cancer patients. Int J Cancer 2004;111(4):611–6.
38. Stehr SN, Heller AR. Omega-3 fatty acid effects on biochemical indices following cancer surgery. Clin Chim Acta 2006;373(1–2):1–8.

Chapter 7
ω-3 PUFAs, Breast and Prostate Cancer: Experimental Studies

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Abstract Although human epidemiological and clinical studies to date have failed to provide conclusive data on a protective effect of ω-3 polyunsaturated fatty acids (PUFAs) on breast and prostate cancer, cell culture and animal studies present a more positive story. Experimental models investigated include various human cancer cell lines, rats with chemically induced tumors, mice with transplantable and human xenograft tumors, and, more recently, transgenic models. They have suggested a number of biological targets for ω-3 PUFAs that impact cell proliferation, survival, apoptosis, angiogenesis, invasiveness, and metastasis, i.e., ω-3 PUFAs may have multifactorial properties in preventing and inhibiting cancer. These models have uncovered numerous mechanisms for the anti-cancer activity of ω-3 PUFAs with the most frequently cited being their ability to block the metabolism of ω-6 PUFAs into agents that promote many facets of the malignant behavior of cancer cells.

Keywords Animal models · Breast cancer · Eicosanoids · ω-3 fatty acids · Prostate cancer

Abbreviations

ALA	α-linolenic acid
AA	arachidonic acid
COX	cyclooxygenase
$cPLA_2$	cytosolic phospholipase A_2
DHA	docosahexaenoic acid
DMBA	dimethylbenz(a)anthracene
EPA	eicosapentaenoic acid
ELOVL	elongase
HER2	human epidermal growth factor receptor 2

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LA	linoleic acid
LDL	low-density lipoproteins
LOX	lipoxygenase
PSA	prostate-specific antigen
PTEN	phosphatase and tensin homolog deleted on chromosome 10
PUFA	polyunsaturated fatty acid
RBC	red blood cells
SNIP	single-nucleotide polymorphism
VEGF	vascular endothelial growth factor

7.1 Introduction

A significant body of evidence compiled over 50 years has supported a role for high-fat diets in promoting cancer. For the past 25 years, experimental studies have addressed the effects of specific dietary fatty acids. A primary interest has been on a tumor-protective role for ω-3 polyunsaturated fatty acids (PUFAs) versus a tumor-promoting role for ω-6 PUFAs. At this time, a large body of evidence indicates that dietary manipulation of tissue PUFAs in experimental models has a powerful impact on the progression of both breast and prostate cancer but a clear understanding of the mechanisms responsible for this is still to emerge. However, one particularly attractive and concise mechanism, supported by a broad body of experimental data, holds that these cancers metabolize ω-6 PUFAs into cancer-promoting products while ω-3 PUFAs inhibit this metabolism.

7.2 Fatty Acid Sources and Metabolism

The hydrocarbon chains of fatty acids can be saturated (no double bonds), mono-unsaturated (one double bond), or polyunsaturated (multiple double bonds). Saturated and monounsaturated fatty acids are synthesized de novo in mammals or may be obtained from diet. Due to the lack of the required desaturase enzymes, mammals cannot synthesize PUFAs and they must be obtained from dietary sources. The ω-3 or ω-6 designation is based on the position of the double bond closest to the ω carbon at the methyl end of the chain. Although shorter chain PUFAs can be metabolized to longer chain PUFAs, there is no conversion in mammalian cells between ω-6 and ω-3 species of PUFAs. The shortest ω-6 PUFA, linoleic acid (LA, 18:2, ω-6), is the most abundant source of dietary PUFAs in most human societies and it can be converted to the longer chain arachidonic acid (AA, 20:4, ω-6) through a series of elongation and desaturase reactions, as depicted in Fig. 7.1 [1]. The major dietary source of LA is plants, and it is abundant in oils such as corn, safflower, sunflower, and olive oil. Likewise the shortest ω-3 PUFA, α-linolenic acid (ALA, 18:2, ω-3), is also a plant product found in leafy vegetables such as kale, broccoli, spinach, and Brussels sprouts as well as in walnuts and seeds such as flax and mustard. ALA can be converted to eicosapentaenoic acid (EPA, 20:5, ω-3) and docosahexaenoic

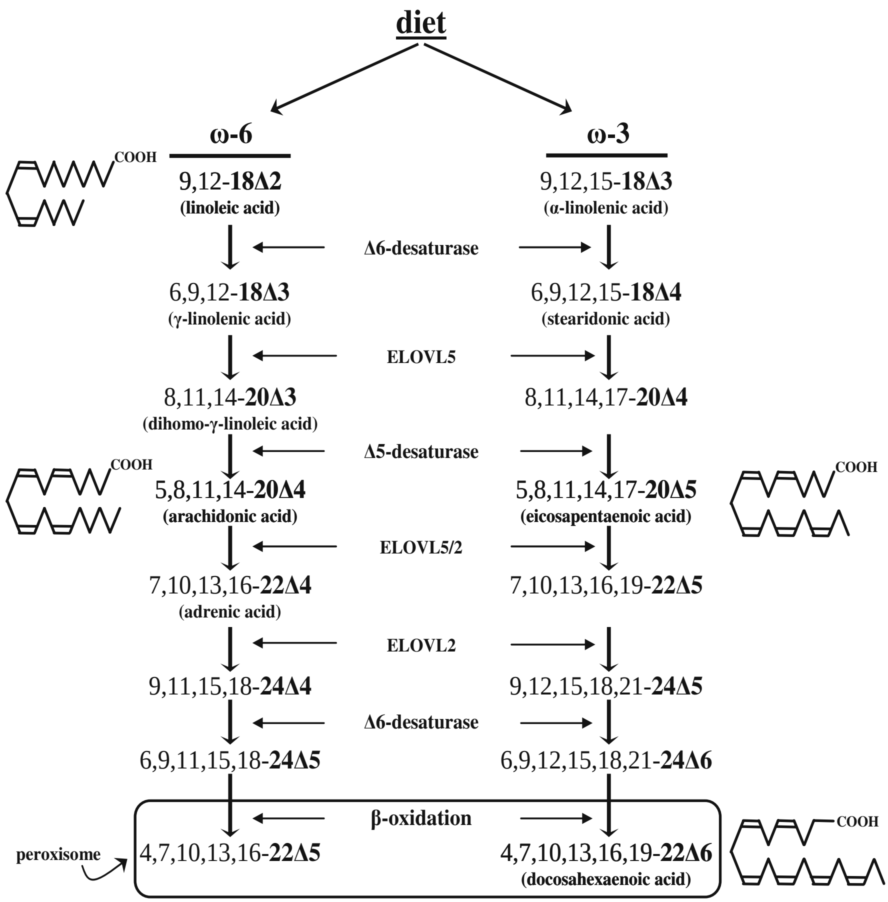

Fig. 7.1 PUFA elongation and desaturation. Short-chain ω-6 and ω-3 PUFAs are lengthened and further desaturated by a common set of elongases (ELOVL) and desaturases. C24 PUFA is converted to C22 PUFA by β-oxidation in peroxisomes. Also shown are the structures of LA and AA, which are metabolized to pro-malignant factors, and EPA and DHA, which bear much of the anti-malignant activity of ω-3 PUFAs

acid (DHA, 22:6, ω-3). However, this conversion is extremely inefficient in humans and the predominant source of EPA and DHA is from dietary intake of fish that are rich in these exact PUFAs [2].

7.3 PUFA Studies in Animals

7.3.1 PUFAs in Animal Breast Cancer

Animal studies provide convincing evidence of a negative relationship with ω-3 PUFA diets and a positive relationship with ω-6 PUFA diets for breast cancer. In studies of breast cancer induced by chemical carcinogens in rats [3–7] and human cancer cell xenografts in nude mice [8–10], tumor growth rate, size, and metastases were all suppressed by dietary ω-3 PUFA supplementation. In one of the earliest studies, Jurkowski and Cave [4] examined the effect of quantitative differences in dietary menhaden oil (i.e., oil from any genus of menhaden ocean fish and containing varying amounts of EPA and DHA) on mammary tumor development in female rats treated with the carcinogen *N*-methyl-*N*-nitrosourea. Animals fed menhaden oil demonstrated a dose-dependent decrease in tumor incidence, tumor number, and tumor size compared to those fed a similar percent of corn oil. Similar results were reported with 7,12-dimethylbenz(a)anthracene (DMBA)-induced tumors in rats with menhaden oil that was reported to contain 17% EPA and 9% DHA [5]. An alternative ω-3 PUFA-enriched oil, MaxEPA that contained approximately 18% EPA and 17% DHA, was also shown to inhibit R3230AC (rat) mammary tumor growth in rats [11]. Because these fish oil preparations had only 2% LA, it was suggested that the suppression of mammary tumor growth by fish oils in the rodent models was due to a reduction in this essential fatty acid. In support of this, mice fed a hydrogenated cotton seed diet that contained only 10% LA and no marine fatty acids showed lower growth of transplantable adenocarcinomas compared to those fed a corn oil diet containing 60% LA [12]. However, when a high menhaden oil diet was supplemented with 5% corn oil in a mouse xenograft model, a significant inhibitory effect on tumor growth was observed in menhaden + corn oil compared to the control corn oil group [13]. Thus, ω-6 PUFAs apparently promote breast cancer while ω-3-PUFAs oppose this promotion.

During the 1990s, human breast cancer cells growing as xenografts in athymic nude mice provided an important model in the study of dietary fatty acids and cancer. MDA-MB-435 cells were injected into the mammary fat pads of mice consuming diets enriched in specific fatty acid ethyl esters rather than fish oil preparations [8, 10, 14]. This allowed the potential resolution of whether the tumor-inhibitory properties of fish oil were due to its EPA, DHA, or some other component. EPA at 8% and DHA at 4 or 8% of a diet that contained 20% fat were shown to be effective inhibitors of xenograft growth [10]. In addition to the studies with marine PUFAs, the xenograft model was used by Chen et al. [15] to demonstrate a significant reduction in tumor growth when mice were fed an ALA-enriched diet.

In addition to focusing on primary tumors of the breast, animal studies have also examined the impact of PUFAs on metastasis. An important property of MDA-MB-435 xenografts for the study of tumor-inhibitory agents is their ability to form solid tumor metastases. The studies of Rose et al. [9, 10, 14] showed that in a 20% fat diet containing as little as 4% of the test PUFAs, EPA and DHA were more effective than LA in inhibiting the incidence and severity of lung metastases. Further studies by this group examined the use of EPA and DHA as neoadjuvant or post-operative adjuvant therapy after surgical removal of the primary tumor [14]. Female nude mice received a 20% fat, 8% LA diet to promote xenograft development and 7 days prior to surgery either continued on the LA diet or switched to one in which LA was replaced by EPA or DHA. Not only did the ω-3 PUFA diets reduce the growth of the primary tumors, but incidence, number, and volume of metastases were significantly lower in the animals receiving 8% EPA or DHA compared to those remaining on the LA diet. Interestingly, analysis of the tumor phospholipids showed that 7 days of feeding the ω-3 fatty acid diets was sufficient to produce significant increases in EPA and DHA content and corresponding decreases in AA in tumor tissue. In the post-operative adjuvant groups, ω-3 PUFA diets were initiated and fed for 7 days post-excision. In these animals the effect of EPA was less pronounced, but DHA inhibition of metastasis was comparable to that seen in the neoadjuvant group. Like the marine fatty acids, ALA was also shown to inhibit metastasis when a 7 week feeding of dietary flaxseed resulted in a 45% reduction in total incidence of breast tumor metastasis to lungs and lymph nodes [15].

In spite of the widely reported effects of ω-3 PUFAs on immune function [16], studies suggest that suppression of tumor growth and metastasis by ω-3 PUFAs does not rely on alterations in host immune responses. First, the athymic mice used in so many studies, showing the anti-cancer effects of ω-3 PUFAs, have B cells that generate antibodies to support humoral responses but lack T cells [17] and murine NK cell activity was shown to be insensitive to dietary LA [18]. Second, different strains of mice with varying levels of immune competency were similarly sensitive to the inhibitory effects of dietary ω-3 PUFAs on breast xenograft growth [19]. Third, the effects of EPA and DHA ethyl esters on breast cancer in immunocompetent animals were studied in rats with DMBA-induced tumors [20]. In this model, DHA reduced tumor incidence by 69% and EPA by 47%. These studies imply that the inhibitory effects of ω-3 PUFAs are directed to the tumor cells themselves, a conclusion amply supported by studies of cancer cell explants containing the *fat-1* gene (Section 7.3.2) and the in vitro studies reviewed in Section 7.4.

More recent studies have taken advantage of transgenic mouse models of breast cancer to measure the effects of ω-3 PUFAs on spontaneous breast cancer development. HER-2/neu is a cell surface receptor tyrosine kinase that is overexpressed in 15–40% of aggressive human breast cancers [21] and is associated with a poor prognosis [22]. Mice bearing the activated neu protooncogene in mammary epithelium provide important models of spontaneous tumor development in an immunocompetent host. Female murine mammary tumor virus (MMTV)-HER-2/neu transgenic mice [23] fed a high ω-3 PUFA diet (ω-6 to ω-3 ratio of 0.16) developed mammary tumors 15 weeks later than those fed a high ω-6 PUFA diet (ω-6 to ω-3 ratio of 35.56) [24]. In similar (MMTV)-HER-2/neu transgenic mice [25], a fish oil diet compared to a corn oil diet increased the latency time to mammary tumor development, reduced the number of tumors, and was associated with a lower grade of mammary gland histopathology [26]. These studies highlight the ability of ω-3 PUFAs to suppress tumor growth in a mouse model driven by a tumor promoting gene, HER-2/neu, involved in human breast cancers.

7.3.2 PUFAs in Animal Prostate Cancer

In vivo studies addressing a tumor-suppressive role for ω-3 PUFAs in prostate cancer models have been sparse but the results are consistent with those in breast cancer.

The androgen-independent human prostate cancer cell line, DU145, was used as a xenograft in athymic nude mice fed fish oil- or corn oil-based diets [27, 28]. In both studies, tumor growth was significantly retarded in the fish oil group. By contrast, the shorter chain ω-3 PUFA, ALA, was not effective in suppressing xenograft growth in this model [29]. The growth of an androgen-sensitive cell line, LAPC-4, injected subcutaneously into severe combined immunodeficiency mice was inhibited by a diet with ω-3 PUFAs (from fish oil) to ω-6 PUFAs (from corn oil) of 1:1 compared to a diet with a ω-6:ω-3 ratio of 26:1 [30]. In that study, serum prostate-specific antigen (PSA) levels (which index tumor load) were reduced in the ω-3 compared to ω-6 PUFA-fed group. Using this same model, Kelavkar et al. [31] demonstrated that when animals were fed a diet enriched in stearidonic acid (SDA, 18:4, ω-3) or LA following surgical removal of the primary tumor, tumor recurrence was reduced by 50%, and size of recurrent tumors was eight-fold smaller in the SDA compared to LA diet groups. PSA was again reduced in the ω-3 diet group.

Androgen ablation therapy is typically used to halt the progression of prostate cancer due to hormone dependence of primary tumors. However, tumors commonly convert to castration-resistant and then continue to progress in the hormone-depleted environment. The time to this conversion may take many years in humans, thus providing an important window for dietary intervention. McEntee et al. [32], using castration-sensitive or castration-resistant human prostatic xenografts in castrated, testosterone-supplemented athymic nude mice, showed that a diet enriched in EPA significantly enhanced the sensitivity of castration-sensitive prostate cancer to androgen ablation therapy and delayed the progression to castration-resistant tumors.

The *PTEN* (phosphatase and tensin homolog deleted on chromosome 10) tumor suppressor gene is one of the most frequently altered genes in human prostate cancer with 30% of primary [33] and 63% of metastatic [34] prostate tissues showing *PTEN* deletions or mutations. A prostate-specific *PTEN* knock-out mouse model has been developed in which 100% of males exhibit tumors in a progression that mimics human prostate cancer [35]. Like the HER-2/neu transgenic mouse model for breast cancer, the prostate-specific *PTEN* mouse has advantages of a defined and relevant genetic alteration, an intact immune system, and an orthotopic site of tumor development. We used *PTEN* knock-out mice to demonstrate that a dietary ratio of ω-6 to ω-3 PUFAs lower than 5 was effective in suppressing tumor growth, retarding histopathological progression, and extending animal lifespan [36]. A second transgenic mouse containing the *fat-1* transgene encoding an ω-3 desaturase from *Caenorhabditis elegans* that converts ω-6 into ω-3 fatty acids [37] was bred with the *PTEN* knock-out mice. $PTEN^{-/-}$ mice with and without the *fat-1* transgene were fed an ω-6 PUFA-enriched diet for 8 weeks. This resulted in an ω-6/ω-3 PUFAs ratio of 4.4 in prostate tissue of the *PTEN* knock-out, *fat-1* transgenic animals ($PTEN^{-/-}$, $fat\text{-}1^{T/-}$) compared to 45.9 in $PTEN^{-/-}$ animals without the transgene. This ratio in the $PTEN^{-/-}$, $fat\text{-}1^{T/-}$ mice was similar to that achieved in $PTEN^{-/-}$, $fat\text{-}1^{-/-}$ mice by an ω-3-enriched diet. Prostate tumor weights were significantly lower in the $PTEN^{-/-}$, $fat\text{-}1^{T/-}$ than in $PTEN^{-/-}$, $fat\text{-}1^{-/-}$ mice [36]. Thus this genetic approach was important in confirming the tumor-protective role for ω-3 PUFAs that had been suggested by the dietary studies. In a recent study, the *fat-1* transgene was expressed in PC3 and DU145 human prostate cancer cells and the transfected cells were used as xenografts in severe combined immunodeficient mice [38]. Expression of the *fat-1* gene markedly reduced the xenograft growth. This result indicates that it is the PUFA levels within the cancer cells rather than elsewhere (such as within inflammatory cells) that are cancer-regulatory, although one cannot exclude the possibility that manipulating PUFAs in immune cells might have an effect as well.

7.4 Mechanisms for the Anti-cancer Activity of PUFAs

7.4.1 PUFA Mechanisms in Breast Cancer

Insight into the mechanisms responsible for the anti-breast cancer properties of ω-3 PUFAs has been provided by in vitro investigations using cultured human cancer cell lines. As reviewed recently [39, 40], at least seven mechanisms have been proposed. PUFAs (1) decrease the activation and thereby cell survival function of NFκB; (2) increase the levels, and thereby support the tumor suppressor action of PKCδ or λ/ξ; (3) decrease membrane-associated PKCβ2 to reduce its cell-proliferating effect; and (4) suppress nitric oxide production to inhibit tumor invasion and vascularization. In addition and as reviewed by Welsch [41], (5) PUFAs in cell membranes increase the susceptibility of these membranes to peroxidation with the propensity to peroxidation higher in longer chain PUFAs. In a xenograft mouse model of breast cancer, tumor growth inhibition by fish oil diets led to a dose-related accumulation of lipid peroxidation products (TBARS) in the tumors. Supplementation of the fish oil with large amounts of antioxidants reduced the TBARS and reversed the fish oil inhibition of tumor growth [42].

Studies also implicate (6) PPARγ in the effects of ω-3 PUFAs in cancer cells. Ligand activation of the PPARγ transcription factor was shown to induce a more differentiated, less malignant state in cultured breast cancer cells [43] and, in accordance with this action, PPARγ agonists were shown to reduce carcinogen-induced mammary tumors in rats [44, 45] and mice [46]. While in HER-2/neu mice, a fish oil diet suppressed breast cancer, a PPARγ activator of the glitazone family, rosiglitazone, did not and there was no interaction between the diet and glitazone [26]. This result suggests that ω-3 PUFAs do not suppress this cancer via PPARγ. However, DHA is a known activator of PPARγ and a more effective one than synthetic glitazones [47, 48]. Moreover, our in vitro studies with human breast cancer cells have shown DHA to be particularly active in inhibiting cell growth and evoking apoptosis in a pathway involving PPARγ activation [49–51]. We found that the molecular target for both DHA and PPARγ in these cells was the heparan sulfate proteoglycan, syndecan-1. Syndecan-1 itself was effective in apoptosis induction and when syndecan-1 was silenced, the ability of DHA to induce apoptosis was lost, as it was in the presence of dominant-negative PPARγ [50]. siRNA for syndecan-1 also blocked troglitazone-induced apoptosis. Thus, a novel pathway linking ω-3 PUFAs to apoptosis in tumor cells is as follows: DHA activates PPARγ, which results in transcriptional up-regulation of the syndecan-1 target gene, and the syndecan-1 protein induces apoptosis. This pathway has been confirmed in human prostate cancer cells [52]. Interestingly, PPARγ was not a target for EPA in either breast or prostate cancer cells. Although there are conflicting reports of a role for syndecan-1 in cancer, the importance of these studies is the identification of a PPARγ molecular target that is regulated by an ω-3 PUFA that results in a functional response in the tumor cells. It will be important to conduct further experiments to determine if these in vitro studies translate to animals and humans.

The most frequently cited mechanism for the anti-cancer activity of ω-3 PUFAs, however, is (7) their ability to block the metabolism of AA and LA into agents that promote the malignant phenotype of cancer cells [39, 40, 53–56]. As shown in Fig. 7.2, ω-3 PUFAs tie up the desaturating and elongating enzymes that convert shorter chain ω-6 PUFAs to AA; displace LA and AA from their storage sites in phospholipids; compete with ω-6 PUFAs for the oxygenases; and, when acylated into phospholipids in place of LA and AA, inhibit cytosolic phospholipase A_2 ($cPLA_2$), thereby inhibiting the release of all PUFAs but in particular ω-6 PUFAs from storage. ω-3 PUFAs thus reduce the ability of cells, including breast cancer cells [53], to make, mobilize, and metabolize LA and AA. The consequential decrease in ω-6 PUFA metabolites is particularly relevant to cancer.

PUFAs are metabolized by various oxygenases to products that modulate many cell processes including proliferation, differentiation, survival, apoptosis, angiogenesis, invasiveness, and metastasis. In particular, 6-series PUFAs are metabolized

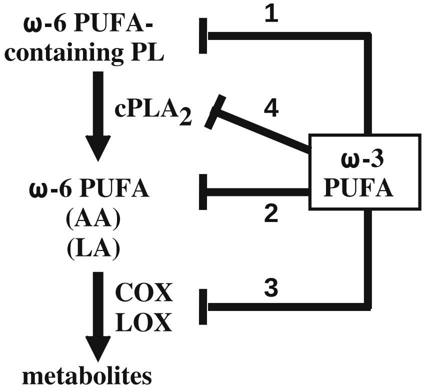

Fig. 7.2 Inhibition of ω-6 PUFA metabolism and activity by ω-3 PUFA. (1) Shorter chain ω-3 PUFAs compete with the ω-6 PUFAs for elongases and desaturases (Fig. 7.1) to reduce the conversion of dietary LA to AA and thereby the amounts of AA-containing phospholipids. (2) ω-3 PUFAs displace ω-6 PUFAs from phospholipids. (3) The ω-3 PUFAs, EPA and DHA, compete with AA and LA as substrates for the oxygenases to form less active or inactive analogs of ω-6 PUFA metabolites. (4) EPA and to a greater extent DHA, when acylated into phospholipids, inhibit $cPLA_2$

to products that promote these processes while their 3-series counterparts inhibit this metabolism, being metabolized by the same oxygenases to products with little or no cancer-promoting activity [7, 57–60]. In other words, COX and LOX utilization of ω-3 PUFAs results in a reduction in the highly reactive products generated from 6-series PUFAs [61, 62] in favor of less active or inactive 3-series products [63]. Table 7.1 lists the human and orthologous murine oxygenases and accompanying products that have been repeatedly implicated in breast and prostate cancer promotion as well as the preferences of these oxygenases for LA versus AA. All of the oxygenases can metabolize EPA (indeed, 5-LOX prefers EPA to AA) and, except perhaps for COX-1, have been shown or suggested to also metabolize DHA [64, 65].

Studies on the effects of ω-6 PUFA metabolites and of anti-oxygenase drugs on the growth and death of cultured human breast cancer cells have implicated essentially all the oxygenases along with their PGE_2, 5-HETE, 12-HETE, 15-HETE, and 13-HODE metabolites in influencing breast cancer growth. However, these studies often disagree as to which oxygenase–metabolite pairings are the critical participants in the malignant function of breast cancer cells [66–70]. Studies in animals and humans have continued this pattern [56–58]. COX-2 and its PGE_2 product are elevated in mouse models of breast cancer; COX-2 overexpression induces breast tumors and *Cox-2* knockout suppresses development of these tumors [71, 72]. Indeed, drugs that block COX-2 are associated with a reduced incidence of human breast cancer in epidemiological studies [73]. COX-1 may contribute to these effects or, in the absence of COX-2, mediate them [67, 72]. On the other hand, 5-LOX inhibitors are particularly effective in inhibiting the growth of cultured human breast cancer cells, and the 5-LOX metabolites, 5-HETE, and 5-oxo-ETE (made from 5-HETE by a specific dehydrogenase; both act on cells through a common receptor, OXE [74]), are more effective than PGE_2 in stimulating cultured human breast cancer cells to proliferate [75–79]. Furthermore, mRNA for 5-LOX and its activating protein (FLAP) are increased in human malignant as opposed to normal breast tissue and in node (+) compared to node (–) disease; FLAP message levels correlate negatively with overall and disease-free survival [80, 81]. Studies focusing on the 12-LOX/12-HETE axis have found that 12-HETE is somewhat weaker than 5-HETE in stimulating the proliferation of human breast cancer cells [79], but forced expression of 12-LOX was shown to stimulate the proliferation of human breast cancer cells in vitro and in mice [82]. 12-LOX message overexpression was demonstrated in malignant compared to normal human breast tissue and cell lines [69, 80, 83] and, when coupled with FLAP mRNA levels, provided a better prognostic indicator of overall and disease-free survival in breast cancer than FLAP mRNA alone [81]. Studies of 15-LOX have found that 15-HETE and its hydroperoxy precursor, 15-HPETE, may or may not inhibit the proliferation of breast cancer cells [78, 84] but mRNA and protein levels of their parent oxygenases 15-LOX-1 and 15-LOX-2 are decreased in malignant compared to normal human breast tissue [80, 85]. Patients with low levels of message and protein for these oxygenases or low 15-LOX-1/15-LOX-2 message or protein ratios were shown to have higher recurrence rates and shortened survivals [85]. This suggests that a 15-LOX,

Table 7.1 Human and mouse oxygenases along with the predominant products they from and their preferences for AA and LA[1,2,3]

Human			Mouse		
Type	Gene	Products	Type	Gene	Products
COX-1	**PTGS1**	PGs and TXA_2 9(R)-HODE>13-HODE	COX-1	*Ptgs-1*	PGs and TXA_2 9(R)-HODE>13-HODE
COX-2	**PTGS2**	PGs, PGI_2, and TXA_2 9(R)-HODE>13-HODE	COX-2	*Ptgs-2*	PGs, PGI_2, and TXA_2 9(R)-HODE>13-HODE
5-LOX	**ALOX5**	5-HETE and leukotrienes	5-LOX	*Alox5*	5-HETE and leukotrienes
12-LOX-p (platelet-type 12-LOX)	**ALOX12**	12-HETE and 15-HETE in a ratio of 98:2 LA is not a substrate	12-LOX-p (platelet-type 12-LOX)	*Alox12 (Alox12p)*	12-HETE, does not make 15-HETE little or no activity on LA
15-LOX-1(leukocytes or reticulocyte type 15-LOX)	**ALOX15**	15-HETE and 12-HETE in a ratio of 89:11 13-HODE prefers LA to AA	12-LOX-1 (12/15-LOX, leukocyte type 12-LOX)	*Alox12l (Alox15)*	15-HETE:12-HETE in a ratio of 1:6; metabolizes LA and AA equally
15-LOX-2(epidermis type 15-LOX)	**ALOX15B**	15-HETE does not make 12-HETE 13-HODE Prefers AA to LA	8-LOX (15-LOX type II)	*Alox15b*[1] *(Alox8)*	8-HETE and 9-HODE links functionally to e-LOX-3 Prefers AA to LA
12R-LOX (12-LOX 12R type)	**ALOX12B**	12R-HETE>8-HETE	e-LOX-2 (epidermis type LOX-12)	*Alox12b (Aloxe2)*	12-HETE>11R-HETE>8R-HETE 13-HODE Prefers PUFA methyl esters
e-LOX-3(epidermis type LOX-3)	**ALOXE3**[2]	Inactive on PUFA	e-LOX-3 (epidermis type LOX-3)	*Aloxe3*	Inactive on PUFA
			12-LOX-e (epidermis type 12-LOX	*Alox12e*	12-HETE, 13-HODE prefers PUFA methyl esters and LA to AA

[1]The human genome also has two unprocessed LOX pseudogenes, ALOX12P1 and ALOX15P.
[2]8-LOX is as efficient as 5-LOX in converting 5-hydroperoxy-ETE to LTB_4.
[3]Shaded boxes indicate oxygenases that because of their restriction to the epidermis as well as substrate preferences have not been implicated in breast or prostate cancer.

particularly 15-LOX-1, serves to suppress breast tumor growth. On the other hand, human breast cancer cells metabolize AA to 15-HETE [86] but also LA to 13-HODE. 13-HODE, which is a product of COX-1/2 as well as 15-LOX-1/2 [87] (Table 7.1), has been implicated in the proliferation, invasiveness, and metastasis of breast cancer [86, 88].

7.4.2 PUFA Mechanisms in Prostate Cancer

The in vitro growth of cultured human prostate cancer cell lines is promoted by ω-6 PUFAs and suppressed by ω-3 PUFAs [49, 89–93]. Furthermore, the expression of the *fat-1* gene in these cells was shown to induce apoptosis by elevating caspase-3 activity and to decrease these cells' capacity for in vitro invasion [38]. Many of these effects are attributable to the metabolism of ω-6 PUFAs to pro-malignancy products and the action of ω-3 PUFAs to inhibit this production. Experimental studies on this are more extensive than those in breast cancer, but yield similar results by implicating the same oxygenases and ω-6 PUFA metabolites in prostate cancer cell proliferation, differentiation, survival, angiogenicity, tissue invasion, and metastasis.

COX-1/2: Prostate cancer cell lines overexpress COX-2, overproduce PGE_2, and may overexpress COX-1; PGE_2 stimulates these cells to proliferate and increase their angiogenicity; and COX-2 suppression causes these cells to stop proliferating and become apoptotic in vitro or when explanted into rodents [80, 94–120]. Many of these effects are mediated by PGE_2's EP1, EP2, and EP4 receptors [118, 119] and involve the activation of protein kinase A and c-fos pathways of gene induction [114], down-regulation of apoptotic pathways by suppressing Bcl-2, p53, and/or Akt [103, 106, 116], activation of the interleukin-6/GP130/ STAT-3 growth-signaling pathway [112], or stimulation of the release of growth cytokines as well as metalloproteinases and vascular endothelial growth factor (VEGF) [111, 120]. It is interesting to note that PGE_2 induces COX-2, thereby potentially participating in a positive feedback loop [110], and that DHA cooperates with COX-1 inhibitors to suppress the in vitro growth of prostate cancer cells [95, 121]. These results translate to in vivo studies. The growth of prostate cancer xenografts was blocked by inhibiting COX-2 or PGE_2 [27, 30]. For example, a selective COX-2 inhibitor, NS398, blocked the growth of PC3 xenografts in mice up to 93% by inducing apoptosis and decreasing angiogenesis rather than decreasing proliferation [105]. In addition, treatment of transgenic adenocarcinoma of the mouse prostate with a COX-2 inhibitor, celecoxib, resulted in not only significant reduction in tumor development and metastases, but also, when given to mice with established tumors, prolonged survival [122]. Human prostate cancer epithelial cells overexpress COX-2, with the extent of this expression correlating positively with disease severity as defined by Gleason scores [98]. Finally, Fradet et al. [123] identified a variant single-nucleotide polymorphism (SNP) in the *COX-2* gene that modifies the response to ω-3 PUFAs. Although carriers of this SNP had an increased risk of aggressive disease, this was detrimental only in those with diets low in ω-3 PUFAs; the association was reversed by a high ω-3 PUFA diet.

5-LOX: Prostate cancer cell lines overexpress 5-LOX; 5-HETE stimulates these cells to proliferate; 5-LOX suppression causes these cells to stop proliferating and become apoptotic; and 5-LOX is overexpressed in human prostate cancer, with the extent of its expression correlating positively with disease severity [89–91, 93, 124–135]. 5-HETE, along with its structural analogs 5-oxo-ETE and 5-oxo-15-hydroxy-ETE, acts through the OXE G protein-coupled receptor, which is present in, necessary for the survival of, and activates Akt and MAPK survival/proliferation pathways in these cells [91, 134]. 5-HETE also forces the Bcl-2/BAX ratio to values favoring apoptosis [115]. It is interesting to note that high concentrations of the two oxo-ETE compounds, but not 5-HETE, activate PPARγ and cause prostate as well as breast cancer cells to become apoptotic [78] and that the 5-LOX gene has a response element for, and is down-regulated by, the orphan nuclear receptor RORα. Ligand-induced activation of RORα inhibited the proliferative effect of ω-6 PUFAs in PC3 and DU145 cells [90].

12-HETE: Prostate cancer cell lines express 12-LOX; 12-HETE stimulates these cells' proliferation and in vitro invasiveness; 12-LOX suppression causes these cells to stop proliferating and become apoptotic; forced expression of 12-LOX increases these cells in vitro proliferation and in vivo malignancy; 12-LOX and 12-HETE are overexpressed in a genetic model of breast cancer; and prostate cancer overexpresses 12-LOX, with the extent of its expression correlating positively with disease severity [136–144]. 12-HETE decreases the levels of cyclin D and Rb proteins in prostate cancer cells [137, 138], promotes the S6 proteasome-dependent degradation of IκB to thereby activate nuclear translocation of NFκB [141], and stimulates VEGF production [143]. The mechanism (i.e., acting through receptors or other means) by which 12-HETE elicits these responses is as yet undefined.

15-LOX-1/2: 15-LOX-1 is highly expressed in prostate cancer cell lines as well as in prostate cancer, where its level of expression correlates positively with disease severity; forced expression of 15-LOX-1 increases the in vitro proliferation and in vivo growth of these cells; and mice with genetically engineered prostatic cancer show progressive increases in 12/15-LOX (the mouse ortholog of 15-LOX-1-see Table 7.1) with progressive disease [145–149]. The metabolite responsible for these effects does not appear to be 15-HETE but rather 13-HODE. Prostate cell lines produce 13-HODE and this LA metabolite promotes the proliferation of prostate cancer cells by an as yet undefined mechanism that involves the promotion of insulin-like growth factor receptor-1 and epidermal growth factor signaling through the Akt and MAPK pathways [145, 150]. 13-HODE also appears to be involved in the 15-LOX-1-dependent growth of genetically engineered prostate cancer in mice [148]. 15-LOX-1 promotes the secretion of VEGF in prostate cancer cells [147, 148]. In contrast to 15-LOX-1, 15-LOX-2 is severely reduced or undetectable in prostate cancer cell lines and prostate cancer tissue, yet readily detected in normal prostate tissue [149–152]. This malignancy-related loss of 15-LOX-2 expression is due to transcriptional silencing [152]. Furthermore, forced expression of 15-LOX-2 induces prostate cancer cell lines to stop proliferating, take on a senescent morphology, and become apoptotic [151, 152]. 15-HETE, particularly at high concentrations, also induces these changes and, in direct opposition to 13-HODE, inhibits insulin-like growth factor-1 and epidermal growth factor signaling through Akt and MAPK [150–152]. However, part of 15-LOX-2's effects may not be mediated by 15-HETE or other metabolites, since forced expression of metabolically *inactive* 15-LOX-2 produces changes in prostate cancer cells similar to those caused by forced expression of active 15-LOX-2 [151, 152]. 15-HETE may achieve its effects at least in part by activating PPARγ [153–155], but the mechanism(s) behind 13-HODE's activity is unclear. In any case, the data suggest that the two 15-LOXs have opposing regulatory roles in prostate cancer, with 15-LOX-1 causing tumor growth by producing 13-HODE and 15-LOX-2 causing tumor suppression in part by producing 15-HETE but also by a path not involving ω-6 PUFA metabolism.

7.5 Conclusions

Cell culture and animal studies support conclusions that ω-3 PUFAs suppress the development of breast and prostate cancer and do so in part by inhibiting the generation of ω-6 PUFA-derived metabolites that stimulate the proliferation, survival, angiogenic capability, invasiveness, and metastatic potential of these cancers. COX-2, 5-LOX, 15-LOX-1, and 12-LOX overproduce these metabolites in both cancers and thereby serve as tumor promoters whereas 15-LOX-2 forms an anti-cancer metabolite, 15-HETE, and serves in other ways as a tumor suppressor that is silenced in prostate and perhaps breast cancer (although, in this latter disease, 15-LOX-1 may mimic certain effects of 15-LOX-2). The oxygenases likely have secondary rather than primary roles in instigating malignancy. The in vitro, in vivo, and clinical studies indicate that essentially all of the oxygenase–metabolite pairings run awry in breast cancer and prostate cancer. It therefore may be best to view these oxygenases and metabolites along with the elements such as $cPLA_2$ that supply substrate to the oxygenase as a network whose components interact in various feedback loops. This feedback is illustrated by the ability of PGE_2 to induce COX-2 [110, 156, 157], of 5-HETE to activate $cPLA_2$ [158] and thereby of both metabolites to promote further production of ω-6 PUFA metabolites. The network can shift to a phenotype favoring malignancy. This shift may occur incrementally, as exemplified in prostate cancer where $cPLA_2$ is increasingly activated as cancer cells become more malignant [159]. Overactive $cPLA_2$ may also occur in breast cancer based on a finding that transfection of fluorescent-tagged $cPLA_2$ results in its abnormal deposition as an active, membrane-bound state in breast as well as prostate cancer but not in benign cell lines [160]. While this system can be forced to a more cancer-promoting state by many perturbations, one prominent cause of this shift may be dietary ω-6 PUFAs which serve to fuel it. Two features of this network may be therapeutically relevant. First, low-density lipoprotein (LDL) receptors are dramatically up-regulated in many cancer cells including those of breast [161] and prostate [92, 162]. This could facilitate the rapid, sustained, and targeted delivery of dietary ω-3 PUFAs to tumor cells via the LDL receptor pathway. Second, ω-3 PUFAs, as opposed to pharmacological interventions with anti-oxygenase drugs, have the potential to inhibit the formation of all the ω-6 PUFA metabolites. Dietary ω-3 PUFAs may therefore shift the PUFA metabolizing network back to lower levels of ω-6 PUFA metabolites, oxygenases, and activated $cPLA_2$ to exert rapid and selective anti-growth effects on breast, prostate, and perhaps other cancer cells.

References

1. Sprecher H. Metabolism of highly unsaturated n-3 and n-6 fatty acids. *Biochim Biophys Acta* 2000; 1486:219–31.
2. Kris-Etherton PM, Taylor DS, Yu-Poth S, et al. Polyunsaturated fatty acids in the food chain in the United States. *Am J Clin Nutr* 2000; 71:179S–88S.
3. Reddy BS, Cohen LA, McCoy GD, Hill P, Weisburger JH, Wynder EL. Nutrition and its relationship to cancer. *Adv Cancer Res* 1980; 32:237–345.
4. Jurkowski JJ, Cave WT, Jr. Dietary effects of menhaden oil on the growth and membrane lipid composition of rat mammary tumors. *J Natl Cancer Inst* 1985; 74:1145–50.
5. Braden LM, Carroll KK. Dietary polyunsaturated fat in relation to mammary carcinogenesis in rats. *Lipids* 1986; 21:285–8.
6. Abou-el-Ela SH, Prasse KW, Carroll R, Wade AE, Dharwadkar S, Bunce OR. Eicosanoid synthesis in 7,12-dimethylbenz(a)anthracene-induced mammary carcinomas in Sprague-Dawley rats fed primrose oil, menhaden oil or corn oil diet. *Lipids* 1988; 23: 948–54.
7. Abou-el-Ela SH, Prasse KW, Farrell RL, Carroll RW, Wade AE, Bunce OR. Effects of D,L-2-difluoromethylornithine and indomethacin on mammary tumor promotion in rats fed high n–3 and/or n–6 fat diets. *Cancer Res* 1989; 49:1434–40.
8. Rose DP, Hatala MA, Connolly JM, Rayburn J. Effect of diets containing different levels of linoleic acid on human breast cancer growth and lung metastasis in nude mice. *Cancer Res* 1993; 53:4686–90.
9. Rose DP, Connolly JM. Effects of dietary omega-3 fatty acids on human breast cancer growth and metastases in nude mice. *J Natl Cancer Inst* 1993; 85:1743–7.
10. Rose DP, Connolly JM, Rayburn J, Coleman M. Influence of diets containing eicosapentaenoic or docosahexaenoic acid on growth and metastasis of breast cancer cells in nude mice. *J Natl Cancer Inst* 1995; 87:587–92.
11. Karmali RA, Marsh J, Fuchs C. Effect of omega-3 fatty acids on growth of a rat mammary tumor. *J Natl Cancer Inst* 1984; 73:457–61.
12. Gabor H, Hillyard LA, Abraham S. Effect of dietary fat on growth kinetics of transplantable mammary adenocarcinoma in BALB/c mice. *J Natl Cancer Inst* 1985; 74:1299–305.
13. Shao Y, Pardini L, Pardini RS. Dietary menhaden oil enhances mitomycin C antitumor activity toward human mammary carcinoma MX-1. *Lipids* 1995; 30:1035–45.
14. Rose DP, Connolly JM, Coleman M. Effect of omega-3 fatty acids on the progression of metastases after the surgical excision of human breast cancer cell solid tumors growing in nude mice. *Clin Cancer Res* 1996; 2:1751–6.
15. Chen J, Stavro PM, Thompson LU. Dietary flaxseed inhibits human breast cancer growth and metastasis and downregulates expression of insulin-like growth factor and epidermal growth factor receptor. *Nutr Cancer* 2002; 43:187–92.
16. Riediger ND, Othman RA, Suh M, Moghadasian MH. A systemic review of the roles of n-3 fatty acids in health and disease. *J Am Diet Assoc* 2009; 109:668–79.

17. Giovanella BC, Fogh J. The nude mouse in cancer research. *Adv Cancer Res* 1985; 44: 69–120.
18. Erickson KL, Schumacher LA. Lack of an influence of dietary fat on murine natural killer cell activity. *J Nutr* 1989; 119:1311–7.
19. Welsch CW, Oakley CS, Chang CC, Welsch MA. Suppression of growth by dietary fish oil of human breast carcinomas maintained in three different strains of immune-deficient mice. *Nutr Cancer* 1993; 20:119–27.
20. Noguchi M, Minami M, Yagasaki R, et al. Chemoprevention of DMBA-induced mammary carcinogenesis in rats by low-dose EPA and DHA. *Br J Cancer* 1997; 75:348–53.
21. Gooch JL, Yee D. Strain-specific differences in formation of apoptotic DNA ladders in MCF-7 breast cancer cells. *Cancer Lett* 1999; 144:31–7.
22. Slamon DJ, Clark GM, Wong SG, Levin WJ, Ullrich A, McGuire WL. Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. *Science* 1987; 235:177–82.
23. Guy CT, Webster MA, Schaller M, Parsons TJ, Cardiff RD, Muller WJ. Expression of the neu protooncogene in the mammary epithelium of transgenic mice induces metastatic disease. *Proc Natl Acad Sci USA* 1992; 89:10578–82.
24. Luijten M, Verhoef A, Dormans JA, et al. Modulation of mammary tumor development in Tg.NK (MMTV/c-neu) mice by dietary fatty acids and life stage-specific exposure to phytoestrogens. *Reprod Toxicol* 2007; 23:407–13.
25. Muller WJ, Sinn E, Pattengale PK, Wallace R, Leder P. Single-step induction of mammary adenocarcinoma in transgenic mice bearing the activated c-neu oncogene. *Cell* 1988; 54:105–15.
26. Yee LD, Young DC, Rosol TJ, Vanbuskirk AM, Clinton SK. Dietary (n–3) polyunsaturated fatty acids inhibit HER-2/neu-induced breast cancer in mice independently of the PPARgamma ligand rosiglitazone. *J Nutr* 2005; 135:983–8.
27. Karmali RA, Reichel P, Cohen LA, et al. The effects of dietary omega-3 fatty acids on the DU-145 transplantable human prostatic tumor. *Anticancer Res* 1987; 7:1173–9.
28. Rose DP, Cohen LA. Effects of dietary menhaden oil and retinyl acetate on the growth of DU 145 human prostatic adenocarcinoma cells transplanted into athymic nude mice. *Carcinogenesis* 1988; 9:603–5.
29. Rose DP. Effects of dietary fatty acids on breast and prostate cancers: evidence from in vitro experiments and animal studies. *Am J Clin Nutr* 1997; 66:1513S–22S.
30. Kobayashi N, Barnard RJ, Henning SM, et al. Effect of altering dietary omega-6/omega-3 fatty acid ratios on prostate cancer membrane composition, cyclooxygenase-2, and prostaglandin E2. *Clin Cancer Res* 2006; 12:4662–70.
31. Kelavkar UP, Hutzley J, Dhir R, Kim P, Allen KG, McHugh K. Prostate tumor growth and recurrence can be modulated by the omega-6:omega-3 ratio in diet: athymic mouse xenograft model simulating radical prostatectomy. *Neoplasia* 2006; 8:112–24.
32. McEntee MF, Ziegler C, Reel D, et al. Dietary n-3 polyunsaturated fatty acids enhance hormone ablation therapy in androgen-dependent prostate cancer. *Am J Pathol* 2008; 173:229–41.
33. Dahia PL. PTEN, a unique tumor suppressor gene. *Endocr Relat Cancer* 2000; 7:115–29.
34. Suzuki H, Freije D, Nusskern DR, et al. Interfocal heterogeneity of PTEN/MMAC1 gene alterations in multiple metastatic prostate cancer tissues. *Cancer Res* 1998; 58: 204–9.
35. Wang S, Gao J, Lei Q, et al. Prostate-specific deletion of the murine Pten tumor suppressor gene leads to metastatic prostate cancer. *Cancer Cell* 2003; 4:209–21.
36. Berquin IM, Min Y, Wu R, et al. Modulation of prostate cancer genetic risk by omega-3 and omega-6 fatty acids. *J Clin Invest* 2007; 117:1866–75.
37. Kang JX, Wang J, Wu L, Kang ZB. Transgenic mice: fat-1 mice convert n-6 to n-3 fatty acids. *Nature* 2004; 427:504.

38. Lu Y, Nie D, Witt WT, et al. Expression of the fat-1 gene diminishes prostate cancer growth in vivo through enhancing apoptosis and inhibiting GSK-3 beta phosphorylation. *Mol Cancer Ther* 2008; 7:3203–11.
39. Berquin IM, Edwards IJ, Chen YQ. Multi-targeted therapy of cancer by omega-3 fatty acids. *Cancer Lett* 2008; 269:363–77.
40. Larsson SC, Kumlin M, Ingelman-Sundberg M, Wolk A. Dietary long-chain n-3 fatty acids for the prevention of cancer: a review of potential mechanisms. *Am J Clin Nutr* 2004; 79:935–45.
41. Welsch CW. Review of the effects of dietary fat on experimental mammary gland tumorigenesis: role of lipid peroxidation. *Free Radic Biol Med* 1995; 18:757–73.
42. Gonzalez MJ, Schemmel RA, Dugan L, Jr., Gray JI, Welsch CW. Dietary fish oil inhibits human breast carcinoma growth: a function of increased lipid peroxidation. *Lipids* 1993; 28:827–32.
43. Mueller E, Sarraf P, Tontonoz P, et al. Terminal differentiation of human breast cancer through PPAR gamma. *Mol Cell* 1998; 1:465–70.
44. Pighetti GM, Novosad W, Nicholson C, et al. Therapeutic treatment of DMBA-induced mammary tumors with PPAR ligands. *Anticancer Res* 2001; 21:825–9.
45. Suh N, Wang Y, Williams CR, et al. A new ligand for the peroxisome proliferator-activated receptor-gamma (PPAR-gamma), GW7845, inhibits rat mammary carcinogenesis. *Cancer Res* 1999; 59:5671–3.
46. Yin Y, Russell RG, Dettin LE, et al. Peroxisome proliferator-activated receptor delta and gamma agonists differentially alter tumor differentiation and progression during mammary carcinogenesis. *Cancer Res* 2005; 65:3950–7.
47. Gani OA, Sylte I. Ligand-induced stabilization and activation of peroxisome proliferator-activated receptor gamma. *Chem Biol Drug Des* 2008; 72:50–7.
48. Itoh T, Yamamoto K.Peroxisome proliferator activated receptor gamma and oxidized docosahexaenoic acids as new class of ligand. *Naunyn Schmiedebergs Arch Pharmacol* 2008; 377:541–7.
49. Edwards IJ, Berquin IM, Sun H, et al. Differential effects of delivery of omega-3 fatty acids to human cancer cells by low-density lipoproteins versus albumin. *Clin Cancer Res* 2004; 10:8275–83.
50. Sun H, Berquin IM, Owens RT, O'Flaherty JT, Edwards IJ. Peroxisome proliferator-activated receptor gamma-mediated up-regulation of syndecan-1 by n-3 fatty acids promotes apoptosis of human breast cancer cells. *Cancer Res* 2008; 68:2912–9.
51. Sun H, Berquin IM, Edwards IJ. Omega-3 polyunsaturated fatty acids regulate syndecan-1 expression in human breast cancer cells. *Cancer Res* 2005; 65:4442–7.
52. Edwards IJ, Sun H, Hu Y,et al. In vivo and in vitro regulation of syndecan 1 in prostate cells by N-3 polyunsaturated fatty acids. *J Biol Chem* 2008; 283:18441–49.
53. Noguchi M, Earashi M, Minami M, Kinoshita K, Miyazaki I. Effects of eicosapentaenoic and docosahexaenoic acid on cell growth and prostaglandin E and leukotriene B production by a human breast cancer cell line (MDA-MB-231). *Oncology* 1995; 52:458–64.
54. Shikano M, Masuzawa Y, Yazawa K, Takayama K, Kudo I, Inoue K. Complete discrimination of docosahexaenoate from arachidonate by 85 kDa cytosolic phospholipase A2 during the hydrolysis of diacyl- and alkenylacylglycerophosphoethanolamine. *Biochim Biophys Acta* 1994; 1212:211–6.
55. Simopoulos AP. The importance of the omega-6/omega-3 fatty acid ratio in cardiovascular disease and other chronic diseases. *Exp Biol Med (Maywood)* 2008; 233:674–88.
56. Rose DP, Connolly JM. Omega-3 fatty acids as cancer chemopreventive agents. *Pharmacol Ther* 1999; 83:217–44.
57. Rose DP, Connolly JM. Effects of fatty acids and inhibitors of eicosanoid synthesis on the growth of a human breast cancer cell line in culture. *Cancer Res* 1990; 50:7139–44.
58. Brown MD, Hart CA, Gazi E, Bagley S, Clarke NW. Promotion of prostatic metastatic migration towards human bone marrow stoma by Omega 6 and its inhibition by Omega 3 PUFAs. *Br J Cancer* 2006; 94:842–53.

59. McCarty MF. Fish oil may impede tumour angiogenesis and invasiveness by down-regulating protein kinase C and modulating eicosanoid production. *Med Hypotheses* 1996; 46:107–15.
60. Rose DP, Connolly JM. Regulation of tumor angiogenesis by dietary fatty acids and eicosanoids. *Nutr Cancer* 2000; 37:119–27.
61. Caughey GE, Mantzioris E, Gibson RA, Cleland LG, James MJ. The effect on human tumor necrosis factor alpha and interleukin 1 beta production of diets enriched in n–3 fatty acids from vegetable oil or fish oil. *Am J Clin Nutr* 1996; 63:116–22.
62. Sperling RI, Benincaso AI, Knoell CT, Larkin JK, Austen KF, Robinson DR. Dietary omega-3 polyunsaturated fatty acids inhibit phosphoinositide formation and chemotaxis in neutrophils. *J Clin Invest* 1993; 91:651–60.
63. Needleman P, Raz A, Minkes MS, Ferrendelli JA, Sprecher H. Triene prostaglandins: prostacyclin and thromboxane biosynthesis and unique biological properties. *Proc Natl Acad Sci USA* 1979; 76:944–8.
64. Serhan CN, Gotlinger K, Hong S, Arita M. Resolvins, docosatrienes, and neuroprotectins, novel omega-3-derived mediators, and their aspirin-triggered endogenous epimers: an overview of their protective roles in catabasis. *Prostaglandins Other Lipid Mediat* 2004; 73:155–72.
65. Serhan CN, Chiang N. Endogenous pro-resolving and anti-inflammatory lipid mediators: a new pharmacologic genus. *Br J Pharmacol* 2008; 153(Suppl 1):S200–15.
66. Cuendet M, Pezzuto JM. The role of cyclooxygenase and lipoxygenase in cancer chemoprevention. *Drug Metabol Drug Interact* 2000; 17:109–57.
67. Furstenberger G, Krieg P, Muller-Decker K, Habenicht AJ. What are cyclooxygenases and lipoxygenases doing in the driver's seat of carcinogenesis? *Int J Cancer* 2006; 119:2247–54.
68. Shureiqi I, Lippman SM. Lipoxygenase modulation to reverse carcinogenesis. *Cancer Res* 2001; 61:6307–12.
69. Steele VE, Holmes CA, Hawk ET, et al. Lipoxygenase inhibitors as potential cancer chemopreventives. *Cancer Epidemiol Biomarkers Prev* 1999; 8:467–83.
70. You J, Mi D, Zhou X, et al. A Positive Feedback between Activated ERK and COX/LOX Maintains Proliferation and Migration of Breast Cancer Cells. *Endocrinology* 2009; 150:1607–17.
71. Howe LR. Inflammation and breast cancer. Cyclooxygenase/prostaglandin signaling and breast cancer. *Breast Cancer Res* 2007; 9:210.
72. Singh-Ranger G, Salhab M, Mokbel K. The role of cyclooxygenase-2 in breast cancer: review. *Breast Cancer Res Treat* 2008; 109:189–98.
73. Takkouche B, Regueira-Mendez C, Etminan M. Breast cancer and use of nonsteroidal anti-inflammatory drugs: a meta-analysis. *J Natl Cancer Inst* 2008; 100:1439–47.
74. O'Flaherty JT, Taylor JS, Thomas MJ. Receptors for the 5-oxo class of eicosanoids in neutrophils. *J Biol Chem* 1998; 273:32535–41.
75. Avis I, Hong SH, Martinez A, et al. Five-lipoxygenase inhibitors can mediate apoptosis in human breast cancer cell lines through complex eicosanoid interactions. *Faseb J* 2001; 15:2007–9.
76. Hammamieh R, Sumaida D, Zhang X, Das R, Jett M. Control of the growth of human breast cancer cells in culture by manipulation of arachidonate metabolism. *BMC Cancer* 2007; 7:138.
77. Kim JH, Hubbard NE, Ziboh V, Erickson KL. Conjugated linoleic acid reduction of murine mammary tumor cell growth through 5-hydroxyeicosatetraenoic acid. *Biochim Biophys Acta* 2005; 1687:103–9.
78. O'Flaherty JT, Rogers LC, Paumi CM, et al. 5-Oxo-ETE analogs and the proliferation of cancer cells. *Biochim Biophys Acta* 2005; 1736:228–36.
79. Tong WG, Ding XZ, Adrian TE. The mechanisms of lipoxygenase inhibitor-induced apoptosis in human breast cancer cells. *Biochem Biophys Res Comm* 2002; 296:942–8.
80. Jiang WG, Douglas-Jones A, Mansel RE. Levels of expression of lipoxygenases and cyclooxygenase-2 in human breast cancer. *Prostaglandins Leukot Essent Fatty Acids* 2003; 69:275–81.

81. Jiang WG, Douglas-Jones AG, Mansel RE. Aberrant expression of 5-lipoxygenase-activating protein (5-LOXAP) has prognostic and survival significance in patients with breast cancer. *Prostaglandins Leukot Essent Fatty Acids* 2006; 74:125–34.
82. Liu XH, Connolly JM, Rose DP. The 12-lipoxygenase gene-transfected MCF-7 human breast cancer cell line exhibits estrogen-independent, but estrogen and omega-6 fatty acid-stimulated proliferation in vitro, and enhanced growth in athymic nude mice. *Cancer Lett* 1996; 109:223–30.
83. Natarajan R, Nadler J. Role of lipoxygenases in breast cancer. *Front Biosci* 1998; 3:E81–8.
84. Najid A, Beneytout JL, Tixier M. Cytotoxicity of arachidonic acid and of its lipoxygenase metabolite 15-hydroperoxyeicosatetraenoic acid on human breast cancer MCF-7 cells in culture. *Cancer Lett* 1989; 46:137–41.
85. Jiang WG, Watkins G, Douglas-Jones A, Mansel RE. Reduction of isoforms of 15-lipoxygenase (15-LOX)-1 and 15-LOX-2 in human breast cancer. *Prostaglandins Leukot Essent Fatty Acids* 2006; 74:235–45.
86. Nony PA, Kennett SB, Glasgow WC, Olden K, Roberts JD. 15S-Lipoxygenase-2 mediates arachidonic acid-stimulated adhesion of human breast carcinoma cells through the activation of TAK1, MKK6, and p38 MAPK. *J Biol Chem* 2005; 280:31413–9.
87. Reddy N, Everhart A, Eling T, Glasgow W. Characterization of a 15-lipoxygenase in human breast carcinoma BT-20 cells: stimulation of 13-HODE formation by TGF alpha/EGF. *Biochem Biophys Res Comm* 1997; 231:111–6.
88. Pasqualini ME, Heyd VL, Manzo P, Eynard AR. Association between E-cadherin expression by human colon, bladder and breast cancer cells and the 13-HODE:15-HETE ratio. A possible role of their metastatic potential. *Prostaglandins Leukot Essent Fatty Acids* 2003; 68:9–16.
89. Ghosh J, Myers CE. Arachidonic acid stimulates prostate cancer cell growth: critical role of 5-lipoxygenase. *Biochem Biophys Res Comm* 1997; 235:418–23.
90. Moretti RM, Montagnani Marelli M, Sala A, Motta M, Limonta P. Activation of the orphan nuclear receptor RORalpha counteracts the proliferative effect of fatty acids on prostate cancer cells: crucial role of 5-lipoxygenase. *Int J Cancer* 2004; 112:87–93.
91. Sundaram S, Ghosh J. Expression of 5-oxoETE receptor in prostate cancer cells: critical role in survival. *Biochem Biophys Res Comm* 2006; 339:93–8.
92. Hughes-Fulford M, Chen Y, Tjandrawinata RR. Fatty acid regulates gene expression and growth of human prostate cancer PC-3 cells. *Carcinogenesis* 2001; 22:701–7.
93. Ghosh J, Myers CE. Inhibition of arachidonate 5-lipoxygenase triggers massive apoptosis in human prostate cancer cells. *Proc Natl Acad Sci USA* 1998; 95:13182–7.
94. Yoshimura R, Sano H, Masuda C, et al. Expression of cyclooxygenase-2 in prostate carcinoma. *Cancer* 2000; 89:589–96.
95. Narayanan NK, Narayanan BA, Reddy BS. A combination of docosahexaenoic acid and celecoxib prevents prostate cancer cell growth in vitro and is associated with modulation of nuclear factor-kappaB, and steroid hormone receptors. *Int J Oncol* 2005; 26: 785–92.
96. Gamradt SC, Feeley BT, Liu NQ, et al. The effect of cyclooxygenase-2 (COX-2) inhibition on human prostate cancer induced osteoblastic and osteolytic lesions in bone. *Anticancer Res* 2005; 25:107–15.
97. Hussain T, Gupta S, Mukhtar H. Cyclooxygenase-2 and prostate carcinogenesis. *Cancer Lett* 2003; 191:125–35.
98. Hong SH, Avis I, Vos MD, Martinez A, Treston AM, Mulshine JL. Relationship of arachidonic acid metabolizing enzyme expression in epithelial cancer cell lines to the growth effect of selective biochemical inhibitors. *Cancer Res* 1999; 59:2223–8.
99. Gupta S, Srivastava M, Ahmad N, Bostwick DG, Mukhtar H. Over-expression of cyclooxygenase-2 in human prostate adenocarcinoma. *Prostate* 2000; 42:73–8.
100. Kirschenbaum A, Liu X, Yao S, Levine AC. The role of cyclooxygenase-2 in prostate cancer. *Urology* 2001; 58:127–31.

101. Hubbard WC, Alley MC, McLemore TL, Boyd MR. Profiles of prostaglandin biosynthesis in sixteen established cell lines derived from human lung, colon, prostate, and ovarian tumors. *Cancer Res* 1988; 48:4770–5.
102. Palayoor ST, Bump EA, Calderwood SK, Bartol S, Coleman CN. Combined antitumor effect of radiation and ibuprofen in human prostate carcinoma cells. *Clin Cancer Res* 1998; 4: 763–71.
103. Liu XH, Yao S, Kirschenbaum A, Levine AC. NS398, a selective cyclooxygenase-2 inhibitor, induces apoptosis and down-regulates bcl-2 expression in LNCaP cells. *Cancer Res* 1998; 58:4245–9.
104. Lim JT, Piazza GA, Han EK, et al. Sulindac derivatives inhibit growth and induce apoptosis in human prostate cancer cell lines. *Biochem Pharmacol* 1999; 58:1097–107.
105. Liu XH, Kirschenbaum A, Yao S, Lee R, Holland JF, Levine AC. Inhibition of cyclooxygenase-2 suppresses angiogenesis and the growth of prostate cancer in vivo. *J Urol* 2000; 164:820–5.
106. Hsu AL, Ching TT, Wang DS, Song X, Rangnekar VM, Chen CS. The cyclooxygenase-2 inhibitor celecoxib induces apoptosis by blocking Akt activation in human prostate cancer cells independently of Bcl-2. *J Biol Chem* 2000; 275:11397–403.
107. Kamijo T, Sato T, Nagatomi Y, Kitamura T. Induction of apoptosis by cyclooxygenase-2 inhibitors in prostate cancer cell lines. *Int J Urol* 2001; 8:S35–9.
108. Ye F, Xui L, Yi J, Zhang W, Zhang DY. Anticancer activity of Scutellaria baicalensis and its potential mechanism. *J Altern Complement Med* 2002; 8:567–72.
109. Andrews J, Djakiew D, Krygier S, Andrews P. Superior effectiveness of ibuprofen compared with other NSAIDs for reducing the survival of human prostate cancer cells. *Cancer Chemother Pharmacol* 2002; 50:277–84.
110. Tjandrawinata RR, Dahiya R, Hughes-Fulford M. Induction of cyclo-oxygenase-2 mRNA by prostaglandin E2 in human prostatic carcinoma cells. *Br J Cancer* 1997; 75: 1111–8.
111. Attiga FA, Fernandez PM, Weeraratna AT, Manyak MJ, Patierno SR. Inhibitors of prostaglandin synthesis inhibit human prostate tumor cell invasiveness and reduce the release of matrix metalloproteinases. *Cancer Res* 2000; 60:4629–37.
112. Liu XH, Kirschenbaum A, Lu M, et al. Prostaglandin E(2) stimulates prostatic intraepithelial neoplasia cell growth through activation of the interleukin-6/GP130/STAT-3 signaling pathway. *Biochem Biophys Res Comm* 2002; 290:249–55.
113. Shappell SB, Manning S, Boeglin WE, et al. Alterations in lipoxygenase and cyclooxygenase-2 catalytic activity and mRNA expression in prostate carcinoma. *Neoplasia* 2001; 3:287–303.
114. Chen Y, Hughes-Fulford M. Prostaglandin E2 and the protein kinase A pathway mediate arachidonic acid induction of c-fos in human prostate cancer cells. *Br J Cancer* 2000; 82:2000–6.
115. Angelucci A, Garofalo S, Speca S, et al. Arachidonic acid modulates the crosstalk between prostate carcinoma and bone stromal cells. *Endocr Relat Cancer* 2008; 15:91–100.
116. Liu XH, Kirschenbaum A, Yu K, Yao S, Levine AC. Cyclooxygenase-2 suppresses hypoxia-induced apoptosis via a combination of direct and indirect inhibition of p53 activity in a human prostate cancer cell line. *J Biol Chem* 2005; 280:3817–23.
117. Mukherjee R, Edwards J, Underwood MA, Bartlett JM. The relationship between angiogenesis and cyclooxygenase-2 expression in prostate cancer. *BJU Int* 2005; 96:62–6.
118. Jain S, Chakraborty G, Raja R, Kale S, Kundu GC. Prostaglandin E2 regulates tumor angiogenesis in prostate cancer. *Cancer Res* 2008; 68:7750–9.
119. Miyata Y, Kanda S, Maruta S, et al. Relationship between prostaglandin E2 receptors and clinicopathologic features in human prostate cancer tissue. *Urology* 2006; 68: 1360–5.
120. Sahin M, Sahin E, Gumuslu S. Cyclooxygenase-2 in cancer and angiogenesis. *Angiology* 2009; 60:242–53.

121. Narayanan NK, Narayanan BA, Bosland M, Condon MS, Nargi D. Docosahexaenoic acid in combination with celecoxib modulates HSP70 and p53 proteins in prostate cancer cells. *Int J Cancer* 2006; 119:1586–98.
122. Gupta S, Adhami VM, Subbarayan M, et al. Suppression of prostate carcinogenesis by dietary supplementation of celecoxib in transgenic adenocarcinoma of the mouse prostate model. *Cancer Res* 2004; 64:3334–43.
123. Fradet V, Cheng I, Casey G, Witte JS. Dietary omega-3 fatty acids, cyclooxygenase-2 genetic variation, and aggressive prostate cancer risk. *Clin Cancer Res* 2009; 15:2559–66.
124. Gupta S, Srivastava M, Ahmad N, Sakamoto K, Bostwick DG, Mukhtar H. Lipoxygenase-5 is overexpressed in prostate adenocarcinoma. *Cancer* 2001; 91:737–43.
125. Anderson KM, Harris JE. 5,8,11,14-eicosatetraynoic acid inhibits PC3 DNA synthesis and cellular proliferation, in part due to its 5′-lipoxygenase activity. *Clin Physiol Biochem* 1990; 8:308–13.
126. Anderson KM, Ondrey F, Harris JE. ETYA, a pleiotropic membrane-active arachidonic acid analogue affects multiple signal transduction pathways in cultured transformed mammalian cells. *Clin Biochem* 1992; 25:1–9.
127. Anderson KM, Seed T, Ondrey F, Harris JE. The selective 5-lipoxygenase inhibitor A63162 reduces PC3 proliferation and initiates morphologic changes consistent with secretion. *Anticancer Res* 1994; 14:1951–60.
128. Hussey HJ, Tisdale MJ. Mechanism of the anti-tumour effect of 2,3,5-trimethyl-6-(3-pyridylmethyl) 1,4-benzoquinone (CV-6504). *Br J Cancer* 1997; 75:845–9.
129. Anderson KM, Seed T, Vos M, et al. 5-Lipoxygenase inhibitors reduce PC-3 cell proliferation and initiate nonnecrotic cell death. *Prostate* 1998; 37:161–73.
130. Harris JE, Alrefai WA, Meng J, Anderson KM. Five-lipoxygenase inhibitors reduce Panc-1 survival: synergism of MK886 with gamma linolenic acid. *Adv Exp Med Biol* 1999; 469:505–10.
131. Miller TA, Ghosh J, Myers CE, Macdonald TL. 5-HETE congeners as modulators of cell proliferation. *Bioorg Med Chem Lett* 2000; 10:1913–6.
132. Lieberman R, Nelson WG, Sakr WA, et al. Executive Summary of the National Cancer Institute Workshop: Highlights and recommendations. *Urology* 2001; 57:4–27.
133. Yang P, Collin P, Madden T, et al. Inhibition of proliferation of PC3 cells by the branched-chain fatty acid, 12-methyltetradecanoic acid, is associated with inhibition of 5-lipoxygenase. *Prostate* 2003; 55:281–91.
134. O'Flaherty JT, Rogers LC, Chadwell BA, et al. 5(S)-Hydroxy-6,8,11,14-E,Z,Z,Z-eicosatetraenoate stimulates PC3 cell signaling and growth by a receptor-dependent mechanism. *Cancer Res* 2002; 62:6817–9.
135. Ezekwudo DE, Wang RC, Elegbede JA. Methyl jasmonate induced apoptosis in human prostate carcinoma cells via 5-lipoxygenase dependent pathway. *J Exp Ther Oncol* 2007; 6:267–77.
136. Gao X, Grignon DJ, Chbihi T, et al. Elevated 12-lipoxygenase mRNA expression correlates with advanced stage and poor differentiation of human prostate cancer. *Urology* 1995; 46:227–37.
137. Yang P, Cartwright C, Chan D, Vijjeswarapu M, Ding J, Newman RA. Zyflamend-mediated inhibition of human prostate cancer PC3 cell proliferation: effects on 12-LOX and Rb protein phosphorylation. *Cancer Biol Ther* 2007; 6:228–36.
138. Pidgeon GP, Kandouz M, Meram A, Honn KV. Mechanisms controlling cell cycle arrest and induction of apoptosis after 12-lipoxygenase inhibition in prostate cancer cells. *Cancer Res* 2002; 62:2721–7.
139. Pidgeon GP, Tang K, Cai YL, Piasentin E, Honn KV. Overexpression of platelet-type 12-lipoxygenase promotes tumor cell survival by enhancing alpha(v)beta(3) and alpha(v)beta(5) integrin expression. *Cancer Res* 2003; 63:4258–67.
140. Nie D, Nemeth J, Qiao Y, et al. Increased metastatic potential in human prostate carcinoma cells by overexpression of arachidonate 12-lipoxygenase. *Clin Exp Metastasis* 2003; 20:657–63.

141. Kandouz M, Nie D, Pidgeon GP, Krishnamoorthy S, Maddipati KR, Honn KV. Platelet-type 12-lipoxygenase activates NF-kappaB in prostate cancer cells. *Prostaglandins Other Lipid Mediat* 2003; 71:189–204.
142. Timar J, Raso E, Dome B, et al. Expression, subcellular localization and putative function of platelet-type 12-lipoxygenase in human prostate cancer cell lines of different metastatic potential. *Int J Cancer* 2000; 87:37–43.
143. Nie D, Krishnamoorthy S, Jin R, et al. Mechanisms regulating tumor angiogenesis by 12-lipoxygenase in prostate cancer cells. *J Biol Chem* 2006; 281:18601–9.
144. Endsley MP, Aggarwal N, Isbell MA, et al. Diverse roles of 2-arachidonoylglycerol in invasion of prostate carcinoma cells: Location, hydrolysis and 12-lipoxygenase metabolism. *Int J Cancer* 2007; 121:984–91.
145. Kelavkar UP, Cohen C. 15-lipoxygenase-1 expression upregulates and activates insulin-like growth factor-1 receptor in prostate cancer cells. *Neoplasia* 2004; 6:41–52.
146. Kelavkar UP, Parwani AV, Shappell SB, Martin WD. Conditional expression of human 15-lipoxygenase-1 in mouse prostate induces prostatic intraepithelial neoplasia: the FLiMP mouse model. *Neoplasia* 2006; 8:510–22.
147. Kelavkar UP, Nixon JB, Cohen C, Dillehay D, Eling TE, Badr KF. Overexpression of 15-lipoxygenase-1 in PC-3 human prostate cancer cells increases tumorigenesis. *Carcinogenesis* 2001; 22:1765–73.
148. Kelavkar UP, Glasgow W, Olson SJ, Foster BA, Shappell SB. Overexpression of 12/15-lipoxygenase, an ortholog of human 15-lipoxygenase-1, in the prostate tumors of TRAMP mice. *Neoplasia* 2004; 6:821–30.
149. Kelavkar UP, Cohen C, Kamitani H, Eling TE, Badr KF. Concordant induction of 15-lipoxygenase-1 and mutant p53 expression in human prostate adenocarcinoma: correlation with Gleason staging. *Carcinogenesis* 2000; 21:1777–87.
150. Hsi LC, Wilson LC, Eling TE. Opposing effects of 15-lipoxygenase-1 and -2 metabolites on MAPK signaling in prostate. Alteration in peroxisome proliferator-activated receptor gamma. *J Biol Chem* 2002; 277:40549–56.
151. Tang S, Bhatia B, Maldonado CJ, et al. Evidence that arachidonate 15-lipoxygenase 2 is a negative cell cycle regulator in normal prostate epithelial cells. *J Biol Chem* 2002; 277:16189–201.
152. Tang DG, Bhatia B, Tang S, Schneider-Broussard R. 15-lipoxygenase 2 (15-LOX-2) is a functional tumor suppressor that regulates human prostate epithelial cell differentiation, senescence, and growth (size) . *Prostaglandins Other Lipid Mediat* 2007; 82:135–46.
153. Chen GG, Xu H, Lee JF, et al. 15-hydroxy-eicosatetraenoic acid arrests growth of colorectal cancer cells via a peroxisome proliferator-activated receptor gamma-dependent pathway. *Int J Cancer* 2003; 107:837–43.
154. Subbarayan V, Xu XC, Kim J, et al. Inverse relationship between 15-lipoxygenase-2 and PPAR-gamma gene expression in normal epithelia compared with tumor epithelia. *Neoplasia* 2005; 7:280–93.
155. Subbarayan V, Krieg P, Hsi LC, et al. 15-Lipoxygenase-2 gene regulation by its product 15-(S)-hydroxyeicosatetraenoic acid through a negative feedback mechanism that involves peroxisome proliferator-activated receptor gamma. *Oncogene* 2006; 25:6015–25.
156. Maldve RE, Kim Y, Muga SJ, Fischer SM. Prostaglandin E(2) regulation of cyclooxygenase expression in keratinocytes is mediated via cyclic nucleotide-linked prostaglandin receptors. *J Lipid Res* 2000; 41:873–81.
157. Bagga D, Wang L, Farias-Eisner R, Glaspy JA, Reddy ST. Differential effects of prostaglandin derived from omega-6 and omega-3 polyunsaturated fatty acids on COX-2 expression and IL-6 secretion. *Proc Natl Acad Sci USA* 2003; 100:1751–6.
158. Wijkander J, O'Flaherty JT, Nixon AB, Wykle RL. 5-Lipoxygenase products modulate the activity of the 85-kDa phospholipase A2 in human neutrophils. *J Biol Chem* 1995; 270:26543–9.
159. Patel MI, Singh J, Niknami M, et al. Cytosolic phospholipase A2-alpha: a potential therapeutic target for prostate cancer. *Clin Cancer Res* 2008; 14:8070–9.

160. Wooten RE, Willingham MC, Daniel LW, et al. Novel translocation responses of cytosolic phospholipase A2alpha fluorescent proteins. *Biochim Biophys Acta* 2008; 1783:1544–50.
161. Rudling MJ, Stahle L, Peterson CO, Skoog L. Content of low density lipoprotein receptors in breast cancer tissue related to survival of patients. *Br Med J (Clin Res Ed)* 1986; 292:580–2.
162. Chen Y, Hughes-Fulford M. Human prostate cancer cells lack feedback regulation of low-density lipoprotein receptor and its regulator, SREBP2. *Int J Cancer* 2001; 91:41–5.

Part IV
ω-3 PUFAs and Other Cancers

Chapter 8
ω-3 PUFAs and Other Cancers

Kyu Lim and Tong Wu

Abstract ω-3 Polyunsaturated fatty acids (ω-3 PUFAs)-rich fish oil is thought to suppress the pathogenesis of several human diseases including cancer. Although systemic reviews of epidemiological and cohort studies on cancer prevention of fish oil have provided controversial results, many preclinical studies convincingly demonstrate preventive and therapeutic efficacies of ω-3 PUFAs on cancer growth, angiogenesis, and metastasis through diverse mechanisms. This chapter summarizes the mechanisms of anticancer action of ω-3 PUFAs in pancreatic cancer, lung cancer, skin cancer, cholangiocarcinoma, and leukemia. By understanding these mechanisms we will be able to recommend the utilization of ω-3 PUFAs, alone or in combination with standard anticancer drugs, as an effective and safe therapeutic approach for the chemoprevention and treatment of these human cancers.

Keywords ω-3 PUFAs · Pancreatic cancer · Lung cancer · Skin cancer · Cholangiocarcinoma · Leukemia

8.1 ω-3 Polyunsaturated Fatty Acids (PUFAs) and Pancreatic Cancer

Pancreatic cancer is the fourth most common cause of cancer mortality in the United States and still remains an almost fatal disease, with mortality rates approaching the number of new diagnosed cases. An estimated around 42,470 new people will be diagnosed with pancreatic cancer in 2009 in the United States [1]. The relative 1-year survival rate from the time of diagnosis for all stages of pancreatic cancer is only 26%, with a 5-year survival rate of 5%. Although gemcitabine either as

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monotherapy or in combination with cytotoxic agents has been used in the treatment of pancreatic cancer for the past decade, the effect reveals only survival benefits of 1–2 months [2]. This sub-chapter summarizes the mechanisms of anticancer action of ω-3 PUFAs in pancreatic cancer.

Although the controversial results on fish oil-dependent chemoprevention and therapy of cancer have been reported in epidemiological and cohort studies [3, 4], fish oil (mainly ω-3 PUFAs) is thought to suppress cancer development and growth [5–7]. In chemically induced pancreatic carcinogenesis models, fish oils were capable of reducing the incidence of pancreatic cancers and hepatic metastases [8, 9]. Fish oil intake prevented L-azaserine-induced preneoplastic lesions in male Wistar rats [10–12]. As the ratio of dietary ω-3 to ω-6 PUFAs increased in a diet, the development of preneoplastic atypical acinar cell nodules after azaserine treatment decreased dramatically [10–12]. When patients with unresectable pancreatic adenocarcinoma received an EPA-enriched supplement, performance status and appetite were significantly improved at 3 weeks, demonstrating that EPA-enriched supplement may reverse cachexia in advanced pancreatic cancers [13, 14].

Many papers have been published demonstrating that ω-3 PUFAs (EPA and DHA) also suppress pancreatic cancer cell growth in vitro. Merendino et al. have shown that ω-3 PUFAs, including DHA, also induce apoptosis [15, 16] and Swamy et al. reported that ω-3 PUFAs enhance the efficiency of chemotherapeutic agents [17, 18] in some human pancreatic cancer cells. Consistently with previous reports, the treatment with ω-3 PUFAs of SW1990 and PANC-1 pancreatic cancer cells potently inhibited the proliferation and also induced apoptosis, as indicated by increased sub-G_1 population, caspase-3 activity, and PARP cleavage [19].

Moreover, preclinical studies demonstrate preventive and therapeutic efficacies of ω-3 PUFAs on pancreatic cancer [20, 21]. It was recently reported that *fat-1* transgenic mice had both lower incidence and growth rate of colitis-associated colon cancer [22, 23] and melanoma [24]. The *fat-1* transgenic mice are capable of producing ω-3 PUFAs from ω-6 PUFAs, leading to abundant ω-3 PUFAs in their organs and tissues without the need of a dietary supply [25]. In our in vivo experiments, tumor size and volume of implanted pancreatic cancer cells were significantly reduced in *fat-1* transgenic mice compared with wild type, and apoptotic (TUNEL positive) cells were also increased in immunohistochemical analysis from tissue of *fat-1* mice tumors compared with wild-type tumors [26].

In order to explain the anticancer action of ω-3 PUFAs in pancreatic cancer, several mechanisms have been proposed. One of the mechanisms is the ω-3 PUFA-dependent inhibition of cyclooxygenase-2 (COX-2) signaling pathway. The growth-promoting effects of arachidonic acid (AA) were specifically mediated by the COX-2 enzyme. The effect of a selective COX-2 inhibitor on AA-induced cell proliferation was completely restored by the addition of prostaglandin E_2 (PGE_2), suggesting that PGE_2 is the key prostaglandin species mediating the growth effects of AA in pancreatic cancer cells [27]. In contrast, Funahashi et al. reported that EPA decreased cell growth of both COX-2-positive and COX-2-negative pancreatic cancer cell lines, strongly suggesting the existence of COX-2-dependent and COX-2-independent mechanisms [28]. The growth-inhibitory effects of EPA are clearly demonstrated through COX-independent mechanisms in COX-2-negative/COX-1-positive pancreatic cancer cells because selective COX-1 inhibitors failed to attenuate the effects of EPA and no changes in prostanoid levels in the culture medium of these cells were detected when exposed to EPA. Besides being substrates for COX-2, ω-3 PUFAs are also capable of activating nuclear receptors (such as peroxisome proliferator-activated receptors [PPARs]) [29–31]. In this context, EPA may inhibit the growth of COX-2-negative pancreatic cancer cells by activating PPAR-γ, which would be concordant to previous data showing pro-apoptotic effects of PPAR-γ ligands [32]. They also suggested that the COX-2-dependent effects of EPA were mediated by PGE_3 because PGE_3 completely reversed the partial effects of the selective COX-2 inhibitor on the EPA-induced decrease in cell growth.

Interestingly, exposure of COX-2-positive pancreatic cancer cells to EPA for 1 hour not only led to the generation of PGE_3 but also decreased PGE_2 levels below baseline. This may be caused by partial replacement of AA in membrane phospholipids by EPA or by competition of EPA with AA at the COX-2 active site. This is in accordance with a recent report showing that addition of EPA reduced the AA-induced increase in PGE_2 production in lung cancer cells [33]. More detailed biochemical analyses suggest that the effects of both AA and EPA in COX-2-positive pancreatic cancer cells are transduced by the E-prostanoids (EP) EP_2 and EP_4 receptors through formation of PGE_2 and PGE_3. However, the exact signaling pathways responsible for the opposite effects of AA and EPA on pancreatic cancer cell growth are still elusive. Fujino et al. reported that the EP_4 but not the EP_2 isoform is able to couple to a pertussis toxin-sensitive G_i protein that can inhibit cAMP-dependent signaling and activate phosphatidylinositol 3-kinase/extracellular signal-regulated kinase-dependent signaling [34]. Their data showed that AA and EPA elicited their effects on cell growth through binding of PGE_2 and PGE_3, respectively, to the EP_2 and EP_4 receptors, but accumulation of intracellular cAMP was observed with AA (and PGE_2) but not with EPA (and PGE_3), suggesting the activation of differential signaling molecules.

Preclinical studies using xenograft mouse models (*fat-1* transgenic mouse and nude mouse) demonstrate that a low ω-6/ω-3 PUFA ratio reduces the incidence and growth of various cancers [24, 35–40]. Fundahashi et al. also confirmed the in vitro results through a xenograft mouse model experiment, which demonstrated antitumor efficacy of a diet enriched in ω-3 PUFAs [28]. The findings were in accordance with other mouse models illustrating therapeutic efficacy of ω-3 PUFA-enriched diets [35–37]. Dietary intake of ω-3 PUFAs will modify the fatty acid composition of membrane phospholipids and hence increase the availability of ω-3 PUFAs as substrates of COX-2 for the formation of 3-series prostaglandins. This notion is supported by a recent study of experimental prostate cancer that described an eightfold higher ω-6/ω-3 PUFA ratio in tumor membranes in animals fed an ω-6 PUFA-rich diet compared to an ω-3 PUFA-rich diet [37]. In contrast, several other reports demonstrated that an ω-3 PUFA-enriched diet had no effect on intratumoral COX-2 protein levels [37, 41, 42]. Nevertheless, dietary intake of ω-3 PUFAs in this study decreased PGE_2 and increased PGE_3 levels in pancreatic cancers, suggesting that the antitumor effect of the ω-3 PUFA-enriched diet involves a COX-2-dependent mechanism. The decrease in tumor growth with the ω-3 PUFA-rich diet correlated with an increased apoptosis in the tumors, which has also been observed in other models [33, 36, 43, 44].

β-catenin signaling pathway also plays a crucial role in the development and progression of pancreatic cancer. Several studies have reported a large percentage of pancreatic tumors overexpressing or stabilizing β-catenin [45], consistently observing an increased β-catenin level in tumor tissue from human pancreatic cancer compared with normal tissue [19]. Some investigators have shown that DHA reduces β-catenin levels in colorectal [46] and cholangiocarcinoma cells [47]. We also observed that DHA significantly reduces the level of β-catenin protein and nuclear translocation in human pancreatic cancer cells. Moreover, T-cell-specific factor/lymphoid enhancer binding factor (Tcf/Lef) reporter activity was also reduced by DHA treatment, and DHA-induced inhibition of cell growth was partially recovered by Wnt3a-condition medium (CM). These results indicate that this is one of the COX-2-independent mechanisms of ω-3 PUFA anticancer actions in pancreatic cancer. In our in vivo experiments using *fat-1* transgenic mice, we found decreased β-catenin expression in immunohistochemical analysis of tumor tissue from *fat-1* mice compared with wild-type tumors, indicating that aberrant activation of the Wnt/β-catenin pathway is associated with pancreatic cancer cell growth and that ω-3 PUFAs can suppress Wnt/β-catenin signaling pathway both in vitro and in vivo [26].

In addition, other mechanisms of anticancer action of ω-3 PUFAs have also been reported in pancreatic cancer. For example, EPA induces apoptosis through blockade of cell cycle progression in S-phase and G2/M-phase [48], ω-3 PUFAs interfere with EGF-induced pancreatic cell growth [49] and ω-3 PUFAs perturb the NF-kB pathway in pancreatic cancer.

Tumor angiogenesis is required for continuous tumor growth and metastasis [50]. Therefore, angiogenesis also represents a promising target for cancer treatment. Anti-angiogenic effect of ω-3 PUFAs is one of the mechanisms of ω-3 PUFA-dependent suppression of tumor growth in pancreatic cancer. Numerous studies have demonstrated ω-3 PUFA anti-angiogenic effects in several cancer cells [41, 46, 51–58]. Rose and Connolly showed that a DHA dietary supplement inhibited the growth of human breast cancer cells in a nude mouse model, and that such an effect was correlated with a reduced number of microvessels in tumors [52]. The anti-angiogenic effect of ω-3 PUFAs was confirmed in mouse mammary tumor cells transplanted in syngeneic Balb/c mice supplemented with a diet containing fish oil [53]. In tumors of fish oil-treated mice, a reduction of the expression of CD31 (a marker of vascular endothelial cells) and of VEGF was found [53]. We also observed that pancreatic tumors from *fat-1* mice have a lower vascular density (CD31 intensity) compared to wild-type mice [26]. These data indicate that ω-3 PUFAs exert anti-angiogenic effects in pancreatic cancer and that their anticancer actions may be partially mediated through inhibition of angiogenesis.

A variety of factors such as growth factors, chemokines, and cytokines are implicated in angiogenesis [59]. Among the proangiogenic factors, VEGF is the main positive regulator of angiogenesis. VEGF secreted from tumor cells and tumor-associated stromal cells binds to its receptor, resulting in endothelial cell proliferation, migration, and survival [60]. In patients with pancreatic cancer, increased expression of VEGF is correlated with increased microvessel density and poor prognosis [61]. DHA reduced VEGF reporter activity in a dose-dependent manner in PANC-1 human pancreatic cancer cells. Furthermore, VEGF-induced migration of HUVEC cells was significantly inhibited by DHA treatment and implanted matrigel in *fat-1* mice had lower hemoglobin contents than wild-type mice in an in vivo VEGF-induced matrigel plug assay [26]. These results strongly suggest that ω-3 PUFAs may inhibit VEGF signaling in endothelial cells as well as the production of VEGF in pancreatic cancer cells.

Suzuki et al. have shown that DHA inhibits the invasion and metastatic spreading of cancer cells through down-regulation of proteolytic enzymes such as matrix metalloproteinase (MMP)-2 and MMP-9 [62]. In our data, the promoter activities of MMP-2 and MMP-9 were also reduced by DHA treatment in pancreatic cancer cells. In addition, the invasiveness of human pancreatic cancer cells was inhibited by DHA treatment [26]. Accumulating evidences suggest that DHA may modulate inflammatory response by attenuating the production of pro-inflammatory cytokines and eicosanoids [63, 64]. The production of pro-inflammatory cytokines is controlled by the transcription factor nuclear factor-κB (NF-κB) [65] and eicosanoids such as PGE_2 [66]. It has been reported that DHA may inhibit the activity of NF-κB, COX-2, and the production of PGE_2 [47, 67]. Consistently, NF-κB and COX-2 promoter activities were inhibited by DHA treatment in pancreatic cancer cells [26]. Taken together, the anti-angiogenic effect of ω-3 PUFAs may be mediated by suppression of vascular endothelial growth factor (VEGF)/MMPs/COX-2 signaling through down-regulation of NF-κB in pancreatic cancer.

Recently, ω-3 PUFAs have been reported to enhance arsenic trioxide (As_2O_3)- [18] and curcumine-mediated cytotoxicity [17] in pancreatic cancer cells. In addition, ω-3 PUFAs may improve chemoresistant pancreatic cancer treatment outcomes and may slow or prevent recurrence of cancer. It has been demonstrated that gemcitabin-resistant pancreatic cancer is associated with enhanced NF-κB activation and antiapoptotic protein synthesis. Hering et al. found that the ω-3 PUFA-dependent inhibition of NF-κB activation and the ω-3 PUFA-induced apoptosis had the potential to restore or facilitate gemcitabin chemosensitivity in pancreatic cancer [68, 69].

In conclusion, these findings provide important preclinical evidences and molecular insights for the use of ω-3 PUFAs in the chemoprevention and treatment of human pancreatic cancer. ω-3 PUFA supplements may represent a useful alternative therapy for patients who are not candidates for standard toxic cancer therapies or may be used in combination therapy with standard chemotherapeutic agents in pancreatic cancer.

8.2 ω-3 PUFAs and Lung Cancer

Lung and bronchus cancers are the first most common causes of cancer mortality in United States for men and women. An estimated around 219,440 new people will be diagnosed with lung and bronchus cancers in 2009 in the United States [1].

Although currently available drug therapies may temporarily slow tumor growth, they lose their effectiveness with prolonged use and the tumors become resistant. As a result, some therapies may not be suitable for long-term use due to severe side effects. Therefore, it is important to develop additional effective and safe drugs for the treatment of lung cancer. Recently, several studies have shown a role for the ω-6/ω-3 PUFA ratio in the inhibition of cancer progression and suggest that modulating the tissue ω-6/ω-3 PUFA ratio may represent a new therapeutic strategy for patients with lung cancer. In support of this, it has been shown that excess of ω-6 PUFAs and high ω-6/ω-3 ratio increase the incidence of modern diseases including cancer [70]. This sub-chapter summarizes the mechanisms of anticancer actions of ω-3 PUFAs in lung cancer.

Numerous studies have documented the anti-proliferative, anti-invasive, and anti-angiogenic effects of ω-3 PUFAs in several cancers including lung cancer. Both intrinsic and extrinsic apoptosis pathways have been found to be activated by ω-3 PUFAs in different types of tumor cells. In particular, ω-3 PUFAs were reported to alter the expression and activity of caspase-3 and -9 in the lymphoma Ramos cell line [71] and HL-60 leukemia cells [71, 72]. Moreover, dietary ω-3 PUFAs were able to modulate Bcl-2 and FAS-ligand expression in normal splenic lymphocytes of Balb/c mice [73]. Alteration of Bcl-2 family proteins, cytochrome *c* release, and mitochondrial membrane depolarization were induced by EPA and DHA in colon cancer cells [41, 73, 74]. Also the ceramide pathway of apoptosis was reported to be up-regulated by ω-3 PUFAs in MDA-MB-231 breast cancer cells [39] and by DHA in Jurkat leukemia cells [75]. Recently, Serini et al. have found that DHA may also induce apoptosis in lung cancer cells, either of human or of animal origin [76]. This is in agreement with the growth-inhibiting and pro-apoptotic effects of DHA recently observed by Trombetta et al. [77] in A549 lung adenocarcinoma cells. We also observed that DHA induced the cleavage of PARP and the activation of caspase-3 and caspase-9 in A549 cells, indicating the induction of apoptosis (unpublished data).

One of the most significant approaches so far is to identify risk factors that are associated with the development of lung cancer and to eliminate the impact of these risk factors. One such potential target is COX, which catalyzes the synthesis of PGs, which appear to be involved in cell proliferation, migration, angiogenesis, and tumor metastasis [78]. Increased expression of COX-2 protein and the consequent production of PGE_2 have been implicated in the pathogenesis of several types of cancer, including colon, breast, and lung [79]. Numerous studies have suggested that altered production of PGs might be associated with the beneficial effects of ω-3 PUFAs [80]. In contrast to the compelling data from numerous studies with PGE_2, little is known about the pharmacology of and cellular response to PGE_3 [33, 81–84]. Yang et al. previously found, by applying LC-MS/MS method, that PGE_3 is formed within a few minutes after EPA exposure of A549 cells [85]. In addition, PGE_3 inhibited the proliferation of A549 human lung cancer cells but did not alter the growth of normal human bronchial epithelial cells. Furthermore, addition of PGE_3 to PGE_2-treated cells resulted in an inhibition of cell proliferation in A549 cells.

It was previously reported that EPA and AA induced the expression of COX-2 protein and mRNA in keratinocytes [86] and macrophages [87]. Yang et al. also observed that all of EPA, DHA, and AA were capable of inducing the expression of COX-2 protein in A549 cells [33]. In contrast, our previous study in cholangiocarcinoma demonstrated that ω-3 PUFAs inhibit COX-2 expression, which is consistent with the recently reported down-regulation of COX-2 by ω-3 PUFAs in other cancer cells [46, 47]. Yang et al. suggested that the rapid increase of PGE_3 in A549 cells may result from the metabolism of EPA by both COX-1 and COX-2 enzymes in the cells, and the addition of AA does not alter the formation of PGE_3 by EPA in A549 cells [33]. Even though the levels of COX-2 protein in A549 cells were increased by exposure to EPA, the relative formation of PGE_2 from AA in those cells was actually reduced by exposure to EPA. A number of in vivo studies have shown that ingestion of fish oil supplements (containing both EPA and DHA) inhibited colon or breast tumor growth and that this anticancer effect may be associated with the inhibition of COX-2 protein expression at the tumor site [88, 89]. Thus, even though the expression of COX-2 protein may be increased within lung cancer cells, the beneficial effects of EPA may involve both a decrease in the COX-2-mediated formation of PGE_2 and the unaltered formation of PGE_3, resulting in consequent inhibition of tumor cell growth. However, PGE_3 inhibited the proliferation of A549 human lung cancer cells, and PGE_2 did not block the inhibitory effect of PGE_3 on the proliferation of these cells. We have examined the effect of DHA on COX-2 reporter activity in A549 cells and found that the COX-2 reporter activity was reduced by DHA in a dose- and time-dependent manner (unpublished data). Therefore, regulation of COX-2 expression by ω-3 PUFAs in lung cancer cells remains controversial.

15-Hydroxyprostaglandin dehydrogenase (15-PGDH) catalyzes the rate-limiting step of PG catabolism and thus represents a physiological antagonist of COX-2 [90, 91]. Tong et al. reported that COX-2 and 15-PGDH were reciprocally regulated in lung cancer cells [92]. Elevated levels of PGE_2 in cancers may be the result of an enhanced COX-2-mediated PGE_2 synthesis as well as a reduced 15-PGDH-mediated degradation of PGE_2. Ding et al. [93] demonstrated that A549 cells infected with adenovirus encoding 15-PGDH showed retardation of the tumor growth in athymic nude mouse model. These authors also found that overexpression of 15-PGDH induced apoptosis in A549 cells. These evidences support the hypothesis that 15-PGDH may function like a tumor suppressor in lung cancer. Recently, it was reported that ω-3 PUFAs induced 15-PGDH expression in cholangiocarcinoma cell lines [47]. We also found that DHA treatment enhanced the expression of 15-PGDH in a dose-dependent manner in A549 cells (unpublished data). These data are consistent with the observation that DHA and EPA also inhibited PGE_2 production in Hep3B cells [94].

It is not yet clear how a decreased ω-6/ω-3 PUFA ratio causes inhibition of lung cancer cell invasion. The mechanisms underlying the ω-3 PUFA-dependent inhibition of invasion may be complex and probably involve multiple pathways. It is possible that the effects are mediated by a decreased production of ω-6 PUFA-derived eicosanoids and/or increased generation of ω-3 PUFA metabolites. Xia et al. demonstrated that gene transfer of ω-3 desaturase into human A549 lung cancer cells decreases the ratio of ω-6 to ω-3 PUFAs in these cells [44, 95]. In this study, they used a genetic approach to modify the cellular ω-6/ω-3 PUFA ratio by converting the endogenous ω-6 to ω-3 PUFAs, without the need for supplementation with exogenous fatty acids. The change of the cellular ω-6/ω-3 ratio from a relatively high to a low/balanced one exhibited an inhibitory effect on the invasive potential of A549 cells. These findings indicate differential/opposing effects of ω-6 and ω-3 PUFAs on cancer metastasis and pointed to the importance of ω-6/ω-3 PUFA ratio in the control of cancer. Inhibition of invasion of ω-3 desaturase-transfected A549 cells was correlated with down-regulation of the expression of MMP-1, integrin-α2 (ITG-α2), and nucleoside diphosphate kinase D (NM23-H4) in lung cancer cells, suggesting that a decreased ω-6/ω-3 PUFA ratio reduces the invasive potential of A549 cells through down-regulation of the adhesion/invasion-related genes [44]. In addition, we also found that ω-3 PUFAs inhibit MMP-2, MMP-9, and VEGF promoter activities in A549 cells (unpublished data).

Recently, another mechanism for the anticancer actions of ω-3 PUFAs was reported in lung cancer cell lines by Serini et al. MAPK-phosphatase-1 (MKP-1) has been recently considered as a possible independent positive prognostic factor in non-small cell lung cancer patients [96, 97]. Serini et al. reported that DHA may exert its pro-apoptotic effect in lung cancer cells by induction of MKP-1 expression and by dephosphorylation of the MAPKs through MKP-1 [76]. DHA-dependent increase of MKP-1 expression was related to increased transcription of MKP-1 gene by DHA in A549 cells. On the other hand, MKP-1 expression was not induced when lung cancer cells were treated with ω-6 PUFAs (AA); these observations suggest a specific action of ω-3 PUFAs [76]. Therefore, DHA-induced up-regulation of MKP-1 and reduction of MAPK phosphorylation may represent crucial steps in the pro-apoptotic effect of ω-3 PUFAs in lung cancer cells. However, further work is needed to verify the mechanism(s) through which DHA induces MKP-1 expression and apoptosis in lung cancer cells. Taken together, these observations document that ω-3 PUFAs exert pro-apoptotic actions in lung cancer cells and these effects likely involve the induction of MKP-1 expression and the resulting dephosphorylation of MAPKs, especially ERK1/2 and p38.

In vivo animal experiments on the anti-tumor effect of ω-3 PUFAs in lung cancer cell growth are relatively few. Maehle et al. reported that subcutaneously transplanted A427 lung adenocarcinoma cells in athymic nude mice were inhibited after feeding diets supplemented with a mixture of ethyl esters of ω-3 PUFAs (mainly EPA and DHA). Hardman et al. recently reported that increased consumption of ω-3 PUFAs using fish oil sensitized the A549 lung xenografts in mice to doxorubicin chemotherapy [98]. Recently, we also examined the effect of ω-3 PUFAs on lung cancer cell growth in vivo by using the *fat-1* transgenic mouse model. This model was selected because it provides a balanced ratio of ω-6 to ω-3 fatty acids in mouse tissues and eliminates the potential dietary variation associated with long-term feeding of PUFAs. When we implanted murine Lewis lung cancer cells (LLC1) into the syngeneic *fat-1* transgenic (heterozygous) and control mice (C57BL/6 genetic background), a significant reduction of LLC1 tumor size and tumor volume was observed in the *fat-1* transgenic mice (unpublished data). These findings provide important in vivo evidence for the inhibition of lung cancer by ω-3 PUFAs.

It is believed that chemoprevention has the potential to be a major component of cancer control in the development of several types of neoplasm, including lung cancer [99]. In this context, it is encouraging that the efficacy of ω-3 PUFAs has been documented in ω-3 PUFA nutritional intervention [100] in a mouse model of lung cancer treated with DHA-conjugated paclitaxel [101] and in patients with advanced non-small cell lung cancer [102].

In conclusion, these findings provide important preclinical evidences and molecular insights for the use of ω-3 PUFAs in the chemoprevention and treatment of human lung cancer. Therefore, ω-3 PUFAs alone or in combination with anticancer drugs may be useful for chemoprevention and therapy of human lung cancer.

8.3 ω-3 PUFAs and Skin Cancer

Skin cancer is the most common type of cancer in the United States and is presently a major cause of morbidity and mortality in our society [103]. An estimated 42,920 men and 31,690 women will be diagnosed with skin cancer in the year 2009 in the United States [1]. Moreover, the incidence of skin cancer will be sustained due to the aging of the population, the greater amounts of UV radiation reaching the surface of the earth because of depletion of ozone layer, and the extensive use of sun tanning devices for cosmetic purposes [104–107]. Actually, epidemiological, clinical, and laboratory studies have implicated solar ultraviolet radiation as a tumor initiator, tumor promoter, and complete carcinogen, and their excessive exposure can lead to the development of various skin disorders including melanoma and nonmelanoma skin cancers [107–110]. Although the sunscreens are useful, their protection is not adequate to prevent the risk of UV-induced skin cancer [111–113]. Therefore, new chemopreventive methods are necessary to protect the skin from photodamaging effects of solar UV radiation. This sub-chapter summarizes possible mechanisms of anticancer action of ω-3 PUFAs such as DHA and EPA against skin cancer in vitro and in vivo.

8.3.1 Nonmelanoma Skin Cancer (NMSC)

Nonmelanoma skin cancers, comprising of squamous cell carcinoma (SCC) and basal cell carcinoma (BCC), represent the most common malignant neoplasms in Caucasians [107]. Although the development of skin cancer has been associated with sun and UV radiation exposure, it is also related to dietary factors, including dietary lipids [114]. Specifically, ω-3 PUFAs such as DHA and EPA are chemopreventive whereas ω-6 PUFAs, including linoleic acid (LA) and AA, are chemopromotive on cancers [115–117]. For example, mice fed with corn oil (containing approximately 50% of ω-6 PUFAs) have shown nearly a linear relationship between ω-6 PUFA intake and UV radiation-induced carcinogenic expression [118–120]. On the other hand, a diet rich in fish oil has been shown to protect against photocarcinogenesis in mice [121]. In hairless mice, 2 weeks of fish oil feeding resulted in a significant reduction of the inflammatory response in skin and increased skin repair after UV exposure, compared with animals raised on corn oil high in LA [122]. Moreover, a case–control study of males with NMSC showed an inverse relationship between skin cancer and dietary fish intake in epidemiological studies [123]. Another study showed tendency toward a lower risk of SCC with higher intakes of ω-3 PUFAs [124]. These results suggest that ω-3 PUFAs have beneficial effects on the treatment and prevention of skin cancer.

Several mechanisms of ω-3 PUFA action on the treatment and chemoprevention of skin cancer have been proposed by in vitro and in vivo studies. ω-3 PUFAs are mainly known for their anti-inflammatory effects that are partially due to their competition with ω-6 PUFAs as substrates for COX and lipooxygenase (LOX) enzymes, resulting in the formation of less active prostaglandins and leukotrienes. UV radiation-induced inflammatory responses including hyperplasia are considered to play a crucial role in skin tumor promotion [125]. PGE_2 is produced by COX from ω-6 PUFA AA. Generally, PGE_2 has been shown to promote cancer development, whereas PGE_3 derived from ω-3 PUFAs has been found to exert anticancer effect [126]. There are two major isoforms of COX, COX-1 and COX-2. COX-1 is constitutively expressed in most tissues and its expression usually does not vary greatly in the adult animal [127]. On the other hand, COX-2 expression is generally undetectable in most unperturbed adult epithelial tissues, but can be highly induced by inflammatory cytokines and UV radiation [128]. Indeed, COX-2 has been shown to be transiently induced in human and mouse skin after a single acute exposure to UV radiation [128–130]. Coincident with the above reported studies, the expression of COX-2 has been shown to be elevated in human SCCs and BCCs [130–132]. Dietary fish oil supplementation in humans reduced UVB erythemal sensitivity, probably by reducing UV radiation-induced PGE_2 levels in the skin [133, 134]. PGE_2 is not only a major mediator of UV radiation-induced erythema [135], but has also been implicated in UV radiation-induced immunosuppression of the TH1 response by inducing IL-4 and IL-10, two cytokines important in the TH2-mediated immune response [136]. Topical and dietary application of EPA in mice protects against immunosuppression induced by UV radiation [137]. In response to UV radiation keratinocytes release a variety of cytokines and PGs, including IL-1α, TNF-α, IL-6, and PGE_2. Recently, EPA, an ω-3 PUFA, has been reported to differentially modulate the expression of these cytokines in UV-irradiated normal human keratinocytes [138]. Moreover, it has been shown that DHA and EPA reduce UV- and TNF-α-induced expression of IL-8, which is a pro-inflammatory and chemotactic cytokine in keratinocytes [139]. These results suggest that ω-3 PUFAs may increase the immune response and decrease inflammation by modulating eicosanoid biosynthesis and cytokine expression.

UV radiation also affects the regulation of signaling molecules such as mitogen-activated protein kinases (MAPKs) and interrelated inflammatory cytokines as well as NF-κB and activator protein-1 (AP-1) [140]. AP-1 is a transcription factor that regulates the expression and function of a number of cell cycle regulatory proteins such as cyclin D1, p53, p21, p19, and p16. AP-1 is a protein dimer consisting of either heterodimers between *fos* and *jun* family proteins or homodimers of *jun* family proteins [140]. UV radiation strongly induces c-jun and c-fos in human primary keratinocytes as well as in rat skin, suggesting that AP-1 activation may play a key role in UV-induced development of cancer. Liu et al. demonstrated that ω-3 PUFAs efficiently inhibit tumor promoter-induced AP-1 transactivation and subsequent cell transformation in mouse epidermal JB6 cells [114].

8.3.2 Melanoma Skin Cancer

An estimated around 68,720 new people will be diagnosed with melanoma of the skin in 2009 in the United States [1]. Although early-stage melanoma that is confined to the epidermis or superficial dermis is curable, the prognosis for individuals with deep invasion and metastases of the dermis is dismal, with a 5-year survival rate of only ~10% [141, 142]. Murine melanoma cells treated with EPA showed a dose-dependent decrease in invasiveness, collagenase IV production, and ability to metastasize to the lung after tail vein injection [143]. In another study, feeding an ω-3 PUFA-rich fish oil diet to mice implanted with B-16 melanoma cells showed more than 50% reduction in lung metastases compared with mice fed an ω-6 PUFA-rich corn oil diet [144]. Moreover, DHA induced cell cycle arrest and apoptosis by alterations in the phosphorylation status of pRb in cultured human metastatic melanoma cells [145]. Recently, Xia et al. reported that B-16 melanoma growth was significantly reduced in *fat-1* transgenic mice, which convert endogenous ω-6 PUFAs to ω-3 PUFAs in multiple tissues [146]. One of the notable changes accompanying the reduction of melanoma formation was the up-regulation of phosphatase and tensin homolog (PTEN) in the tumor and surrounding tissues of *fat-1* mice compared with wild-type mice. *Fat-1* transgenic mice showed higher levels of PTEN and caspase-3 with a lower level of Akt compared with wild-type animals in the tumor and surrounding tissues. PTEN has been shown to be a critical tumor suppressor in melanoma tumorigenesis [147–149]. PTEN has at least two biochemical functions: it has both lipid phosphatase and protein phosphatase activities. Activation of the lipid phosphatase of PTEN decreases intracellular PIP_3 level and downstream Akt acitivity [147]. Some studies have shown PTEN mutations in ~30–40% of melanoma cell lines and ~10% of primary melanomas [150, 151]. In PTEN-deficient mice, ectopic expression of PTEN was able to reduce melanoma tumorigenicity and metastasis. PTEN expression was induced by PGE_3 (EPA metabolite) treatment in B16 melanoma cells, indicating that ω-3 PUFAs have an anti-melanoma effect in part through PGE_3/PTEN pathway [152]. Recently, exposure of cells to DHA or EPA was shown to down-regulate COX-2 expression and to decrease PGE_2 production, thereby inhibiting in vitro tumor cell invasion in brain-metastatic melanoma [153]. One of the mechanisms proposed to explain these actions may include the competition of ω-3 with ω-6 PUFAs as substrates for COX-2 and the subsequent generation of anti-inflammatory and anti-mitogenic prostaglandins such as PGE_3.

In conclusion, the utilization of ω-3 PUFAs may enhance the endogenous immune system and reduce inflammatory response, thereby contributing to the chemoprevention and treatment of skin cancer.

8.4 ω-3 PUFAs and Cholangiocarcinoma

Liver and intrahepatic bile duct cancer is the sixth most common cause of cancer mortality in men and ninth in women in the United States [1]. The incidence of hepatocellular carcinoma (HCC) is rising worldwide, especially in the United States. An estimated 16,140 men and 6,210 women will be diagnosed with liver and intrahepatic bile duct cancer in the year 2009 in United States alone [1]. Cholangiocarcinoma is a highly malignant neoplasm of the biliary tree. Although it has a high rate of mortality, there is no effective chemoprevention and treatment. It is reported that ω-3 PUFAs, DHA, and EPA prevent tumor growth in vitro and in vivo in several cancers. This sub-chapter summarizes possible mechanisms of anticancer action of ω-3 PUFAs in cholangiocarcinoma.

Cholangiocarcinoma is a malignant epithelial neoplasm of the biliary tree and its incidence and mortality is rising [154–159]. Early diagnosis of cholangiocarcinoma is difficult and there is currently no effective chemoprevention or treatment. The tumor often arises from background conditions that cause chronic inflammation, injury, and reparative biliary epithelial cell proliferation such as primary sclerosing cholangitis, clonorchiasis, hepatolithiasis, or complicated fibropolycystic diseases [154–159].

Although ω-3 PUFAs rich in fish oil, such as DHA and EPA, have been reported to suppress the development and progression of several cancer types, the effect of ω-3 PUFAs in cholangiocarcinoma is relatively less studied [51, 160]. Recently, we reported that ω-3 PUFAs inhibited the growth of human cholangiocarcinoma cell lines (CCLP1, SG231 and HuCCT1). Treatment of these cholangiocarcinoma cells with two ω-3 PUFAs, DHA and EPA, resulted in a time- and dose-dependent reduction of cell viability; in contrast AA had no significant effect. The cells treated with DHA and EPA exhibited morphological features of cell death, characterized by shrunken, round, and detached cells. In contrast, AA treatment did not significantly alter cell morphology. DHA activated the pro-apoptotic molecules, caspase-3, caspase-9, and PARP cleavage in CCLP1 cells and concomitantly induced the release of cytochrome *c* from mitochondria to cytosol, indicating that ω-3 PUFAs inhibit human cholangiocarcinoma cell growth through the induction of apoptosis [47].

Consistent with the strong association between bile duct chronic inflammation and cholangiocarcinoma, recent studies have documented an important role of COX-2-derived PGE_2, a potent lipid inflammatory mediator, in cholangiocarcinogenesis [154, 155, 157, 159]. For example, increased COX-2 expression has been documented in cholangiocarcinoma cells and pre-cancerous bile duct lesions but not in normal bile duct epithelial cells [161–163]. Overexpression of COX-2 in cultured human cholangiocarcinoma cells enhances PGE_2 production and promotes tumor growth, whereas depletion of COX-2 attenuates tumor growth [164, 165]. Treatment of cholangiocarcinoma cells with exogenous PGE_2 increases tumor cell growth and prevents apoptosis [164–169]. Consistent with these findings, selective COX-2 inhibitors prevent cholangiocarcinoma cell growth and invasion, in vitro and in vivo [157, 164, 165, 168–170], although their effect may be mediated through COX-2-dependent or COX-2-independent mechanisms. We also found increased cytoplasmic staining for COX-2 in cholangiocarcinoma cells compared with the non-neoplastic bile duct epithelium of human cholangiocarcinoma tissues. COX-2 is expressed exclusively in the cytoplasm of cholangiocarcinoma cells and to a less degree in bile duct epithelial cells. The average staining intensity for COX-2 in cholangiocarcinoma cells is significantly higher than that in non-neoplastic bile duct epithelium [47]. These findings indicate over-expression of COX-2 in human cholangiocarcinomas. A prominent mechanism for the chemopreventive action of ω-3 PUFAs is their suppressive effect on the production of AA-derived prostanoids, particularly PGE_2 [160, 171]. This is important since PGE_2 is implicated in different stages of tumorigenesis, including modulation of inflammation, cancer cell proliferation, differentiation, apoptosis, angiogenesis, metastasis, and host immune response to cancer cells [157]. Treatment of CCLP1 cells with DHA inhibited COX-2 promoter activity in a time- and dose-dependent manner. DHA treatment also inhibited the expression of COX-2 protein in cholangiocarcinoma cells [47], suggesting that DHA inhibits the expression of COX-2 through suppression of gene transcription.

Recent evidences suggest that alteration of other growth-regulatory molecular pathways such as Wnt/β-catenin pathway is also implicated in cholangiocarcinogenesis [172–175]. β-catenin is a key mediator in Wnt regulation of multiple cellular functions in embryogenesis and tumorigenesis [176–179]. In adult tissues, β-catenin is a component of stable cell adherent complexes whereas its free form functions as a co-activator for a family of transcription factors termed T-cell factor/lymphoid enhancer factor (TCF/LEF). In the nucleus, β-catenin associates with TCF/LEF thus stimulating the transcription of target genes important for proliferation, differentiation, and apoptosis [176–179]. When we first performed immunohistochemical stains for β-catenin in human cholangiocarcinoma tissues, staining for β-catenin was increased in nuclei of cholangiocarcinoma cells compared with the non-neoplastic bile duct epithelium. Different β-catenin expression patterns were observed between cholangiocarcinoma cells and interlobular bile ducts. In the interlobular bile duct epithelial cells, β-catenin is expressed exclusively in the plasma membrane with no significant cytoplasmic staining and absence of nuclear staining. In cholangiocarcinoma cells, there is evident cytoplasmic staining with decreased plasma membrane staining. Nuclear staining for β-catenin was observed in cholangiocarcinoma tissues (approximately 30%), but not in non-neoplastic bile duct epithelial cells. DHA or EPA also reduced the level of β-catenin protein as well as c-Met, one of the β-catenin-controlled downstream genes and DHA treatment significantly inhibited the TCF/LEF reporter activity, indicating activation of β-catenin in human cholangiocarcinomas [47].

The level of β-catenin in cells is tightly controlled by its degradation complex composed of Axin, GSK-3β, and β-catenin, in which GSK-3β phosphorylates β-catenin and thus triggers its ubiquitination and subsequent proteosomal degradation. DHA treatment reduced GSK-3β phosphorylation, whereas it had no effect on the protein level of total GSK-3 and Akt phosphorylation in CCLP1 cells [47]. Thus, DHA most likely inhibited GSK-3β phosphorylation through a mechanism independent of Akt. In addition, treatment of CCLP1 cells with DHA promoted the association of Axin with GSK-3β as well as β-catenin. In contrast, AA, an ω-6 PUFA, reduced the association of Axin with β-catenin and GSK-3β. These findings indicate that DHA induces the formation of β-catenin destruction complex [47]. Thus, ω-3 PUFAs may induce β-catenin degradation through dephosphorylation of GSK-3β and formation of β-catenin destruction complex, inducing apoptosis in human cholangiocarcinoma cells.

Recent studies have shown that the accumulation of nuclear β-catenin is induced by PGE_2, in addition to the canonical Wnt/Frizzled signaling, in human colon cancer cells. Castellone et al. reported that PGE_2 activates its G protein-coupled receptor EP_2, resulting in direct association of the G protein alpha subunit with the regulator of G protein signaling (RGS) domain of axin; this results in release of GSK-3β from its complex with axin, thus leading to β-catenin accumulation [180]. Shao et al. showed the involvement of cAMP/protein kinase A pathway in PGE_2-induced β-catenin accumulation in colon cancer cells [181].

Overexpression of COX-2 increased the TCF/LEF reporter activity in the human cholangiocarcinoma cell line CCLP1 and PGE_2 treatment of CCLP1 cells also increased TCF/LEF reporter activity. In contrast, PGE_3, an ω-3 PUFA metabolite, exhibited no significant effect. Moreover, PGE_2 treatment also increased GSK-3β phosphorylation (thus inactivation) and caused GSK-3β dissociation from axin (thus preventing the formation of β-catenin destruction complex) in a time- and dose-dependent manner. DHA may also block PGE_2 signaling pathway. DHA prevented PGE_2-induced phosphorylation of GSK3β as well as dissociation of GSK-3β/Axin complex, and thereby inhibited PGE_2-induced TCF/LEF reporter activity [47].

15-PGDH catalyzes the rate-limiting step of prostaglandin catabolism and thus represents a physiological antagonist of COX-2 [90, 91]. Recent emerging evidence suggests that elevated PGE_2 in cancers may be the result of enhanced COX-2-mediated PGE_2 synthesis as well as reduced 15-PGDH-mediated degradation of PGE_2. DHA treatment enhanced the expression of 15-PGDH in CCLP1 cells, whereas AA exhibited no significant effect in human cholangiocarcinoma cells (CCLP1, etc.) [47]. The latter observations are novel and noteworthy, since 15-PGDH is a prostaglandin-degrading enzyme that physiologically antagonizes COX-2 and suppresses tumor growth.

In conclusion, ω-3 PUFAs may inhibit cholangiocarcinoma cell growth by simultaneously inhibiting β-catenin and COX-2 signaling pathways and these findings provide important preclinical evidence and molecular insight for utilization of ω-3 PUFAs in the chemoprevention and treatment of cholangiocarcinoma.

8.5 ω-3 PUFAs and Leukemia

Leukemia is a type of cancer in which the body produces a large number of abnormal blood cells. An estimated 25,630 men and 19,610 women will be diagnosed with leukemia in the year 2009 in the United States alone [1]. There are several types of leukemia, grouped as either acute (the disease progresses rapidly) or chronic (the disease progresses slowly) [182]. In this sub-chapter, the effects of ω-3 PUFAs on the acute and chronic leukemia are summarized.

Acute myeloid leukemia (AML) is the most common leukemia diagnosed in adults, with two-thirds of the new cases being diagnosed in patients over the age of 60 [183, 184]. However, less than 30% of adult patients are cured by current existing therapies, with the elderly population being linked with a poorer outcome [185]. In some studies, ω-3 PUFAs inhibited proliferation dose-dependently in some human leukemia cell lines including HL-60 cells. DHA treatment induced apoptosis through up-regulation of Bax, which is a pro-apoptotic protein, in KG1a acute myeloid leukemia cell line. Importantly, DHA did not have a pronounced effect on normal hematopoiesis [185]. In other studies, EPA also inhibited the growth of HL-60 cells and induced apoptosis through modulation of mitochondrial functions, thereby releasing cytochrome *c* from the mitochondria into the cytosol. Caspase-8 activation and Bid cleavage in response to EPA was shown in HL-60 cells [72]. Moreover, EPA also induced the membrane permeability transition and swelling of mitochondria isolated from HL-60 cells [72], indicating that EPA affects directly the mitochondrial membrane.

Reactive oxygen species (ROS) production also represents another mechanism by which ω-3 PUFAs may induce apoptosis. ω-3 PUFAs are incorporated into membrane phospholipids and thereby change the physical properties of membranes [186, 187]. ω-3 PUFAs are able to stimulate the generation of ROS through lipid peroxidation [188–190]. EPA has been reported to induce apoptosis of RBL2H3 rat basophilic leukemia cells through the generation of hydroperoxide in mitochondria, whereas linoleic acid (LA) had no significant effect [191].

In numerous studies, ω-3 PUFAs have been shown to enhance the efficacy and minimize the toxicity of anticancer drugs against several cancers including breast and prostate. For example, exogenous supplementation with DHA synergistically enhanced taxane cytotoxicity and down-regulated Her-2/neu oncogene expression in human breast cancer cells [192]. DHA in combination with celecoxib modulated HSP70 and p53 protein in prostate cancer cells [193]. In leukemia, DHA also enhanced arsenic trioxide (As_2O_3)-mediated apoptosis in As_2O_3-resistant HL-60 cells [189]. As_2O_3 was first introduced into the treatment of acute promyelocytic leukemia (APL) with remarkable clinical success. Generally, As_2O_3-induced apoptosis is related to a ROS-dependent mechanism. Accumulation of ROS by As_2O_3 led to disruption of the mitochondrial membrane potential, release of cytochrome *c* with consequent activation of the caspase cascade, and ultimately to programmed cell death through apoptosis [194–197]. DHA markedly reduced the viability of HL-60 cells when combined with As_2O_3 and enhanced As_2O_3-induced apoptosis with increased intracellular concentration of ROS and lipid peroxidation as well as an up-regulation of Bax. Furthermore, treatment with oleic acid, a monounsaturated fatty acid that is not susceptible to lipid peroxidation, did not enhance the effect of As_2O_3 treatment [189]. It has been also found that DHA has a synergistic effect when administered in combination with As_2O_3 and IFN-α in the human T-cell leukemia virus type I (HTLV-I) cells [198]. The mechanism underlying these effects was in part linked to ROS production and inhibition of the AP-1 and Akt pathways. DHA also has a synergistic effect on chronic myeloid leukemia (CML). Most patients with CML present the Philadelphia chromosome (Ph), a reciprocal translocation involving the long arms of chromosomes 9 and 22 [199, 200]. This translocation results in the Bcr-Abl fusion protein, a constitutively active cytoplasmic tyrosine kinase that does not block the differentiation, but enhances the proliferation and viability of myeloid lineage cells [201]. Although imatinib mesylate (IM) was known as an inhibitor of Abl tyrosine kinase and thereby as an inhibitor of cell proliferation, clinical studies have shown that varying proportions of patients do not achieve clinical responses to IM [202–204]. Long-term pre-treatment with DHA made Bcr-Abl

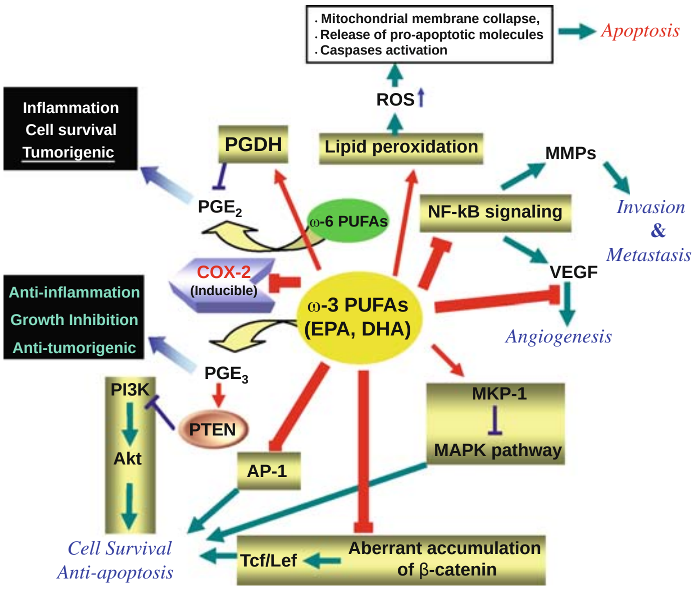

Fig. 8.1 Proposed molecular mechanisms of the anticancer action of ω-3 PUFAs in cancer. ω-3 PUFAs may induce apoptosis and suppress proliferation, angiogenesis, invasion, and metastasis of cancers including pancreatic cancer through modulation of a variety of molecular pathways including eicosanoids synthesis, transcription factors, signaling molecules, gene expression, and lipid peroxidation. ROS, reactive oxygen species; VEGF, vascular endothelial growth factor; MMPs, matrix metalloproteinases; PGDH, 15-hydroxyprostaglandin dehydrogenase; COX-2, cyclooxygenase-2; PTEN, phosphatase and tensin homolog; MAPKs, mitogen-activated protein kinases; MKP-1, MAPK-phosphatase-1; Tcf/Lef, T-cell-specific factor/lymphoid enhancing factor; PI3K, phosphoinositide 3-kinase; AP-1, activator protein-1; NF-κB, nuclear factor-κB

expressing HL-60 cells more susceptible to the toxic effect of IM [205]. Moreover, conjugated drugs including DHA-paclitaxel and DHA-HCPT have shown significantly higher therapeutic index on diverse types of cancer such as colon, lung, and leukemia [101, 206]. At an equimolar dose, DHA-10-hydroxycamptothecin (DHA-HCPT) had a much higher therapeutic efficacy than HCPT. DHA-HCPT had a greatly decreased toxicity compared to HCPT against L1210 leukemia in mice [206].

In conclusion, these findings provide important evidence for utilization of ω-3 PUFAs in the chemoprevention and treatment of leukemia. Therefore, ω-3 PUFAs can be used as direct therapy and a nontoxic adjunct in leukemia.

8.6 Conclusions

Many epidemiological and biochemical studies provide evidences that ω-3 PUFAs suppress development and progression of cancer in vitro and in vivo. This chapter summarized the anticancer effects of ω-3 PUFAs and their possible molecular mechanisms in pancreatic cancer, lung cancer, skin cancer, cholangiocarcinoma, and leukemia. ω-3 PUFAs modulate a variety of molecular pathways involved in cell proliferation, apoptosis, angiogenesis, invasion, and metastasis of cancers including pancreatic cancer (Fig. 8.1). ω-3 PUFAs are easily available as supplements and are effective and nontoxic in the dosage needed to inhibit cancer progression. Thus, ω-3 PUFAs alone or in combination with standard anticancer drugs could result in better outcome or allow a lower drug dosage. Further studies should be performed to establish the optimal ω-3 PUFA dosage.

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References

1. Jemal A, Siegel R, Ward E, Hao Y, Xu J, Thun MJ. Cancer statistics, 2009. CA Cancer J Clin 2009; 59(4):225–49.
2. Hochster HS, Haller DG, de Gramont A, et al. Consensus report of the international society of gastrointestinal oncology on therapeutic progress in advanced pancreatic cancer. Cancer 2006; 107(4):676–85.
3. Chen YQ, Berquin IM, Daniel LW, et al. Omega-3 fatty acids and cancer risk. JAMA 2006; 296(3):282; author reply.
4. MacLean CH, Newberry SJ, Mojica WA, et al. Effects of omega-3 fatty acids on cancer risk: A systematic review. JAMA 2006; 295(4):403–15.
5. Granados S, Quiles JL, Gil A, Ramirez-Tortosa MC. Dietary lipids and cancer. Nutr Hosp 2006; 21(Suppl 2):42–52, 44–54.
6. Simopoulos AP. Evolutionary aspects of diet, the omega-6/omega-3 ratio and genetic variation: Nutritional implications for chronic diseases. Biomed Pharmacother 2006; 60(9):502–7.

7. Mozaffarian D, Rimm EB. Fish intake, contaminants, and human health: evaluating the risks and the benefits. JAMA 2006; 296(15):1885–99.
8. Gregor JI, Heukamp I, Kilian M, et al. Does enteral nutrition of dietary polyunsaturated fatty acids promote oxidative stress and tumor growth in ductal pancreatic cancer? Experimental trial in Syrian Hamster. Prostaglandins Leukot Essent Fatty Acids 2006; 74(1):67–74.
9. Heukamp I, Gregor JI, Kilian M, et al. Influence of different dietary fat intake on liver metastasis and hepatic lipid peroxidation in BOP-induced pancreatic cancer in Syrian hamsters. Pancreatology 2006; 6(1–2):96–102.
10. O'Connor TP, Roebuck BD, Peterson F, Campbell TC. Effect of dietary intake of fish oil and fish protein on the development of L-azaserine-induced preneoplastic lesions in the rat pancreas. J Natl Cancer Inst 1985; 75(5):959–62.
11. O'Connor TP, Roebuck BD, Peterson FJ, Lokesh B, Kinsella JE, Campbell TC. Effect of dietary omega-3 and omega-6 fatty acids on development of azaserine-induced preneoplastic lesions in rat pancreas. J Natl Cancer Inst 1989; 81(11):858–63.
12. Cave WT, Jr. Omega 3 fatty acid diet effects on tumorigenesis in experimental animals. World Rev Nutr Diet 1991; 66:462–76.
13. Barber MD, Ross JA, Voss AC, Tisdale MJ, Fearon KC. The effect of an oral nutritional supplement enriched with fish oil on weight-loss in patients with pancreatic cancer. Br J Cancer 1999; 81(1):80–6.
14. Barber MD, Fearon KC. Tolerance and incorporation of a high-dose eicosapentaenoic acid diester emulsion by patients with pancreatic cancer cachexia. Lipids 2001; 36(4):347–51.
15. Merendino N, Loppi B, D'Aquino M, et al. Docosahexaenoic acid induces apoptosis in the human PaCa-44 pancreatic cancer cell line by active reduced glutathione extrusion and lipid peroxidation. Nutr Cancer 2005; 52(2):225–33.
16. Zhang W, Long Y, Zhang J, Wang C. Modulatory effects of EPA and DHA on proliferation and apoptosis of pancreatic cancer cells. J Huazhong Univ Sci Technolog Med Sci 2007; 27(5):547–50.
17. Swamy MV, Citineni B, Patlolla JM, Mohammed A, Zhang Y, Rao CV. Prevention and treatment of pancreatic cancer by curcumin in combination with omega-3 fatty acids. Nutr Cancer 2008; 60(Suppl 1):81–9.
18. Baumgartner M, Sturlan S, Roth E, Wessner B, Bachleitner-Hofmann T. Enhancement of arsenic trioxide-mediated apoptosis using docosahexaenoic acid in arsenic trioxide-resistant solid tumor cells. Int J Cancer 2004; 112(4):707–12.
19. Song KS Yun EJ, Kim JS, Heo JY, et al. Docosahexaenoic acid-induced apoptotic cell death is correlated with inhibition of β-catenin/wnt signaling pathway and Cox-2 in human pancreatic cancer cells. In the 99th Annual Meeting of the American Association Cancer Research 2008; Abstract # 2705.
20. Barber MD, Fearon KC, Tisdale MJ, McMillan DC, Ross JA. Effect of a fish oil-enriched nutritional supplement on metabolic mediators in patients with pancreatic cancer cachexia. Nutr Cancer 2001; 40(2):118–24.
21. Moses AW, Slater C, Preston T, Barber MD, Fearon KC. Reduced total energy expenditure and physical activity in cachectic patients with pancreatic cancer can be modulated by an energy and protein dense oral supplement enriched with n-3 fatty acids. Br J Cancer 2004; 90(5):996–1002.
22. Nowak J, Weylandt KH, Habbel P, et al. Colitis-associated colon tumorigenesis is suppressed in transgenic mice rich in endogenous n-3 fatty acids. Carcinogenesis 2007; 28(9):1991–5.
23. Jia Q, Lupton JR, Smith R, et al. Reduced colitis-associated colon cancer in Fat-1 (n-3 fatty acid desaturase) transgenic mice. Cancer Res 2008; 68(10):3985–91.
24. Xia S, Lu Y, Wang J, et al. Melanoma growth is reduced in fat-1 transgenic mice: Impact of omega-6/omega-3 essential fatty acids. Proc Natl Acad Sci USA 2006; 103(33):12499–504.
25. Kang JX, Wang J, Wu L, Kang ZB. Transgenic mice: Fat-1 mice convert n-6 to n-3 fatty acids. Nature 2004; 427(6974):504.
26. Song KS, Kim JS, Yun EJ, Park HD, et al. Anti-angiogenic effect of w3-polyunsaturated fatty acids is mediated through suppression of ELR+ CXC chemokines expression and VEGF

signaling in pancreatic cancer. In the 100th Annual Meeting of the American Association Cancer Research 2009: Abstract # 131.
27. Eibl G, Bruemmer D, Okada Y, et al. PGE(2) is generated by specific COX-2 activity and increases VEGF production in COX-2-expressing human pancreatic cancer cells. Biochem Biophys Res Commun 2003; 306(4):887–97.
28. Funahashi H, Satake M, Hasan S, et al. Opposing effects of n-6 and n-3 polyunsaturated fatty acids on pancreatic cancer growth. Pancreas 2008; 36(4):353–62.
29. Abedin M, Lim J, Tang TB, Park D, Demer LL, Tintut Y. N-3 fatty acids inhibit vascular calcification via the p38-mitogen-activated protein kinase and peroxisome proliferator-activated receptor-gamma pathways. Circ Res 2006; 98(6):727–9.
30. Sun H, Berquin IM, Edwards IJ. Omega-3 polyunsaturated fatty acids regulate syndecan-1 expression in human breast cancer cells. Cancer Res 2005; 65(10):4442–7.
31. Xu HE, Lambert MH, Montana VG, et al. Molecular recognition of fatty acids by peroxisome proliferator-activated receptors. Mol Cell 1999; 3(3):397–403.
32. Eibl G, Wente MN, Reber HA, Hines OJ. Peroxisome proliferator-activated receptor gamma induces pancreatic cancer cell apoptosis. Biochem Biophys Res Commun 2001; 287(2):522–9.
33. Yang P, Chan D, Felix E, et al. Formation and antiproliferative effect of prostaglandin E(3) from eicosapentaenoic acid in human lung cancer cells. J Lipid Res 2004; 45(6):1030–9.
34. Fujino H, Regan JW. EP(4) prostanoid receptor coupling to a pertussis toxin-sensitive inhibitory G protein. Mol Pharmacol 2006; 69(1):5–10.
35. Kato T, Hancock RL, Mohammadpour H, et al. Influence of omega-3 fatty acids on the growth of human colon carcinoma in nude mice. Cancer Lett 2002; 187(1–2):169–77.
36. Kelavkar UP, Hutzley J, Dhir R, Kim P, Allen KG, McHugh K. Prostate tumor growth and recurrence can be modulated by the omega-6: Omega-3 ratio in diet: Athymic mouse xenograft model simulating radical prostatectomy. Neoplasia 2006; 8(2):112–24.
37. Kobayashi N, Barnard RJ, Henning SM, et al. Effect of altering dietary omega-6/omega-3 fatty acid ratios on prostate cancer membrane composition, cyclooxygenase-2, and prostaglandin E2. Clin Cancer Res 2006; 12(15):4662–70.
38. Reddy BS, Patlolla JM, Simi B, Wang SH, Rao CV. Prevention of colon cancer by low doses of celecoxib, a cyclooxygenase inhibitor, administered in diet rich in omega-3 polyunsaturated fatty acids. Cancer Res 2005; 65(17):8022–7.
39. Wu M, Harvey KA, Ruzmetov N, et al. Omega-3 polyunsaturated fatty acids attenuate breast cancer growth through activation of a neutral sphingomyelinase-mediated pathway. Int J Cancer 2005; 117(3):340–8.
40. Yee LD, Young DC, Rosol TJ, Vanbuskirk AM, Clinton SK. Dietary (n-3) polyunsaturated fatty acids inhibit HER-2/neu-induced breast cancer in mice independently of the PPARgamma ligand rosiglitazone. J Nutr 2005; 135(5):983–8.
41. Calviello G, Di Nicuolo F, Gragnoli S, et al. n-3 PUFAs reduce VEGF expression in human colon cancer cells modulating the COX-2/PGE2 induced ERK-1 and -2 and HIF-1alpha induction pathway. Carcinogenesis 2004; 25(12):2303–10.
42. Horia E, Watkins BA. Complementary actions of docosahexaenoic acid and genistein on COX-2, PGE2 and invasiveness in MDA-MB-231 breast cancer cells. Carcinogenesis 2007; 28(4):809–15.
43. Shirota T, Haji S, Yamasaki M, et al. Apoptosis in human pancreatic cancer cells induced by eicosapentaenoic acid. Nutrition 2005; 21(10):1010–7.
44. Xia SH, Wang J, Kang JX. Decreased n-6/n-3 fatty acid ratio reduces the invasive potential of human lung cancer cells by downregulation of cell adhesion/invasion-related genes. Carcinogenesis 2005; 26(4):779–84.
45. Zeng G, Germinaro M, Micsenyi A, et al. Aberrant Wnt/beta-catenin signaling in pancreatic adenocarcinoma. Neoplasia 2006; 8(4):279–89.
46. Calviello G, Resci F, Serini S, et al. Docosahexaenoic acid induces proteasome-dependent degradation of beta-catenin, down-regulation of survivin and apoptosis in human colorectal cancer cells not expressing COX-2. Carcinogenesis 2007; 28(6):1202–9.

47. Lim K, Han C, Xu L, Isse K, Demetris AJ, Wu T. Cyclooxygenase-2-derived prostaglandin E2 activates beta-catenin in human cholangiocarcinoma cells: evidence for inhibition of these signaling pathways by omega-3 polyunsaturated fatty acids. Cancer Res 2008; 68(2):553–60.
48. Lai PB, Ross JA, Fearon KC, Anderson JD, Carter DC. Cell cycle arrest and induction of apoptosis in pancreatic cancer cells exposed to eicosapentaenoic acid in vitro. Br J Cancer 1996; 74(9):1375–83.
49. Jordan A, Stein J. Modulation of epidermal growth factor-induced cell proliferation by an omega-3 fatty-acid-containing lipid emulsion on human pancreatic cancer cell line Mia Paca-2. Nutrition 2001; 17(6):474–5.
50. Carmeliet P, Jain RK. Angiogenesis in cancer and other diseases. Nature 2000; 407(6801):249–57.
51. Hardman WE. (n-3) Fatty acids and cancer therapy. J Nutr 2004; 134(12 Suppl):3427S–30S.
52. Rose DP, Connolly JM. Antiangiogenicity of docosahexaenoic acid and its role in the suppression of breast cancer cell growth in nude mice. Int J Oncol 1999; 15(5):1011–5.
53. Mukutmoni-Norris M, Hubbard NE, Erickson KL. Modulation of murine mammary tumor vasculature by dietary n-3 fatty acids in fish oil. Cancer Lett 2000; 150(1):101–9.
54. Wen B, Deutsch E, Opolon P, et al. n-3 polyunsaturated fatty acids decrease mucosal/ epidermal reactions and enhance antitumour effect of ionising radiation with inhibition of tumor angiogenesis. Br J Cancer 2003; 89(6):1102–7.
55. Hardman WE, Sun L, Short N, Cameron IL. Dietary omega-3 fatty acids and ionizing irradiation on human breast cancer xenograft growth and angiogenesis. Cancer Cell Int 2005; 5(1):12.
56. Sheehan KM, O'Connell F, O'Grady A, et al. The relationship between cyclooxygenase-2 expression and characteristics of malignant transformation in human colorectal adenomas. Eur J Gastroenterol Hepatol 2004; 16(6):619–25.
57. Szymczak M, Murray M, Petrovic N. Modulation of angiogenesis by omega-3 polyunsaturated fatty acids is mediated by cyclooxygenases. Blood 2008; 111(7):3514–21.
58. Mannini A, Kerstin N, Calorini L, Mugnai G, Ruggieri S. An enhanced apoptosis and a reduced angiogenesis are associated with the inhibition of lung colonization in animals fed an n-3 polyunsaturated fatty acid-rich diet injected with a highly metastatic murine melanoma line. Br J Nutr 2009; 101(5):688–93.
59. Risau W. Mechanisms of angiogenesis. Nature 1997; 386(6626):671–4.
60. Donovan EA, Kummar S. Targeting VEGF in cancer therapy. Curr Probl Cancer 2006; 30(1):7–32.
61. Ikeda N, Adachi M, Taki T, et al. Prognostic significance of angiogenesis in human pancreatic cancer. Br J Cancer 1999; 79(9–10):1553–63.
62. Suzuki I, Iigo M, Ishikawa C, et al. Inhibitory effects of oleic and docosahexaenoic acids on lung metastasis by colon-carcinoma-26 cells are associated with reduced matrix metalloproteinase-2 and -9 activities. Int J Cancer 1997; 73(4):607–12.
63. Simopoulos AP. Omega-3 fatty acids in inflammation and autoimmune diseases. J Am Coll Nutr 2002; 21(6):495–505.
64. Noguchi M, Earashi M, Minami M, Kinoshita K, Miyazaki I. Effects of eicosapentaenoic and docosahexaenoic acid on cell growth and prostaglandin E and leukotriene B production by a human breast cancer cell line (MDA-MB-231). Oncology 1995; 52(6):458–64.
65. Blackwell TS, Christman JW. The role of nuclear factor-kappa B in cytokine gene regulation. Am J Respir Cell Mol Biol 1997; 17(1):3–9.
66. Yoshimura R, Sano H, Masuda C, et al. Expression of cyclooxygenase-2 in prostate carcinoma. Cancer 2000; 89(3):589–96.
67. Schley PD, Jijon HB, Robinson LE, Field CJ. Mechanisms of omega-3 fatty acid-induced growth inhibition in MDA-MB-231 human breast cancer cells. Breast Cancer Res Treat 2005; 92(2):187–95.

68. Hering J, Garrean S, Dekoj TR, et al. Inhibition of proliferation by omega-3 fatty acids in chemoresistant pancreatic cancer cells. Ann Surg Oncol 2007; 14(12):3620–8.
69. Persaud R. Inhibition of proliferation to omega-3 fatty acids in chemoresistant pancreatic cancer cells: mechanism of action may be more complex. Ann Surg Oncol 2008; 15(7):2057.
70. Chajes V, Bougnoux P. Omega-6/omega-3 polyunsaturated fatty acid ratio and cancer. World Rev Nutr Diet 2003; 92:133–51.
71. Heimli H, Giske C, Naderi S, Drevon CA, Hollung K. Eicosapentaenoic acid promotes apoptosis in Ramos cells via activation of caspase-3 and -9. Lipids 2002; 37(8):797–802.
72. Arita K, Kobuchi H, Utsumi T, et al. Mechanism of apoptosis in HL-60 cells induced by n-3 and n-6 polyunsaturated fatty acids. Biochem Pharmacol 2001; 62(7):821–8.
73. Avula CP, Zaman AK, Lawrence R, Fernandes G. Induction of apoptosis and apoptotic mediators in Balb/C splenic lymphocytes by dietary n-3 and n-6 fatty acids. Lipids 1999; 34(9):921–7.
74. Danbara N, Yuri T, Tsujita-Kyutoku M, et al. Conjugated docosahexaenoic acid is a potent inducer of cell cycle arrest and apoptosis and inhibits growth of colo 201 human colon cancer cells. Nutr Cancer 2004; 50(1):71–9.
75. Siddiqui RA, Jenski LJ, Harvey KA, Wiesehan JD, Stillwell W, Zaloga GP. Cell-cycle arrest in Jurkat leukemic cells: a possible role for docosahexaenoic acid. Biochem J 2003; 371 (Pt 2):621–9.
76. Serini S, Trombino S, Oliva F, et al. Docosahexaenoic acid induces apoptosis in lung cancer cells by increasing MKP-1 and down-regulating p-ERK1/2 and p-p38 expression. Apoptosis 2008; 13(9):1172–83.
77. Trombetta A, Maggiora M, Martinasso G, Cotogni P, Canuto RA, Muzio G. Arachidonic and docosahexaenoic acids reduce the growth of A549 human lung-tumor cells increasing lipid peroxidation and PPARs. Chem Biol Interact 2007; 165(3):239–50.
78. Dannenberg AJ, Lippman SM, Mann JR, Subbaramaiah K, DuBois RN. Cyclooxygenase-2 and epidermal growth factor receptor: Pharmacologic targets for chemoprevention. J Clin Oncol 2005; 23(2):254–66.
79. Prescott SM, Fitzpatrick FA. Cyclooxygenase-2 and carcinogenesis. Biochim Biophys Acta 2000; 1470(2):M69–M78.
80. Tapiero H, Ba GN, Couvreur P, Tew KD. Polyunsaturated fatty acids (PUFA) and eicosanoids in human health and pathologies. Biomed Pharmacother 2002; 56(5):215–22.
81. Rose DP, Connolly JM. Regulation of tumor angiogenesis by dietary fatty acids and eicosanoids. Nutr Cancer 2000; 37(2):119–27.
82. Karmali RA. Eicosanoids in neoplasia. Prev Med 1987; 16(4):493–502.
83. Nie D, Tang K, Szekeres K, Trikha M, Honn KV. The role of eicosanoids in tumor growth and metastasis. Ernst Schering Res Found Workshop 2000; 31:201–17.
84. Ge Y, Chen Z, Kang ZB, Cluette-Brown J, Laposata M, Kang JX. Effects of adenoviral gene transfer of *Caenorhabditis elegans* n-3 fatty acid desaturase on the lipid profile and growth of human breast cancer cells. Anticancer Res 2002; 22(2A):537–43.
85. Yang P, Felix E, Madden T, Fischer SM, Newman RA. Quantitative high-performance liquid chromatography/electrospray ionization tandem mass spectrometric analysis of 2- and 3-series prostaglandins in cultured tumor cells. Anal Biochem 2002; 308(1):168–77.
86. Maldve RE, Kim Y, Muga SJ, Fischer SM. Prostaglandin E (2) regulation of cyclooxygenase expression in keratinocytes is mediated via cyclic nucleotide-linked prostaglandin receptors. J Lipid Res 2000; 41(6):873–81.
87. Lo CJ, Chiu KC, Fu M, Lo R, Helton S. Fish oil augments macrophage cyclooxygenase II (COX-2) gene expression induced by endotoxin. J Surg Res 1999; 86(1):103–7.
88. Hardman WE. Omega-3 fatty acids to augment cancer therapy. J Nutr 2002; 132(11 Suppl):3508S–12S.
89. Rao CV, Hirose Y, Indranie C, Reddy BS. Modulation of experimental colon tumorigenesis by types and amounts of dietary fatty acids. Cancer Res 2001; 61(5):1927–33.

90. Backlund MG, Mann JR, Holla VR, et al. 15-Hydroxyprostaglandin dehydrogenase is down-regulated in colorectal cancer. J Biol Chem 2005; 280(5):3217–23.
91. Myung SJ, Rerko RM, Yan M, et al. 15-Hydroxyprostaglandin dehydrogenase is an in vivo suppressor of colon tumorigenesis. Proc Natl Acad Sci USA 2006; 103(32):12098–102.
92. Tong M, Ding Y, Tai HH. Reciprocal regulation of cyclooxygenase-2 and 15-hydroxyprostaglandin dehydrogenase expression in A549 human lung adenocarcinoma cells. Carcinogenesis 2006; 27(11):2170–9.
93. Ding Y, Tong M, Liu S, Moscow JA, Tai HH. NAD+-linked 15-hydroxyprostaglandin dehydrogenase (15-PGDH) behaves as a tumor suppressor in lung cancer. Carcinogenesis 2005; 26(1):65–72.
94. Lim K, Han C, Xu L, Wu T. Omega-3 polyunsaturated fatty acids inhibit hepatocellular carcinoma cell growth through downregulation of beta-catenin/wnt signaling pathway. In the 97th Annual Meeting of the American Association Cancer Research 2006: Abstract # 630.
95. Kang ZB, Ge Y, Chen Z, et al. Adenoviral gene transfer of *Caenorhabditis elegans* n–3 fatty acid desaturase optimizes fatty acid composition in mammalian cells. Proc Natl Acad Sci USA 2001; 98(7):4050–4.
96. Tsujita E, Taketomi A, Gion T, et al. Suppressed MKP-1 is an independent predictor of outcome in patients with hepatocellular carcinoma. Oncology 2005; 69(4):342–7.
97. Vicent S, Garayoa M, Lopez-Picazo JM, et al. Mitogen-activated protein kinase phosphatase-1 is overexpressed in non-small cell lung cancer and is an independent predictor of outcome in patients. Clin Cancer Res 2004; 10(11):3639–49.
98. Hardman WE, Moyer MP, Cameron IL. Dietary fish oil sensitizes A549 lung xenografts to doxorubicin chemotherapy. Cancer Lett 2000; 151(2):145–51.
99. Narayanan BA. Chemopreventive agents alters global gene expression pattern: Predicting their mode of action and targets. Curr Cancer Drug Targets 2006; 6(8):711–27.
100. Pardini RS, Wilson D, Schiff S, Bajo SA, Pierce R. Nutritional intervention with omega-3 Fatty acids in a case of malignant fibrous histiocytoma of the lungs. Nutr Cancer 2005; 52(2):121–9.
101. Bradley MO, Webb NL, Anthony FH, et al. Tumor targeting by covalent conjugation of a natural fatty acid to paclitaxel. Clin Cancer Res 2001; 7(10):3229–38.
102. Payne M, Ellis P, Dunlop D, et al. DHA-paclitaxel (Taxoprexin) as first-line treatment in patients with stage IIIB or IV non-small cell lung cancer: Report of a phase II open-label multicenter trial. J Thorac Oncol 2006; 1(9):984–90.
103. Diepgen TL, Mahler V. The epidemiology of skin cancer. Br J Dermatol 2002; 146(Suppl 61):1–6.
104. Urbach F. Incidence of nonmelanoma skin cancer. Dermatol Clin 1991; 9(4):751–5.
105. Miller DL, Weinstock MA. Nonmelanoma skin cancer in the United States: Incidence. J Am Acad Dermatol 1994; 30(5 Pt 1):774–8.
106. Johnson TM, Dolan OM, Hamilton TA, Lu MC, Swanson NA, Lowe L. Clinical and histologic trends of melanoma. J Am Acad Dermatol 1998; 38 (5 Pt 1):681–6.
107. Baliga MS, Katiyar SK. Chemoprevention of photocarcinogenesis by selected dietary botanicals. Photochem Photobiol Sci 2006; 5(2):243–53.
108. Huang XX, Bernerd F, Halliday GM. Ultraviolet A within sunlight induces mutations in the epidermal basal layer of engineered human skin. Am J Pathol 2009; 174(4):1534–43.
109. Gruber F, Kastelan M, Brajac I, et al. Molecular and genetic mechanisms in melanoma. Coll Antropol 2008; 32(Suppl 2):147–52.
110. Ullrich SE. Sunlight and skin cancer: Lessons from the immune system. Mol Carcinog 2007; 46(8):629–33.
111. Green A, Williams G, Neale R, et al. Daily sunscreen application and betacarotene supplementation in prevention of basal-cell and squamous-cell carcinomas of the skin: A randomised controlled trial. Lancet 1999; 354(9180):723–9.

112. Weinstock MA. Do sunscreens increase or decrease melanoma risk: An epidemiologic evaluation. J Investig Dermatol Symp Proc 1999; 4(1):97–100.
113. Haywood R, Wardman P, Sanders R, Linge C. Sunscreens inadequately protect against ultraviolet-A-induced free radicals in skin: implications for skin aging and melanoma? J Invest Dermatol 2003; 121(4):862–8.
114. Liu G, Bibus DM, Bode AM, Ma WY, Holman RT, Dong Z. Omega 3 but not omega 6 fatty acids inhibit AP-1 activity and cell transformation in JB6 cells. Proc Natl Acad Sci USA 2001; 98(13):7510–5.
115. Potter JD, Slattery ML, Bostick RM, Gapstur SM. Colon cancer: A review of the epidemiology. Epidemiol Rev 1993; 15(2):499–545.
116. Parkinson AJ, Cruz AL, Heyward WL, et al. Elevated concentrations of plasma omega-3 polyunsaturated fatty acids among Alaskan Eskimos. Am J Clin Nutr 1994; 59(2):384–8.
117. Sauer LA, Dauchy RT, Blask DE. Mechanism for the antitumor and anticachectic effects of n-3 fatty acids. Cancer Res 2000; 60(18):5289–95.
118. Black HS, Lenger W, Phelps AW, Thornby JI. Influence of dietary lipid upon ultraviolet-light carcinogenesis. Nutr Cancer 1983; 5(2):59–68.
119. Black HS, Lenger WA, Gerguis J, Thornby JI. Relation of antioxidants and level of dietary lipid to epidermal lipid peroxidation and ultraviolet carcinogenesis. Cancer Res 1985; 45 (12 Pt 1):6254–9.
120. Black HS, Rhodes LE. The potential of omega-3 fatty acids in the prevention of non-melanoma skin cancer. Cancer Detect Prev 2006; 30(3):224–32.
121. Black HS, Thornby JI, Gerguis J, Lenger W. Influence of dietary omega-6, -3 fatty acid sources on the initiation and promotion stages of photocarcinogenesis. Photochem Photobiol 1992; 56(2):195–9.
122. Yen A, Black HS, Tschen J. Effect of dietary omega-3 and omega-6 fatty acid sources on PUVA-induced cutaneous toxicity and tumorigenesis in the hairless mouse. Arch Dermatol Res 1994; 286(6):331–6.
123. Kune GA, Bannerman S, Field B, et al. Diet, alcohol, smoking, serum beta-carotene, and vitamin A in male nonmelanocytic skin cancer patients and controls. Nutr Cancer 1992; 18(3):237–44.
124. Hakim IA, Harris RB, Ritenbaugh C. Fat intake and risk of squamous cell carcinoma of the skin. Nutr Cancer 2000; 36(2):155–62.
125. Mukhtar H, Elmets CA. Photocarcinogenesis: Mechanisms, models and human health implications. Photochem Photobiol 1996; 63(4):356–7.
126. Wang MT, Honn KV, Nie D. Cyclooxygenases, prostanoids, and tumor progression. Cancer Metastasis Rev 2007; 26(3–4):525–34.
127. Smith WL, DeWitt DL, Garavito RM. Cyclooxygenases: structural, cellular, and molecular biology. Annu Rev Biochem 2000; 69:145–82.
128. Rundhaug JE, Fischer SM. Cyclo-oxygenase-2 plays a critical role in UV-induced skin carcinogenesis. Photochem Photobiol 2008; 84(2):322–9.
129. An KP, Athar M, Tang X, et al. Cyclooxygenase-2 expression in murine and human non-melanoma skin cancers: Implications for therapeutic approaches. Photochem Photobiol 2002; 76(1):73–80.
130. Tripp CS, Blomme EA, Chinn KS, Hardy MM, LaCelle P, Pentland AP. Epidermal COX-2 induction following ultraviolet irradiation: suggested mechanism for the role of COX-2 inhibition in photoprotection. J Invest Dermatol 2003; 121(4):853–61.
131. Akunda JK, Chun KS, Sessoms AR, Lao HC, Fischer SM, Langenbach R. Cyclooxygenase-2 deficiency increases epidermal apoptosis and impairs recovery following acute UVB exposure. Mol Carcinog 2007; 46(5):354–62.
132. Fischer SM, Pavone A, Mikulec C, Langenbach R, Rundhaug JE. Cyclooxygenase-2 expression is critical for chronic UV-induced murine skin carcinogenesis. Mol Carcinog 2007; 46(5):363–71.

133. Rhodes LE, Durham BH, Fraser WD, Friedmann PS. Dietary fish oil reduces basal and ultraviolet B-generated PGE2 levels in skin and increases the threshold to provocation of polymorphic light eruption. J Invest Dermatol 1995; 105(4):532–5.
134. Rhodes LE, O'Farrell S, Jackson MJ, Friedmann PS. Dietary fish-oil supplementation in humans reduces UVB-erythemal sensitivity but increases epidermal lipid peroxidation. J Invest Dermatol 1994; 103(2):151–4.
135. Gresham A, Masferrer J, Chen X, Leal-Khouri S, Pentland AP. Increased synthesis of high-molecular-weight cPLA2 mediates early UV-induced PGE2 in human skin. Am J Physiol 1996; 270(4 Pt 1):C1037–50.
136. Shreedhar V, Giese T, Sung VW, Ullrich SE. A cytokine cascade including prostaglandin E2, IL-4, and IL-10 is responsible for UV-induced systemic immune suppression. J Immunol 1998; 160(8):3783–9.
137. Moison RM, Steenvoorden DP, Beijersbergen van Henegouwen GM. Topically applied eicosapentaenoic acid protects against local immunosuppression induced by UVB irradiation, cis-urocanic acid and thymidine dinucleotides. Photochem Photobiol 2001; 73(1):64–70.
138. Pupe A, Moison R, De Haes P, et al. Eicosapentaenoic acid, a n-3 polyunsaturated fatty acid differentially modulates TNF-alpha, IL-1alpha, IL-6 and PGE2 expression in UVB-irradiated human keratinocytes. J Invest Dermatol 2002; 118(4):692–8.
139. Storey A, McArdle F, Friedmann PS, Jackson MJ, Rhodes LE. Eicosapentaenoic acid and docosahexaenoic acid reduce UVB- and TNF-alpha-induced IL-8 secretion in keratinocytes and UVB-induced IL-8 in fibroblasts. J Invest Dermatol 2005; 124(1):248–55.
140. Svobodova A, Walterova D, Vostalova J. Ultraviolet light induced alteration to the skin. Biomed Pap Med Fac Univ Palacky Olomouc Czech Repub 2006; 150(1):25–38.
141. Dreiling L, Hoffman S, Robinson WA. Melanoma: Epidemiology, pathogenesis, and new modes of treatment. Adv Intern Med 1996; 41:553–604.
142. Atkins MB. The treatment of metastatic melanoma with chemotherapy and biologics. Curr Opin Oncol 1997; 9(2):205–13.
143. Reich R, Royce L, Martin GR. Eicosapentaenoic acid reduces the invasive and metastatic activities of malignant tumor cells. Biochem Biophys Res Commun 1989; 160(2):559–64.
144. Abbott WG, Tezabwala B, Bennett M, Grundy SM. Melanoma lung metastases and cytolytic effector cells in mice fed antioxidant-balanced corn oil or fish oil diets. Nat Immun 1994; 13(1):15–28.
145. Albino AP, Juan G, Traganos F, et al. Cell cycle arrest and apoptosis of melanoma cells by docosahexaenoic acid: Association with decreased pRb phosphorylation. Cancer Res 2000; 60(15):4139–45.
146. Xia S, Lu Y, Wang J, et al. Melanoma growth is reduced in fat-1 transgenic mice: impact of omega-6/omega-3 essential fatty acids. Proc Natl Acad Sci USA 2006; 103(33):12499–504.
147. Wu H, Goel V, Haluska FG. PTEN signaling pathways in melanoma. Oncogene 2003; 22(20):3113–22.
148. Sansal I, Sellers WR. The biology and clinical relevance of the PTEN tumor suppressor pathway. J Clin Oncol 2004; 22(14):2954–63.
149. Stokoe D. Pten. Curr Biol 2001; 11(13):R502.
150. Guldberg P, thor Straten P, Birck A, Ahrenkiel V, Kirkin AF, Zeuthen J. Disruption of the MMAC1/PTEN gene by deletion or mutation is a frequent event in malignant melanoma. Cancer Res 1997; 57(17):3660–3.
151. Tsao H, Zhang X, Benoit E, Haluska FG. Identification of PTEN/MMAC1 alterations in uncultured melanomas and melanoma cell lines. Oncogene 1998; 16(26):3397–402.
152. Hwang PH, Yi HK, Kim DS, Nam SY, Kim JS, Lee DY. Suppression of tumorigenicity and metastasis in B16F10 cells by PTEN/MMAC1/TEP1 gene. Cancer Lett 2001; 172(1):83–91.
153. Denkins Y, Kempf D, Ferniz M, Nileshwar S, Marchetti D. Role of omega-3 polyunsaturated fatty acids on cyclooxygenase-2 metabolism in brain-metastatic melanoma. J Lipid Res 2005; 46(6):1278–84.

154. Gores GJ. Cholangiocarcinoma: Current concepts and insights. Hepatology (Baltimore, MD) 2003; 37(5):961–9.
155. Sirica AE. Cholangiocarcinoma: Molecular targeting strategies for chemoprevention and therapy. Hepatology (Baltimore, MD) 2005; 41(1):5–15.
156. Berthiaume EP, Wands J. The molecular pathogenesis of cholangiocarcinoma. Semin Liver Dis 2004; 24(2):127–37.
157. Wu T. Cyclooxygenase-2 and prostaglandin signaling in cholangiocarcinoma. Biochim Biophys Acta 2005; 1755(2):135–50.
158. Lazaridis KN, Gores GJ. Cholangiocarcinoma. Gastroenterology 2005; 128(6):1655–67.
159. Malhi H, Gores GJ. Cholangiocarcinoma: Modern advances in understanding a deadly old disease. J Hepatol 2006; 45(6):856–67.
160. Larsson SC, Kumlin M, Ingelman-Sundberg M, Wolk A. Dietary long-chain n-3 fatty acids for the prevention of cancer: a review of potential mechanisms. Am J Clin Nutr 2004; 79(6):935–45.
161. Endo K, Yoon BI, Pairojkul C, Demetris AJ, Sirica AE. ERBB-2 over expression and cyclooxygenase-2 up-regulation in human cholangiocarcinoma and risk conditions. Hepatology (Baltimore, MD) 2002; 36(2):439–50.
162. Hayashi N, Yamamoto H, Hiraoka N, et al. Differential expression of cyclooxygenase-2 (COX-2) in human bile duct epithelial cells and bile duct neoplasm. Hepatology (Baltimore, MD) 2001; 34 (4 Pt 1):638–50.
163. Chariyalertsak S, Sirikulchayanonta V, Mayer D, et al. Aberrant cyclooxygenase isozyme expression in human intrahepatic cholangiocarcinoma. Gut 2001; 48(1):80–6.
164. Han C, Leng J, Demetris AJ, Wu T. Cyclooxygenase-2 promotes human cholangiocarcinoma growth: evidence for cyclooxygenase-2-independent mechanism in celecoxib-mediated induction of p21waf1/cip1 and p27kip1 and cell cycle arrest. Cancer Res 2004; 64(4):1369–76.
165. Wu T, Leng J, Han C, Demetris AJ. The cyclooxygenase-2 inhibitor celecoxib blocks phosphorylation of Akt and induces apoptosis in human cholangiocarcinoma cells. Mol Cancer Ther 2004; 3(3):299–307.
166. Wu T, Han C, Lunz JG, 3rd, Michalopoulos G, Shelhamer JH, Demetris AJ. Involvement of 85-kd cytosolic phospholipase A(2) and cyclooxygenase-2 in the proliferation of human cholangiocarcinoma cells. Hepatology (Baltimore, MD) 2002; 36(2):363–73.
167. Nzeako UC, Guicciardi ME, Yoon JH, Bronk SF, Gores GJ. COX-2 inhibits Fas-mediated apoptosis in cholangiocarcinoma cells. Hepatology (Baltimore, MD) 2002; 35(3):552–9.
168. Zhang Z, Lai GH, Sirica AE. Celecoxib-induced apoptosis in rat cholangiocarcinoma cells mediated by Akt inactivation and Bax translocation. Hepatology (Baltimore, Md) 2004; 39(4):1028–37.
169. Lai GH, Zhang Z, Sirica AE. Celecoxib acts in a cyclooxygenase-2-independent manner and in synergy with emodin to suppress rat cholangiocarcinoma growth in vitro through a mechanism involving enhanced Akt inactivation and increased activation of caspases-9 and -3. Mol Cancer Ther 2003; 2(3):265–71.
170. Sirica AE, Lai GH, Endo K, Zhang Z, Yoon BI. Cyclooxygenase-2 and ERBB-2 in cholangiocarcinoma: potential therapeutic targets. Semin Liver Dis 2002; 22(3):303–13.
171. Smith WL. Cyclooxygenases, peroxide tone and the allure of fish oil. Curr Opin Cell Biol 2005; 17(2):174–82.
172. Ashida K, Terada T, Kitamura Y, Kaibara N. Expression of E-cadherin, alpha-catenin, beta-catenin, and CD44 (standard and variant isoforms) in human cholangiocarcinoma: an immunohistochemical study. Hepatology (Baltimore, MD) 1998; 27(4):974–82.
173. Sugimachi K, Taguchi K, Aishima S, et al. Altered expression of beta-catenin without genetic mutation in intrahepatic cholangiocarcinoma. Mod Pathol 2001; 14(9):900–5.
174. Tokumoto N, Ikeda S, Ishizaki Y, et al. Immunohistochemical and mutational analyses of Wnt signaling components and target genes in intrahepatic cholangiocarcinomas. Int J Oncol 2005; 27(4):973–80.

175. Settakorn J, Kaewpila N, Burns GF, Leong AS. FAT, E-cadherin, beta catenin, HER 2/neu, Ki67 immuno-expression, and histological grade in intrahepatic cholangiocarcinoma. J Clin Pathol 2005; 58(12):1249–54.
176. Hoppler S, Kavanagh CL. Wnt signalling: variety at the core. J Cell Sci 2007; 120(Pt 3):385–93.
177. Clevers H. Wnt/beta-catenin signaling in development and disease. Cell 2006; 127(3):469–80.
178. Gordon MD, Nusse R. Wnt signaling: multiple pathways, multiple receptors, and multiple transcription factors. J Biol Chem 2006; 281(32):22429–33.
179. Moon RT, Kohn AD, De Ferrari GV, Kaykas A. WNT and beta-catenin signalling: Diseases and therapies. Nat Rev Genet 2004; 5(9):691–701.
180. Castellone MD, Teramoto H, Williams BO, Druey KM, Gutkind JS. Prostaglandin E2 promotes colon cancer cell growth through a Gs-axin-beta-catenin signaling axis. Science 2005; 310(5753):1504–10.
181. Shao J, Jung C, Liu C, Sheng H. Prostaglandin E2 stimulates the beta-catenin/T cell factor-dependent transcription in colon cancer. J Biol Chem 2005; 280(28):26565–72.
182. Mayani H, Flores-Figueroa E, Chavez-Gonzalez A. In vitro biology of human myeloid leukemia. Leuk Res 2009; 33(5):624–37.
183. Lu C, Hassan HT. Human stem cell factor-antibody [anti-SCF] enhances chemotherapy cytotoxicity in human CD34+ resistant myeloid leukemia cells. Leuk Res 2006; 30(3):296–302.
184. Krause DS, Van Etten RA. Right on target: Eradicating leukemic stem cells. Trends Mol Med 2007; 13(11):470–81.
185. Yamagami T, Porada CD, Pardini RS, Zanjani ED, Almeida-Porada G. Docosahexaenoic acid induces dose dependent cell death in an early undifferentiated subtype of acute myeloid leukemia cell line. Cancer Biol Ther 2009; 8(4):331–7.
186. Stulnig TM, Huber J, Leitinger N, et al. Polyunsaturated eicosapentaenoic acid displaces proteins from membrane rafts by altering raft lipid composition. J Biol Chem 2001; 276(40):37335–40.
187. Li Q, Tan L, Wang C, et al. Polyunsaturated eicosapentaenoic acid changes lipid composition in lipid rafts. Eur J Nutr 2006; 45(3):144–51.
188. Colquhoun A, Schumacher RI. Gamma-Linolenic acid and eicosapentaenoic acid induce modifications in mitochondrial metabolism, reactive oxygen species generation, lipid peroxidation and apoptosis in Walker 256 rat carcinosarcoma cells. Biochim Biophys Acta 2001; 1533(3):207–19.
189. Sturlan S, Baumgartner M, Roth E, Bachleitner-Hofmann T. Docosahexaenoic acid enhances arsenic trioxide-mediated apoptosis in arsenic trioxide-resistant HL-60 cells. Blood 2003; 101(12):4990–7.
190. Leonardi F, Attorri L, Benedetto RD, et al. Docosahexaenoic acid supplementation induces dose and time dependent oxidative changes in C6 glioma cells. Free Radic Res 2007; 41(7):748–56.
191. Koumura T, Nakamura C, Nakagawa Y. Involvement of hydroperoxide in mitochondria in the induction of apoptosis by the eicosapentaenoic acid. Free Radic Res 2005; 39(3):225–35.
192. Menendez JA, Lupu R, Colomer R. Exogenous supplementation with omega-3 polyunsaturated fatty acid docosahexaenoic acid (DHA; 22:6n-3) synergistically enhances taxane cytotoxicity and downregulates Her-2/neu (c-erbB-2) oncogene expression in human breast cancer cells. Eur J Cancer Prev 2005; 14(3):263–70.
193. Narayanan NK, Narayanan BA, Bosland M, Condon MS, Nargi D. Docosahexaenoic acid in combination with celecoxib modulates HSP70 and p53 proteins in prostate cancer cells. Int J Cancer 2006; 119(7):1586–98.
194. Ho SY, Chen WC, Chiu HW, Lai CS, Guo HR, Wang YJ. Combination treatment with arsenic trioxide and irradiation enhances apoptotic effects in U937 cells through increased mitotic arrest and ROS generation. Chem Biol Interact 2009; 179(2–3):304–13.

195. Han YH, Kim SZ, Kim SH, Park WH. Arsenic trioxide inhibits the growth of Calu-6 cells via inducing a G2 arrest of the cell cycle and apoptosis accompanied with the depletion of GSH. Cancer Lett 2008; 270(1):40–55.
196. Kim HR, Kim EJ, Yang SH, et al. Combination treatment with arsenic trioxide and sulindac augments their apoptotic potential in lung cancer cells through activation of caspase cascade and mitochondrial dysfunction. Int J Oncol 2006; 28(6):1401–8.
197. Woo SH, Park IC, Park MJ, et al. Arsenic trioxide induces apoptosis through a reactive oxygen species-dependent pathway and loss of mitochondrial membrane potential in HeLa cells. Int J Oncol 2002; 21(1):57–63.
198. Brown M, Bellon M, Nicot C. Emodin and DHA potently increase arsenic trioxide interferon-alpha-induced cell death of HTLV-I-transformed cells by generation of reactive oxygen species and inhibition of Akt and AP-1. Blood 2007; 109(4):1653–9.
199. Nowell PC, Hungerford DA. Chromosome studies on normal and leukemic human leukocytes. J Natl Cancer Inst 1960; 25:85–109.
200. Rowley JD. Letter: A new consistent chromosomal abnormality in chronic myelogenous leukemia identified by quinacrine fluorescence and Giemsa staining. Nature 1973; 243(5405):290–3.
201. Sattler M, Griffin JD. Molecular mechanisms of transformation by the BCR-ABL oncogene. Semin Hematol 2003; 40(2 Suppl 2):4–10.
202. Cortes J, Kim DW, Raffoux E, et al. Efficacy and safety of dasatinib in imatinib-resistant or -intolerant patients with chronic myeloid leukemia in blast phase. Leukemia 2008; 22(12):2176–83.
203. Franceschino A, Tornaghi L, Piazza R, Pogliani E, Gambacorti-Passerini C. Imatinib failed to eradicate chronic myeloid leukemia in a patient with minimal residual disease. Haematologica 2006; 91(6 Suppl):ECR14.
204. Gambacorti-Passerini C, Piazza R, D'Incalci M. Bcr-Abl mutations, resistance to imatinib and imatinib plasma levels. Blood 2003; 102(5):1933–4; author reply 4–5.
205. de Lima TM, Amarante-Mendes GP, Curi R. Docosahexaenoic acid enhances the toxic effect of imatinib on Bcr-Abl expressing HL-60 cells. Toxicol In Vitro 2007; 21(8):1678–85.
206. Wang Y, Li L, Jiang W, Larrick JW. Synthesis and evaluation of a DHA and 10-hydroxycamptothecin conjugate. Bioorg Med Chem 2005; 13(19):5592–9.

Chapter 9
The Influence of ω-3 PUFAs on Chemo- or Radiation Therapy for Cancer

W. Elaine Hardman

Abstract The efficacy of ω-3 fatty acids against cancer has been demonstrated in preclinical in vitro and in vivo studies. However, for most clinical trials, it is unlikely that ω-3 supplementation would be used alone but rather it would be combined with standard chemo- or radiation therapy. Thus, it is important to know the effects of ω-3 fatty acids on the efficacy of standard therapies. Preclinical studies using numerous cancer models with multiple chemotherapeutic agents have demonstrated the enhanced efficacy of the standard therapy when ω-3 fatty acids are incorporated in the culture media of cells or in the diet of the animal. These studies will be summarized in this chapter.

Keywords ω-3 fatty acids · Cancer · Chemotherapy · Oxidative stress · PUFA

9.1 Introduction

Previous chapters have provided evidence of the effects of ω-3 polyunsaturated fatty acids (PUFAs) on cancer growth and/or survival both in vitro and in vivo. Multiple mechanisms for tumor growth suppression by ω-3 PUFAs have been identified to date and were presented in Chapter 1, thus these mechanisms will not be described in detail in this chapter. Addition of ω-3 PUFAs to the diet, as done in in vivo studies, might be recommended for patients for whom, for reasons of age or other health problems, treatment with standard chemo- or radiation therapy is not an option. To date, clinical trials have not been performed to test whether the benefit against cancer from consumption of ω-3 PUFAs, as seen in animal studies, would be seen in humans. However, the reality is that for most patients treatment of cancer with ω-3 PUFAs alone would not be considered; thus, it is important to know how ω-3 PUFAs

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interact with various cancer therapeutic modalities. In addition, cancer patients frequently are very interested in what diets or other supplements they might consume to increase the benefit of standard cancer therapy. One study reported that 86% of breast cancer patients used some type of complementary or alternative medicine in the time immediately following diagnosis of cancer [1]. Unfortunately, there have been few controlled clinical trials using complementary and alternative supplements, including ω-3 PUFAs, in combination with standard therapies. In this chapter, the in vitro and in vivo studies of combinations of ω-3 PUFAs and chemo- or radiation therapy will be briefly summarized and the results of the few available human trials will be presented.

9.2 In Vitro Studies Using ω-3 PUFAs and Chemotherapeutic Agents

Numerous cell culture studies indicate that, with all cancer cell types tested, supplementing the culture media with ω-3 PUFAs, usually either EPA (eicosapentaenoic acid, 20:5*n*–3) or DHA (docosahexaenoic acid, 22:6*n*–3), increased the sensitivity of the cells to all chemotherapies tested. The increased sensitivity to chemotherapy has been associated with incorporation of the ω-3 PUFAs into the phospholipids of cell membranes. Activation of multiple mechanisms to suppress cancer cell growth and/or increase sensitivity to chemotherapy has been reported.

For example, many cancer cell lines representing a broad range of cancer types have been utilized for testing ω-3 PUFAs in combination with various chemotherapeutic drugs. Cancer types that have been tested include L1210 [2] and P388 [3] mouse leukemia; L5178 mouse lymphoma [4]; A427 lung cancer [5]; human glioblastoma [5]; small cell lung cancer [6]; both estrogen-independent MDA-MB-231 [7–9] and estrogen-dependent MCF 7 [9–11] or T47D [12] human breast cancers; PC-3 [13] or LnCap [14] human prostate cancer; SKOV-3 [13] and A-2780 [15] human ovarian cancer; and Caco-2 [16], LS-174T [17], or HCA-7 [18] colon cancer. In every instance, ω-3 PUFAs enhanced the sensitivity of cancer cells to the chemotherapeutic agent. The results of these studies have also provided insights into multiple mechanisms of action for increasing the sensitivity of cells to the drug. Of interest is that the mechanisms do not appear to be mutually exclusive, that is, it is likely that more than one mechanism contributed to the enhanced sensitivity to chemotherapy after treatment with ω-3 PUFAs.

9.3 Mechanisms of Action for Increased Sensitivity to Drug

Cell culture studies have allowed identification of diverse mechanisms for the action of ω-3 PUFAs to enhance cancer sensitivity to chemotherapy. Since cancer cells, even within the same tumor, demonstrate heterogeneity and chemotherapeutic agents have multiple mechanisms of action, it seems that targeting multiple mechanisms could allow for greater cancer cell kill.

One of the first mechanisms identified for the action of ω-3 PUFAs was that of increasing lipid peroxidation in cell membranes [7, 19, 20]. The long, polyunsaturated hydrocarbon chains of EPA and DHA are easily oxidized. Increased lipid peroxidation, alone, following incorporation of long-chain ω-3 PUFAs in the cells has been associated with increased cell kill or decreased tumor growth [7, 19, 21, 22]. Thus, it seems reasonable that the efficacy of drugs that induce oxidative stress should be increased if the fractions of EPA and/or DHA are increased in cell membranes. This mechanism will be addressed first.

9.3.1 Oxidative Stress

A common drug that induces oxidative stress is doxorubicin (adriamycin). The efficacy of doxorubicin has been increased by ω-3 PUFAs and increased cell kill has been associated with increased oxidative stress in lymphoma cells [4], breast cancer cells [7], leukemia cells [23, 24], ovarian cancer cells [15], and glioblastoma [5].

Ether lipid drugs intercalate into and damage cell membranes. It seems logical that increasing the susceptibility of the membrane to oxidative damage could increase the efficacy of ether lipid drugs. ω-3 PUFA enhancement of cell kill associated with increased oxidative damage has been reported in murine leukemia cells with BM-41440 [24] and in human breast cancer cells with ET-18-OCH3 (edelfosine) [8]. The mechanism of action of arsenic trioxide is not well understood but the drug induces apoptosis, promotes cell differentiation, and suppresses cell proliferation in many cell types (NIH Drug Dictionary, www.cancer.gov). The activity of arsenic trioxide has been increased by ω-3 PUFAs; an associated mechanism was an increase in oxidative damage in colon, breast, prostate, ovarian, and melanoma cancer cells [13] and in human leukemia cells [25]. Clearly increasing oxidative damage is a mechanism by which ω-3 PUFAs can increase the efficacy of chemotherapy.

9.3.2 Altered Membrane Permeability, Fluidity, or Drug Uptake

The altered structure of the membrane due to the presence of long-chain ω-3 PUFAs changes the permeability and fluidity of cell membrane [26–29]. Such changes have the potential to modify the activity of transmembrane receptors and are also associated with changes in chemotherapeutic drug uptake [30]. The uptake of doxorubicin in mouse leukemia [2, 30, 31], human lung cancer [5, 6], cervical cancer [32], breast cancer [33], nasopharyngeal [33], and human glioblastoma cells [5] was increased by incorporation of ω-3 PUFAs into the cells. ω-3 PUFA treatment increased *cis*-platin uptake in small cell lung cancer [34] and cervical cancer cells [32] and vincristine uptake in neuroblastoma [35] or cervical cancer cells [32].

9.3.3 Changes in Apoptosis Susceptibility

Most chemotherapeutic drugs ultimately function by inducing apoptosis in cancer cells [36]. However, cancers are frequently resistant to apoptosis [37]; any mechanism to overcome that resistance can be beneficial to cancer therapy by restoring sensitivity to drug. Multiple reports indicate that ω-3 PUFAs can alter the expression or activity of a number of genes associated with apoptosis. DHA has been shown to downregulate the expression of the anti-apoptotic Bcl-2 in colon cancer cells without changing the expression of pro-apoptotic Bax [38]. EPA downregulated Bcl-2 in HL-60 human leukemic cells [39], whereas DHA upregulated the expression of Bax [40] in the same cell type. In human pancreatic cancer [41], human breast cancer, and bladder cancer cells [42], induction of apoptosis was associated with caspase-3 activation. Sensitivity to arsenic trioxide (a drug which induces apoptosis) was restored in leukemia [25], colon, breast, ovarian, prostate and melanoma cancer cells [13] by treatment with ω-3 PUFAs. Sensitivity of doxorubicin-resistant mouse leukemia [3] or small cell lung cancer [6] has also been restored by ω-3 PUFA treatment of the cells. It is likely that altering the expression of genes associated with apoptosis is part of the mechanism by which ω-3 PUFAs can increase the efficacy of or restore sensitivity to chemotherapy.

9.3.4 Modulation of the Transcription Factor, Nuclear Factor Kappa B (NF-κB)

NF-κB is an oxidant stress-sensitive [43] transcription factor that is often deregulated in cancers [44, 45]. NF-κB has been extensively studied and its contribution to carcinogenesis has been well documented [46]. Activation of NF-κB has been linked to inflammation, proliferation, apoptosis inhibition, angiogenesis, cell proliferation, cell transformation, and metastasis [46]. NF-κB has been identified as a good target for chemotherapy [47] and as a molecule that is often activated by chemotherapy leading to chemoresistance [48–51]. It is clear that ω-3 PUFAs can reduce the activation of NF-κB [50, 52, 53] and that the expression of cancer promoting genes downstream of NF-κB is altered [54] following incorporation of ω-3 PUFAs into cells.

Inhibition of NF-κB has been identified as a mechanism of action for the increased efficacy of chemotherapy with 5-FU (5-fluorouracil) or CPT-11 (irinotecan) in gastric cancer [51]. The ability of genistein to suppress invasion of MDA-231 human breast cancer cells was increased [55] by DHA; the associated mechanism was suppression of COX-2 (cyclooxygenase-2), a gene downstream of NF-κB. DHA with celecoxib suppressed NF-κB and prevented prostate cell growth [14] whereas DHA enhanced the efficacy of docetaxel against prostate cancer cells [56].

9.3.5 Altered Activity of Membrane-Associated Receptors

Altering the activity of membrane-associated receptors has been identified as a mechanism for ω-3 PUFAs and this activity may be a mechanism for increasing the efficacy of chemotherapy. Taxane [both paclitaxel (Taxol) and docetaxel (Taxotere)] efficacy against MDA-231 breast cancer was increased by ω-3 PUFAs; the implicated mechanism was the suppression of Her-2/neu oncoprotein activity [12]. Especially important, the sensitivity to taxanes of taxane-resistant SK-Br3 and BT-474 breast cancer cells was restored by DHA treatment [12] and the Her-2/neu expression was significantly decreased.

9.3.6 Altered Cell Signaling

ω-3 PUFAs have been shown to influence the activity of multiple cell-signaling cascades. Several studies have associated an altered activity of cell-signaling cascades with increased efficacy of chemotherapy or restored sensitivity to the therapy.

EPA has been shown to restore tamoxifen sensitivity to insensitive breast cancer cells via suppression of Akt activity [11]. DHA treatment was associated with the normalization of β-catenin activity in colon cancer cells also treated with a synthetic selenocyanate [16].

Lipid rafts are rich in signaling molecules and both EPA and DHA have been reported to affect the localization and the activity of signaling molecules in lipid rafts [57–59]. Treatment of cells with EPA+DHA has been shown to reduce epidermal growth factor receptor (EGFR) localization in lipid rafts and to increase phosphorylation of both EGFR and p38 mitogen-activated protein kinase resulting in apoptosis [60]. This would be expected to increase/restore the sensitivity of cells to chemotherapy.

9.3.7 In Vivo Studies on Enhanced Chemotherapy with ω-3 PUFAs

There are multiple reports of studies on the ability of ω-3 PUFAs to suppress growth of xenografts in nude mice or the development and growth of cancer in carcinogen-treated or transgenic animals. There are fewer studies that report the results of combinations of ω-3 PUFAs and chemotherapy in vivo. In these studies, ω-3 PUFAs are usually administered mixed into the diet. Usually fish oil or a fish oil concentrate is the source of the EPA and DHA. The results of these studies support the benefit of ω-3 PUFAs with chemotherapy as was seen in cell culture. Additionally, some of the studies not only report increased efficacy of the drug but also report reduced side effects from the chemotherapy when ω-3 PUFAs were included into the diet.

ω-3 PUFAs increased the response of human breast cancers in nude mice to mitomycin C [61], cyclophosphamide [61], doxorubicin [62, 63], edelfosine [8], and irinotecan [64]. In rats, the efficacy of irinotecan against a Ward colon tumor [65], of epirubicin against NMU (nitrosomethylurea)-induced mammary cancer [66, 67], of araC against fibrosarcoma [68] was increased by ω-3 PUFAs in the diet. In mice being treated with *cis*-platin, Lewis lung carcinoma growth was suppressed in mice consuming ω-3 PUFAs compared to mice that did not consume the fish oil [69]. In adult dogs with lymphoma, adding ω-3 PUFAs to doxorubicin therapy extended survival time compared to dogs that did not consume ω-3 PUFAs [70].

Several of these studies also assessed the side effects to the animal of combining chemotherapy and ω-3 PUFAs. Even though the efficacy of the drug was increased, the toxic side effects of the drug were frequently decreased. The dose-limiting toxicity of irinotecan is often diarrhea. In mice, which do not develop diarrhea but do exhibit the intestinal damage associated with diarrhea, consumption of ω-3 PUFAs completely ameliorated the damage due to irinotecan treatment [64]. In araC-treated rats that also consumed ω-3 PUFAs, white cell precursors were increased and intestinal crypt depth was maintained [68] compared to rats that did not receive ω-3 PUFAs. In rats treated with doxorubicin, bone marrow cellularity was increased when ω-3 PUFAs were included in the diet [71]. ω-3 PUFAs have been demonstrated to protect rats from 5-FU-induced intestinal damage [72].

The ability of ω-3 PUFAs to reduce the side effects of chemotherapy would allow greater tolerance of the drug with the potential of increasing treatment. Increased treatment time combined with increasing the efficacy of therapy by the multiple mechanisms described in this and other chapters could have a dramatic effect on clinical outcome.

9.4 ω-3 PUFAs and Radiation Therapy

Many cancer patients receive radiation therapy concurrent with their chemotherapy, thus it is important to know how consuming ω-3 PUFAs could interact with the radiation treatment. A few studies have reported these interactions.

In cell culture, incubation with 30–100 μM of ω-3 PUFAs prior to a single 4 Gy radiation exposure resulted in a dose-dependent decrease in cell survival of LS174T, CO112, and Caco-2 colon cancer cells [73]. Addition of vitamin E partially abolished the benefit of ω-3 PUFAs. Induction of apoptosis was implicated as a mechanism [73]. Addition of polyunsaturated fatty acids, gamma linoleic acid (ω-3), and EPA and DHA enhanced radiation-induced killing of astrocytoma cells [74]. These data implicate enhanced susceptibility to oxidative damage as part of the mechanism for enhanced sensitivity of cancer cells to radiation following treatment with ω-3 PUFAs.

Peroxisome proliferator-activated receptors (PPARs) are a family of nuclear receptors that can be activated by fatty acids, prostaglandins, and other xenobiotics [75]. PPAR-γ activation can cause growth arrest, stimulate expression of p53 and of Fas ligand-inducing apoptosis [76]. In Ramos cells, activation of PPAR-γ and suppression of NF-κB were implicated as important to DHA-induced sensitizing of cells to radiation [77].

Flaxseed oil contains about 50% alpha linolenic acid, an 18 carbon ω-3 PUFA. In animals this 18 C fatty acid can be elongated and desaturated to EPA and DHA. Addition of flaxseed oil to the diet has reduced the liver toxicity of radiation [78] and oxidative lung damage [79] in mice.

The sensitivity of rat mammary tumors to radiation has been increased by consumption of DHA [80]. In this study, as in cell culture, addition of vitamin E (an antioxidant that should decrease oxidative damage) partially abrogated the benefit of DHA implicating that increased oxidative damage contributed to the mechanism of action. Consumption of a fish oil concentrate increased the efficacy of radiation treatment of implanted MDA-231 human breast cancer in nude mice [81]. Suppression of angiogenesis was implicated as part of the mechanism. As for chemotherapy, the enhanced efficacy of radiation therapy following treatment with ω-3 PUFAs can be linked to multiple mechanisms including oxidative stress, suppression of NF-κB, and activation of PPARs.

The toxic side effects of radiation have been reduced by pretreatment with ω-3 PUFAs. Radiation damages intestinal epithelium [82] but mucosal/epidermal reactions to radiation have been reduced by ω-3 PUFA treatment [83]. Taken as a whole, consumption of ω-3 PUFAs should be beneficial during radiation therapy for cancer.

9.5 ω-3 PUFAs in Clinical Trials

The NIH web site for clinical trials (http://nccam.nih.gov/research/clinicaltrials) lists 12 trials of use of ω-3 PUFAs. Under the heading 'omega 3 fatty acids,' there was one study on the use of ω-3 PUFAs for cancer cachexia and esophageal cancer. Under the heading 'cancer' there were two studies assessing the effects of ω-3 PUFAs plus other components on polycystic ovary syndrome. None of the trials that are in progress are studies to assess the effects of ω-3 PUFAs on cancer tumor either with or without chemotherapy.

There are several trials that have reported the use of ω-3 PUFAs on cancer cachexia; cachexia will be covered in the next chapter. A small trial assessed short-term supplementation with ω-3 PUFAs to modulate ω-3/ω-6 ratio in plasma, gluteal fat, and prostate of men with prostate cancer. The authors reported that the 3 months intervention increased the ω-3 PUFA fat content as expected and reduced the expression of COX-2 in the prostate [84]. This result would be expected to reduce tumor growth. Another small study reported the modulation of systemic syndrome in patients with advanced lung cancer [85]. In this study, there was improved body weight and muscle strength in patients who took ω-3 PUFAs and celecoxib (COX-2 inhibitor). Patients who took fish oil + placebo had improved appetite, less fatigue, and lower C-reactive protein than their baseline values. The level of DHA in adipose tissue, indicating higher consumption of ω-3 PUFA-containing foods, in breast cancer patients who achieved complete remission following standard primary chemotherapy (mitoxantrone, vindesine, cyclophosphamide, and 5-fluorouracil) was significantly higher than in those who did not achieve remission [86].

As previously noted, dogs with lymphoma that were supplemented with fish oil and arginine during the doxorubicin treatment had a longer survival time than those that did not receive the supplement [70].

There are so few clinical trials to date that it is difficult to determine whether ω-3 PUFA supplements could benefit cancer patients or not. However, the little information that is available agrees with the preclinical data. ω-3 PUFAs as an adjuvant to cancer therapy may well be very beneficial to patients. It is clear that ω-3 PUFAs function by multiple mechanisms, reducing the likelihood of cancer cells mutating around all the mechanisms. There is no evidence that supplementing with ω-3 PUFAs will harm patients even if they do not provide benefit. It seems that it is time for well-designed trials to test whether ω-3 PUFAs can provide benefits to humans with cancer.

References

1. Greenlee H, Kwan ML, Ergas IJ, Sherman KJ, Krathwohl SE, Bonnell C, et al. Complementary and alternative therapy use before and after breast cancer diagnosis: the Pathways Study. Breast Cancer Res Treat 2009;117:653–65.
2. Guffy MM, North JA, Burns CP. Effect of cellular fatty acid alteration on adriamycin sensitivity in cultured L1210 murine leukemia cells. Cancer Res 1984;44:1863–6.
3. Liu QY, Tan BK. Effects of cis-unsaturated fatty acids on doxorubicin sensitivity in P388/DOX resistant and P388 parental cell lines. Life Sci 2000;67(10):1207–18.
4. Kinsella JE, Black JM. Effects of polyunsaturated fatty acids on the efficacy of antineoplastic agents toward L5178Y lymphoma cells. Biochem Pharmacol 1993 May 5;45(9):1881–7.
5. Rudra PK, Krokan HE. Cell-specific enhancement of doxorubicin toxicity in human tumour cells by docosahexaenoic acid. Anticancer Res 2001 Jan;21(1A):29–38.
6. Zijlstra JG, de Vries EG, Muskiet FA, Martini IA, Timmer-Bosscha H, Mulder NH. Influence of docosahexaenoic acid in vitro on intracellular adriamycin concentration in lymphocytes and human adriamycin-sensitive and -resistant small-cell lung cancer cell lines, and on cytotoxicity in the tumor cell lines. Int J Cancer 1987 Dec 15;40(6):850–6.
7. Germain E, Chajes V, Cognault S, Lhuillery C, Bougnoux P. Enhancement of doxorubicin cytotoxicity by polyunsaturated fatty acids in the human breast tumor cell line MDA-MB-231: relationship to lipid peroxidation. Int J Cancer 1998;75:578–83.
8. Hardman WE, Barnes CJ, Knight CW, Cameron IL. Effects of iron supplementation and ET-18-OCH_3 on MDA-MB 231 breast carcinomas in nude mice consuming a fish oil diet. Br J Cancer 1997;76:347–54.
9. Wang Z, Butt K, Wang L, Liu H. The effect of seal oil on paclitaxel induced cytotoxicity and apoptosis in breast carcinoma MCF-7 and MDA-MB-231 cell lines. Nutr Cancer 2007;58(2):230–8.
10. Nakagawa H, Yamamoto D, Kiyozuka Y, Tsuta K, Uemura Y, Hioki K, et al. Effects of genistein and synergistic action in combination with eicosapentaenoic acid on the growth of breast cancer cell lines. J Cancer Res Clin Oncol 2000 Aug;126(8):448–54.
11. DeGraffenried LA, Friedrichs WE, Fulcher L, Fernandes G, Silva JM, Peralba JM, et al. Eicosapentaenoic acid restores tamoxifen sensitivity in breast cancer cells with high Akt activity. Ann Oncology 2003;14(7):969–70.

12. Menendez JA, Ropero S, Lupu R, Colomer R. Omega-6 polyunsaturated fatty acid gamma-linolenic acid (18:3n-6) enhances docetaxel (Taxotere) cytotoxicity in human breast carcinoma cells: Relationship to lipid peroxidation and HER-2/neu expression. Oncol Rep 2004 Jun;11(6):1241–52.
13. Baumgartner M, Sturlan S, Roth E, Wessner B, Bachleitner-Hofmann T. Enhancement of arsenic trioxide-mediated apoptosis using docosahexaenoic acid in arsenic trioxide-resistant solid tumor cells. Int J Cancer 2004 Nov;112(4):707–12.
14. Narayanan NK, Narayanan BA, Reddy BS. A combination of docosahexaenoic acid and celecoxib prevents prostate cancer cell growth in vitro and is associated with modulation of nuclear factor-kappaB, and steroid hormone receptors. Int J Oncol 2005;26(3):785–92.
15. Plumb JA, Luo W, Kerr DJ. Effect of polyunsaturated fatty acids on the drug sensitivity of human tumour cell lines resistant to either cisplatin or doxorubicin. Br J Cancer 1993;67:728–33.
16. Narayanan BA, Narayanan NK, Desai D, Pittman B, Reddy BS. Effects of a combination of docosahexaenoic acid and 1,4-phenylene bis(methylene) selenocyanate on cyclooxygenase 2, inducible nitric oxide synthase and beta-catenin pathways in colon cancer cells. Carcinogenesis 2004 Dec;25(12):2443–9.
17. Habbel P, Weylandt KH, Lichopoj K, Nowak J, Purschke M, Wang JD, et al. Docosahexaenoic acid suppresses arachidonic acid-induced proliferation of LS-174T human colon carcinoma cells. World J Gastroenterol 2009 Mar 7;15(9):1079–84.
18. Swamy MV, Cooma I, Patlolla JM, Simi B, Reddy BS, Rao CV. Modulation of cyclooxygenase-2 activities by the combined action of celecoxib and decosahexaenoic acid: novel strategies for colon cancer prevention and treatment. Mol Cancer Ther 2004 Feb;3(2):215–21.
19. Das UN. Tumoricidal action of cis-unsaturated fatty acids and their relationship to free radicals and lipid peroxidation. Cancer Lett 1991;56:235–43.
20. Masotti L, Casali E, Galeotti T. Lipid peroxidation in tumor cells. Free Rad Biol Med 1988;4:377–86.
21. Gonzalez MJ, Schemmel RA, Dugan L, Gray JI, Welsch CW. Dietary fish oil inhibits human breast carcinoma growth: a function of increased lipid peroxidation. Lipids 1993;28:827–32.
22. Hardman WE, Munoz J, Cameron IL. Role of lipid peroxidation and antioxidant enzymes in omega-3 fatty acids induced suppression of breast cancer xenograft growth in mice. Cancer Cell Int 2002;2(10):17.
23. de Salis HM, Meckling-Gill KA. EPA and DHA alter nucleoside drug and doxorubicin toxicity in L1210 cells but not in normal murine S1 macrophages. Cellular Pharmacology 1995;2:69–74.
24. Petersen ES, Kelley EE, Modest EJ, Burns CP. Membrane lipid modification and sensitivity of leukemic cells to the thioether lipid analogue BM 41.440. Cancer Res 1992 Nov 15;52(22):6263–9.
25. Sturlan S, Baumgartner M, Roth E, Bachleitner-Hofmann T. Docosahexaenoic acid enhances arsenic trioxide-mediated apoptosis in arsenic trioxide-resistant HL-60 cells. Blood 2003 Jun 15;101(12):4990–7.
26. Fischer MA, Black HS. Modification of membrane composition, eicosanoid metabolism, and immunoresponsiveness by dietary omega-3 and omega-6 fatty acid sources, modulators of ultraviolet-carcinogenesis. Photochem Photobiol 1991;54:381–7.
27. Jenski LJ, Sturdevant LK, Ehringer WD, Stillwell W. Omega-3 fatty acid modification of membrane structure and function. I. Dietary manipulation of tumor cell susceptibility to cell- and complement- mediated lysis. Nutr Cancer 1993;19:135–46.
28. Lund EK, Harvey LJ, Ladha S, Clark DC, Johnson IT. Effects of dietary fish oil supplementation on the phospholipid composition and fluidity of cell membranes from human volunteers. Ann Nutr Metab 1999;43(5):290–300.
29. Stillwell W, Ehringer W, Jenski LJ. Docosahexaenoic acid increases permeability of lipid vesicles and tumor cells. Lipids 1993 Feb;28(2):103–8.

30. Burns CP, Spector AA. Membrane fatty acid modifications in tumor cells: potential therapeutic adjunct. Lipids 1987;22:178–84.
31. Burns CP, North JA. Adriamycin transport and sensitivity in fatty acid-modified leukemia cells. Biochim Biophys Acta 1986 Aug 29;888(1):10–7.
32. Das UN, Madhavi G, Kumar GS, Padma M, Sangeetha P. Can tumour cell drug resistance be reversed by essential fatty acids and their metabolites? Prostaglandins Leukot Essent Fatty Acids 1998;58(1):39–54.
33. Abulrob AN, Mason M, Bryce R, Gumbleton M. The effect of fatty acids and analogues upon intracellular levels of doxorubicin in cells displaying P-glycoprotein mediated multidrug resistance. J Drug Target 2000;8(4):247–56.
34. Timmer-Bosscha H, Hospers GA, Meijer C, Mulder NH, Muskiet FA, Martini IA, et al. Influence of docosahexaenoic acid on cisplatin resistance in a human small cell lung carcinoma cell line. J Natl Cancer Inst 1989 Jul 19;81(14):1069–75.
35. Ikushima S, Fujiwara F, Todo S, Imashuku S. Effects of polyunsaturated fatty acids on vincristine-resistance in human neuroblastoma cells. Anticancer Res 1991 May;11(3): 1215–20.
36. Kim R, Tanabe K, Uchida Y, Emi M, Inoue H, Toge T. Current status of the molecular mechanisms of anticancer drug-induced apoptosis. The contribution of molecular-level analysis to cancer chemotherapy.Cancer Chemother Pharmacol 2002 Nov;50(5):343–52.
37. Fulda S. Tumor resistance to apoptosis. Int J Cancer 2009 Feb 1;124(3):511–5.
38. Chen ZY, Istfan NW. Docosahexaenoic acid is a potent inducer of apoptosis in HT-29 colon cancer cells. Prostaglandins Leukot Essent Fatty Acids 2000 Nov;63(5):301–8.
39. Chiu LC, Wan JM. Induction of apoptosis in HL-60 cells by eicosapentaenoic acid (EPA) is associated with downregulation of bcl-2 expression. Cancer Lett 1999 Oct 18;145(1–2): 17–27.
40. Chiu LC, Wong EY, Ooi VE. Docosahexaenoic acid modulates different genes in cell cycle and apoptosis to control growth of human leukemia HL-60 cells. Int J Oncol 2004 Sep;25(3):737–44.
41. Shirota T, Haji S, Yamasaki M, Iwasaki T, Hidaka T, Takeyama Y, et al. Apoptosis in human pancreatic cancer cells induced by eicosapentaenoic acid. Nutrition 2005 Oct;21(10): 1010–7.
42. Colquhoun A. Mechanisms of action of eicosapentaenoic acid in bladder cancer cells in vitro: alterations in mitochondrial metabolism, reactive oxygen species generation and apoptosis induction. J Urol 2009 Apr;181(4):1885–93.
43. Kabe Y, Ando K, Hirao S, Yoshida M, Handa H. Redox regulation of NF-kappa B activation: distinct redox regulation between the cytoplasm and the nucleus. Antioxid Redox Signal 2005;7(3–4):395–403.
44. Shen HM, Tergaonkar V. NF-kappa B signaling in carcinogenesis and as a potential molecular target for cancer therapy. Apoptosis 2009 Apr;14(4):348–63.
45. Dolcet X, Llobet D, Pallares J, Matias-Guiu X. NF-κB in development and progression of human cancer. Virchows Arch 2005;446(5):475–82.
46. Aggarwal BB. Nuclear factor-kappaB: the enemy within. Cancer Cell 2004;6(3):203–8.
47. Kim HJ, Hawke N, Baldwin AS. NF-kappa B and IKK as therapeutic targets in cancer. Cell Death Differ 2006 May;13(5):738–47.
48. Cilloni D, Messa F, Arruga F, Defilippi I, Morotti A, Messa E, et al. The NF-kappa B pathway blockade by the IKK inhibitor PS1145 can overcome imatinib resistance. Leukemia 2006 Jan;20(1):61–7.
49. Endo T, Nishio M, Enzler T, Cottam HB, Fukuda T, James DF, et al. BAFF and APRIL support chronic lymphocytic leukemia B-cell survival through activation of the canonical NF-kappa B pathway. Blood 2007 Jan 15;109(2):703–10.
50. Hering J, Garrean S, Dekoj TR, Razzak A, Saied A, Trevino J, et al. Inhibition of proliferation by omega-3 fatty acids in chemoresistant pancreatic cancer cells. Ann Surg Oncol 2007 Dec;14(12):3620–8.

51. Hochwald SN, Li J, Copeland EM, III, Moldawer LL, Lind DS, Mackay SL. Inhibition of NF-κB activation potentiates the cytotoxicity of 5-FU and CPT-11 chemotherapy in human gastric cancer cells. Proc Am Assoc Can Res 2002;43(abstr.).
52. Ross JA, Maingay JP, Fearon KC, Sangster K, Powell JJ. Eicosapentaenoic acid perturbs signalling via the NF-kappa B transcriptional pathway in pancreatic tumour cells. Int J Oncol 2003 Dec;23(6):1733–8.
53. Schley PD, Jijon HB, Robinson LE, Field CJ. Mechanisms of omega-3 fatty acid-induced growth inhibition in MDA-MB-231 human breast cancer cells. Breast Cancer Res Treat 2005 Jul;92(2):187–95.
54. Narayanan BA, Narayanan NK, Simi B, Reddy BS. Modulation of inducible nitric oxide synthase and related proinflammatory genes by the omega-3 fatty acid docosahexaenoic acid in human colon cancer cells. Cancer Res 2003 Mar 1;63(5):972–9.
55. Horia E, Watkins BA. Complementary actions of docosahexaenoic acid and genistein on COX-2, PGE2 and invasiveness in MDA-MB-231 breast cancer cells. Carcinogenesis 2007 Apr;28(4):809–15.
56. Shaikh IA, Brown I, Schofield AC, Wahle KW, Heys SD. Docosahexaenoic acid enhances the efficacy of docetaxel in prostate cancer cells by modulation of apoptosis: the role of genes associated with the NF-kappa B pathway. Prostate 2008 Nov 1;68(15): 1635–46.
57. Stillwell W, Shaikh SR, Zerouga M, Siddiqui R, Wassall SR. Docosahexaenoic acid affects cell signaling by altering lipid rafts. Reprod Nutr Dev 2005 Sep;45(5):559–79.
58. Li Q, Tan L, Wang C, Li N, Li Y, Xu G, et al. Polyunsaturated eicosapentaenoic acid changes lipid composition in lipid rafts. Eur J Nutr 2006 Mar;45(3):144–51.
59. Kim W, Fan YY, Barhoumi R, Smith R, McMurray DN, Chapkin RS. n-3 polyunsaturated fatty acids suppress the localization and activation of signaling proteins at the immunological synapse in murine CD4+ T cells by affecting lipid raft formation. J Immunol 2008 Nov 1;181(9):6236–43.
60. Schley PD, Brindley DN, Field CJ. (n-3) PUFA alter raft lipid composition and decrease epidermal growth factor receptor levels in lipid rafts of human breast cancer cells. J Nutr 2007 Mar;137(3):548–53.
61. Shao Y, Pardini L, Pardini RS. Dietary menhaden oil enhances mitomycin C antitumor activity toward human mammary carcinoma MX-1. Lipids 1995;30:1035–45.
62. Borgeson CE, Pardini L, Pardini RS, Reitz R.C. Effects of dietary fish oil on human mammary carcinoma and on lipid-metabolizing enzymes. Lipids 1989;24:290–5.
63. Hardman WE, Avula CPR, Fernandes G, Cameron IL. Three percent dietary fish oil concentrate increased efficacy of doxorubicin against MDA-MB 231 human breast cancer xenografts. Clin Cancer Res 2001;7:2041–9.
64. Hardman WE, Moyer MP, Cameron IL. Fish oil supplementation enhanced CPT-11 (Irinotecan) efficacy against MCF7 breast carcinoma xenografts and ameliorated intestinal side effects. Br J Cancer 1999;81:440–8.
65. Xue H, Sawyer MB, Field CJ, Dieleman LA, Baracos VE. Nutritional modulation of antitumor efficacy and diarrhea toxicity related to irinotecan chemotherapy in rats bearing the ward colon tumor. Clin Cancer Res 2007 Dec 1;13(23):7146–54.
66. Colas S, Maheo K, Denis F, Goupille C, Hoinard C, Champeroux P, et al. Sensitization by dietary docosahexaenoic acid of rat mammary carcinoma to anthracycline: a role for tumor vascularization. Clin Cancer Res 2006 Oct 1;12(19):5879–86.
67. Germain E, Lavandier F, Chaiés V, Schubnel V, Bonnet P, Lhuillery C, Bougnoux P. Dietary n-3 polyunsaturated fatty acids and oxidants increase ret mammary tumor sensitivity to epirubicin without change in cardiac toxicity. Lipids 1999;34 Suppl:S203.
68. Atkinson TG, Murray L, Berry DM, Ruthig DJ, Meckling-Gill KA. DHA feeding provides host protection and prevents fibrosarcoma-induced hyperlipidemia while maintaining the tumor response to AraC in Fischer 344 rats. Nutr Cancer 1997;28(3):225–35.

Chapter 10
ω-3 PUFAs and Cachexia

Michael J. Tisdale

Abstract Patients with cancer, especially of the lung and gastrointestinal tract, exhibit a wasting syndrome called cachexia, in which there is extensive loss of both adipose tissue and skeletal muscle mass, leading to weakness and death. Although anorexia is invariably present, nutritional supplementation is unable to reverse the changes in body composition, especially the loss of lean body mass. ω-3 polyunsaturated fatty acids (PUFAs), in particular eicosapentaenoic acid (EPA), have been shown to attenuate the development of cachexia in murine models with retention of both fat and muscle protein. EPA has been shown to attenuate both the action and production of a lipolytic factor, zinc-α_2-glycoprotein (ZAG), involved in loss of adipose tissue in cachexia. EPA preserves skeletal muscle in cachexia by attenuating the increased protein degradation, although it has no effect on the depression of protein synthesis. The effect on protein degradation is due to downregulation of the increased expression and activity of the ubiquitin-proteasome proteolytic pathway by preventing nuclear binding of the transcription factor nuclear factor-κB (NF-κB). Clinical studies confirm the ability of EPA to attenuate weight loss in cachectic cancer patients with resulting weight stabilisation. However, when used in combination with a high calorie, high protein nutritional supplement weight gain was seen, although there have been problems with compliance in large-scale placebo-controlled trials. Further clinical studies are warranted to confirm activity and whether it is equally effective in the treatment of cachexia in different cancer types.

Keywords Appetite stimulation · Cachexia · Clinical trials · Eicosapentaenoic acid (EPA) · Muscle atrophy

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10.1 What Is Cachexia?

Cachexia literally means bad condition, but it is generally recognised as the wasting condition seen in cancer patients, which often precedes death. Cachexia is highly prevalent among patients with gastrointestinal tumours such as pancreas and stomach, as well as lung cancer, and is an important prognostic factor, since the greater the extent of weight loss and total weight loss, the shorter the survival time [1]. Patients with cancer cachexia show a progressive loss of both adipose tissue and skeletal muscle mass, while visceral proteins are preserved. Loss of adipose tissue and skeletal muscle can be extensive, such that patients with loss of 30% of body weight, which is close to death, had lost 85% of their adipose tissue and 75% of their skeletal muscle mass [2].

Cachexia also occurs in other conditions, e.g. in patients with chronic heart failure, about 16% of patients suffer from cardiac cachexia [3]. As with cancer patients this is associated with a poor prognosis (18-month mortality of 50% compared with non-cachectic subjects of 17%). Patients with acquired immunodeficiency syndrome (AIDS) also suffer from cachexia, and as with cancer patients death is imminent when they have lost 34% of their ideal body weight [4]. Loss of skeletal muscle is probably the most important factor governing survival in patients with cachexia. Loss of lean tissue eventually leads to loss of function of respiratory muscles resulting in death from hypostatic pneumonia [5]. A shorter survival time of patients with cancer of the lung, breast and gastrointestinal tract was independently associated with a weight loss of greater than 8.1 kg in the previous 6 months and serum albumin levels of less than 35 g/L [6]. In addition patients with non-small cell lung cancer (NSCLC) and mesothelioma with weight loss failed to complete at least three cycles of chemotherapy and had fewer symptomatic responses [7].

10.2 Anorexia and Cachexia

Anorexia is common in cancer patients, even if they are not receiving chemotherapy. It results from an imbalance of the neural signalling pathways regulating appetite and satiety. Although anorexia is frequently associated with cachexia a study of 297 cancer patients with solid tumours found that weight loss could not be accounted for by diminished dietary intake [8]. In fact dietary intake in the weight losing subjects was not different from non-weight losing controls, and the intake per kg body weight was actually higher in the weight losing patients. Weight losing cancer patients have been shown to have an elevated resting energy expenditure (REE) compared with undernourished non-cancer patients [9], although the total energy expenditure (TEE) and physical activity level (PAL) were reduced [10]. It seems that in cachexia weight loss and hypermetabolism are not compensated for by an increase in spontaneous food intake [8]. Nutritional supplementation is unable to counteract the decrease in lean body mass in cachectic subjects. Thus when individuals with HIV were given total parenteral nutrition (TPN), an increase in body weight was observed, but this did not represent accrual of lean body mass [4]. A similar situation is seen in sepsis where aggressive nutritional support results in weight gain, but body composition analysis showed this to be adipose tissue and not lean body mass [11]. In cancer patients TPN also failed to reverse the wasting process, and any weight gain was transitory and was due to accumulation of fat and water and not lean body mass [12]. Pharmacological manipulation of appetite has a similar effect. Thus treatment of cancer patients with either megestrol acetate (Megace) [13] or medroxyprogesterone acetate (Depo-Provera) [14] resulted in weight gain, but body composition analysis showed that the gained weight was due to adipose tissue and water, the latter being responsible for the majority of the weight gained, without a significant effect on lean body mass. Indomethacin has also been shown to produce a significant increase in food intake in a mixed group of cancer patients, but again this resulted in an increase in fat, but not lean body mass [9]. These results suggest that nutritional supplementation alone will not counteract loss of lean body mass, and that it is important to understand the mechanism of loss of adipose tissue and skeletal muscle in cachexia in order to devise effective therapy. Malnutrition resulting from therapy is different from that induced by the tumour itself, since it responds to nutritional supplementation.

10.3 Mechanism of Loss of Adipose Tissue in Cachexia

In cachectic mice there is extensive loss of white adipose tissue (WAT) resulting in shrunken adipocytes with severe delipidation and modification in cell membrane conformation [15]. In addition the mitochondria differ from typical WAT mitochondria, are electron dense and have increased cristae. These changes have been attributed to an increased lipolysis independent of malnutrition [16]. Both the expression and activity of the rate-limiting enzyme of the lipolytic pathway hormone-sensitive lipase (HSL) have been shown to be increased in cancer patients [17] resulting in a two- to threefold increase in the lipolytic effects of catecholamines and natriuretic peptide [16]. Patients who had lost weight through factors other than cancer cachexia showed no change in adipocyte lipolysis. There have been no reports to changes in expression of adipocyte triglyceride lipase (ATGL), which may be of less importance in regulating catecholamine-induced lipolysis. Although some studies have reported a decrease in plasma lipoprotein lipase (LPL), the enzyme responsible for the movement of fatty acids from the blood into adipocytes for triglyceride synthesis [18], other studies [17] have reported no change in either total LPL or the relative level of LPL mRNA in adipose tissue of cancer patients. This suggests that agents which are thought to provoke a decrease in triglyceride storage in WAT through inhibition of LPL (the majority of the cytokines) do not play a major role in loss of fat in cachexia.

10.4 Possible Mediators of Fat Loss in Cachexia

In addition to an increased expression, HSL is activated by phosphorylation through protein kinase A (PKA), which in turn is activated by cyclic AMP. Hormone production of cyclic AMP is stimulated as a consequence of GTP-binding protein

(G protein)-coupled receptors acting through adenylyl cyclase, which converts ATP into cyclic AMP. G protein-coupled receptors can also activate mitogen-activated protein kinase (MAPK) and extracellular signal-regulated kinase (ERK) pathways. Activated ERK phosphorylates HSL at Ser^{600}, one of the sites phosphorylated by PKA, increasing its activity [19].

Two factors have been identified that stimulate lipolysis through an elevation of cyclic AMP and which could play a role in loss of adipose tissue in cachexia. One of these is zinc-α_2-glycoprotein (ZAG) identified as a lipid-mobilising factor (LMF) associated with cachexia-inducing human cancers, and which stimulates lipolysis through activation of adenylyl cyclase in a GTP-dependent manner [20]. The other is tumour necrosis factor-α (TNF-α), originally suggested as a possible cachectic factor as a result of studies on the mechanism of weight loss in rabbits infected with *Trypanosoma brucei brucei*. TNF-α stimulates lipolysis in human white adipocytes through stimulation of MAPK and ERK [21]. In addition TNF-α has been shown to decrease the expression of cyclic AMP phosphodiesterase by 50%, suggesting another mechanism by which it could increase cyclic AMP levels. Both TNF-α and ZAG may also stimulate thermogenesis by increasing the expression of uncoupling proteins (UCP) in brown fat (BAT) and skeletal muscle [22, 23]. This would provide an energy sink for the released fatty acids from WAT, preventing resynthesis of triglycerides and driving the process of triglyceride hydrolysis.

Both ZAG and TNF-α are produced by adipose tissue, but while expression of ZAG has been shown to be elevated 10-fold during the process of wasting in cachexia [24] there have been no measurements of changes in TNF-α, although many studies have found no correlation between circulating TNF-α and cachexia [25]. Treatment of overweight or obese mice with ZAG causes weight loss through selective depletion of fat mass, while lean body tissues are preserved [20]. In addition ZAG-knockout mice were found to be more susceptible to weight gain, especially on a high-fat diet, while adipocytes from these animals showed a decreased lipolysis in response to a range of stimuli [26].

10.5 Mechanism of Loss of Skeletal Muscle in Cachexia

Skeletal muscle mass is governed by both protein synthesis and degradation, and decreases in protein synthesis or increases in protein degradation, or a combination of both result in atrophy during the process of cachexia. Protein synthesis requires the full complement of amino acids and may be depressed in cachexia due to decreases in the plasma levels of amino acids, particularly the branched-chain amino acids, leucine, isoleucine and valine [27]. In addition protein synthesis in skeletal muscle is reduced in an experimental model of cachexia through regulation at the initiation stage of translation. In particular phosphorylation of eukaryotic initiation factor 2 (eIF2) on the α-subunit will lead to inhibition of translation initiation by competition with the guanine-nucleotide exchange factor, eIF2B, preventing the conversion of eIF2 from its GDP-bound state to its active GTP-bound form, and so blocking the first step of translation, the binding of initiator methionyl-tRNA to the 40S ribosomal subunit [28]. Levels of phospho eIF2α, together with its kinase, dsRNA-dependent protein kinase (PKR), have also been shown to be enhanced in cancer patients with weight loss irrespective of the amount [29]. In addition binding of mRNA to the 43S ribosomal subunit is also inhibited due to decreased levels of the eIF4G–eIF4E complex through hypophosphorylation of eIF4E-binding protein 1 (4E-BP1) leading to an increased binding of eIF4E [28]. There was a fivefold increase in the phosphorylation of eukaryotic elongation factor 2 (eEF2), which would result in an inhibition of elongation by decreasing its affinity for the ribosome. This would act to depress global protein synthesis.

Protein degradation in skeletal muscle has been attributed to an increased expression and activity of the ubiquitin-proteasome proteolytic pathway [30]. Expression of mRNA for proteasome subunits C2 and C5 has been shown to be increased about threefold in the skeletal muscle of cachectic patients [30], although the rate-limiting step of this pathway is considered to be ubiquitin conjugation to the substrate through the two ubiquitin ligases MuRF1 and atrogin-1/MAFbx. There is evidence

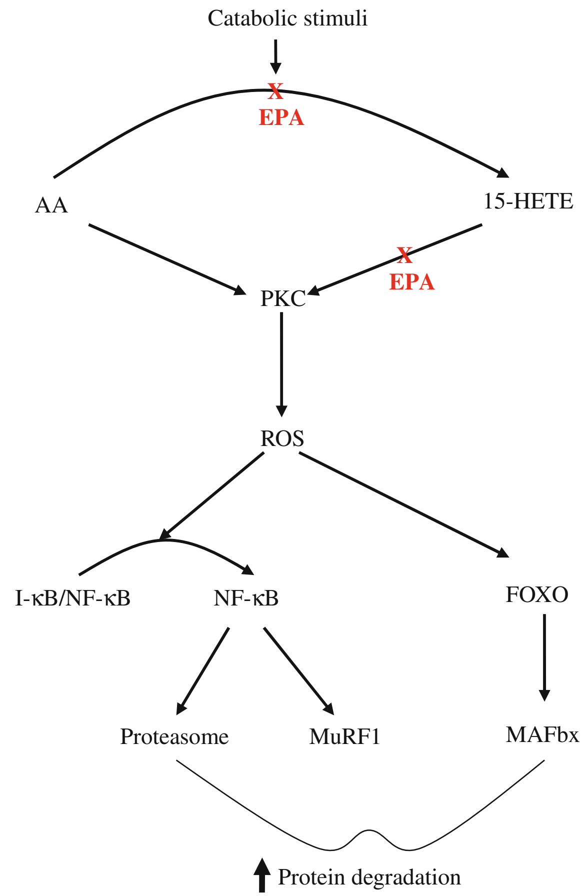

Fig. 10.1 Potential mechanism by which EPA attenuates cellular signalling pathways leading to protein degradation

that different components of the ubiquitin-proteasome pathway are induced by different transcription factors. Thus induction of proteasome subunits and MuRF1 occurs through nuclear factor-κB (NF-κB) [31], while the forkhead type transcription factors (FOXO) are involved in the induction of MAFbx [32]. The involvement of NF-κB in muscle protein degradation would provide the link between weight loss and systemic inflammation [33]. Muscle catabolic factors such as proteolysis-inducing factor (PIF) [34], angiotensin II [35] and TNF-α [36] have been shown to signal increased expression and activity of the ubiquitin-proteasome pathway through activation of NF-κB.

All three agents appear to signal activation of NF-κB through the formation of reactive oxygen species (ROS), leading to increased expression and activity of the ubiquitin-proteasome system [37] (Fig. 10.1). Generation of ROS by TNF-α increases expression of atrogin-1/MAFbx by activation of p38 mitogen-activated protein kinase (p38MAPK) [38], while MuRF1 expression is mediated by ROS activation of NF-κB.

10.6 Effect of ω-3 PUFAs on Experimental Cachexia Models

The first report that ω-3 PUFAs attenuated the development of cachexia was in mice transplanted with an experimental colon adenocarcinoma (MAC16), which induces weight loss without a reduction in food or water intake, and with small tumour burdens [39]. When the diet was enriched in fish oil, which is rich in both eicosapentaenoic (20:5, *n*–3; EPA) and docosahexaenoic (22:6, *n*–3; DHA) acids, weight loss was attenuated without an alteration in total calorie consumption, or nitrogen intake. There was also a significant reduction in tumour growth rate, although the reduction in host weight loss was greater than that might be expected from a smaller tumour burden. The reduction in host body weight loss was associated with retention of both adipose tissue and skeletal muscle. The effect could be mimicked by pure EPA, but not DHA, which was ineffective in attenuating cachexia in the MAC16 model [40]. The antiproliferative, but not the anticachectic, effect of EPA could be reversed by linoleic acid, suggesting that the two effects were produced by different mechanisms [41]. Both EPA and DHA were found to be effective in attenuating weight loss in mice bearing the Lewis lung carcinoma transfected with interleukin-6 (IL-6) cDNA, without an effect on serum levels of IL-6 [42]. Dietary fish oil had a similar effect in rats bearing a methylcholanthrene-induced sarcoma, where it restored a normal eating pattern, delayed the onset of anorexia, tumour appearance and growth, and prevented loss of body weight [43]. In rats bearing the Walker 256 tumour dietary fish oil decreased tumour growth and partially attenuated the development of weight loss [44]. These results show ω-3 PUFAs, in particular EPA, to be effective in attenuating the development of cachexia in experimental animal models.

10.7 Mechanism of Effect of ω-3 PUFAs on Adipose Tissue

EPA, but not DHA, has been shown to directly inhibit the induction of lipolysis by a tumour LMF, with a K_i value of 104 μM (40). The elevation of cyclic AMP in response to LMF as well as lipolytic hormones was attenuated by EPA, suggesting that the effect was exerted on a step central to the process of lipolysis. In plasma membranes of adipocytes from mice treated with EPA stimulation of adenylyl cyclase by forskolin, which acts directly on the enzyme without the involvement of a receptor, was decreased, suggesting a direct interaction between EPA and adenylyl cyclase [45]. The effect appears to be mediated, at least partially, through the inhibitory GTP-binding protein Gαi, because pertussis toxin blocked both the inhibition of lipolysis and the activation of adenylyl cyclase by EPA. Pertussis toxin has been shown to catalyse the NAD-dependent ADP ribosylation of the αi subunit of Gi blocking the inhibitory process. The mechanism by which EPA interacts with Gαi is not known.

In addition to direct inhibition of the lipolytic effect of LMF/ZAG, EPA has been shown to downregulate its expression in both white and brown adipose tissues [46]. ZAG expression in adipose tissue of cachectic mice is upregulated by glucocorticoids, which are elevated in serum during the process of weight loss, and this process is attenuated by EPA, although the mechanism is not known.

10.8 Mechanism of Effect of ω-3 PUFAs on Skeletal Muscle

Preservation of lean body mass by EPA in cachectic mice bearing the MAC16 tumour has been shown to arise from a significant reduction in protein degradation, without an effect on protein synthesis [47]. However, when combined with the amino acids leucine, arginine and methionine there was almost a doubling of protein synthesis, and this combination had a more significant effect on weight loss than EPA alone. These results suggested that combination therapy of cancer cachexia involving both inhibition of the enhanced protein degradation and stimulation of the reduced protein synthesis would be more effective than either treatment alone, and this combination has been found to be most effective clinically.

The mechanism by which EPA attenuates the increased protein degradation in skeletal muscle in cancer cachexia has been shown to be due to downregulation of the increased expression of the ubiquitin-proteasome proteolytic pathway [48]. Thus treatment of mice bearing the MAC16 tumour with EPA completely suppressed functional proteasome activity in gastrocnemius muscle, together with attenuation of the expression of the 20S proteasome α-subunits and the p42 regulator, and resulted in an increased expression of the myofibrillar protein myosin. EPA has also been shown to attenuate protein degradation and induction of the ubiquitin-proteasome pathway in other catabolic conditions including acute starvation [49], sepsis [50] and hyperthermia [51]. These results suggest a common mechanism of protein degradation through upregulation of the ubiquitin-proteasome pathway in a range of catabolic conditions and suggest that EPA may prove useful in the preservation of lean body mass in conditions other than cancer cachexia.

To understand the mechanism by which EPA attenuates protein degradation in cachexia, experiments have been carried out in vitro using either murine myoblasts or myotubes exposed to PIF. Under such conditions PIF mimics the effect of cachexia on skeletal muscle, inhibiting protein synthesis and increasing protein degradation through the ubiquitin-proteasome pathway. Also, as in vivo, EPA has no effect on the depression of protein synthesis by PIF, but attenuates the increased protein degradation [52]. At a concentration of 50 μM, EPA was found to completely block the release of arachidonic acid (AA) from membrane phospholipids in the presence of PIF and attenuate its conversion to prostaglandins E_2 and $F_{2\alpha}$ (PGE_2; $PGF_{2\alpha}$) and to 5-, 12- and 15-hydroxyeicosatetraenoic acids (HETE) (Fig. 10.1). Of all of the metabolites of arachidonic acid only 15-HETE was found to induce protein degradation in murine myoblasts [52] and myotubes [53], and this occurred through the ubiquitin-proteasome pathway. Both the induction of protein degradation and the induction of the ubiquitin-proteasome pathway by 15-HETE were attenuated by EPA, suggesting that it also acts downstream of the formation of 15-HETE (Fig. 10.1). Interestingly an inhibitor of 15-lipoxygenase (15-LOX) also attenuated muscle protein catabolism, proteasome ‘chymotrypsin-like’ enzyme activity and expression of proteasome 20S α-subunits in skeletal muscle from acute fasted mice, suggesting that the cellular signalling pathway was the same as that induced by PIF [49].

This signalling pathway involves activation of NF-κB by both PIF [54] and 15-HETE [53] by increasing degradation of the inhibitory protein I-κB releasing free NF-κB from the NF-κB/I-κB complex, which migrates into the nucleus, causing an increased activity of the ubiquitin-proteasome pathway by increasing expression of proteasome subunits and MuRF1 [31], by binding to the promoter. Depletion of I-κB from the cytosol in response to either PIF [54] or 15-HETE [55] was not seen in myotubes pretreated with EPA. EPA also acts by a similar mechanism in other cell systems. Thus in phorbol 12-myristate 13-acetate (PMA) activated Jurkat T-cells, EPA inhibited the expression of the gene for interleukin-2 (IL-2) by blocking nuclear accumulation of NF-κB [55]. The mechanism by which this occurs is not known, but it may arise from attenuation of upstream kinases such as protein kinase C (PKC) (Fig. 10.1). PKC has been suggested as being an upstream activator of the I-κB kinase complex (IKK), leading to I-κB phosphorylation, ubiquitination and subsequent processing by the 26S proteasome followed by translocation of NF-κB into the nucleus. In murine myotubes, in the presence of PIF, PKC was shown to be responsible for the degradation of I-κB and nuclear binding of NF-κB, suggesting that EPA may attenuate activation of PKC [56]. In addition to attenuation of the ubiquitin-proteasome pathway, EPA has also been shown to attenuate the activity of the extralysosomal peptidase tripeptidyl-peptidase II (TPP II) which cleaves oligopeptides released from the proteasome into tripeptides [57]. In murine myotubes treatment with PIF produced an increase in activity of both proteasome and TPP II, with an identical dose–response curve, and both activities were inhibited by EPA.

In addition to protein degradation through the ubiquitin-proteasome pathway, apoptosis of muscle cells may play a contributory part to muscle atrophy. As well as increasing proteasome expression and activity PIF has been shown to induce apoptosis in murine myotubes, possibly through the common intermediate arachidonic acid [58]. EPA has been shown to attenuate the apoptotic process initiated by PIF, possibly by preventing release of arachidonic acid from membrane phospholipids. Thus the PIF-induced increase in the cytosolic content of cytochrome *c*, as well as expression of the pro-apoptotic protein Bax, was both attenuated by EPA, as was the increase in free nucleosome formation and increased DNA fragmentation [58]. These results suggest that EPA will act to conserve muscle mass in cancer cachexia by attenuating both protein degradation through the ubiquitin-proteasome pathway and apoptosis.

10.9 Clinical Studies of ω-3 PUFAs in Cachexia

The first clinical study to evaluate ω-3 PUFAs in the treatment of cachexia was carried out in patients with unresectable pancreatic cancer and with a median weight loss of 2.9 kg/month [59]. Patients received fish oil capsules containing 18% EPA and 12% DHA with a median of 12 g/day fish oil. After 3 months patient's weight loss was stabilised, such that they had a median weight gain of 0.3 kg/month, accompanied by a temporary, but significant reduction in acute phase protein production and by stabilisation of REE. Some patients received the related PUFA, gamma-linolenic acid (GLA), as a continuous i.v. infusion at a dose of 7.6 g/day for 10 days, followed initially by oral GLA in gelatine capsules, initially at a dose of 3 g/day, increasing to a maximum of 6 g/day if tolerated. After 3 months the majority of patients receiving GLA continued to lose weight. As with the studies in mice [40] these clinical studies show that attenuation of cachexia is specifically achieved by ω-3 PUFAs.

To confirm whether the anticachectic effect of the fish oil resided in EPA, a further study was undertaken in weight losing patients with pancreatic cancer, who received 95% pure EPA as the free acid, as in the animal experiments [40], at a starting dose of 1 g/day, increasing to 6 g/day [60]. As with the study using fish oil capsules, EPA supplementation stabilised the weight of the cancer patients, although they were losing weight at a median rate of 2 kg/month before starting therapy with EPA. Weight stabilisation was observed within 4 weeks of administration of EPA and persisted over the 12-week study period. In both of these studies supplementation with EPA was well tolerated. In separate studies [61] the maximum tolerated dose of fish oil was found to be 0.3 g/kg per day, meaning that a 70 kg person could tolerate up to twenty-one 1 g capsules /day containing 13.1 g of EPA + DHA. The dose-limiting toxicity was gastrointestinal, mainly diarrhoea. However, another study [62], in a group of weight losing (>5% pre-illness body weight) patients with advanced cancer, found that the majority of patients were not able to swallow more than 10 fish oil capsules per day, mainly because of burping and after-taste. This provided 1.8 g EPA and 1.2 g DHA, but unfortunately the study was only carried out for 14 days when there was no effect of the fish oil supplement on weight, appetite or fatigue compared with placebo. When patients were given higher doses of ω-3 PUFAs (equivalent to 4.7 g EPA and 2.8 g DHA for a 70 kg patient) for longer periods of time (>1 month) the majority did not lose weight [63]. Thus of 36 patients 24 had weight stabilisation, 6 gained >5% of their body weight and 6 patients lost ≥5% of their body weight. This shows that it is necessary to maintain high dose levels for longer periods of time in order to see a beneficial effect of ω-3 PUFAs in the treatment of cachexia.

High doses of ω-3 PUFAs (3.06 g EPA and 2.07 g DHA/day) administered to patients with generalised solid tumours until death were found to improve survival irrespective of weight loss [64]. ω-3 PUFAs also had an immunomodulating effect, increasing the ratio of T-helper cells to T-suppressor cells, which was significantly lower in malnourished patients. Animal studies show a significant decrease in the splenocyte proliferation index after feeding an EPA-enriched diet for 10 and 24 days, when stimulated by concanavalin A, but no differences when cells were stimulated with anti-CD3 or IL-2 [65]. Interestingly this study found a significant increase in the production of TNF-α in response to lipopolysaccharide when animals were fed the EPA-enriched diet [65], while studies in humans [64] found a decrease in TNF-α production in malnourished cancer patients which was restored by ω-3 PUFAs.

As predicted from the animal studies [47, 48], where EPA has been shown to attenuate the increased protein degradation in skeletal muscle without increasing the depressed rate of protein synthesis, treatment of cancer patients with EPA has been shown to stabilise the loss of body weight, but rarely are weight increases seen. However, certain amino acids, primarily the branched-chain amino acids, in addition to acting as substrates for protein synthesis, can initiate signal transduction pathways that modulate translation initiation [66]. The mechanism for stimulation is through activation of the mRNA-binding steps in translation initiation through hyperphosphorylation of 4E-BP1, resulting in the release of eIF4E from the inactive 4E-BP1-eIF4E complex, which then associates with eIF4G to form the active eIF4F complex [28]. This may be due to a reduction in phosphorylation of the mammalian target of rapamycin (mTOR), which may also be responsible for the decreased phosphorylation of the 70 kDa ribosomal S6 kinase ($p70^{S6k}$). There was also a decreased phosphorylation of eukaryotic elongation factor 2 (eEF2). All of these changes would be expected to stimulate protein synthesis.

In view of this EPA has been combined with a conventional oral nutritional supplement, energy dense and high in protein, and providing 620 kcal, 32.2 g protein, 2.2 g EPA and 0.96 g DHA per day in two cans [67]. In a preliminary study involving 20 patients with unresectable pancreatic adenocarcinoma, and who were losing weight at a median rate of 2.9 kg/month, the median consumption of the supplement was 1.9 cans/day. Such patients had a significant weight gain of 1 kg at 3 weeks and 2 kg at 7 weeks of initiation of therapy. Moreover body composition changes indicated that the weight that was gained was lean body mass [67], in contrast with nutritional supplementation [12], or pharmacological manipulation of appetite [13, 14], where it was fat and water. Both performance status and appetite were significantly improved by 3 weeks and dietary intake increased significantly by almost 400 kcal/day. These results suggest that in addition to the attenuation of the loss of lean body mass, the EPA-enriched supplement had a profound effect on the anorexia.

The effect of EPA supplement on appetite improvement was confirmed in a randomised clinical trial involving 421 assessable patients with cancer-associated wasting [68]. Here the supplement was tested against Megace, and a combination with Megace, and it was found that the percentage of patients with appetite improvement was the same as Megace, although 4 week Functional Assessment of Anorexia/Cachexia Therapy suggested that megestrol acetate-containing arms experienced superior appetite stimulation compared with the EPA arm. However, a smaller percentage taking the EPA supplement gained $\geq$10% of baseline weight compared with those taking Megace, although when weight gain was evaluated with increments of less than 10% weight increase there was no significant differences among the arms. These lower weight gains are more relevant to the EPA-containing supplement in view of the study showing that it increased lean body mass [67], in contrast with Megace which increases fat and water [13], which would be expected to accumulate more rapidly.

A randomised double-blind trial of the EPA-enriched supplement in 200 patients with pancreatic cancer employing an isocaloric isonitrogenous control supplement resulted in some confusion over the efficacy of EPA in the treatment of cancer cachexia [69]. An intention to treat group analysis showed that the ω-3 PUFA supplement did not provide a therapeutic advantage over the control supplement in attenuating weight loss, even when the mean dose taken was the same. However, the mean dose of the EPA supplement taken was only 1.4 cans/day, rather than the recommended 2 cans, and this would provide only 1.54 g EPA/day rather than the recommended 2.2 g. In addition the measurement of compliance based on plasma phospholipid EPA levels showed derivation from the protocol in both the control and experimental groups. Patients were recruited based on their non-use of fish oil or any ω-3 PUFA preparations. However, there were patients in both groups with high levels of EPA (14%) at baseline, suggesting possible undisclosed prior supplementation with ω-3 PUFAs. In addition 18% of control patients had high levels of EPA at weeks 4 and/ or 8, suggesting that they had been taking an ω-3 PUFA supplement, while 26% of the experimental group had little, or no, elevation of EPA, even though they had reported taking the EPA supplement. With this level of derivation from the protocol it is not surprising that there was no difference in weight loss between control and experimental groups. It also underlines the importance of an independent measurement of compliance rather than relying on what the patient reports. When a secondary analysis of the data was carried out using plasma EPA phospholipid levels as a measure of compliance, there was a strong correlation between increased levels and gain of body weight and lean body mass [69].

There is evidence to indicate that the EPA-enriched nutritional supplement may improve the quality of life of the cancer patients. Thus after 8 weeks the REE, TEE and PAL of patients who received the control supplement did not change, while in those who received the EPA-enriched supplement, although REE did not change, TEE and PAL significantly increased [10], and the increase in PAL may reflect an improved quality of life.

While it has generally been assumed that cachexia associated with different types of cancer will respond equally to EPA, a placebo-controlled randomised study of EPA diester suggested that this was not correct [70]. Patients with a clinical diagnosis of gastrointestinal and lung cancer, and with an average weight loss between 16 and 18%, were recruited into the study and were randomised to receive either placebo or EPA diester at 2 or 4 g daily. Over the 8-week treatment period there was no statistically significant improvement in survival, weight or other nutritional variables, although there was a trend in favour of EPA. However, when the type of cancer was factored in, it was found that the weight of patients with gastrointestinal cancer, who received EPA, increased significantly compared with placebo, while the weight of patients with lung cancer showed no significant response [69]. This difference could reflect differences between cachectic mediators in the different tumour types, although none has been found to date.

10.10 Conclusion

Problems that have arisen during the clinical evaluation of EPA make it essential to independently measure compliance by plasma phospholipid composition during any further clinical investigations. Both animal and human studies have shown that EPA increases lean body mass and clinical trials should be factored to allow time for this to happen (minimum of 3 weeks). Animal studies have shown an increase in lean body mass to be due to the ability of EPA to attenuate protein degradation through the ubiquitin-proteasome pathway, without an effect on protein synthesis [47], leading to an increase in the myofibrillar protein myosin. An increase in structural proteins in muscle will increase muscle strength, as well as mass, and would counteract general muscle weakness (asthenia), which is seen in cachectic cancer patients. This would explain how treatment with EPA increased PAL [10].

Part of the problem of treating cachectic patients with EPA is the current formulation, which requires patients to consume 474 ml/day of a nutritional supplement, which is clearly higher than most patients can manage [69]. One way around this problem would be to provide gelatine capsules containing either pure EPA [60] or fish oil highly enriched with EPA. In view of the uncertainty and confusion surrounding the double-blind placebo-controlled trials of EPA [69, 70] further studies are warranted with careful monitoring of compliance. Also with the possibility that cachexia in different tumour types may respond differently to EPA [70], patients should all have the same cancer type, or the group size should be large enough to readily separate the different cancer types. Only then we will be sure of the anticachectic properties of ω-3 PUFAs.

References

1. DeWys WD, Begg C, Lavin PT, Band PR, Bennett JM, Bertino JR, et al. Prognostic effect of weight loss prior to chemotherapy in cancer patients. Am. J. Med. 1980; 69: 491–497.
2. Fearon KCH. The mechanisms and treatment of weight loss in cancer. Proc. Nutr. Soc. 1992; 51: 251–260.
3. Anker SD, Sharma R. The syndrome of cardiac cachexia. Int. J. Cardiol. 2002; 85: 51–66.
4. Kotter DP, Tierney AR, Culpepper-Morgan JA, Wang J, Pierson RW. Effect of home total parenteral nutrition on body composition in patients with acquired immunodeficiency syndrome. J. Parent. Enteral. Nutr. 1990; 14: 454–458.
5. Windsor JA, Hill GL. Risk factors for postoperative pneumonia. The importance of protein depletion. Ann. Surg. 1998; 208: 209–217.
6. Vigano A, Bruera E, Jhangri GS, Newman SC, Fields AL, Suarez-Almazor ME. Clinical survival predictors in patients with advanced cancer. Arch. Int. Med. 2000; 160: 861–868.
7. Ross PJ, Ashley S, Norton A, Prost K, Waters JS, Eisen T, et al. Do patients with weight loss have a worse outcome when undergoing chemotherapy for lung cancers? Br. J. Cancer 2004; 90: 1905–1911.
8. Bosaeus I, Daneryd P, Svanberg E, Lundholm K. Dietary intake and resting energy expenditure in relation to weight loss in unselected cancer patients. Int. J. Cancer 2001; 93: 380–383.
9. Lundholm K, Daneryd P, Korner U, Bosaeus I. Evidence that long-term COX-treatment improves energy homeostasis and body composition in cancer patients with progressive cachexia. Int. J. Oncol. 2004; 24: 505–512.
10. Moses AGW, Slater C, Preston T, Barber MD, Fearon KCH. Reduced total energy expenditure and physical activity in cachectic patients with pancreatic cancer can be modulated by an energy and protein dense oral supplement enriched with n–3 fatty acids. Br. J. Cancer 2004; 90: 991–1002.
11. Strent SJ, Beddoe AH, Hill GL. Aggressive nutritional support does not prevent protein loss despite fat gain in septic intensive care patients. J. Trauma 1987; 27: 262–266.
12. Evans WK, Makuch R, Clamon GH, Feld K, Weiner RS, Moran E, et al. Limited impact of total parenteral nutrition on nutritional status during treatment for small cell lung cancer. Cancer Res. 1985; 45: 3347–3352.
13. Loprinzi CL, Schaid DJ, Dose AM, Burnam NL, Jensen MD. Body composition changes in patients who gain weight while receiving megestrol acetate. J. Clin. Oncol. 1993; 11: 152–154.
14. Simons JPFHA, Schols AMJ, Hoefnagels JMJ, Westerterp KR, ten Velde GPM, Wouters EFM. Effects of medoxyprogesterone acetate on food intake, body composition and resting energy expenditure in patients with advanced, nonhormone-sensitive cancer. Cancer 1989; 82: 553–560.
15. Bing C, Russell S, Becket E, Pope M, Tisdale MJ, Trayhurn P, et al. Adipose atrophy in cancer cachexia: morphological and molecular analysis of adipose tissue in tumour-bearing mice. Br. J. Cancer 2006; 95: 1028–1037.
16. Agustsson T, Ryden M, Hoffstedt J, van Harmelen V, Dicker A, Laurencikiene J, et al. Mechanism of increased lipolysis in cancer cachexia. Cancer Res. 2007; 67: 5531–5537.
17. Thompson MP, Cooper ST, Parry BR, Tuckey JA. Increased expression of the mRNA for hormone sensitive lipase in adipose tissue of cancer patients. Biochim. Biophys. Acta 1993; 1180: 236–242.
18. Vlassura H, Spiegel RJ, Doval DS, Cerami A. Reduced plasma lipoprotein lipase activity in patients with malignancy-associated weight loss. Horm. Metab. Res. 1986; 18: 698–703.
19. Greenberg AS, Shen W-J, Muliro K, Patel S, Souza SC, Roth AR, et al. Stimulation of lipolysis and hormone-sensitive lipase via the extracellular signal-regulated kinase pathway. J. Biol. Chem. 2001; 276: 45456–45461.

20. Hirai K, Hussey HJ, Barber MD, Price SA, Tisdale MJ. Biological evaluation of a lipid-mobilizing factor isolated from the urine of cancer patients. Cancer Res. 1998; 58: 2359–2365.
21. Zhang HH, Halbleib M, Ahmad F, Manganiello VC, Greenberg AS. Tumor necrosis factor-α stimulates lipolysis in differentiated human adipocytes through activation of extracellular signal-related kinase and elevation of intracellular cAMP. Diabetes 2002; 51: 2929–2935.
22. Busquets S, Sanchis D, Alvarez B, Ricquier D, Lopez-Soriano FJ, Argiles JM. In the rat, tumor necrosis factor alpha administration results in an increase in both UCP2 and UCP3 mRNAs in skeletal muscle: a possible mechanism for cytokine-induced thermogenesis? FEBS Lett. 1998; 40: 348–350.
23. Sanders PM, Tisdale MJ. Effect of zinc-α_2-glcycoprotein (ZAG) on expression of uncoupling proteins in skeletal muscle and adipose tissue. Cancer Lett. 2004; 212: 71–81.
24. Bing C, Bao Y, Jenkins J, Sanders P, Manieri M, Cinti S, et al. Zinc-α-glycoprotein, a lipid mobilizing factor, is expressed in adipocytes and is upregulated in mice with cancer cachexia. Proc. Natl. Acad. Sci. USA 2004; 101: 2500–2505.
25. Maltoni M, Fabbri L, Nanni O, Scarpi E, Pezzi L, Flamini E, et al. Serum levels of tumour necrosis factor alpha and other cytokines do not correlate with weight loss and anorexia in cancer patients. Support Care Cancer 1997; 5: 130–135.
26. Rolli V, Radosavljevic M, Astier V, Macquin C, Castan-Laurell I, Visentin V, et al. Lipolysis is altered in MHC class I zinc-α_2-glycoprotein deficient mice. FEBS Lett. 2007; 581: 394–400.
27. Norton JA, Gorschboth CM, Wesley RA. Fasting plasma amino acid levels in cancer patients. Cancer 1985; 56: 1181–1186.
28. Eley HL, Russell ST, Tisdale MJ. Effect of branched-chain amino acids on muscle atrophy in cancer cachexia. Biochem. J. 2007; 407: 113–120.
29. Eley HL, Skipworth RJE, Deans DAC, Fearon KCH, Tisdale MJ. Increased expression of phosphorylated forms of RNA-dependent protein kinase and eukaryotic initiation factor 2α may signal skeletal muscle atrophy in weight-losing cancer patients. Br. J. Cancer 2008; 98: 443–449.
30. Khal J, Hine AV, Fearon KCH, Dejong CHC, Tisdale MJ. Increased expression of proteasome subunits in skeletal muscle of cancer patients with weight loss. Int. J. Biochem. Cell Biol. 2005; 37: 2196–2206.
31. Cai D, Frantz JD, Tawa Jr NE, Melendez PA, Oh B-C, Lidov HGW. IKKβ/NF-κB activation causes severe muscle wasting in mice. Cell 2004; 119: 285–298.
32. Sandri M, Sandri C, Gilbert A, Skurk C, Calabria E, Picard A. FOXO transcription factors induce the atrophy-related ubiquitin ligase atrogin-1 and cause skeletal muscle atrophy. Cell 2004; 117: 399–412.
33. Dejong CHC, Busquets S, Moses AGN, Schrauwen P, Ross JA, Argiles JM, et al. Systemic inflammation correlates with increased expression of skeletal muscle ubiquitin but not uncoupling proteins in cancer cachexia. Oncol. Rep. 2005; 14: 257–263.
34. Whitehouse AS, Tisdale MJ. Increased expression of the ubiquitin-proteasome pathway in murine myotubes by proteolysis-inducing factor (PIF) is associated with activation of the transcription factor NF-κB. Br. J. Cancer 2003; 89: 1116–1122.
35. Russell ST, Wyke SM, Tisdale MJ. Mechanism of induction of muscle protein degradation by angiotensin II. Cell Sig. 2006; 18: 1087–1096.
36. Li YP, Reid MB. NF-αB mediates the protein loss induced by TNF-α in differentiated skeletal muscle myotubes. Am. J. Physiol. 2000; 279: R1165–R1170.
37. Russell ST, Eley H, Tisdale MJ. Role of reactive oxygen species in protein degradation in murine myotubes induced by proteolysis-inducing factor and angiotensin II. Cell. Sig. 2007; 19: 1797–1806.
38. Li Y-P, Chen Y, John J, Moylan J, Jin B, Mann DL, Reid MB. TNF-α acts via p38MAPK to stimulate expression of the ubiquitin ligase atrogin-1/MAFbx in skeletal muscle. FASEB J. 2005; 19: 362–370.
39. Tisdale MJ, Dhesi JC. Inhibition of weight loss by ω-3 fatty acids in an experimental cachexia model. Cancer Res. 1990; 50: 5022–5026.

40. Tisdale MJ, Beck SA. Inhibition of tumour-induced lipolysis in vitro and cachexia and tumour growth in vivo by eicosapentaenoic acid. Biochem. Pharmacol. 1991; 41: 103–107.
41. Hudson EA, Beck SA, Tisdale MJ. Kinetics of the inhibition of tumour growth in mice by eicosapentaenoic acid-reversal by linoleic acid. Biochem. Pharmacol. 1993; 45: 2189–2194.
42. Ohira T, Nishio K, Ohe Y, Arioka H, Nishio M, Funayama Y, et al. Improvement by eicosanoids in cancer cachexia induced by LLC-IL6 transplantation. Cancer Res. Clin. Oncol. 1996; 122: 711–715.
43. Ramos EJB, Middleton FA, Laviano A, Sato T, Romanova I, Das UN, et al. Effects of omega-3 fatty acid supplementation on tumor-bearing rats. J. Am. Coll. Surg. 2004; 199: 716–723.
44. Pinto Jr JA, Folador A, Bonato SJ, Aikawa J, Yamzaki RK, Pizato N, et al. Fish oil supplementation in F1 generation associated with naproxen, clenbuterol, and insulin administration reduce tumour growth and cachexia in Walker 256 tumor-bearing rats. J. Nutr. Biochem. 2004; 15: 358–365.
45. Price SA, Tisdale MJ. Mechanism of inhibition of a tumor lipid-mobilizing factor by eicosapentaenoic acid. Cancer Res. 1998; 58: 4827–4831.
46. Russell ST, Tisdale MJ. Effect of eicosapentanoic acid (EPA) on expression of a lipid mobilizing factor in adipose tissue in cancer cachexia. Prostaglandins Leukot. Essent. Fatty Acids 2005;72: 409–414.
47. Smith HJ, Greenberg NA, Tisdale MJ. Effect of eicosapentaenoic acid, protein and amino acids on protein synthesis and degradation in skeletal muscle of cachectic mice. Br. J. Cancer 2004; 91: 408–412.
48. Whitehouse AS, Smith HJ, Drake JL, Tisdale MJ. Mechanism of attenuation of skeletal muscle protein catabolism in cancer cachexia by eicosapentaenoic acid. Cancer Res. 2001; 61: 3604–3609.
49. Whitehouse AS, Tisdale MJ. Downregulation of muscle protein degradation by eicosapentaenoic acid in acute starvation. Biochem. Biophys. Res. Comm. 2001; 285: 598–602.
50. Khal J, Tisdale MJ. Downregulation of muscle protein degradation by eicosapentaenoic acid (EPA) . Biochem. Biophys. Res. Comm. 2008; 375: 238–240.
51. Smith HJ, Khal J, Tisdale MJ. Downregulation of ubiquitin-dependent protein degradation in murine myotubes during hyperthermia by eicosapentaenoic acid. Biochem. Biophys. Res. Comm. 2005; 332: 83–88.
52. Smith HJ, Lorite MJ, Tisdale MJ. Effect of a cancer cachectic factor on protein synthesis/degradation in murine C_2C_{12} myoblasts: Modulation by eicosapentaenoic acid. Cancer Res. 1999; 59: 5507–5513.
53. Whitehouse AS, Khal J, Tisdale MJ. Induction of protein catabolism in myotubes by 15 (S)-hydroxyeicosatetraenoic acid through increased expression of the ubiquitin-proteasome pathway. Br. J. Cancer 2003; 89: 737–745.
54. Whitehouse AS, Tisdale MJ. Increased expression of the ubiquitin-proteasome pathway in murine myotubes by proteolysis-inducing factor (PIF) is associated with activation of the transcription factor NF-κB. Br. J. Cancer 2003; 89: 1116–1122.
55. Denys A, Hichami A, Khan NA. n-3 PUFAs modulate T-cell activation via protein kinase C-α and ϵ and the NF-κB signaling pathway. J. Lipid. Res. 2005; 46: 752–758.
56. Smith HJ, Wyke SM, Tisdale MJ. Role of protein kinase C and NF-κB in proteolysis-inducing factor-induced proteasome expression in C_2C_{12} myotubes. Br. J. Cancer 2004; 90: 1850–1857.
57. Chand A, Wyke SM, Tisdale MJ. Effect of cancer cachexia on the activity of tripeptidyl-peptidase II in skeletal muscle. Cancer Lett. 2005; 218: 215–222.
58. Smith HJ, Tisdale MJ. Induction of apoptosis by a cachectic-factor in murine myotubes and inhibition by eicosapentaenoic acid. Apoptosis 2003; 8: 161–169.
59. Wigmore SJ, Ross JA, Falconer JS, Plester CE, Tisdale MJ, Carter DC, et al. The effect of polyunsaturated fatty acids on the progress of cachexia in patients with pancreatic cancer. Nutrition 1996; 12: S27–S30.

60. Wigmore SJ, Barber MD, Ross JA, Tisdale MJ, Fearon KCH. Effect of oral eicosapentaenoic acid on weight loss in patients with pancreatic cancer. Nutr. Cancer 2000; 36: 177–184.
61. Burns CP, Halabi S, Clamon GH, Hars V, Wagner BA, Hohl RJ, et al. Phase I clinical study of fish oil fatty acid capsules for patients with cancer cachexia: Cancer and Leukemia Group B study 9473. Clin. Cancer Res. 1999; 5: 3942–3947.
62. Bruera E, Strasser F, Palmer JL, Wiley J, Colder K, Amyotte G, et al. Effect of fish oil on appetite and other symptoms in patients with advanced cancer and anorexia/cachexia: A double-blind, placebo-controlled study. J. Clin. Oncol. 2003; 21: 129–134.
63. Burns CP, Halabi S, Clamon G, Kaplan E, Hohl RJ, Atkins JN, et al. Phase II study of high-dose fish oil capsules for patients with cancer-related cachexia. Cancer 2004; 101: 370–378.
64. Gogos CA, Ginopoulos P, Salsa B, Apostolidou E, Zoumbos NC, Kalfarentzos F. Dietary omega-3 polyunsaturated fatty acids plus vitamin E restore immunodeficiency and prolong survival for severely ill patients with generalized malignancy. Cancer 1998; 82: 395–402.
65. Barber MD, Fearon KCH, Ross JA. Eicosapentaenoic acid modulates the immune response but has no effect on a mimic of antigen-specific responses. Nutrition 2005; 21: 588–593.
66. Yoshizawa F. Regulation of protein synthesis by branched-chain amino acids in vivo. Biochem. Biophys. Res. Comm. 2004; 313: 417–422.
67. Barber MD, Ross JA, Voss AC, Tisdale MJ, Fearon KCH. The effect of an oral nutritional supplement enriched with fish oil in weight loss in patients with pancreatic cancer. Br. J. Cancer 1999; 81: 80–86.
68. Jatoi A, Rowland K, Loprinzi CL, Sloan JA, Dakhil SR, MacDonald N, et al. An eicosapentaenoic acid supplement versus megestrol acetate versus both for patients with cancer-associated wasting: A North Central Cancer Treatment Group and National Cancer Institute of Canada collaborative effect. J. Clin. Oncol. 2004; 22: 2469–2476.
69. Fearon KCH, von Meyenfeldt MF, Moses AGW, van Geenen R, Roy A, Gouma DJ, et al. Effect of a protein and energy dense n–3 fatty acid enriched oral supplement on loss of weight and lean tissue in cancer cachexia: A randomised double blind trial. Gut 2003; 52: 1479–1486.
70. Fearon KCH, Barber MD, Moses AG, Ahmedzai SH, Taylor GS, Tisdale MJ, et al. Double-blind, placebo-controlled randomised study of eicosapentaenoic acid diester in patients with cancer cachexia. J. Clin. Oncol. 2006; 24: 3401–3407.

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